Hart County
Emergency Medical Services
Prehospital Clinical Operating Protocols and Standing Orders

Hart County EMS Medical Director – Morgan Wood, MD
Hart County EMS Director – Mike Adams

2026

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Hart County EMS Administrative

Approval Statement

These medical protocols and standing orders are approved by **Morgan Wood, MD, Medical Director for Hart County Emergency Medical Services.They are reviewed periodically to ensure continued compliance with the standard of care for Pre-hospital medicine.**

These protocols and standing orders are valid until such time that they may be rescinded by the medical director. Any changes, additions, or deletions will be approved by the medical director prior to publication. There will be no changes/modifications to these protocols and (or) standing orders without prior approval from the EMS Director and (or) the medical director.

These protocols are templates for treatment, not education or training, and are designed and written to rely heavily on the training and good judgment of the individuals using them. These standardized guidelines may be modified to HCEMS-specific recommendations at the judgment and discretion of the medical director.

These protocols contain treatment guidelines that are appropriate and specific for the condition identified in the protocol, i.e., hypoglycemia. These standing orders are authorized by the HCEMS Medical Director and are to be utilized only when on duty and acting as a duly authorized representative of the HCEMS. Treatments and interventions listed in the specified boxes that represent **"CONTACT MEDICAL CONTROL" will not be performed without on-line medical control authorization.**

7/29/25

Morgan C. Wood, MD
Hart County EMS Medical Director

Date

7-29-25

Mike Adams
Hart County EMS Director

Date

General Information

Purpose

The purpose of this document is to provide:

1) Guidelines regarding permissible and appropriate emergency medical services procedures and treatment modalities which may be rendered by emergency medical personnel to a patient not in the hospital, and
2) Communications protocols regarding which medical situations require direct voice communications between medical personnel and a physician (or his/her designee) prior to those emergency medical personnel rendering specified emergency medical services procedures to a patient not in a hospital.

This document is divided into basic general guidelines, pediatric, and adult clinical guidelines (Medical and Trauma), information regarding medications commonly administered by emergency medical personnel and important resource documents.

Recognizing the wide variety in Georgia EMS medical direction, call volume, resources for training and quality assurance, the many different levels of EMS licensure, and the importance of providing pre-hospital care that is consistent with national standards and evidence-based medicine, these documents reflect the best effort of the Guidelines Revision Group to provide the most important basic information.

Professional Judgement

Since each medical emergency must be dealt with on an individual basis and appropriate care determined accordingly, professional judgement is mandatory in determining treatment modalities with the parameters of these Guidelines.

Authority

The authority for implementing these guidelines for care of pre-hospital patients is found in state law OCGA 31-11-60.1 (b) and (c), OCGA 31-11-50 (b) and the Rules of the Department of Public Health Chapter 511-9-2.

It is the responsibility of each emergency medical service personnel to be familiar with the laws, rules and regulations, and guidelines and adhere to them. Even an order by a Physician does not justify procedures not in accordance with the laws, rules and regulations.

Legend

The following chart provides explanations for icons and graphics found throughout the protocols/guidelines. An explanation of abbreviations utilized in specific protocols can be found in the resources section of this document.

	Emphasizes important points and reminders within a particular protocol/guideline
STOP Contact Medical Control	Indicates the point within each guideline at which contact with medical control should be made. Treatments provided beyond this point should be performed in conjunction with online medical direction or autho0rized by local agency protocol.
	Emphasizes the important points to be included in patient care documentation
CONTINUED ON NEXT PAGE	Indicates that the guideline or resource provided continues on the next page

Hart Co. EMS General Guidelines

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Hart Co. EMS Medication Administration and Disposal Protocol

Administration of Medication

1. Medications shall only be administered as per protocol and/or orders from Medical Control.
2. You may only administer medications that fall within your SCOPE of practice.
3. Remember to confirm that you ensure the patient's safety and effective medication delivery by confirming all of the following:
   1) Right Patient
   2) Right Drug
   3) Right Dose
   4) Right Route
   5) Right Time
   6) Right Reason
   7) Right Documentation
   8) Right Response
   9) Right to Refuse
4. After administering the medication make sure to properly document the medication given, what you gave the medication for, the dose you gave, and the concentration of medication.
5. If the protocols call for a medication to be given and you chose not to, make sure you document the reason you did not administer the medication.
6. Even though you may have standing orders, you must still obtain a physician's signature for the narcotic medications administered and all medication orders received from online Medical Control in order to get the medication replaced at the pharmacy. This is a pharmacy requirement.

Disposal of Medication

1. After you have completed the call, any remaining medication must be properly disposed of.
2. Have your partner or supervisor witness the disposal of the remaining medication.
3. If the medication is a narcotic, you must also complete the narcotic sheet for the pharmacy.
4. In your patient care report, make sure to document how much of the medication was given and how much of the medication was wasted.

Communications

When contacting Medical Control and receiving facilities, patient reports should be brief and clear. 10 Codes and other signals should NOT be used. A typical patient radio report should Include the following:

- Identify Unit by service, number, and level of capability (BLS or ALS)
- Specify enroute or on scene (state whether Emergency or Non-Emergency)
- Identify patient age and sex
- Chief Complaint/Mechanism of Injury
- LOC: AVPU
- Vital Signs, pertinent clinical findings
- Pertinent patient medications, allergies, past medical history
- Care and treatment given
- Patient's response to treatment
- Request for any orders
- ETA

According to local protocol and capability of receiving facility, patient report may be called in as an Alert or Activation of Trauma, Cardiac, or Stroke Team.

The following information **SHOULD NOT** be transmitted over the radio:

- Patient name
- Patient race
- Personal or Sensitive patient information (e.g., Social Security Number, History of AIDS, etc.)

Control of Patient Care: Physician on Scene

Control of patient care at the scene of an emergency shall be the responsibility of the individual in attendance most appropriately trained and knowledgeable in providing pre-hospital emergency stabilization and transport. When EMS personnel arrive at the scene of a medical emergency, and contact is made with medical control by EMS personnel, a physician/patient relationship is established between the patient and the physician providing medical control. The physician is responsible for the management of the patient and EMS personnel act as an agent of medical control unless the patient's physician is present. When a physician, other than the patient's physician, on the scene of a medical emergency, properly identifies himself and demonstrates his willingness to assume responsibility for patient management and documents his interventions by signing the patient care report, EMS personnel should place the intervening physician in communication with medical control. If there is a disagreement between the intervening physician and the medical control physician, or if the intervening physician refuses to speak with medical control, EMS personnel should continue to take orders from the medical control physician.

Reference: *DPH Rule 511-9-2-.07 (6) (i) Control of Patient care at the scene*

Intervener Physician

- An Intervener physician is a physician on the scene who has no previous connection with the patient. For the Good Samaritan physician to assume control of the patient, he/she must provide proof of licensure in Georgia
- Be willing to assume responsibility for the patient at the scene and during patient transportation to the hospital. This includes accompanying the patient during transportation. (except multi-casualty situations)
- Perform procedures outside the scope of EMS protocol his or herself

If the physician is unwilling to comply with these requirements, then his/her assistance should be respectfully declined.

Physician in His/her Office

1. EMS shall perform its duties per protocol
2. The physician may elect to supervise care provided by EMS.
3. If the physician directs the EMS personnel to perform a procedure or administer a medication which is not covered by EMS Scope of Practice or local protocol, then EMS will advise the physician of such. **The EMS provider will not perform this procedure.** The EMS provider may assist the physician in performing said procedure. If the physician initiates a medication which is to be continued during patient transport, which is not covered by scope or protocol, then the physician is expected to accompany the patient to the hospital.

Crime Scene

If you believe a crime has been committed, immediately contact law enforcement. Scene safety is paramount. **Protect yourself and other EMS personnel.** Once a crime scene is deemed safe by law enforcement, initiate patient contact and medical care.

- Do not touch or move anything at a crime scene unless it is necessary to do so for patient care.
- Have all EMS providers use the same path of entry and exit.
- Do not walk through fluids on the floor.
- Observe and document original location of items moved by crew.
- When removing patient clothing, leave it intact as possible.
- Do not cut through clothing holes made by gunshot or stabbing.
- If you remove any items from the scene, such as impaled objects or medication bottles, document your actions and advise investigating officers
- Do not sacrifice patient care to preserve evidence.
- Do not go through the patient's personal effects.
- If transporting, inform staff at the receiving hospital that this is a "crime scene" patient.
- If the patient is obviously dead, contact Medical Control for directions to withhold resuscitative measures, and do not touch the body.
- If family is present, instruct them to move outside if weather is permissible. If not instruct them to a room away from the scene and remain with them.
- **DO NOT** let any other family member enter the residence.
    - In the event of unruly family member(s) that refuse to follow verbal directions to stay clear of the scene, HCEMS personnel should defer to law enforcement on scene for scene control.
    - If law enforcement is not present, request law enforcement response and exit the scene safely and quickly.

Destination Decision

Hospital Destination of Prehospital Patients

1. When a patient requires initial transport to a hospital, the patient shall be transported by the ambulance service to the hospital of his or her choice provided:
   i. The hospital chosen can meet the patient's immediate needs
   ii. The hospital chosen is within a reasonable distance as determined by the EMS personnel's assessment in collaboration with medical control to not further jeopardize the patient's health or compromise the ability of the EMS system to function in a normal manner.
   iii. The hospital chosen is within a usual and customary patient transport or referral are as determined by the local medical director; and
   iv. The patient does not, in the judgement of the medical director or an attending physician, lack sufficient understanding or capacity to make a reasonable decision regarding the choice of hospital.
2. If the patient's choice of hospital is not appropriate or if the patient does not, cannot, or will not express a choice, the patient's destination will be determined by pre-established guidelines. If for any reason the pre-established guidelines are unclear or not applicable to the specific case, then medical control shall be consulted for a definitive decision.
3. If the patient continues to insist on being transported to the hospital he or she has chosen, and it is within a reasonable distance as determined by the local medical director, then the patient shall be transported to that hospital after notifying local medical control of the patient's decision.
4. If the patient does not, cannot, or will not express a choice of hospitals, the ambulance service shall transport the patient to the nearest hospital believed capable of meeting the patient's immediate medical needs without regard to other factors, e.g., patient's ability to pay, hospital charges, county or city limits, etc.

Reference: *DPH Rule 511-9-2-.07 (6) (k)*

Reasonable Distances can be established based on:

1. The patient's emergency
2. Resources at the local and surrounding facilities
3. Geographic location of the various facilities
4. EMS agency resources
5. Obligation to provide emergency services in the assigned ambulance zone
6. Availability of mutual aid
7. The usual and customary hospital destinations of that EMS agency

Documentation

Documentation information becomes the legal record of a patient's history and treatment by pre-hospital personnel. It may be used as defense or prosecution if an EMS provider is charged with medical negligence.

For every patient contact, the following must be documented:

- A clear history of the present illness, including chief complaint, time of onset, associated complaints, pertinent negatives and mechanism of injury.
- A complete physical exam appropriate for the emergency condition.
- Level of consciousness using the AVPU method.
- At least two sets of vital signs
- Patients transported shall have at a minimum two complete sets of vital signs. On longer transports vital signs should be documented every 15 minutes or less.
- Vital signs should be repeated after every drug administration.
- For drug administration, note patient weight, dosage of the drug, route of administration, time of administration and response.
- A complete listing of treatments performed in chronological order.
- For extremity injuries, neurovascular status must be noted before and after immobilization.
- For potential spinal injuries, document motor function before and after immobilization.
- For IV or IO administration, note the size of the catheter, placement location, number of attempts, type of IV fluid or medication, and flow rate.
- An ECG lead II strip (at a minimum) shall be documented for all patients placed on the cardiac monitor.
- Any significant rhythm changes should be noted.
- For cardiac arrest, document the initial ECG, ending ECG, pre and post defibrillation, pacing attempts or code summary report
- For intubation, document the centimeter mark at teeth, methods which confirm placement (tube visualized passing through vocal cords, equal breath sounds, chest wall movement, absent gastric sounds, CO2 detector or End Tidal CO2 Monitoring/Waveform), size of ET tube and number of attempts.
- Any Medical Control orders requested and whether approved or denied.

Emergency Vehicle Operations

The driver of any authorized emergency vehicle must always drive with **due regard** for the safety of all persons, including the patient being transported, the transport crew and the public.

When operating a vehicle as “an authorized emergency vehicle”, both the warning lights and audible signal must be in use. Operating a vehicle with only one of these two warning devices in use does not satisfy the requirements of OCGA 40-6-6 (see below)

There are certain medical conditions that may require the rapid transport of the patient, without the use of an audible warning device due to the patient’s condition (e.g. acute MI, pre-eclampsia). **In circumstances where lights only are being used for transport, the vehicle cannot proceed as “an authorized emergency vehicle” under the conditions set forth in OCGA 40-6-6.** The operator of the ambulance using lights only without the use of an audible warning device must proceed in complete compliance with the “Rules of the Road”.

Similarly, there may be environmental conditions (e.g. traffic, weather), in which operating as an emergency vehicle (lights and siren) introduces unreasonable risk and provides minimal opportunity to arrive at the destination sooner.

The Law

OCGA § 40-6-6. Authorized Emergency Vehicles

a) The driver of an authorized emergency vehicle or law enforcement vehicle, when responding to an emergency call, when in pursuit of an actual or suspected violator of the law, or when responding to but not upon returning from a fire alarm, may exercise the privileges set forth in this Code section.
b) The driver of an authorized emergency vehicle or law enforcement vehicle may:
    1) Park or stand, irrespective of the provisions of this chapter.
    2) Proceed past a red or stop signal or stop sign, but only after slowing down as may be necessary for safe operation.
    3) Exceed the maximum speed limits so long as he or she does not endanger life or property; and
    4) Disregard regulations governing direction of movement or turning in specified directions.
c) The exceptions granted by this Code section to an authorized emergency vehicle shall apply only when such vehicle is making use of an audible signal and use of a flashing or revolving red light visible under normal atmospheric conditions from a distance of 500 feet to the front of such vehicle, except that a vehicle belonging to a federal, state, or local law enforcement agency and operated as such shall be making use of an audible signal and a flashing or revolving blue light with the same visibility to the front of the vehicle.
d) The foregoing provisions shall not relieve the driver of an authorized emergency vehicle from the duty to drive with due regard for the safety of all persons.

Handling Patient Personal Property

EMS Personnel's first responsibility is to treat the patient. Handling a patient's valuables or personal property is secondary to proper pre-hospital emergency care. However, special attention needs to be paid to how a patient's personal property is handled by EMS personnel (when handling it cannot be avoided). In "load and go" situations, patient care and transport are the priority.

Proper procedure is determined by location of the patient (at home, incident scene, etc.) whether family members or friends of the patient are present, whether law enforcement personnel is present, and other factors. Every situation cannot be described here, but the following will serve as a guideline.

Patient's personal property could include but not be limited to: glasses, dentures, wallets, money, watches, jewelry, clothing, medications and keys.

Patient at Home or a Residence

Advise and encourage the patient to leave all unnecessary personal items and valuables at home or with a trusted family member or friend.

Patient medications, in most cases, will need to go to the hospital either with the patient or carried by a family member or friend. If it is necessary to transport these medications with the patient, they should be treated like any other patient's valuables.

Do not remove a watch, jewelry, or wallet from a patient unless it is necessary to treat the patient, (e.g. start an IV). If it is necessary to do so, tell the patient you are removing the item. Give the item to the patient (if conscious and alert) or to a family member (if present). Document the item(s) and disposition of personal property on the patient care report. If possible, have another medic or law enforcement officer witness what you did with the patient's personal property.

If the patient insists on taking personal items with him/her, the patient must be alert enough to maintain possession of the items.

If you are concerned about the security of the patient's home or the premises you are leaving, notify law enforcement.

Patient at Incident Scene or Not at Home

If the patient is conscious, encourage the patient to give personal property and valuables to a responsible person of choice. If you have to remove any items from the patient (e.g. watch, jewelry, etc.) to treat the patient, return the items to the patient if possible. Document the details of the situation on the patient care report and have a third-party witness via signature.

If law enforcement presents you with a patient's personal items, request that law enforcement present the items to the patient (if conscious and alert), to the patient's family (if available), or to hospital staff.

If personal items or valuables are handled by first responders or bystanders before they were presented to you, document this with as much detail as possible on the patient care report.

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Handling Patient Personal Property (cont.)

If a patient is disoriented or unconscious, give the patient's personal items to a family member or law enforcement officer if possible. Document any incident involving valuables on the patient care report and obtain a signature from the person receiving the valuables. If a family member or law enforcement officer is unavailable, transport the valuables with the patient.

Taking Responsibility of the Patient's Personal Property

When the medic finds themselves in possession of a patient's personal property, carefully document the individual items and what was done with those items. If possible, place the item(s) in a container (such as a zip lock bag or plastic garbage bag) used for the sole purpose of containing the patient's personal property. Make a list of the items placed in each bag, and secure the list in, or on, the bag. Medications should be listed separately. Currency should be listed by amount. Have your partner or a law enforcement officer verify (by signature) the list of items included in the container. Upon arrival at the hospital, turn the container over to the appropriate hospital staff member, verify content, and obtain a signature from the individual receiving the property.

Retain a copy of the list (with the signature) and attach to the patient care report.

If EMS personnel locate any patient's property within the ambulance after care of the patient has been transferred to the receiving facility, EMS personnel should make every effort to ensure the patient's property is returned to the patient. Any property that has been located after transfer of care should be returned to the patient, witnessed by hospital staff, and documented on an addendum to the patient care report.

Safe Transport of Pediatric Patients

Children are at risk of injury during transport by EMS. Appropriate protection must be provided for all pediatric patients. The National Highway and Traffic Safety Administration recently published Best Practice Recommendations for Safe Transport of Children.

See: http://www.nhtsa.gov/staticfiles/nti/pdf/811677.pdf

- No child or infant should ever be held in the arms or lap of parent, caregiver, or medic during transport **EVER.**
- All monitoring devices and equipment should be tightly secured.
- All personnel should be secure.
- Children who are not patients should be transported, properly restrained, in an alternate passenger vehicle, whenever possible.
- Available child restraint devices should be used for all pediatric patients less than 40 pounds, according to manufacturer's instructions, if the patient is not secured by other means as part of patient care.
- Do not transport a pediatric patient who meets **CDC Field Triage Criteria** in a child seat that was involved in an MVC.
- While manufacturers do not recommend using a child's own car seat for transportation post-accident, this may be better than no restraint during transport. Use of child safety seat after involvement in a minor MVC may be allowed if **ALL** of the following apply:
  - Visual inspection, including under moveable seat padding, does not reveal cracks or deformity.
  - Vehicle in which safety seat was installed was capable of being driven from the scene.
  - Vehicle door nearest the child safety seat was undamaged.
  - Air bags (if any) did not deploy.

Patients with Special Healthcare Needs

Children and adults with special healthcare needs include those on ventilators, tracheotomies, indwelling catheters, gastrostomy tubes, left ventricular assist devices (LVAD), and bariatric patients. As medical technology advances and home health capabilities increase, new equipment and patient needs will appear.

General Considerations:

- Do not be overwhelmed by equipment. Treat the ABC's first. **Treat the patient, not the equipment.** If the emergency is due to an equipment malfunction, manage the patient using your own equipment.
- Parents and caregivers are usually trained in device management and can assist EMS personnel. Ask for guidance.
- When moving a special needs patient, use slow, careful transfer. Transfer of bariatric patients will require additional manpower. Do not use excessive force to straighten or manipulate contracted extremities, as this may cause injury or pain to the patient.
- Transfer the patient, if possible, to their medical "home" hospital. This may involve bypassing the closest facility.
- Ask for the **"go bag"**, which generally has the patient's spare equipment and supplies, and bring this with you during transport.
- Physical handicaps do not necessarily imply mental deficits. Remember to communicate with the patient.
- Find out the patient's baseline vital signs, medications, allergies, and other medical information, which may not be typical.

Suspected Abuse

All healthcare providers are obligated by law to report cases of suspected child, elder, or vulnerable adult abuse.

Report all alleged or suspected abuse or neglect to the appropriate agency. Georgia Code requires providers to report incidents of abuse to their county's public children services agency, municipal or county peace officer.

Simply notifying hospital personnel about concerns of maltreatment do not meet mandated EMS reporting responsibilities. If any maltreatment is suspected, the EMS provider MUST, by law, notify the local public children's services agency or law enforcement as soon as possible.

Physical abuse and neglect are often difficult to determine – the following indicators of possible abuse:

- Injuries scattered on many areas of the body.
- Malnutrition or lack of cleanliness
- Any fracture in a child under 2 years of age
- Injuries in various stages of healing
- More injuries than are usually seen in other children of the same age

Initial Management

- **DO NOT** confront or become hostile to the parent or caregiver
- Treat any obvious injuries
- In cases of suspected sexual abuse or assault:
  - Discourage patient from washing and/or using the restroom
  - If the child/patient has not changed clothes, transport patient in these clothes.
  - If clothes have been removed but unwashed, bring clothes and underwear with patient in a paper (not plastic) bag.
  - Do not delay transport to search for evidence
- If caregivers refuse to let you transport the child/patient after treatment, remain at the scene and notify law enforcement.
- Contact Medical Control and advise of questionable injures, but **DO NOT** report abuse and/or neglect over the radio.

Reporting

- Report your suspicions to the ED physician
- Notify the local public children services agency or law enforcement as soon as possible. **You are legally responsible to report your suspicions.**
- **DO NOT** initiate the report in front of the patient or caregiver.

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Suspected Abuse (Continued)

Documentation:

- Document any statement the child/adult patient makes in their own words. All verbatim statements made by the patient, the parent, or caregiver should be placed in quotation marks along with the source of the statement
- Document unexplained injuries, discrepant history, delays in seeking medical care, and repeated episodes of suspicious injuries.
- Document history, physical exam findings, environmental surroundings, and ED notification on the pre-hospital patient care report.

The Law:

TITLE 19: DOMESTIC RELATIONS

CHAPTER 7: PARENT AND CHILD RELATIONSHIP GENERALLY

ARTICLE 1: GENERAL PROVISIONS

OCGA § 19-7-5 (2024)

(e)(2) *With respect to reporting required by subsection (c) of this Code section, an oral report by telephone or other oral communication or a written report by electronic submission or facsimile shall be made immediately, but in no case later than 24 hours from the time there is reasonable cause to believe that suspected child abuse has occurred. When a report is being made by electronic submission or facsimile to the Division of Family and Children Services of the Department of Human Services, it shall be done in the manner specified by the division. Oral reports shall be followed by a later report in writing, if requested, to a child welfare agency providing protective services, as designated by the Division of Family and Children Services of the Department of Human Services, or, in the absence of such agency, to an appropriate police authority or district attorney. Such report shall be provided to military law enforcement, if applicable. If a report of child abuse is made to the child welfare agency or independently discovered by the agency, and the agency has reasonable cause to believe such report is true or the report contains any allegation or evidence of child abuse, then the agency shall immediately notify the appropriate police authority or district attorney and notify military law enforcement, if applicable. Such reports shall contain the names and addresses of the child and the child's parents or caretakers, if known, the child's age, the nature and extent of the child's injuries, including any evidence of previous injuries, and any other information that the reporting person believes might be helpful in establishing the cause of the injuries and the identity of the perpetrator. Photographs of the child's injuries to be used as documentation in support of allegations by hospital employees or volunteers, physicians, law enforcement personnel, school officials, or employees or volunteers of legally mandated public or private child protective agencies may be taken without the permission of the child's parent or guardian. Such photographs shall be made available as soon as possible to the chief welfare agency providing protective services, the appropriate police authority, and military law enforcement.*

(f) Any person or persons, partnership, firm, corporation, association, hospital, or other entity participating in the making of a report or causing a report to be made, and individuals who otherwise provide information or assistance, including, but not limited to, medical evaluations or consultations, in connection with a report made to a child welfare agency providing protective services, an appropriate police authority, or military law enforcement pursuant to this Code section or any other law or participating in any judicial proceeding or any other proceeding resulting therefrom shall in so doing be immune from any civil or criminal liability that might otherwise be incurred or imposed, provided that such participation pursuant to this Code section or any other law is made in good faith. Any person making a report, whether required by this Code section or not, shall be immune from liability as provided in this subsection.

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Suspected Abuse (Continued)

TITLE 31: HEALTH

CHAPTER 8: CARE AND PROTECTION OF INDIGENT AND ELDERLY PATIENTS

ARTICLE 4: REPORTING ABUSE OR EXPLOTATION OF RESIDENTS IN LONG-TERM CARE FACILITIES

OCGA § 31-8-82 (2024) REPORTING ABUSE OR EXPLOITIATION; RECORDS

(a) *Any of the following people who have reasonable cause to believe that any resident or former resident has been abused or exploited while residing in a long-term care facility shall immediately make a report as described in subsection (d) of this Code section by telephone or in person to the department and shall make the report to the appropriate law enforcement agency or prosecuting attorney:*

1. ***(1)*** *Any person required to report child abuse as provided in subsection (c) of Code Section 19-7-5;*
2. ***(2)*** *Administrators, managers, or other employees of hospitals or long-term care facilities;*
3. ***(3)*** *Physical therapists;*
4. ***(4)*** *Occupational therapists;*
5. ***(5)*** *Day-care personnel;*
6. ***(6)*** *Coroners;*
7. ***(7)*** *Medical examiners;*
8. ***(8)*** *Emergency medical services personnel, as defined in Code Section 31-11-49;*
9. ***(9)*** *Any person who has been certified as an emergency medical technician, cardiac technician, paramedic, or first responder pursuant to Chapter 11 of Title 31;*
10. ***(10)*** *Employees of a public or private agency engaged in professional health related services to residents; and*
11. ***(11)*** *Clergy members.*

(b) *Persons required to make a report pursuant to subsection (a) of this Code section shall also make a written report to the department within 24 hours after making the initial report.*
(c) *Any other person who has knowledge that a resident or former resident has been abused or exploited while residing in a long-term care facility may report or cause a report to be made to the department or the appropriate law enforcement agency.*
(d) *A report of suspected abuse or exploitation shall include the following:*

(1) *The name and address of the person making the report unless such person is not required to make a report;*
(2) *The name and address of the resident or former resident;*
(3) *The name and address of the long-term care facility;*
(4) *The nature and extent of any injuries or the condition resulting from the suspected abuse or exploitation;*
(5) *The suspected cause of the abuse or exploitation; and*
(6) *Any other information which the reporter believes might be helpful in determining the cause of the resident's injuries or condition and in determining the identity of the person or persons responsible for the abuse or exploitation.*

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Suspected Abuse (Continued)

TITLE 30: HANDICAPPED PERSONS
CHAPTER 5: PROTECTION OF DISABLED ADULST AND ELDER PERSONS
OCGA § 30-5-4 (2024)

(b)

(1)

(A) *A report that a disabled adult or elder person is in need of protective services or has been the victim of abuse, neglect, or exploitation shall be made to an adult protection agency providing protective services as designated by the department and to an appropriate law enforcement agency or prosecuting attorney. If a report of a disabled adult or elder person abuse, neglect, or exploitation is made to an adult protection agency or independently discovered by the agency, then the agency shall immediately make a reasonable determination based on available information as to whether the incident alleges actions by an individual, other than the disabled adult or elder person, that constitute a crime and include such information in their report. If a crime is suspected, the report shall immediately be forwarded to the appropriate law enforcement agency or prosecuting attorney. During an adult protection agency's investigation, it shall be under a continuing obligation to immediately report the discovery of any evidence that may constitute a crime.*

(B) *If the disabled adult or person is 65 years of age or older and is a resident, a report shall be made in accordance with Article 4 of Chapter 8 of Title 31. If a report made in accordance with the provisions of this Code section alleges that the abuse or exploitation occurred within a long-term care facility, such report shall be investigated in accordance with Articles 3 and 4 of Chapter 8 of Title 31.*

(2) *Reporting required by subparagraph (A) of paragraph (1) of this subsection may be made by oral or written communication. Such report shall include the name and address of the disabled adult or elder person and should include the name and address of the disabled adult's or elder person's caretaker, the age of the disabled adult or elder person, the nature and extent of the disabled adult's or elder person's injury or condition resulting from abuse, exploitation, or neglect, and other pertinent information.*

(3) *When a report of a disabled adult's or elder person's abuse, neglect, or exploitation is originally reported to a law enforcement agency, it shall be forwarded by such agency to the director or his or her designee within 24 hours of receipt.*

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Suspected Abuse (Continued)

TITLE 30: HANDICAPPED PERSONS
CHAPTER 5: PROTECTION OF DISABLED ADULST AND ELDER PERSONS
OCGA § 30-5-4 (2024) (Continued)

(c) *Anyone who makes a report pursuant to this chapter, who testifies in any judicial proceeding arising from the report, who provides protective services, who participates in a required investigation, or who participates on an Adult Abuse, Neglect, and Exploitation Multidisciplinary Team under the provisions of this chapter shall be immune from any civil liability or criminal prosecution on account of such report or testimony or participation, unless such person acted in bad faith, with a malicious purpose, or was a party to such crime or fraud. Any financial institution or investment company, including without limitation officers and directors thereof, that is an employer of anyone who makes a report pursuant to this chapter in his or her capacity as an employee, or who testifies in any judicial proceeding arising from a report made in his or her capacity as an employee, or who participates in a required investigation under the provisions of this chapter in his or her capacity as an employee, shall be immune from any civil liability or criminal prosecution on account of such report or testimony or participation of its employee, unless such financial institution or investment company knew or should have known that the employee acted in bad faith or with a malicious purpose and failed to take reasonable and available measures to prevent such employee from acting in bad faith or with a malicious purpose. The immunity described in this subsection shall apply not only with respect to the acts of making a report, testifying in a judicial proceeding arising from a report, providing protective services, or participating in a required investigation but also shall apply with respect to the content of the information communicated in such acts.*

Patient Initiated Refusal of Care

Persons identified by EMS as a patient may have requested an EMS response or may have had an EMS response requested for them. An assessment must be offered and performed, to the extent permitted, on all patients. For patients initially refusing care, attempt to ask them, “would you allow us to check you out and evaluate whether you’re OK ?”

Adult patients may refuse evaluation, treatment, and/or transport. Similarly, a patient’s parent or guardian my make these same refusals.

Determination of capacity to refuse has 4 components:

- **Legal capacity to refuse.**
  - Age 18 years or older
  - Emancipated minor (married, pregnant, or the parent of a child)
  - No suicidal or homicidal intent expressed.
- **Mental capacity to refuse.**
  - Oriented x 3 (Person, Place, & Time/Event)
  - Any language barrier has been removed.
  - Able to understand and sign the transport refusal form.
- **Medical capacity to refuse.**
  - No altered level of consciousness, by injury, illness, or substance
  - Not under the obvious effects of alcohol and/or drugs
  - No abnormal blood glucose
  - No abnormal pulse oximetry or signs of hypoxia
  - No serious chief complaint (chest pain, difficulty breathing, syncope, etc.,)
  - No serious mechanism of injury
- **Medical Control Contacted**

Documentation should include:

- Signature of patient/parent/guardian. If refusal to sign, documentation of refusal should be signed by a witness, including EMS crew member(s).
- Vital signs and evaluation as allowed by patient.
- Description of possible consequences of refusal, as described to patient.
- Statement that the patient may call 9-1-1 at any time if assistance or treatment is desired or required.
- Patient advised of the risk of refusing medical treatment and patient understands the risk of refusal of transport/medical treatment, including death and permanent disability.

Withholding or Termination of Resuscitation

In all situations where any possibility of life exists, make every effort to resuscitate.

DO NOT TERMINATE resuscitation efforts if:

- Patient is under 18 years of age
- Patient is visibly pregnant
- Arrest may be due to hypothermia, drug overdose, toxins, or electrocution
- Any ROSC or neurological signs

Termination of resuscitative efforts for trauma victims (blunt or penetrating) should be determined by the local EMS Medical Director, based on local resources. Typically, victims of traumatic cardiac arrest do not survive if no EMS witnessed signs of life were ever present and the patient is asystolic.

DO NOT INITIATE resuscitation if:

- Obvious death in the field: absence of vital signs **AND** any of the following:
  - Decapitation
  - Decomposition
  - Rigor Mortis
  - Dependent Lividity
  - Incineration
  - Visual massive trauma to brain or heart, incompatible with life
- The Individual has been pronounced dead by a Georgia Licensed Physician, Medical Examiner, or Coroner.
- Valid DNR order

DNR Orders and other Advanced Directives

- Advanced directives and Living Wills are addressed in State Code OCGA Chapter 32 Title 31
- If a family member does not want the DNR Order to be honored, continue BLS until Medical Control is contacted for guidance.
- If EMS personnel are advised that the patient has an existing DNR Order, but the document **IS NOT PRESENT AT THE SCENE**, initiate BLS and contact Medical Control.
- A living will is intended to address patients who have been admitted to a healthcare facility. Living wills will rarely, if ever, have application in the pre-hospital environment.
- If presented with a document other than a valid DNR Order, such as a Living Will or Durable Power of Attorney, which appears to be an Advanced Directive, regarding resuscitation, initiate BLS until Medical Control is contacted for guidance.

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Withholding or Termination of Resuscitation (Continued)

OCGA § 31-39-6.1 (2024)

- **(a)** In addition to those orders not to resuscitate authorized elsewhere in this chapter, any physician, health care professional, nurse, physician assistant, caregiver, or emergency medical technician shall be authorized to effectuate an order not to resuscitate for a person who is not a patient in a hospital, nursing home, or licensed hospice if the order is evidenced in writing containing the patient's name, date of the form, printed name of the attending physician, and signature of the attending physician on a form substantially similar to the following:

  "DO NOT RESUSCITATE ORDER

  NAME OF PATIENT: ___________________

  THIS CERTIFIES THAT AN ORDER NOT TO RESUSCITATE HAS BEEN ENTERED ON THE ABOVE-NAMED PATIENT.

  SIGNED: ___________________

  ATTENDING PHYSICIAN

  PRINTED OR TYPED NAME OF ATTENDING PHYSICIAN: ___________________

  ATTENDING PHYSICIAN'S TELEPHONE NUMBER: ___________________

  DATE: ___________________"

- **(b)** A person who is not a patient in a hospital, nursing home, or licensed hospice and who has an order not to resuscitate pursuant to this Code section may wear an identifying bracelet on either the wrist or the ankle or an identifying necklace and shall post or place a prominent notice in such person's home. The bracelet shall be substantially similar to identification bracelets worn in hospitals. The bracelet, necklace, or notice shall provide the following information in boldface type:

  "DO NOT RESUSCITATE ORDER

  Patient's name: ___________________

  Authorized person's name and telephone number, if applicable: ___________________

  Patient's physician's printed name and telephone number: ___________________

  Date of order not to resuscitate: ___________________"

Intravenous Access

Indications for IV Access

- Current patient condition requires pre-hospital IV fluids or medication administration.
- Pre-hospital provider believes that a pre-hospital IV is warranted because of a high likelihood for clinical deterioration.

Locate appropriate peripheral insertion site.

- In the event that an IV cannot be established, and the IV is considered critical for the care of the patient, other peripheral sites may be used (e.g. external jugular, feet, legs, etc.)
- The primary peripheral IV site should be the hand and forearm. Large bore catheters may be placed in the antecubital space veins, but placement in this location will require that the arm be maintained in an extended position. Secondary access sites include the leg, foot and the external jugular veins. Lower extremity vein cannulation should be avoided in diabetics and patients with evidence of venous stasis.
- Prepare insertion site using aseptic technique.
- Apply tourniquet proximal to puncture site.
- Prepare appropriately sized over-the-needle catheters(s)
    - Patients with major trauma or evidence of shock should have at least one and preferably two 16-gauge catheters placed.
    - Adult medical patients should have an 18-to-20-gauge catheter placed if indicated.
    - Patients requiring **Dextrose 50%** should have, at a minimum, a 20-gauge catheter placed with an 18-gauge being preferred.
- Stabilize the site and insert catheter.
- Remove the needle from the catheter.
- Confirm placement and attach Saline Lock – Release tourniquet.
- Flush the catheter with 10ml of **Normal Saline**.
- Secure catheter, dress site and secure tubing
- Initiate IV fluid administration or medication if indicated.

External Jugular Veins should never be the first line attempt unless the patient has no limbs for the initial attempts. INTs should not be used in External Jugular Access.

Intraosseous Access

<u>If the patient exhibits one or more of the following:</u>

- An altered mental status (GCS of $\leq$ 8)
- Respiratory compromise (decreased SpO_2 after appropriate O_2 therapy, respiratory rate of <10/min or >40/min.
- Hemodynamic instability (Systolic BP <90 mmHg)

<u>IO can be considered prior to peripheral IV attempts in the following situations:</u>

- Cardiac arrest (medical or trauma)
- Trauma and unable to access peripheral IV sites.

<u>Contraindications to IO access:</u>

- Fracture of the bone selected for IO infusion (Consider alternate site)
- Excessive tissue at insertion site with absence of anatomical landmarks (Consider alternate site)
- Previous significant orthopedic procedures (IO within 24 hours, prosthesis (Consider alternate site)
- Infection at the site selected for insertion (Consider alternate site)

<u>Considerations for IO access:</u>

- **Flow rate**
  - Due to the anatomy of the IO space, flow rates may appear to be slower than those with an IV catheter.
  - Ensure administration of an appropriate rapid syringe bolus (flush) prior to infusion. **(NO FLUSH = NO FLOW)**
  - Rapid syringe bolus (flush) with EZ-IO Pedi with 5ml of Normal Saline
  - Repeat syringe bolus as necessary
- **Pain**
  - Insertion of the EZ-IO in an unconscious patient has been noted to cause mild to moderate discomfort (usually no more painful than a large bore IV). However, IO infusions for conscious patients have been noted to cause severe discomfort.
  - Prior to IO syringe bolus (flush) or continuous infusion in alert patients, slowly administer **<u>Lidocaine 2%</u>** (preservative free through IO hub). Ensure that the patient does not have allergies or sensitivity to Lidocaine.
  - Adult – Slowly administer 20-40mg of **<u>Lidocaine 2%</u>.**
  - Pedi – Slowly administer 0.5mg/kg of **<u>Lidocaine 2%</u>.**
  - EMTs will only use the EZ-IO on unconscious patients due to need for Lidocaine administration for conscious patients.

Hart Co. EMS Pediatric Clinical Protocols

Pediatric Protocols Table of Contents

Contents:

Pediatric Assessment

First Impression (Pediatric Assessment Triangle)

- Appearance **–** Observe muscle tone, interactivity, Consolability, look/gaze, and speech/cry
- Breathing **–** Observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation **–** Observe the skin color for pallor, mottling, or cyanosis.

Based on your first impression, does the child appear sick?

YES

Urgent

- Proceed immediately with evaluation of the Primary Survey (ABCDE).
- If a problem is identified, perform necessary interventions.

Primary Survey (ABCDE)

- **A**irway – assess airway patency.
  - open, clear and maintain airway.
- **B**reathing – assess rate and quality of breathing.
  - Ensure adequate ventilation.
  - Initiate appropriate O2 therapy.
- **C**irculation – assess perfusion status.
  - Control major bleeding.
  - Manage shock appropriately.
- **D**isability – assess LOC.
  - Is the child responding and acting appropriately?
- **E**xposure/**E**nvironment – expose injuries as needed.
  - Evaluate body temperature.
  - Examine the skin for rashes, discoloration, and trauma.
  - Assess the behavior of adults, appearance of other children, and safety of the home.

Secondary Assessment

NO

Not Urgent

- Proceed systematically with the patient assessment.

Though they may not appear sick, trauma patients should receive a rapid primary survey and rapid transport when significant mechanism of injury exist. These patients may appear well despite their emergent condition.

Primary Survey (ABCDE)

- **A**irway
- **B**reathing
- **C**irculation
- **D**isability
- **E**xposure/**E**nvironment

Secondary Assessment

- Vital signs
- Focused History
- Physical Examination

Ongoing Assessment

For the non-urgent child, a head-to-toe assessment, with the least invasive parts of the examination being performed first, will allow the child some time to become accustomed to you and will help maximize the information gained from the assessment.

Document:

- Vital Signs
- OPQRST/SAMPLE
- Environmental observations
- Treatment
- Communications with Medical Control

Pediatric Pain Management

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe the skin color for pallor, mottling, and/or cyanosis.

Primary Survey

- **Ensure airway**- ensure patency and proper positioning.
  - Consider Spinal Motion Restriction if evidence of trauma.
- **Assess Breathing** – give O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq 94\%$.**
  - Assist with BVM if ineffective respiratory effort.
- **Assess Circulation** – manage bleeding and/or shock appropriately.
- **Assess Disability** – assess LOC.
- **Exposure/Environment** – expose injuries as needed.

Secondary Assessment and History

- Monitor vital signs and oxygen saturation.
- Initiate cardiac monitoring.
- OPQRST/SAMPLE history
- Physical Exam
- Place patient in position of comfort.
- Immobilize any obvious injuries.
  - Elevate injured extremities, if possible
  - Consider application of cold pack(s)
- Keep the Patient NPO
- Initiate IV/IO of Normal Saline KVO.
  - . If fluid resuscitation is needed, 20 ml/kg bolus IV/IO. *See* ***Shock Protocol***.

CONTINUED ON NEXT PAGE

Document:

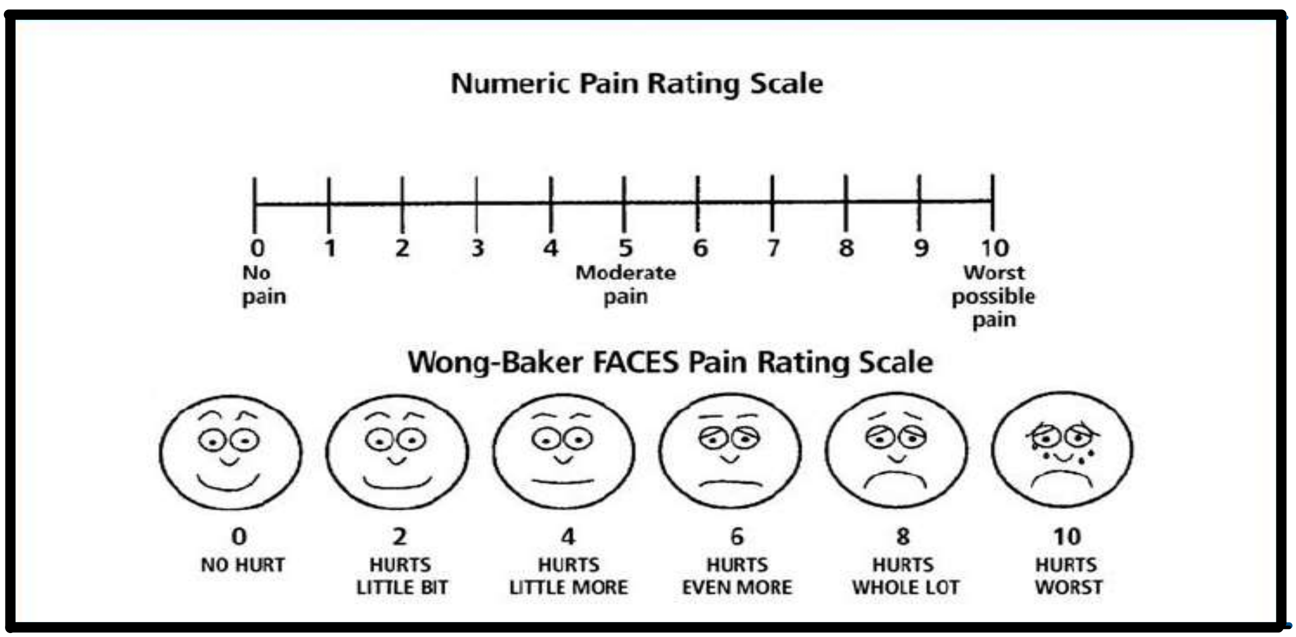
- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment
- Pain Scale
- Communication with Medical Control

Pediatric Pain Management (Continued)

Assess the patient's pain

- Ages 3-8 years – use the Wong-Baker FACES scale (below)
- Ages 8-18 years – use the numerical 1-10 scale.

- If pain scale is < 6, consider ***Toradol*.**
  - *Toradol* – **0.5 mg/kg** IV or IM slow push. **Max Dose – 15 mg.**
- If pain scale is ≥ 6 consider ***Morphine*** or ***Fentanyl***
  - ***Morphine*** – **0.1 mg/kg** IV/IO/IM slow push. **Max Dose – 10 mg.**
  - ***Fentanyl*** – **1 mcg/kg** IV/IO/IM/IN slow push. **Max Dose – 50 mcg.**
- Consider as a third-tier agent: **Contact Medical control prior to administration**.
  - ***Ketamine*** – **0.1 - 0.2** mg/kg slow push. **Max Dose – 25 mg.**
    - Monitor respiratory drive and systolic blood pressure.
- **After intervention, reassess mental status, pain level, and signs of respiratory depression every 5 minutes. Have Naloxone available.**
- If respirations become depressed, administer ***Naloxone*** **0.01mg/kg** IV/IO/IM/IN slow push. (If no clinical improvement, **0.1 mg/kg** may be administered. **Max Dose – 2 mg.**
- If patient becomes nauseated or vomits,
  - administer ***Ondansetron*** (Zofran) **0.1 mg/kg** IV/IO slow push. **Max Dose – 4 mg**



Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

Pediatric Shock Management

This protocol applies to pediatric patients presenting with signs and symptoms consistent with shock. **All forms of shock are associated with inadequate tissue perfusion**.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe the skin color for pallor, mottling, and/or cyanosis.

Primary Survey

- Ensure airway – ensure patency and proper positioning.
  - Consider Spinal Motion Restriction if evidence of trauma.
- Assess Breathing – give O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq$ 94%.**
  - Assist with bag valve mask if ineffective respiratory effort.
- Assess Circulation – control bleeding if present.
- Assess Disability – Assess LOC
- Exposure/Environment – expose injures as needed.
  - take measures to prevent hypothermia.
- **Children possess very strong compensatory mechanisms which allow them to appear relatively well in early shock. However, once these compensatory mechanisms are overwhelmed, they tend to decompensate rapidly. Early manifestations of shock may be subtle. The perfusion status of infants and children must be evaluated carefully and frequently.** See ***Shock Protocol***.

Secondary Assessment and History

- Monitor vital signs and SpO2.
- Perform Blood Glucose analysis – Treat if BGL is **< 60 mg/dl.**
- Initiate Cardiac Monitoring – treat dysrhythmias per appropriate protocol.
- Initiate EtCO2 monitoring – Maintain $EtCO_2$ of 30-35 mmHg.
- OPQRST/SAMPLE history
- Physical Exam
- Keep the patient NPO.
- Advanced airway/ventilatory management as needed.
- Initiate IV/IO
  - For ***Hypovolemic Shock***, administer Lactated Ringers. 20ml/kg bolus
  - For ***Cardiogenic Shock***, administer Normal Saline at KVO rate.
  - For all other types of shock, administer Normal Saline 20ml/kg bolus.
  - Fluid boluses may be repeated x 1 – titrate to clinical effect.
- Attempt to identify cause and treat in accordance with appropriate guideline. (e.g. tension pneumothorax, overdose, trauma, etc.)

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Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
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	Effective Date:	6/8/2026

Pediatric Shock Management (Continued)

- Initiate transport as soon as possible.
- Continue resuscitation and evaluation enroute – Frequently reevaluate ABCs and mental status.

STOP

CONTACT MEDICAL CONTROL

Document:

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communications with Medical Control

- If patient deteriorates or fails to improve, Medical Control may authorize the following:
  - For **Hypovolemic shock**, administer additional 20 ml/kg Lactated Ringers bolus.
  - For **Cardiogenic shock**, ensure rate and rhythm have been treated
    - Consider ***Dopamine*** drip **5-20 mcg/kg/min** – titrate to clinical effect
  - For all other types of shock administer Normal Saline 20ml/kg bolus.

Dopamine Infusion: Standard 1600mcg/ml Concentration

Broselow Color	Weight (kg)	Milliliters hour or drops per minute with micro drip tubing (60gtt/ml)				
		5 mcg/kg/hr	7.5 mcg/kg/hr	10 mcg/kg/hr	15 mcg/kg/hr	20 mcg/kg/hr
Gray	3	0.6*	0.8*	1.1	1.7	2.3
Gray	4	0.8*	1.1	1.5	2.3	3.0
Gray	5	0.9*	1.4	1.9	2.8	3.8
Pink	6-7	1.2	1.9	2.4	3.7	4.9
Red	8-9	1.6	2.4	3.2	4.8	6.4
Purple	10-11	2	3	3.9	5.9	7.9
Yellow	12-14	2.4	3.7	4.9	7.3	9.8
White	15-18	3.1	4.7	6.2	9.3	12.4
Blue	19-23	3.9	5.9	7.9	11.8	15.8
Orange	24-29	5	7.5	9.9	14.9	19.9
Green	30-36	6.2	9.3	12.4	18.6	24.8

*For rates <1 mL/hour, consider using 800 mcg/mL concentration.

Dopamine Infusion: 800 mcg/ml Concentration

Broselow Color	Weight (kg)	Milliliters/hr. or drops per minute with micro drip tubing (60gtt/min)				
		5mcg/kg/hr	7.5mcg/kg/hr.	10mcg/kg/hr.	15mcg/kg/hr.	20mcg/kg/hr.
Gray	3	1.1	1.7	2.3	3.4	4.5
Gray	4	1.5	2.3	3.0	4.5	6
Gray	5	1.9	2.8	3.8	5.6	7.5

Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
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	Effective Date:	6/8/2026

Pediatric Apparent Life-Threatening Event (ALTE)

This protocol applies to the pediatric patient that has experienced an episode that is frightening to the observer and involved some combination of apnea, choking or gagging, color change, and/or marked change in muscle tone (infant is “floppy”). ALTE usually occurs in infants less than 12 months old.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Secondary Assessment and History

- Ensure airway – ensure patency and proper positioning.
  - Consider Spinal Mobile Restriction if evidence of trauma.
- Assess Breathing – give O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq$ 94%.**
  - Assist with Bag Valve Mask if ineffective respiratory effort.
- Assess Circulation – manage shock appropriately. *See **Shock Protocol**.*
- Assess Disability – Assess LOC
- Exposure/Environment – expose injures as needed.
  - Take measures to prevent hypothermia.

Secondary Assessment and History

- Monitor vital signs and O_2 saturation.
- Perform Blood Glucose Analysis – Treat if BGL is **< 60mg/dl.**
  - If BGL is less than 60 mg/dl and the patient is symptomatic
    - < 6 months : **5 ml/kg** of ***D_{10}*** IV/IO.
    - 6 months – 2 years **: 2 ml/kg** of ***D_{25}*** IV/IO.
    - > 2 years – **1 ml/kg** of ***D_{50}*** IV/IO
- Initiate Cardiac Monitoring
- Complete and thorough history and physical
  - Assess for history of apnea, decreased tone, pallor or cyanosis.
  - Obtain history of medication/toxin exposure and/or ingestions.
- Initiate IV/IO – Normal Saline KVO
  - . If fluid resuscitation is needed, 20 ml/kg bolus IV/IO normal saline
- **TRANSPORT !!! Often patients with ALTE are asymptomatic when EMS arrives. Evaluation and transport are still needed. Contact Medical control if parent/guardian is refusing medical care and/or transport.**
- If patient shows continued respiratory depression, consider administration of ***Naloxone*** **0.01 mg/kg** IV/IO/IM. If no clinical improvement is observed with initial dose, **0.1 mg/kg** may be administered. **Max dose - 2mg**.

Document:
- Vital Signs
- Pertinent assessment findings
- Onset/duration of event
- Treatment
- Communications with Medical Control

Childbirth/Labor

This protocol applies to women whose chief complaint is related to labor and/or impending delivery.

Primary Survey

- Assess LOC – AVPU
- Ensure Airway – have suction ready.
- Assess Breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \leq$ 94%.**
- Assess Circulation – manage shock accordingly.
- **Altered LOC and signs of poor perfusion could indicate shock and should be managed appropriately.** *See* ***Shock Protocol***.
- **If patient is unstable, initial resuscitation/stabilization must precede any action specified in this protocol. Resuscitation of the mother is key to survival of both the mother and the fetus.**

Secondary Assessment and History

- Monitor vital signs and SpO_2.
- Perform Blood Glucose Analysis – Treat if BGL is **< 60 mg/dl.**
- Cardiac Monitoring
- OPQRST/SAMPLE, LMP, obstetrics, and gynecological history
- **Determine how many previous deliveries, due date, onset of contractions, if membranes have ruptured, if bleeding or vaginal discharge present, if patient has the urge to push or move bowels, and if pregnancy is high risk.**
- Time contractions – frequency and duration.
- Physical Exam – assess for signs of shock.
- IV/IO access with Normal Saline – initiate 20 ml/kg bolus.
- **If active labor, inspect the perineum for crowning**.
  - If crowning, apply gentle pressure with your gloved hand to the infant's head and prepare for delivery.
  - If no crowning, monitor and reassess frequency and duration of contractions.
- **If feet or buttocks presentation** – **DO NOT** pull on the infant.
  - Support head and trunk.
  - Place your gloved hand inside the vagina and form a "**V**" with first two fingers, place over infant's face – keep vagina wall away from infant's face.
- **If prolapsed umbilical cord**
  - Place mother in knee-chest position to relieve pressure on the umbilical cord.
  - Place your gloved hand inside the vagina and push upwards on the presenting part to further reduce pressure on the umbilical cord.
  - Cover the umbilical cord with moist sterile dressings and avoid manipulating it.
- **Priority symptoms: Crowing < 36 weeks gestation, prolapsed umbilical cord, abnormal presentation, severe vaginal bleeding, multiple gestation, or seizures. If noted, expedite transport and notify Medical Control as early as possible.**

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Childbirth/Labor (Continued)

Delivery and Post Delivery Care of Mother

- Maintain gentle pressure on the infant's head and allow it to deliver in a controlled gradual manner. **Routine suctioning of the oropharynx and nasopharynx as soon as the head is delivered is no longer recommended.**
- Check around the infant's neck for the umbilical cord.
  - If the umbilical cord is looped around the infant's neck, use your finger to hook the umbilical cord and pull it over the infant's head.
  - If you are unable to free the umbilical cord, clamp the umbilical cord in two places and cut the umbilical cord between the two clamps.
- Gently direct the infant's head and body downward to deliver the anterior shoulder and support the rest of the body as it delivers.
- Keep the infant at the level of the vagina and use a gauze pad to wipe any secretions around the mouth and nose.
- Vigorously dry the infant and provide warmth (increasing ambient temperature, cover with a blanket)
- If needed, stimulate breathing by flicking the soles of the infant's feet or rubbing the infant's back.
- Clamp the umbilical cord at 4 and 6 inches and cut the umbilical cord between the clamps.
- Wrap the infant in dry, clean towels or blankets.
- Note time of delivery. Obtain **APGAR** score at 1 and 5 minutes post-delivery. (See APGAR chart below.)
  - **Score $\leq$ 3 = Critical**
  - **Score $\geq$ 7 = Good to Excellent**
- If excessive secretions and signs of compromise are present, clear airway with bulb syringe.
- **If the newborn fails to respond to initial stimulation and are in need of resuscitation efforts, initiate resuscitation and refer to the *Newborn Resuscitation Guideline.***
- Once the placenta delivers, place it in a clean container and transport the placenta to the hospital with the mother and infant.
- After delivery, keep mother warm and watch for signs of shock.
- If excessive blood loss occurs (> 500ml)
  - Apply ABD pad to external vagina area.
  - Consider an additional fluid bolus.
  - Massage the uterus to promote uterine contraction.
  - Consider allowing the mother to breast feed the infant.
- **Transport to a facility capable of handling an obstetrical patient.**

SCORE	0 points	1 point	2 points
Appearance (Skin color)	Cyanotic / Pale all over	Peripheral cyanosis only	Pink
Pulse (Heart rate)	0	<100	100-140
Grimace (Reflex irritability)	No response to stimulation	Grimace or weak cry when stimulated	Cry when stimulated
Activity (Tone)	Floppy	Some flexion	Well flexed and resisting extension
Respiration	Apneic	Slow, irregular breathing	Strong cry

Document:
- Vital Signs
- Obstetric History
- Frequency/duration of contractions.
- Treatment(s)
- Communications with Medical Control.

Newborn Resuscitation

This protocol applies to term and pre-term newborn patients who fail to respond to initial stimulation and are in need of resuscitation efforts. This protocol also applies to all newborns and infants in the first few weeks of life.

Within the first 30 seconds:

- As soon as the baby is born:
  - Vigorously dry the infant and provide warmth (increasing ambient temperature, cover with blankets).
- Position the infant to open the airway.
- Clamp and cut the umbilical cord.
- If excessive secretions and signs of compromise are present, clear airway with bulb syringe.
  - Routine suctioning of the oropharynx and nasopharynx as soon as the head is delivered is no longer recommended.
  - If meconium staining is present AND the newborn is not vigorous (weak or absent respiratory effort, weak or absent muscle tone, heart rate <100 bpm) tracheal suctioning may be considered.
- Stimulate breathing (flicking soles of the newborn's feet or rubbing the newborn's back).

Assess respirations:

- If inadequate or gasping respirations are present, assist ventilations at a rate of **40-60** breaths per minute using a BVM with 100% O_2.
- If respirations are shallow or slow, attempt 1 minute period of stimulation while administering O_2 via blow by.
  - If respirations do not increase, assist ventilations at a rate of **40-60** breaths per minute using a BVM with 100% O_2.

Assess heart rate:

- If less than 60 beats per minute, begin chest compressions.
  - Compression-to-ventilation ratio of 3:1 in neonatal resuscitation. Compress at 120/minute.
  - Compressions should be discontinued when the heart rate is higher than 60 beats per minute.

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Newborn Resuscitation (Continued)

Most neonates transition to post-natal life without difficulty. About 10% of newborns require some assistance to begin breathing at birth. Less than 1% require extensive resuscitative measures.

Advanced Resuscitation:

- Consider advanced airway (one attempt only) for:
  - Persistent apnea
  - Central cyanosis
  - Bradycardia (HR <100 beats per minute)
- If heart rate is persistently < 60 beats per minute:
  - Continue CPR
  - Ensure that optimal ventilation is being provided with 100% O_2.
  - Initiate IV/IO of normal saline.
  - **For persistent HR <60 beats per minute, mix and administer 1:100,000 Epinephrine 0.01mg/kg. IV/IO every 3-5 minutes as needed.**
    - **Mixing Instructions for *Epinephrine* 1:100,000**
      - 1 ml of ***Epinephrine*** 1:10,000 in 9 ml Saline flush. (0.01 mg/ml)
- Obtain blood glucose level: If < 60 mg/dl, administer ***$D_{10}W$*** **0.5g/kg** (5ml/kg) IV/IO
- If no improvement despite adequate ventilation, chest compressions and epinephrine, consider fluid administration (10ml/kg normal saline over 5-10 minutes .

APGAR score to be calculated at 1 and 5 minutes after delivery. Score ≤ 3: Critical. Score ≥ 7 good to excellent.

SCORE	0 points	1 point	2 points
Appearance (Skin color)	Cyanotic / Pale all over	Peripheral cyanosis only	Pink
Pulse (Heart rate)	0	<100	100-140
Grimace (Reflex irritability)	No response to stimulation	Grimace or weak cry when stimulated	Cry when stimulated
Activity (Tone)	Floppy	Some flexion	Well flexed and resisting extension
Respiration	Apneic	Slow, irregular breathing	Strong cry

Document:

- Appearance
- Respirations
- Heart rate/rhythm
- Skin Color
- Resuscitation efforts
- APGAR Scores
- Communication with medical control

Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

Pediatric Bradycardia

This protocol applies to pediatric patients with a heart rate < 60 beats per minute.

First Impression **(Pediatric Assessment Triangle)**

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- Ensure airway – ensure patency and proper positioning
- Assess breathing – give O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq$ 94%.**
  - Ventilate with BVM and 100% O_2 if ineffective respiratory effort.
  - Avoid hyperventilation.
- **The most common cause of bradycardia in an infant/child is hypoxia. Ensure airway is patent and ventilation is adequate.**
- Assess circulation – peripheral pulses, CRT, skin color and temp.
  - If patient is stable, monitor and transport.
  - If heart rate < 60 beats per minute with signs of shock after adequate ventilation, start CPR
- Assess disability – assess LOC.
- Exposure/Environment – expose injuries as needed.

Secondary Survey and History

- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis. – Treat if BGL is **<60 mg/dl.**
- Initiate cardiac monitoring.
- Physical exam and OPQRST/SAMPLE history
- **Signs and symptoms may be non-specific, such as dizzy or weak, or may be dramatic with shock, altered level of consciousness, difficulty breathing or collapse.**
- Initiate IV/IO of normal saline – administer 20ml/kg bolus if signs of shock are present.
- If bradycardia and signs of cardiopulmonary compromise persist:
  - ***Epinephrine*** ***1:10,000***; **0.01 mg/kg** IV/IO **Max Dose – 1 mg**
    - May repeat every 3-5 minutes.
  - ***Atropine*** **0.02 mg/kg** IV/IO – **Minimum Dose**: **0.1 mg** **Max Dose – 0.5 mg**
    - May repeat once in 5 minutes.

CONTINUED ON NEXT PAGE

Pediatric Bradycardia (Continued)

STOP

Contact Medical Control

If no response:

- Consider transcutaneous pacing. *See* ***Transcutaneous Pacing Guideline***.
- If suspected beta-blocker or calcium channel blocker ingestion consult Medical Control for treatment.

Document:

- Vital Signs
- Pertinent assessment findings
- Onset/duration of event
- Treatment
- Communication with medical control

Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

Pediatric Tachycardia

This protocol applies to pediatric patients with a heart rate that is fast compared to normal for patient's age; and too fast for the child's level of activity and clinical condition.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey:

- Ensure airway – ensure patency and proper positioning
- Assess breathing – provide O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq$ 94%.**
  - Ventilate with bag valve mask and 100% O_2 if ineffective respiratory effort.
- Assess circulation – evaluate peripheral pulses, verify heart rate
  - If no pulse, start CPR. Treat according to ***Pulseless Arrest Guidelines***
- Assess disability – assess LOC.
- Expose/Environment – expose injuries as needed.
- **Altered level of consciousness and signs of poor perfusion could be an indication of shock and should be managed appropriately.** See ***Shock Protocol*****.**

Secondary Assessment and History:

- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Initiate cardiac monitoring.
- Record and evaluate 12-Lead ECG – **DO NOT** delay treatment**.**
- Physical exam and OPQRST/SAMPLE history
- **Signs and symptoms: may be non-specific, such as dizzy or weak, or may be dramatic with shock, altered level of consciousness, difficulty breathing and collapse.**

Sinus Tachycardia

- **HR <180 bpm in children**
- **HR < 220 in infants**
- **ORS ≤ 0.08 seconds**

If signs of cardiopulmonary compromise, IV/IO NS 20ml/kg.

Search for and treat causes – hypovolemia, dehydration, etc.

SVT

- **HR > 180 in children**
- **HR > 220 in infants**

If signs of cardiopulmonary compromise, IV/IO 20 ml/kg

- Consider vagal maneuvers
- ***Adenosine*** IV/IO
  - **0.2 mg/kg**

Wide Complex Tachycardia

- **QRS > 0.08 seconds**

If signs of cardiopulmonary compromise, IV/IO 20 ml/kg

- Consider sedation – **DO NOT** delay cardioversion
- Synchronized cardioversion with **0.5 – 1 J/kg**; may repeat with **2 J/kg**.

CONTINUED ON NEXT PAGE

Pediatric Tachycardia (Continued)

Advanced airway management; ventilate with 100% O_2 if indicated.

- If at any time a cardiac rhythm other than tachycardia is noted, treat based on the appropriate protocol.

Sinus Tachycardia

- **HR < 180/min children**
- **HR < 220/min infants**
- **ORS ≤ 0.08 seconds**
- **The most common causes of sinus tachycardia in children are hypovolemia and dehydration.**

SVT

- **HR > 180/min in children**
- **HR > 220/min in infants**
- If ***Adenosine*** is ineffective and shock persist
- Consider sedation – **DO NOT** delay cardioversion
- Synchronized cardioversion with **0.5-1 J/kg**
- May repeat with **2 J/kg**

Wide Complex Tachycardia

- **QRS > 0.08 seconds**
- If cardioversion is ineffective and shock persist consider anti-dysrhythmic.
- ***Amiodarone*** **– 5 mg/kg over 20-60 minutes.**

Document:

- Vital Signs
- History
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control

Pediatric Pulseless Arrest

Resuscitation

- Assess patient for respiratory and cardiac arrest.
- Initiate CPR and AED/Defibrillator using most current American Heart Association guidelines.
- Provide high-quality compressions, minimizing interruptions.
- Compressions should be at a rate of 100-120 compressions per minute.
- Ventilate with bag valve mask and 100% O_2, consider OPA/NPA.
- Consider advanced airway management.
  - **Do Not** delay resuscitation for advanced airway placement.
  - If advanced airway is utilized, initiate $EtCO_2$ monitoring.

Shockable Rhythm

YES

VF/Pulseless VT

- Defibrillate at 2 joules/kg.
- Resume CPR for 2 minutes.
- Initiate IV/IO of normal saline.
- ***1:10,000 Epinephrine*** **0.01 mg/kg (0.1 ml/kg)** IV/IO**, repeat every 3-5 minutes**
- Reassess rhythm every 2 minutes, if rhythm is organized, check pulse.
- Defibrillate at 4 joules/kg.
- Resume CPR for 2 minutes.
- ***Amiodarone**** **5mg/kg bolus,** may repeat up to 2 times for refractory VF/VT.
- Continue CPR/treatment as indicated.
- Consider and treat reversible causes.

NO

Asystole/PEA

- Initiate IV/IO of normal saline.
- ***1:10,000 Epinephrine*** **0.01mg/kg (0.1ml/kg)** IV/IO**, repeat every 3-5 minutes.**
- Reassess rhythm every 2 minutes, if rhythm is organized, check pulse.
- Continue CPR/treatment as indicated
- Consider and treat reversible causes:

-Hypovolemia	-Tension Pneumothorax
-Hypoxia	-Tamponade
-Hydrogen Ion	-Toxins
-Hypoglycemia	-Thrombosis
-Hypokalemia	
-Hyperkalemia	
-Hypothermia	

- **Transport to the closest appropriate facility.**
- **If cardiac rhythm change is noted, treat based on the appropriate guideline.**

Document:

- Events preceding arrest.
- Code summary
- Treatment(s)
- Communication with Medical Control.

*Lidocaine **may be utilized to replace Amiodarone. Dose – 1mg/kg IV/IO.**

Pediatric Abdominal Discomfort

This protocol applies to pediatric patients with pain/discomfort presenting in the abdomen or the flanks with no history or signs of trauma.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- Ensure airway – ensure patency and proper positioning.
- Assess breathing – give O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq$ 94%.**
- Assist with bag valve mask if ineffective respiratory effort.
- Assess circulation – manage shock appropriately
- Assess disability - assess LOC
- Exposure/Environment – expose injuries as needed.
- **Altered LOC and signs of poor perfusion could indicate shock and should be managed appropriately.** *See* ***Shock Protocol***.

Secondary Assessment and History

- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl.**
- Cardiac monitor – record and evaluate 12-Lead ECG.
- OPQRST/SAMPLE history
  - History of blood in vomit or stool? Prior abdominal surgery? Last meal?
  - Physical exam – assess for signs of dehydration and/or shock.
- Consider possible causes; GI, GU, cardiac, meds/toxin ingestion, pregnancy, etc.
- Save emesis or other drainage for signs of GI bleed, etc.
- Keep the patient NPO.
- Initiate IV/IO of normal saline – administer 20 ml/kg bolus if signs of shock.
- If nausea or vomiting are present, see ***Nausea/Vomiting Protocol***.

STOP

Contact Medical Control

For pain management, *See* ***Pain Management Protocol***.

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment
- Communication with Medical Control

Pediatric Allergic Reaction/Anaphylaxis

This protocol applies to pediatric patients presenting with rash, hives, shortness of breath, or other signs and symptoms, up to and including shock, possibly due to an allergic reaction.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- Ensure airway – ensure patency and proper positioning.
- Assess breathing – give O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq 94\%$.**
- Assess circulation – manage shock appropriately. *See **Shock Protocol**.*
- Assess disability – assess LOC
- Exposure/Environment – expose injuries as needed.
- **If respiratory compromise and/or signs of shock are present, treat immediately with epinephrine. All EMS provider levels are authorized to utilize epinephrine auto-injectors. (AEMT, CT and Paramedic providers may give 1:1,000 Epinephrine SQ or IM.)**
- **Isolate the patient from the source of the allergen if possible.**

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Monitor capnography.
- Initiate cardiac monitoring.
- Physical exam and OPQRST/SAMPLE history.
- Advanced airway and ventilatory management as needed.
- Initiate IV/IO of normal saline KVO
  - Administer 20 ml/kg bolus if signs of shock are present.
- If localized reaction (hives):
  - ***Diphenhydramine*** – **1-2 mg/kg** IV slow, or deep IM. **Max Dose - 50mg**
- If patient presents with respiratory distress, in addition to ***Diphenhydramine***:
  - ***1:1,000 Epinephrine*** – **0.01 mg/kg** subcutaneous
  - Nebulize **3 ml** ***DuoNeb***
  - Administer ***Solu-Medrol*** **2 mg/kg** IV for pediatric patients. **Max Dose - 125 mg.**

CONTINUED ON NEXT PAGE

Pediatric Allergic Reaction/Anaphylaxis (Continued)

- If anaphylactic shock is present:
  - **DO NOT** delay ***Epinephrine*** ***1:1,000*** administration attempting IV/IO access.
    - ***1:1,000*** ***Epinephrine*** **0.01 mg/kg** IM (preferred) or SQ. **Max Dose: 0.3mg.**

All levels (AEMT, CT, Paramedic) may repeat 1:1,000 epinephrine IM/SQ/auto-injector dosing every 5 minutes as needed.

STOP

Contact Medical Control

Document:

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control

If no response, within 10 minutes of the IM or SQ *Epinephrine* and fluid bolus, administer 1:10,000 *Epinephrine* 0.01 mg/kg IV/IO. Max Dose: 0.3mg.

1. Severe findings include dyspnea, hypotension, and/or altered mental status. Patients with evidence of facial swelling should be evaluated for the presence of intra-oral swelling. If swelling of the tongue is present, then the patient should be treated for a Severe Reaction. Swelling of the lips and mid-face without evidence of respiratory distress or intra-oral swelling may be treated as a Mild/Moderate Reaction.
2. Epinephrine should be used with caution in patients over 40-years-old or individuals with a history of coronary heart disease or hypertension.
3. ***Diphenhydramine*** **1 mg/kg** may be administered IM if unable to establish an IV. **Max Pediatric Dose – 25 mg.**
4. Most patients will respond to SQ ***Epinephrine***. For severe symptoms that do not improve after SQ ***Epinephrine***, request approval from on-line Medical Control for an ***Epinephrine Drip***. Mix the Epinephrine Drip using the table below.

Pediatric Epinephrine Drip Chart

Pediatric Dosing										
Preferred Method – Mix 2mg in 250cc bag and place on IV pump. 0.1 – 1 mcg/kg/min OR see below only if pump is not available.										
Step 1: Determine starting rate. **Only if pump is not available**			Epi 1 mg/ml (1:1,000)							
2 kg	**4kg**	**6kg**	**8-9kg**	**10-11 kg**	**12-14 kg**	**15-16 kg**	**17-19 kg**	**20-23 kg**	**24-29 kg**	**30-36 kg**
2gtt/min	3gtt/min	5gtt/min	6gtt/min	8gtt/min	9gtt/min	11gtt/min	13gtt/min	15gtt/min	15gtt/min	15gtt/min

Pediatric Altered Level of Consciousness

This protocol applies to pediatric patients who are disoriented, weak, dizzy, confused, agitated, exhibit bizarre behavior, have had a syncopal episode, or are unconscious.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- Ensure airway – ensure patency and proper positioning.
  - Consider spinal motion restriction if evidence of trauma.
- Assess breathing – give O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq$ 94%.**
  - Assist with bag valve mask if ineffective respiratory effort.
- Assess circulation – manage shock appropriately.
- Assess disability – assess LOC
- Exposure/Environment – expose injuries as needed.
- **Altered LOC and signs of poor perfusion could indicate shock and should be managed appropriately.** See ***Shock Protocol*.**

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Initiate cardiac monitoring.
- Physical Exam and OPQRST/SAMPLE history.
- Advanced airway/ventilatory management as needed.
- Initiate IV/IO of normal saline – administer 20 ml/kg bolus if signs of shock are present.
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl.**
  - If the patient is able to protect and maintain their own airway, administer:
    - ***Oral Glucose*** **15g PO** (Use with caution in pediatric patients).
  - If the patient is not able to protect their own airway, given ***Dextrose*** IV/IO.
    - < 6 months old: **5 ml/kg of *$D_{10}W$*** IV/IO
    - 6 months old – 2 yrs: **2 ml/kg of *$D_{25}W$*** IV/IO
    - > 2 yrs old: **1 ml/kg of *$D_{50}W$*** IV/IO
- If IV/IO cannot be established, give ***Glucagon*** **0.1 mg/kg** IM or IN.
- Consider toxic exposure or ingestion: Contact Medical Control and/or the Georgia Poison Center at 1-800-222-1222.

CONTINUED ON NEXT PAGE

Document:

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control

Pediatric Altered Level of Consciousness (Continued)

Secondary Assessment and Treatment (Continued)

If patient shows continued respiratory depression, consider administration of:

- *Naloxone* **0.01 mg/kg** IV/IO slow or IN.
  - If no clinical improvement, **0.1 mg/kg. Max Dose: 2 mg.**
  - Titrate naloxone administration to patient's respiratory status.
- Contact Medical Control for possible orders if patient appears agitated or violent.

STOP
Contact Medical Control
If no change in LOC

Document:

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control

Possible Causes of Altered Mental Status (AEIOU TIPPS)	
A: Alcohol	**T**; Trauma
E: Electrolytes	**I**: Infection
I: Insulin (hypoglycemia)	**P**: Poison
O: Opiates	**P**: Psychogenic
U: Uremia	**S**: Seizure or Shock

Mixing $D_{25}W$ and $D_{10}W$ for Pediatric Patients

$D_{50}W$ to $D_{10}W$	$D_{50}W$ to $D_{25}W$
Expel 40 ml of 50% solution and draw up 40 ml of Normal Saline. = 10% solution	Expel 25 ml of 50% solution and draw up 25 ml of Normal Saline. = 25% solution

Pediatric Cold Related Emergencies

This protocol applies to a pediatric patient having a body temperature below 95° F (35° C) secondary to environmental exposure.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- Ensure airway – use least invasive means possible to secure airway.
  - Intubate only, if necessary, as gently as possible.
- Assess breathing – give O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq$ 94%.**
  - Assist with bag valve mask if ineffective respiratory effort.
- Assess circulation – check for pulse, if no pulse, begin CPR.
- **It may be necessary to assess pulse and respirations for up to 30-45 seconds to confirm arrest.**
- If no pulse, initiate CPR in conjunction with an AED/defibrillator. Use the most current American Heart Association guidelines.
  - If severe hypothermia (< 86° F/30° C) is strongly suspected, limit defibrillation attempts to 1 and withhold medications.
  - If body temperature is >86° F/30° C treat in accordance with the ***Pulseless Arrest Guideline***.
  - Resuscitation efforts should continue until core temperature approaches normal range.
- If pulse is present, **DO NOT** initiate CPR, no matter how slow the rate.
  - Treat ***Bradycardia*** only if patient is hypotensive.
- Assess disability – assess LOC
- Exposure/Environment – carefully move patient to a warm environment, remove all wet clothing, dry the patient and cover with blankets.
- **Avoid any rough movement that may cause cardiac dysrhythmias. It may be beneficial to immobilize the patient on a backboard.**

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Initiate cardiac monitoring.
- Physical exam and OPQRST/SAMPLE history
- Initiate **WARM** IV/IO normal saline – administer 20 ml/kg bolus if signs of shock are present.
- Apply warm packs to groin, axilla, neck and chest.

CONTINUED ON NEXT PAGE

Pediatric Cold Related Emergencies (Continued)

Frostbite

- Protect injured, frostbitten areas, do not rub or place on any heated surface.
    - Remove clothing and jewelry from injured body parts.
    - Do not attempt to thaw injured part with local heat.
    - Severe frostbite injuries should be transported to a trauma center.

Document:

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control

Consider ***Morphine*** for pain relief when the patient is conscious, alert, not hypotensive and complaining of severe pain.

- *See* ***Pediatric Pain Management Protocol***.

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	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

Pediatric Fever

This protocol applies to pediatric patients with a body temperature of 100.4° F (38° C) or greater. Fever may be associated with seizures, hallucinations, and other forms of altered mental status. The febrile patient may be dehydrated.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- Ensure airway – ensure patency and proper positioning.
- Assess breathing – give O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq$ 94%.**
  - Assist with bag valve mask if ineffective respiratory effort.
- Assess circulation – manage shock appropriately.
- Assess disability – assess LOC.
- Exposure/Environment – expose injuries as needed.
- **Altered LOC and signs of poor perfusion could indicate shock and should be managed appropriately.** *See* ***Shock Protocol*****.**

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Initiate cardiac monitoring.
- Perform blood glucose analysis – treat if BGL is < than **60mg/dl.**
  - ***Pediatric Altered Level of Consciousness protocol***.
- Physical exam and OPQRST/SAMPLE history.
  - Document history of fever and record temperature (forehead, ear or tympanic membrane)
- Initiate IV/IO normal saline – administer 20 ml/kg bolus if signs of shock, dehydration, hypotension, and/or sepsis are present. *See* ***Shock Protocol****.*
- If high grade fever (103°/39.5°), initiate gradual cooling.
  - Remove excess clothing.
  - Consider placing moistened towels in axilla and groin.
  - **DO NOT use ice or rubbing alcohol to cool.**
  - Avoid rapid cooling, **DO NOT** allow the patient to shiver**.**
- Administer ***Acetaminophen*** **15 mg/kg PO** if > 4 hours since last antipyretic.

Prepare for seizures, see ***Seizure Protocol*** for management of seizures.

Document:

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control

Pediatric Heat Related Emergencies

This protocol applies to pediatric patients with fatigue or altered level of consciousness secondary to environmental heat exposure.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- Ensure airway – ensure patency and proper positioning.
- Assess breathing – give O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq$ 94%.**
  - Assist with bag valve mask if ineffective respiratory effort.
- Assess circulation – manage shock appropriately.
- Assess disability – assess LOC
- Exposure/Environment – remove the patient from the environment
  - Expose injuries as needed.
- **Altered LOC and signs of poor perfusion could indicate shock and should be managed appropriately.** *See* ***Shock Protocol***.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Initiate cardiac monitoring.
- Perform blood glucose analysis – treat if BGL is **< 60 mg/dl.**
- Physical Exam and OPQRST/SAMPLE history.
- Initiate IV/IO normal saline – administer 20 ml/kg bolus if signs of shock, dehydration and/or hypotension. *See* ***Shock Protocol***.
- If conscious and not vomiting or extremely nauseous provide oral fluids.
- If heat stroke is suspected, active cooling with cold packs water and a fan.
- **Signs and symptoms of a heat stroke may include hot, dry skin (25% of patients will still be moist), seizures, altered mental status, dilated pupils, rapid heart rate, and/or arrythmia.**

Prepare for seizures, see ***Seizure Protocol*** for management of seizures.

Document:

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with medical control

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Pediatric Nausea/Vomiting

This protocol applies to pediatric patients presenting with prolonged vomiting, or those actively vomiting after EMS arrival with no other symptoms or complaints present. Assess any acute abdominal pain prior to resolving nausea/vomiting.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- Ensure airway – ensure patency and positioning
    - Have suction ready
- Assess breathing – Give O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq$ 94%.**
- Assess circulation – if needed, manage shock using the ***Shock Protocol***.
- Assess disability – assess LOC
- Exposure/Environment – expose injuries as needed.

Secondary Assessment and History

- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis – treat if BGL is < **60 mg/dl.**
- Consider cardiac monitoring.
- Physical Exam and OPQRST/SAMPLE history.
- Keep the patient NPO.
- Consider IV normal saline KVO rate.
    - If fluid resuscitation is needed, 20 ml/kg bolus IV/IO normal saline.

Treatment

- Patients 1 year old and up to 18 years old:
    - ***Ondansetron (Zofran)*** **0.15 mg/kg** IV/IO slowly or IM/IN/ODT. **Max Dose: 4 mg**
- Under 1 year of age:
    - Contact Medical Control for orders.

Pediatric Respiratory Distress

This protocol applies to pediatric patients presenting with inadequate ventilation or oxygenation; which may include increased or decreased respirations, cyanosis, nasal flaring, grunting, retractions, absent or diminished breath sounds, or decreased responsiveness.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- Ensure airway – Ensure patency and proper positioning.
- Assess breathing – give O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq 94\%$.**
  - Assist with bag valve mask if ineffective respiratory effort.
- Assess circulation – if needed, manage shock using the ***Shock Protocol***.
- Assess disability – assess LOC.
- Exposure/Environment – expose injuries as needed.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Auscultate breath sounds.
- Apply and monitor capnography.
- Initiate cardiac monitoring
- Physical exam and OPQRST/SAMPLE history.
- **If fever is present with any respiratory signs and symptoms or if the patient is coughing, sneezing or generating airborne droplets, a HEPA mask should be worn by EMS personnel to avoid transmission and infection.**
- Consider IV of normal saline at KVO rate.
- If wheezing or capnography indicates bronchospasm:
  - Nebulize ***DuoNeb*** **1.5-3 ml** (may repeat once after 20 minutes if needed)
  - Administer ***Solu-Medrol*** **2 mg/kg** IV for pediatric patients. **Max Dose: 125 mg**
- If stridor, history and exam are suggestive of croup:
  - Nebulize ***Epinephrine*** **1:1,000 2.5 mg** (in 3 ml of saline) < 15kg, if greater than 15kg – **5 mg**
- If laryngeal edema, obstruction, history and/or exam is suggestive of epiglottitis:
  - Use removal techniques if FBAO
  - Needle cricothyrotomy (only if bag valve mask and/or other non-invasive procedures are ineffective)
- **CROUP signs and symptoms may include: $\leq$ 4 years old, low-grade fever, inspiratory stridor, hoarseness, bark-like cough, recent upper respiratory infection.**
- **EPIGLOTTITIS signs and symptoms may include: $\geq$ 3 years old, high-grade fever, stridor, drooling, sore throat, dysphagia, rapid onset of distress.**

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Pediatric Respiratory Distress (Continued)

Consider advanced airway placement if patient presents with a decreased level of consciousness with respiratory failure or poor ventilatory effort (with hypoxia not responding to supplemental O_2 at 100%.) or unable to maintain a patent airway.

STOP

Contact Medical Control

Document:

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control

- Additional doses of ***Albuterol*** every 5 – 10 minutes or continuous
- Consider CPAP – *See* ***CPAP Protocol*** ***(If you can obtain a good mask seal with adult mask)***

IMPORTANT

- **Patients must be alert and able to maintain their own airway for CPAP**
- **With CPAP, most patients will improve in 5-10 minutes. If no improvement within this time consider ventilation with a BVM.**
- **Indications for CPAP are constantly broadening.**
- **Local medical directors may also authorize the use of steroids, magnesium sulfate for management of respiratory distress.**

Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

Pediatric Seizure

This protocol applies to pediatric patients actively seizing or those that have a history of seizures prior to EMS arrival.

First Impression **(Pediatric Assessment Triangle)**

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- Ensure airway – ensure patency and positioning.
    - Have suction equipment ready.
- Assess breathing – give O_2 as tolerated by mask or blow-by. **Maintain SpO2 $\geq$ 94%.**
- Assess circulation – manage shock appropriately. *See* ***Shock Protocol.***
- Assess disability – assess LOC.
- Exposure/Environment – expose injuries as needed.
- **Altered LOC and signs of poor perfusion could indicate shock and should be managed appropriately.**

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation
- Perform blood glucose analysis – treat if BGL is **< 60 mg/dl.**
- Consider cardiac monitoring
- Physical Exam and OPQRST/SAMPLE history.
    - Obtain a description of seizure activity – duration and severity.
    - Note any history of illness or trauma.
- Advanced airway/ventilatory management as needed.
- Initiate IV normal saline at KVO rate.
    - If fluid resuscitation is needed 20 ml/kg bolus IV/IO
- If evidence of high-grade fever (103° F/39.5° C), initiate ***Pediatric Fever Protocol***.

Active Seizure

- ***Midazolam*** **0.05 mg/kg – 0.1 mg/kg** IV/IO slowly or **0.2 mg/kg** IM/IN. **Max Dose – 5mg.**
    - *See **Versed Dosing Chart.***
- ***Diazepam*** **–** Contact Medical Control for dosing instruction.

STOP

Contact Medical Control for Additional Dosing

Document:

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Pediatric Sickle Cell Crisis

This protocol applies to pediatric patients presenting with sickle cell crisis. The typical sickle cell EMS call in children is severe pain in the abdomen, chest, joints, and/or difficulty breathing with hypoxia. Many of these patients are dehydrated.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- Ensure airway – ensure patency and proper positioning.
- Assess breathing – give O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq 94\%$.**
  - Assist with bag valve mask if ineffective respiratory effort.
- Assess circulation – manage shock appropriately. *See **Shock Protocol**.*
- Assess disability – assess LOC
- Exposure/Environment – expose injuries as needed.

Secondary Assessment and Treatment

- Vital signs and O_2 saturation.
- Initiate cardiac monitoring.
- Physical exam and OPQRST/SAMPLE history.
- Initiate IV/IO of normal saline at KVO rate
  - If signs of shock, administer 20 ml/kg bolus. *See **Shock Protocol**.*
- Comfort measures, such as support for painful joints.
- Assess patient's pain level
  - Ages 3-8 years – use Wong-Baker FACES scale (next page).
  - Ages 8-18 years – use numerical scale.

STOP

Contact Medical Control

CONTINUED ON NEXT PAGE

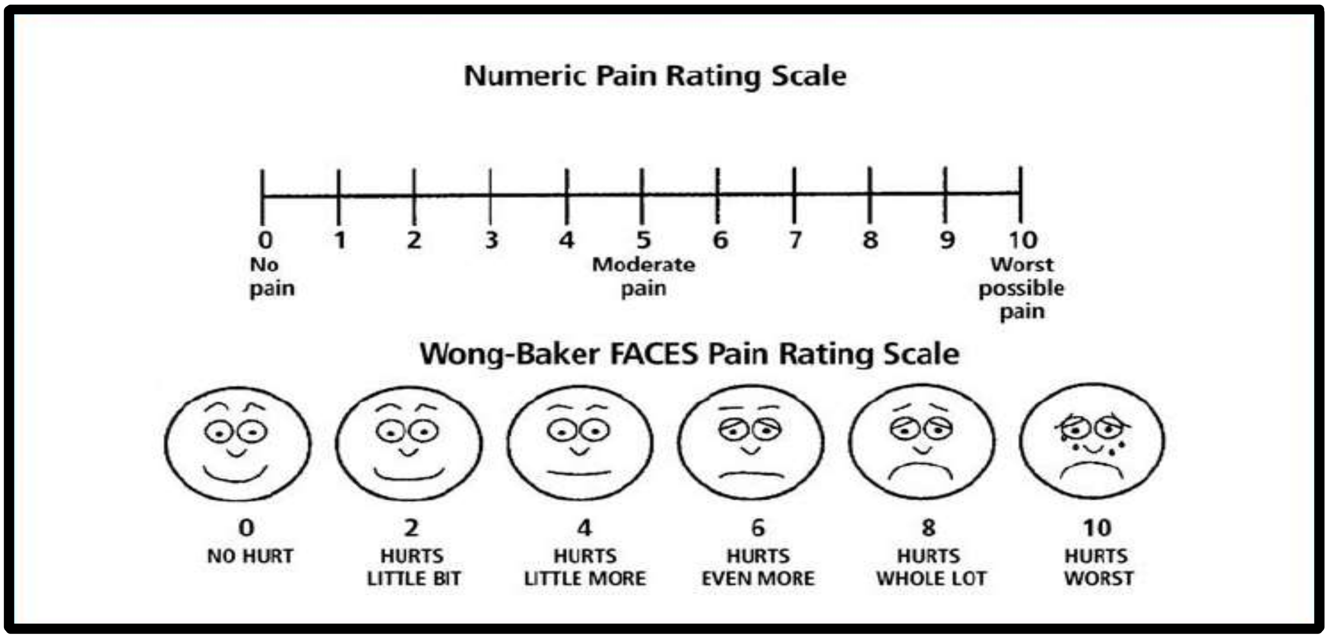
Document:

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Pain scale
- Communication with Medical Control

Pediatric Sickle Cell Crisis (Continued)

Pain Management

- If pain scale is $\geq$ 6, consider ***Morphine***
    - ***Morphine*** **0.1 mg/kg** IV/IO/IM. **Max Dose – 4 mg.**
    - After intervention, reassess mental status, pain level, and signs of respiratory depression every 5 minutes.
    - If respirations become depressed:
        - Administer ***Naloxone*** **0.01 mg/kg** IV/IO slow push or IN.
        - If no clinical improvement, **0.1 mg/kg. Max Dose – 2 mg.**
        - Titrate Naloxone administration to patient's respirations.
- If patient becomes nauseated or vomits, consider administering:
    - ***Ondansetron*** ***(Zofran)*** **0.1 mg/kg** IV/IM/ODT. **Max Dose – 4 mg.**



Pediatric Toxic Exposure

This protocol applies to pediatric patients with toxic exposure secondary to the ingestion, inhalation, contact or intravenous administration of a potentially toxic substance.

Scene Safety and Initial Management

- **Prevent exposure of EMS personnel. Assess and ensure scene security prior to proceeding with this protocol.**
- If patient is still located within toxic environment, have the patient moved to safety by appropriately trained personnel using the proper level of PPE.
- If signs of hazardous materials incident, call for HazMat team, keep patient(s) isolated in contaminated zone until HazMat team arrives.
    - Coordinate efforts with HazMat personnel.
- Identify agent and mechanism/route of exposure. (Inhaled, contact, etc.)
- Decontaminate as appropriate – EMS personnel must be wearing PPE prior to helping with the decontamination process.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- Ensure airway – ensure patency and positioning.
    - Have suction equipment ready.
    - Keep patient NPO.
- Assess breathing – give O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq$ 94%.**
    - Assist with bag valve mask if ineffective respiratory effort.
- Assess circulation – manage shock appropriately. *See **Shock Protocol**.*
- Assess disability – assess LOC.
- Exposure/Environment – expose injuries as needed.
    - Take measures to prevent hypothermia, especially following decontamination.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
    - **Pulse oximetry may not be accurate for toxic inhalation victims.**
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl.**
- Initiate cardiac monitoring
- Physical exam and OPQRST/SAMPLE history.
    - Identify substance/toxin and amount of exposure
    - Determine mechanism, time, and duration of exposure.
    - If ingestion, *See **Poisoning/Overdose Protocol**.*
- Initiate IV/IO of normal saline at KVO rate.
    - If signs of shock, administer 20 ml/kg bolus. *See **Shock Protocol**.*

CONTINUED ON NEXT PAGE

Pediatric Toxic Exposure (Continued)

Secondary Assessment and Treatment (Continued)

- If known or suspected **carbon monoxide** poisoning:
  - Provide 100% O_2, if not yet initiated.
  - Monitor carbon monoxide saturation.
  - Consult Medical Control for destination choice, including consideration of medical facilities equipped with hyperbaric capability.
- If **organophosphate**, **carbamate**, or **nerve agent** poisoning:
  - Administer ***Atropine*** **0.02 mg/kg** IV/IO/IM every 3-5 minutes.
    - Titrate to clinical symptoms (drying of secretions)
  - Contact **Georgia Poison Control** (1-800-222-1222) for consultation and/or **Chempack** deployment. *See **Chempack** in resources.*
- If patient is asymptomatic, monitor for delayed effects.

Frequently reassess patient, manage any presenting respiratory distress, seizures, and/or dysrhythmias in accordance with the appropriate protocol.

Document:

- Vital Signs
- Agent/mechanism of exposure
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control

All suspected suicide attempts must be reported before leaving the scene.

EMS personnel may contact Poison Control directly. EMS personnel are directed to follow the advice offered by the Poison Control Center as if it came directly from medical Control. Georgia Poison Control: 1-800-222-1222

Pediatric Toxic Ingestion

This protocol applies to pediatric patients with an acute overdose and/toxic ingestion.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- Ensure airway – ensure patency and positioning.
  - Have suctioning equipment ready
  - Keep patient NPO
- Assess breathing – give O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq$ 94%.**
  - Assist with bag valve mask if ineffective respiratory effort.
- Assess circulation – manage shock appropriately. *See* ***Shock Protocol*****.**
- Assess disability – assess LOC
- Exposure/Environment – expose injuries as needed.

Secondary Assessment and History

- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl.**
- Consider cardiac monitoring.
- Physical exam and OPQRST/SAMPLE history
  - Identify substance/toxin and amount of exposure.
- Initiate IV/IO of normal saline.
  - If fluid resuscitation is needed, administer 20 ml/kg bolus. *See* ***Shock Protocol*****.**

Treatments

- If **calcium channel blocker** or **beta blocker** overdose:
  - Administer ***Glucagon*** **0.025 – 0.1 mg/kg** IM slowly or IN. **Max Dose – 1mg.**
- If **Narcotic** overdose:
  - Administer ***Naloxone*** **0.01 mg/kg** IV/IO slowly or IN.
    - If no clinical improvement, administer **0.1 mg/kg. Max Dose – 2 mg.**
    - Titrate naloxone administration to patient's respirations
- If **Tricyclic Antidepressants** overdose with wide complex tachycardia:
  - Administer ***Sodium Bicarbonate*** **1 mEq/kg** IV/IO slowly.

CONTINUED ON NEXT PAGE

Pediatric Toxic Ingestion (Continued)

Treatments (Continued)

- If **stimulant/hallucinogen** overdose (cocaine, amphetamine, ecstasy, etc.):
    - Administer ***Midazolam*** **0.2 mg/kg** IV/IO/IM slowly, or IN. **Max Dose – 5 mg.**
    - Cool patient passively but do not allow patient to shiver.
- If patient is asymptomatic, monitor for delayed effects.

Frequently reassess patient, manage any presenting respiratory distress, seizures, and/or dysrhythmias in accordance with the appropriate protocol.

All suspected suicide attempts must be reported before leaving the scene.

EMS personnel may contact Poison Control directly. EMS personnel are directed to follow the advice offered by the Poison Control Center as if it came directly from Medical Control. Georgia Poison Control: 1-800-222-1222.

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control

Pediatric Multiple System Trauma

This protocol applies to pediatric patients presenting with injury to more than one body system.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- **Massive hemorrhage**
  - Treat all life-threatening hemorrhage
    - Tourniquets for massive extremity hemorrhage
    - Wound packing for junctional hemorrhage
- **Airway –** ensure patency, manually stabilize C-Spine.
  - Have suction equipment ready
  - Consider spinal motion restriction if indicated.
- **Respiratory/Breathing** – give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Manage any injuries impairing ventilation. *See **Pediatric Chest Trauma Protocol**.*
    - Chest wounds: Penetrating/Sucking chest wound, impaled objects, bruising
    - Signs of pneumothorax
    - Flail segments
  - Assist with bag valve mask if respiratory effort is ineffective.
- **Circulation** – assess pulses and perfusion status
  - Manage shock appropriately. *See **Shock Protocol**.*
- **Head/Hypothermia**
  - Initial GCS/AVPU
  - Pupils
  - Prevent hypothermia

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl.**
- Consider cardiac monitoring
  - Record and evaluate 12-Lead ECG
- Physical Exam and OPQRST/SAMPLE history
- Perform spinal motion restriction
  - Apply a rigid cervical collar.
  - Secure to long spine board.
- Initiate patient transport to an appropriate level trauma center as soon as possible.
- Advanced airway/ventilatory management as needed. *See **Sedation Assisted Intubation Protocol**.*
  - Initiate $EtCO_2$ monitoring.

CONTINUED ON NEXT PAGE

Pediatric Multiple System Trauma (Continued)

Secondary Assessment and Treatment (Continued)

- Initiate IV/IO of normal saline at KVO rate.
    - If fluid resuscitation is needed, administer 20 ml/kg bolus. *See* ***Shock Protocol*.**
- Perform detailed/focused physical assessment.
    - Re-evaluate ABCs.
    - Repeat neurological exam.
- Continue resuscitation and evaluation enroute.
- Manage any presenting ***Respiratory Distress***, ***Seizures***, and/or dysrhythmias in accordance with the appropriate protocol.

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control

Pediatric Head and Spine Injuries

This protocol applies to pediatric patients presenting with injuries to the head and/or spine.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- **Massive hemorrhage**
  - Treat all life-threatening hemorrhage
    - Tourniquets for massive extremity hemorrhage
    - Wound packing for junctional hemorrhage
- **Airway –** ensure patency, manually stabilize C-Spine.
  - Have suction equipment ready
  - Consider spinal motion restriction if indicated.
- **Respiratory/Breathing** – give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq$ 94%.**
  - Manage any injuries impairing ventilation.
    - Chest wounds: Penetrating/Sucking chest wound, impaled objects, bruising
    - Signs of pneumothorax
    - Flail segments
  - Assist with bag valve mask if respiratory effort is ineffective.
- **Circulation** – assess pulses and perfusion status
  - Manage shock appropriately. *See **Shock Protocol**.*
- **Head/Hypothermia**
  - Initial GCS/AVPU
  - Pupils
  - Prevent hypothermia

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl.**
- Consider cardiac monitoring.
  - Record and evaluate 12-Lead ECG.
- Physical Exam and OPQRST/SAMPLE history.
- Perform spinal motion restriction.
  - Apply a rigid cervical collar.
  - Secure to long spine board.
- Initiate patient transport to an appropriate level trauma center as soon as possible.
- Advanced airway/ventilatory management as needed. *See **Sedation Assisted Intubation Protocol**.*
  - Initiate $EtCO_2$ monitoring.

CONTINUED ON NEXT PAGE

Pediatric Head and Spine Injuries (Continued)

Secondary Assessment and Treatment (Continued)

- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg bolus. *See **Shock Protocol**.*
- Perform detailed/focused physical assessment.
  - Re-evaluate ABCs.
  - Repeat neurological exam.
- Continue resuscitation and evaluation enroute.
- Manage any presenting ***Respiratory Distress***, ***Seizures***, and/or dysrhythmias in accordance with the appropriate protocol.
- **Frequently re-assess for clinical signs of cerebral herniation. Signs and symptoms include dilated, unreactive pupils, asymmetric pupils, extensor posturing or no motor response, decreased GCS > 2 points in patients with initial GCS < 9.**
- Hyperventilation therapy titrated to clinical effect may be necessary for **brief** periods in cases of cerebral herniation or acute neurological deterioration.
  - Hyperventilation is administered as:
    - 25 breaths per minute in a child
    - 30 breaths per minute in an infant less than 1 year-old.
    - Maintain $EtCO_2$ of 30 – 35 mmHg.
- **Children with traumatic brain injury are likely to exhibit post-traumatic seizures.**
  - Manage any presenting seizure activity in accordance with the ***Seizure Protocol***.

Document:

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Pediatric Eye Trauma

This protocol applies to pediatric patients with blunt or penetrating trauma to the eye(s) or who have a chemical substance in the eye.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- **Massive hemorrhage**
  - Treat all life-threatening hemorrhage
    - Tourniquets for massive extremity hemorrhage
    - Wound packing for junctional hemorrhage
- **Airway –** ensure patency, manually stabilize C-Spine.
  - Have suction equipment ready
  - Consider spinal motion restriction if indicated.
- **Respiratory/Breathing** – give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq$ 94%.**
  - Manage any injuries impairing ventilation.
    - Chest wounds: Penetrating/Sucking chest wound, impaled objects, bruising
    - Signs of pneumothorax
    - Flail segments
  - Assist with bag valve mask if respiratory effort is ineffective.
- **Circulation** – assess pulses and perfusion status
  - Manage shock appropriately. *See **Shock Protocol**.*
- **Head/Hypothermia**
  - Initial GCS/AVPU
  - Pupils
  - Prevent hypothermia

Secondary Assessment and Treatment

- Physical exam and OPQRST/SAMPLE history.
  - Establish mechanism and nature of injury.
- **Assess vision, if possible, with injured eye. Can the patient count the number of fingers you hold up ? If not, can the patient perceive light ?**
- **Never apply pressure directly to the eyeball.**
- Monitor vital signs and O_2 saturation.
- If the eye has been avulsed or if the globe has been ruptured:
  - Carefully cover the injured eye to protect it.
  - Prevent conjugated eye movements – also cover the uninjured eye.
  - **Do Not** apply any pressure or absorbent dressing.
- If a foreign body is embedded in the eye:
  - **Do Not** attempt to remove the object.
  - Do attempt to stabilize the object.
  - Carefully cover both eyes.
- If the eyes are injured by chemical exposure, pepper spray or mace:
  - Responders should protect themselves with appropriate PPE.
  - Remove victim from the source of exposure.
  - Remove contaminated clothing and seal in plastic bag(s)
  - Irrigate eyes with copious amounts of water or normal saline.
- Transport patient with head elevated about 30 degrees, and both eyes closed or loosely patched, unless irrigating.
- For pain, contact Medical Control or see ***Pain Management Protocol***.

Pediatric Chest Trauma

This protocol applies to pediatric patients presenting with chest trauma.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- **Massive hemorrhage**
  - Treat all life-threatening hemorrhage
    - Tourniquets for massive extremity hemorrhage
    - Wound packing for junctional hemorrhage
- **Airway –** ensure patency, manually stabilize C-Spine.
  - Have suction equipment ready
  - Consider spinal motion restriction if indicated.
- **Respiratory/Breathing** – give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq$ 94%.**
  - Manage any injuries impairing ventilation.
    - Chest wounds: Penetrating/Sucking chest wound, impaled objects, bruising
    - Signs of pneumothorax
    - Flail segments
  - Assist with bag valve mask if respiratory effort is ineffective.
- **Circulation** – assess pulses and perfusion status
  - Manage shock appropriately. *See Shock Protocol.*
- **Head/Hypothermia**
  - Initial GCS/AVPU
  - Pupils
  - Prevent hypothermia

If at any time during the primary survey or secondary assessment the following chest injuries are identified, treat immediately.

- For **Penetrating Trauma** or **Sucking Chest Wound**
  - Seal initially with a gloved hand
  - Apply occlusive dressing, tape on 4 sides.
  - Monitor for Tension Pneumothorax
- For **Flail Segment**
  - Stabilize with bulky dressing.
    - Gentle pressure, **Do Not impair ventilation.**
  - Provide positive pressure ventilation as needed.
- For **Tension Pneumothorax**
  - Perform needle decompression on the affected side.
    - Preferred location is the 5th intercostal space along the anterior axillary line.
    - Alternate location is the 2nd intercostal space along the midclavicular line.

CONTINUED ON NEXT PAGE

Pediatric Chest Trauma (Continued)

Secondary Assessment and Treatment

- Physical exam and OPQRST/SAMPLE history
    - Examine the chest for bruising, abrasions, instability, crepitus, and/or open wounds.
    - Auscultate breath sounds and heart tones.
- Monitor vital signs and O_2 saturation, determine GCS
- Administer prehospital care and resuscitate as needed.
- Perform spinal motion restriction – apply a rigid cervical collar, secure to long spine board.
- Initiate patient transport as soon as possible to the appropriate facility.
    - Consider transport to trauma center
- Advanced airway/ventilatory management as needed.
-
- Initiate IV/IO of normal saline KVO.
    - If fluid resuscitation is needed, 20 ml/kg bolus. *See* ***Shock Protocol.***
- Re-evaluate ABCs and perform detailed/focused head-to-toe assessment and repeat neurological exam.
    - Continuously monitor patient's respiratory and perfusion status.
    - Auscultate breath sounds
    - Apply capnography
    - Perform blood glucose analysis. Treat if BGL is **< 60 mg/dl.**
- Initiate cardiac monitoring.
    - Treat dysrhythmias in accordance with the appropriate protocol.
- Continue resuscitation and evaluation enroute.

Document;

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

IMPORTANT – NEEDLE CHEST DECOMPRESSION

- **Indications for needle chest decompression**:
    - Peri-arrest, PEA, shock with hypotension, and at least one of the following:
        - Neck vein distension
        - Tracheal deviation away from injured side.
        - Increased resistance when ventilating
        - Hyper-expanded chest with little movement during respiration.

- **Needle chest decompression should never be utilized based solely on the presence of poor or absent breath sounds on one side of the chest. This procedure has complications and should not be used lightly. When used appropriately, it can be life-saving.**

CAUTION: Overly aggressive positive pressure ventilation may cause a pneumothorax or exacerbate an existing pneumothorax.

Pediatric Abdominal and Pelvic Trauma

This protocol applies to pediatric patients presenting with injury to the abdomen and/or pelvis.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- **Massive hemorrhage**
  - Treat all life-threatening hemorrhage
    - Tourniquets for massive extremity hemorrhage
    - Wound packing for junctional hemorrhage
- **Airway –** ensure patency, manually stabilize C-Spine.
  - Have suction equipment ready
  - Consider spinal motion restriction if indicated.
- **Respiratory/Breathing** – give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq$ 94%.**
  - Manage any injuries impairing ventilation.
    - Chest wounds: Penetrating/Sucking chest wound, impaled objects, bruising
    - Signs of pneumothorax
    - Flail segments
  - Assist with bag valve mask if respiratory effort is ineffective.
- **Circulation** – assess pulses and perfusion status
  - Manage shock appropriately. *See **Shock Protocol**.*
- **Head/Hypothermia**
  - Initial GCS/AVPU
  - Pupils
  - Prevent hypothermia

Secondary Assessment and Treatment

- Physical Exam and OPQRST/SAMPLE history
  - Note any abdominal rigidity, distention, tenderness, etc.
  - Note any pelvic instability.
  - Monitor vital signs and O_2 saturation, determine GCS Score
- For **Evisceration – DO NOT** attempt to replace protruding organs.
  - Apply a moistened sterile dressing directly to the site.
  - Cover the moistened sterile dressing with an occlusive dressing.
  - Place the patient on their back, with legs flexed at the knees. This is to reduce pain by relaxing the strain on the abdominal muscles.
- For **Impaled objects – DO NOT** attempt to remove the object.
  - Carefully cut away any clothing that is around the object.
  - Manually stabilize the object – avoid applying any pressure to the impaled object.
  - Use bulky dressing and cravats to stabilize the impaled object.
  - Minimize patient movement.
  - **If the impaled object was removed prior to your arrival, try to bring the object with you to the ER.**
  - Perform spinal motion restriction, apply cervical collar and secure to long spine board.
- Initiate transport as soon as possible.
  - Consider transport to a trauma center.

CONTINUED ON NEXT PAGE

Pediatric Abdominal and Pelvic Trauma (Continued)

Secondary Assessment and Treatment (Continued)

- Advanced airway/ventilatory management as needed.
- Initiate IV/IO of normal saline at KVO rate.
    - If fluid resuscitation is needed, 20 ml/kg bolus. *See **Shock Protocol**.*
- Re-evaluate ABCs and perform detailed/focused head-to-to exam and repeat neurological exam
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl.**
- Consider cardiac monitoring
- Continue resuscitation and evaluation enroute.

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Pediatric Extremity Trauma

This protocol applies to pediatric patients presenting with extremity trauma.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- **Massive hemorrhage**
  - Treat all life-threatening hemorrhage
    - Tourniquets for massive extremity hemorrhage
    - Wound packing for junctional hemorrhage
- **Airway –** ensure patency, manually stabilize C-Spine.
  - Have suction equipment ready
  - Consider spinal motion restriction if indicated.
- **Respiratory/Breathing** – give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Manage any injuries impairing ventilation.
    - Chest wounds: Penetrating/Sucking chest wound, impaled objects, bruising
    - Signs of pneumothorax
    - Flail segments
  - Assist with bag valve mask if respiratory effort is ineffective.
- **Circulation** – assess pulses and perfusion status
  - Manage shock appropriately. *See* ***Shock Protocol****.*
- **Head/Hypothermia**
  - Initial GCS/AVPU
  - Pupils
  - Prevent hypothermia

Secondary Assessment and Treatment

- Physical exam and OPQRST/SAMPLE history
  - Establish the mechanism and nature of injury.
  - Monitor vital signs and O_2 saturation
  - For **Fractures** and/or **Dislocations**
    - Assess distal pulse, motor and sensation before/after splinting and during transport.
    - If open fractures, control bleeding and cover with dry sterile dressing.
    - If the extremity is severely angulated **AND** pulses are absent, apply gentle traction while trying to straighten the extremity.
    - If pulses are present or if resistance is encountered while attempting to straighten the extremity, splint the extremity in the position found.
    - Apply the appropriate splinting device.
    - To reduce swelling, elevate the extremity and apply cold packs.

CONTINUED ON NEXT PAGE

Pediatric Extremity Trauma (Continued)

Secondary Assessment and Treatment (Continued)

- For **Amputation**
    - If limb is located, initiate care for amputated limb after patient has been stabilized.
    - Remove gross contaminants with saline rinse.
    - Wrap in saline moistened gauze and place in plastic bag or container (sterile if available).
    - Seal the bag or container tightly and place in solution of ice water, if available.
    - Transport the amputated limb to the hospital regardless of its condition.
    - If the part cannot be immediately located, transport the patient and have other field providers search for and transport limb as soon as possible.
- Initiate transport as soon as possible.
    - Consider transport to a trauma center.
- Initiate IV/IO of normal saline at KVO rate.
    - If fluid resuscitation is needed, 20 ml/kg bolus. *See **Shock Protocol**.*
- Re-evaluate patient's ABCs and perform detailed focused head-to-toe assessment.
- Cardiac monitoring

- For pain management, *See **Pediatric Pain Management Protocol**.*

Document:

- Vital signs
- Neurovascular status of the extremity before and after management
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Pediatric Trauma Arrest

This protocol applies to pediatric trauma patients with absent vital signs. Patients with injuries incompatible with life are covered under the ***Withholding or Termination of Resuscitation Protocol.***

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis

Primary Survey

- **Massive hemorrhage**
  - Treat all life-threatening hemorrhage
    - Tourniquets for massive extremity hemorrhage
    - Wound packing for junctional hemorrhage
- **Airway –** ensure patency, manually stabilize C-Spine.
  - Have suction equipment ready
  - Consider spinal motion restriction if indicated.
- **Respiratory/Breathing** – give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Manage any injuries impairing ventilation.
    - Chest wounds: Penetrating/Sucking chest wound, impaled objects, bruising
    - Signs of pneumothorax
    - Flail segments
  - Assist with bag valve mask if respiratory effort is ineffective.
- **Circulation** – assess pulses and perfusion status
  - Manage shock appropriately. *See **Shock Protocol**.*
- **Head/Hypothermia**
  - Initial GCS/AVPU
  - Pupils
  - Prevent hypothermia

Airway management, bleeding control and rapid transport are the most important interventions for victims of traumatic arrest. Minimize scene time to 10 minutes or less, (barring extrication time), and perform only critical interventions prior to transport.

Secondary Assessment and Treatment

- Attempt to obtain OPQRST/SAMPLE history prior to transport.
- Begin transport as soon as possible. Minimize scene time **10 minutes or less** if possible
- Continue protocol enroute.
- Move as rapidly and safely as possible toward an appropriate facility.
- Initiate cardiac monitoring
  - Manage dysrhythmias per appropriate protocol.
- Initiate $EtCO_2$ monitoring
- Advanced airway/ventilatory management as needed.
- Continue with compressions until return of adequate pulses.
- Establish IV/IO access using normal saline.
  - Rapid infusion and monitor for the return of palpable pulses
    If pulse is restored, titrate the infusion rate to a Systolic blood pressure of 80-90 mmHg.

CONTINUED ON NEXT PAGE

Pediatric Trauma Arrest (Continued)

Secondary Assessment and Treatment (Continued)

- If mechanism of injury, symptoms, and physical exam suggest a tension pneumothorax, consider needle decompression on the affected side or both sides if unsure.
- Contact Medical Control with patient status and treatment(s) as soon as possible.

Document;

- Vital signs
- Mechanism of Injury
- ROSC/Vital sign changes if relevant
- Cardiac rhythms
- Treatment(s)
- Advanced airway procedures with $EtCO_2$
- Medications administered
- Available history
- Communications with Medical Control.

Pediatric Burns

This protocol applies to pediatric patients who have sustained a thermal, chemical or electrical burn and/or have sustained inhalation injuries. Hypotension is not normally seen with prehospital burn patients. Hypotension suggests other trauma. Refer to the ***Trauma Protocol*** as needed.

- Ensure scene safety
- Remove patient from the burning process if possible but only if properly trained with proper PPE.

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis

Primary Survey

- Ensure airway – ensure patency and proper positioning.
  - Be prepared to manage airway aggressively.
- Assess breathing – give O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq 94\%$.**
- Assess circulation – manage bleeding and shock appropriately. *See* ***Shock Protocol****.*
- Assess disability – assess LOC
- Exposure/Environment – expose injuries as needed.
  - Children are more susceptible to hypothermia; cabin must be warm.
  - Cover exposed areas with dry sterile dressings.
- **Look for evidence of inhalation injury (hoarseness, stridor, sooty sputum, facial burns, or singed nasal or facial hair). Aggressive airway management my be warranted if evidence is present.**
- **Burn victims may have suffered carbon monoxide poisoning and show a false reading on pulse oximetry.**

Initial Burn Management and Treatment

- Initiate spinal motion restriction as needed.
  - If no suspicion of spinal injury, place patient in position of comfort.
  - If evidence of shock, place the patient supine and monitor airway closely. Treat shock according to the ***Shock Protocol****.*
- Remove and secure any jewelry, belts, shoes, etc. from burn injuries.
- Remove burned or singed clothing that did not adhere to skin.
- Initiate care for burn wounds.
  - **Chemical Injury**
    - brush off any excess chemical, flush with water to remove any residual chemical.
  - **Electrical Injury**
    - Treat any dysrhythmias per appropriate protocol.
  - **Thermal Injury**
    - Dry sterile dressings
- Begin transport as soon as possible. Consider Air Transport.
  - If no other trauma mechanism, consider transport to burn center.
  - If trauma mechanism exists, consider transport to trauma center.
  - Transport patients with unmanageable airway or uncontrolled hemorrhage to the closest hospital emergency department.

CONTINUED ON NEXT PAGE

Pediatric Burns (Continued)

Initial Burn Management and Treatment (Continued)

- Advanced airway/ventilatory management as needed.
  - Consider Sedation Assisted Intubation if needed. *See **Sedation Assisted Intubation Protocol**.*
- Secondary Assessment and Treatment.
  - Record and monitor vital signs, O_2 saturation and $EtCO_2$. Consider false pulse oximetry readings caused by carbon monoxide poisoning.
- Initiate cardiac monitoring.
- Assess.
  - Heat inhalation injuries – airway burns require aggressive airway management.
  - Approximate burn surface area – Rule of 9s or Palmar Rule method.
    - Depth of burns – only count 2nd, 3rd and 4th degree burns in % TBSA estimation.
    - Location of burns.
      - Circumferential burns can decrease blood flow to extremities and decrease cardiac and lung function when chest wall is involved.
  - Other injuries and illnesses.
- Initiate IV/IO of ***Lactated Ringers***.
  - **DO NOT** delay transport for IV/IO access.
    - Fluid resuscitation plan(s)
      - **Parkland Formula** – (4ml/kg x pt weight in kilograms x %BSA / 2)
      - **Consensus Formula** – (2-4 ml/kg x pt weight in Kg x % BSA / 2)
      - **Brook Formula** – (2ml/kg x pt weight in Kg x % BSA / 2)
      - **USAISR Rule of 10** - The calculated TBSA is rounded to the nearest 10 (i.e. the TBSA is calculated at 37%, this is then rounded to 40).
- **Pediatric patients have less metabolic glycogen reserves due to their smaller body weight and body surface ratio. Check BGL frequently !!!**
    - Consider ***D_5W*** IV maintenance drip included with fluid resuscitation plan.
      - Consider the **4-2-1 Rule** for patients weighing less than 45 kg.
        - 4 ml/kg/hr. for the first 10kg of patient's weight, + 2 ml/kg/hr. for the next 10 kg of patient weight, +1 ml/kg/hr. for each additional kg of patient weight.

Pain Management

- ***Morphine*** **– 0.1 mg/kg** IV/IO/IM. **Max Dose 5 mg.**
- ***Fentanyl*** **– 1mcg/kg** IV/IO/IM **Max Dose 50mcg.**

*See **Pain Management Protocol**.*

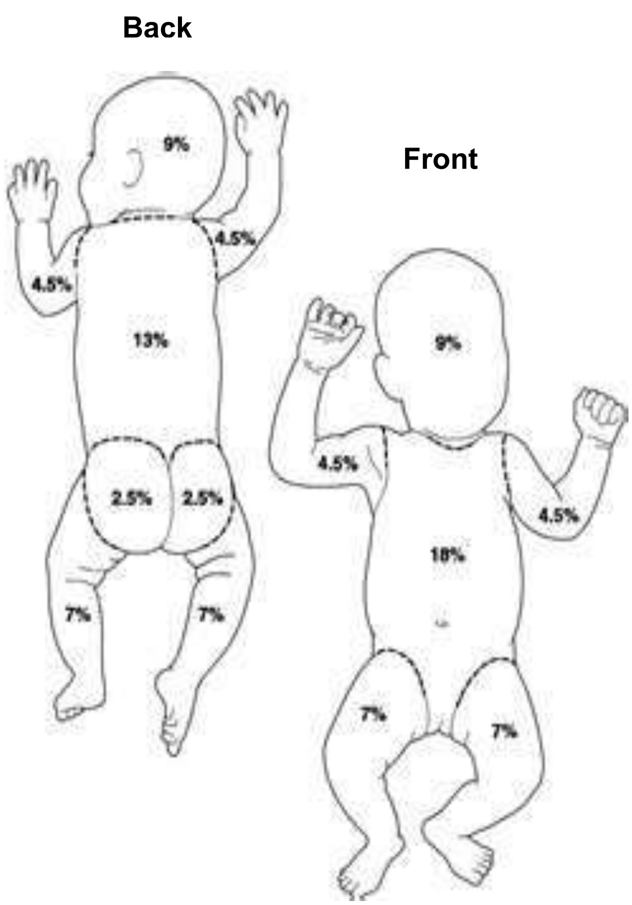
STOP

CONTACT MEDICAL CONTROL

CONTINUED ON NEXT PAGE

Pediatric Burns (Continued)

Pediatric Rule of 9s and Palmar Rule



Palmar, palm + fingers of patient = 1%

Document:

- Vital signs
- Burn type, location, size, and depth
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

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	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

Pediatric Snakebite

This protocol applies to pediatric patients who are suspected of being bitten by a venomous snake.

Special note: The safety of rescue personnel is top priority ! Ensure scene safety and determine location of snake. DO NOT transport snake. (A picture will suffice.) DEAD SNAKES ARE STILL DANGEROUS.

First Impression [(Pediatric Assessment Triangle)]

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- Ensure airway – ensure patency and proper positioning.
- Assess breathing – give O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq$ 94%.**
  - Assist ventilation with bag valve mask if respirations are ineffective.
- Assess circulation – manage shock appropriately. *See* ***Shock Protocol.***
- Exposure/Environment – expose injuries as needed.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl.**
- Consider cardiac monitoring
- Physical exam and OPQRST/SAMPLE history
  - Assess for swelling, skin color changes, shock
- **Mark the leading edge of swelling and erythema and record time, repeat if the leading edge is progressing.**
- If able, safely determine the type, size and length of snake.
- Advanced airway/ventilatory management as needed.
- Place patient in the position of comfort. Minimize movement and exertion.
- **DO NOT** place bitten extremity in an elevated or lowered position.
- Clean wound – apply light dressing, unless wound is bleeding profusely.
  - **DO NOT** apply ice or constricting bands, no cutting of bite injury.
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus IV/IO. *See* ***Shock Protocol.***
- Frequently re-assess patient, manage any presenting ***Respiratory Distress***, ***Seizures***, and/or dysrhythmias in accordance with the appropriate protocol.
- *See* ***Pain Management Protocol.***

Document:

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

Pediatric Submersion Event

This protocol applies to any pediatric patient that has been submerged under water for any period of time. **Special note: Safety of rescue personnel is top priority ! Enter the water only if trained and as a last resort.**

First Impression (Pediatric Assessment Triangle)

- Appearance – observe muscle tone, interactivity, consolability, look/gaze, and speech/cry.
- Breathing – observe chest wall movement, accessory muscle use, abnormal airway sounds.
- Circulation – observe skin color for pallor, mottling and/or cyanosis.

Primary Survey

- Ensure airway – ensure patency and proper positioning.
  - Consider spinal mobile restriction if any evidence of trauma.
- Assess breathing – give O_2 as tolerated by mask or blow-by. **Maintain $SpO_2 \geq$ 94%.**
  - Assist ventilation with bag valve mask if respirations are ineffective.
- Assess circulation – manage shock appropriately. *See **Shock Protocol**.*
- Assess disability – assess LOC.
- Exposure/Environment – take measures to prevent hypothermia. *See **Cold Related Emergencies Protocol**.*
  - Remove wet clothing
  - Cover and warm patient.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl.**
- Consider cardiac monitor
- Physical Exam and OPQRST/SAMPLE history
- Advanced airway/ventilatory management as needed.
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See **Shock Protocol**.*
- If patient is hypothermic, see ***Cold Related Emergencies Protocol***.
- **All submersion victims should be transported even if they appear uninjured or appear to have recovered.**

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Hart Co. EMS Adult Clinical Protocols

Adult Protocols Table of Contents

Contents:

Adult Clinical Protocols

Adult Cardiac Emergencies

Adult Medical Emergencies

Adult Trauma Emergencies

Adult Assessment

Primary Survey

- Assess LOC – determine responsiveness, utilize the AVPU scale.
- Ensure airway – ensure patency and positioning
  - Open, clear and maintain airway.
  - Simultaneously initiate spinal motion restriction if indicated.
- Assess breathing – assess rate and quality of breathing
  - Ensure adequate ventilation
  - Me
- Assess circulation – assess pulse and perfusion status.
  - Control major bleeding
  - Manage shock appropriately. *See* ***Shock Protocol*****.**

Perform necessary interventions for all identified life threats and determine patient priority.

Initiate Secondary Assessment and History

If new life threats are identified, treat immediately

Trauma

If the primary survey is abnormal, minimize scene time. Perform only necessary interventions, such as major bleeding control, spinal motion restriction, initial airway management, bag valve mask ventilation, in the field.

- Continue spinal motion restriction support as indicated. If no spinal injury is suspected, place patient in position of comfort.
- Initiate basic care for specific injuries.
- Perform physical examination.
  - Expose and examine any potentially injured area.
  - Take measures to prevent hypothermia.
- Monitor vital signs and O_2 saturation.
- Consider cardiac monitoring
- Perform blood glucose analysis.
- Obtain OPQRST/SAMPLE history

Onset	**S**igns, Symptoms
Provocation	**A**llergies
Quality	**M**edications
Region, Radiation	**P**ast Medical Hx
Severity	**L**ast Oral Intake
Time	**E**vents proceeding Illness or injury.

- Perform ongoing assessment
  - Repeat primary survey and vital signs
  - Evaluate response to treatment
  - Repeat physical exam.

Medical

If patient is unresponsive or has a diminished LOC, perform a rapid head-to-toe examination to rule out the presence of trauma and identify medically significant phyical findings.

- Obtain OPQRST/SAMPLE history.

Onset	**S**igns, Symptoms
Provocation	**A**llergies
Quality	**M**edications
Region, Radiation	**P**ast Medical Hx
Severity	**L**ast Oral Intake
Time	**E**vents proceeding illness or injury

- Perform physical examination
- Monitor vital signs and O_2 saturation
- Place patient on cardiac monitor
- Consider 12-Lead ECG
- Consider $EtCO_2$
- Perform blood glucose analysis
- Perform ongoing assessment
  - Repeat primary survey and vital signs
  - Evaluate response to treatment
  - Repeat physical exam.

Document:

- Vital Signs
- OPQRST/SAMPLE
- Environmental Observations
- Treatment
- Communication with Medical Control

Adult Pain Management

This protocol applies to adult patients suffering from severe pain or discomfort, including isolated extremity injuries, musculoskeletal or soft tissue injures, flank pain due to suspected kidney stones, sickle cell crisis, burns or any other cause.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See* ***Shock Protocol***.
  - Take measures to prevent hypothermia.

Secondary Assessment and Treatment

- Place the patient in a position of comfort and minimize patient exertion.
  - If patient is hypotensive, place supine and treat accordingly. *See* ***Shock Protocol.***
- Monitor vital signs and O_2 saturation.
- Initiate $EtCO_2$ monitoring.
- Consider cardiac monitoring.
  - Record and evaluate 12 Lead ECG
- Obtain OPQRST/SAMPLE history
- Physical Exam
- Immobilize any obvious injuries.
  - Elevate injured extremities if possible.
  - Consider application of a cold pack.
- Keep the patient NPO
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See* ***Shock Protocol.***
- Assess the patient's pain level
  - Use the numerical scale or the Wong-Baker FACES scale (see next page)

CONTINUED ON NEXT PAGE

Document:

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Pain Scale
- Communication with Medical Control

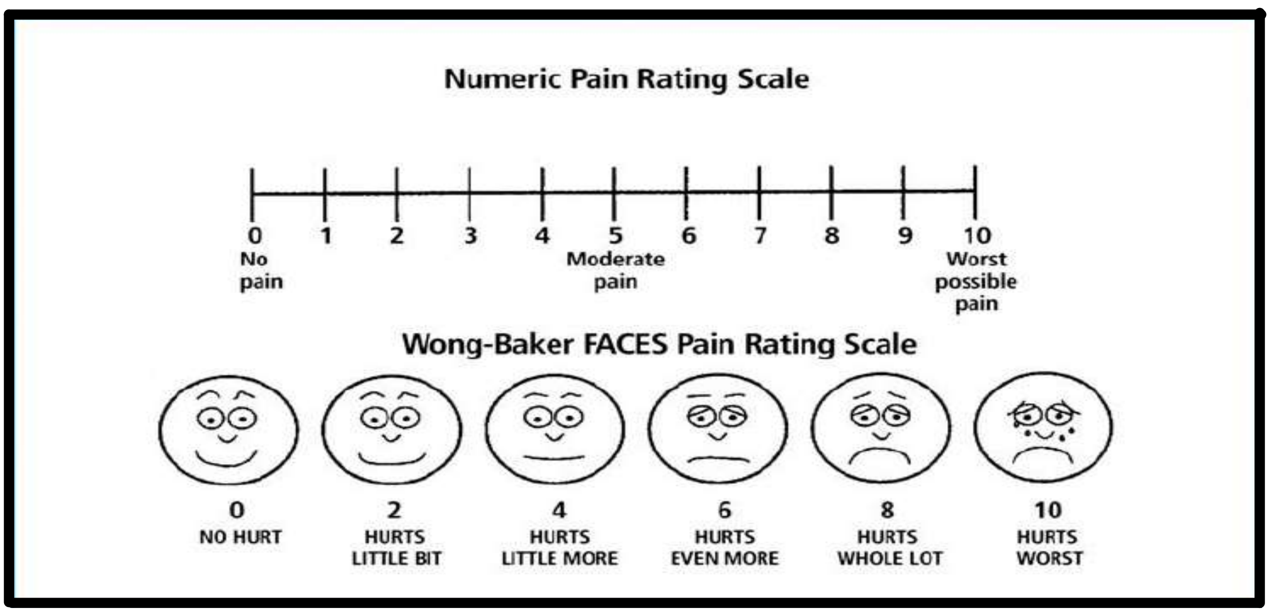
Pain Management (Continued)

- **Consider the administration of ONE of the following:**
  - *Morphine* – **2 mg** increments IV/IO/IM slowly. **Max Dose – 10 mg.**
    - Titrate administration to pain relief.
  - *Fentanyl* - **1 – 2 mcg/kg** IV/IO/IM/IN. **Max Dose – 200 mcg.**
    - Titrate administration to pain relief.
  - *Ketamine* – **0.1 mg/kg – 0.2 mg/kg** IV/IO/IM/IN.
    - Titrate administration to pain relief.
    - Withhold for patients with eye trauma.
    - May repeat every 15 minutes, twice.
  - *Toradol* – **15mg** IV/IM **Max Dose – 15 mg.**
    - **DO NOT** administer if patient has an allergy to aspirin.

⚕ *Toradol* is intended to treat suspected kidney stones or isolated minor extremity injury.

⚕ After any medication administration, you should re-assess the patient's mental status, pain level, blood pressure, and check for signs of respiratory depression every 5 minutes.

- If respirations become depressed, administer *Naloxone* **– 2 mg** IV/IO/IM/IN.
- If patient becomes nauseated or vomits, administer *Ondansetron* **– 4 mg** IV/IO/IM/IN may repeat in 15 minutes if no relief. **Max Dose – 8 mg.**
- May administer ~~*Phenergan*~~.



Shock Management

This protocol applies to adult patients presenting with signs and symptoms consistent with shock. All forms of shock are associated with inadequate tissue perfusion.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq$ 94%.**
  - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately.
  - Take measures to prevent hypothermia.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl.**
- Initiate cardiac monitoring
  - Treat dysrhythmias per appropriate protocol.
- Initiate $EtCO_2$ monitoring.
- Obtain OPQRST/SAMPLE history
- Physical Exam
- Keep patient NPO
- Advanced airway/ventilatory management as needed.
- Initiate IV/IO
  - For **Hypovolemic Shock,** administer normal saline bolus. 20 ml/kg.
  - For **Cardiogenic Shock,** administer normal saline at KVO rate.
  - For all other types of shock, administer normal saline bolus. 20 ml/kg.
  - Fluid boluses may be repeated x 1 – titrate to clinical effect.
- Attempt to identify cause(s) and treat in accordance with the appropriate protocol. (Tension Pneumothorax, overdose, trauma, etc.)
- Initiate transport as soon as possible.
- Continue resuscitation and evaluation enroute.
  - Frequently revaluate ABCs and mental status.

CONTINUED ON NEXT PAGE

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Shock Management (Continued)

Secondary Assessment and Treatment (Continued)

- If patient deteriorates or fails to improve, Medical Control may authorize the following:
  - For **Hypovolemic Shock,** administer additional 20 ml/kg bolus.
  - For **Cardiogenic Shock,** ensure rate and rhythm have been treated.
    - ***Levophed* IV Infusion – 1-10 mcg/min.**
      - 4 mg in 250 ml normal saline = 16 mcg/ml
      - Start drip at calculated rate
      - Monitor patient for deterioration into cardiac arrest.
      - *See* ***Levophed Dosing Chart*** (below)

    May also consider:

      - ***Dopamine* IV Infusion 2 – 20 mcg/kg/min**
      - *See* ***Dopamine Dosing Chart***
        - Titrate to clinical effect.
- For all other types of shock, administer 20 ml/kg normal saline fluid bolus.

AND

- **Consider push dose epinephrine in hypotension or presumed peri-arrest.**
    - ***Epinephrine 1:100,000* – 0.5 – 2ml** every 3-5 minutes.

      - **Mixing instructions for *Epinephrine 1:100,000***
        - Mix 1 ml of 1:10,000 Epinephrine in 9 ml of Normal Saline. 1:100,000 solution.

Levophed Dosing Chart

4 mg in 250 ml of Normal Saline

Concentration = 16 mcg/ml

Mcg/min	Volume/minute with 60 gtts	Micro drops/minute
1	0.25 ml/min	4
2	0.5 ml/min	8
3	0.75 ml/min	11
4	1.0 ml/min	15
5	1.25 ml/min	18
6	1.5 ml/min	22
7	1.75 ml/min	26
8	2.0 ml/min	30
9	2.25 ml/min	34
10	2.5 ml/min	38

CONTINUED ON NEXT PAGE

Shock Management (Continued)

Dopamine Infusion: Standard 1600mcg/ml Concentration												
Weight		Milliliters hour or drops per minute with micro drip tubing (60gtt/ml)										
lbs	kg	2 mcg/kg/hr	3 mcg/kg/hr	4 mcg/kg/hr	5 mcg/kg/hr	6 mcg/kg/hr	7 mcg/kg/hr	8 mcg/kg/hr	9 mcg/kg/hr	10 mcg/kg/hr	15 mcg/kg/hr	20 mcg/kg/hr
77	35	3	4	5	7	8	9	10	11	13	20	26
88	40	3	5	6	8	9	11	12	14	15	23	30
99	45	3	5	7	8	10	12	14	15	17	25	34
110	50	4	6	8	9	11	13	15	17	19	28	38
121	55	4	6	8	10	12	14	17	19	21	31	41
132	60	5	7	9	11	14	16	18	20	23	34	45
143	65	5	7	10	12	15	17	20	22	24	37	49
154	70	5	8	11	13	16	18	21	24	26	39	53
165	75	6	8	11	14	17	20	23	25	28	42	56
176	80	6	9	12	15	18	21	24	27	30	45	60
187	85	6	10	13	16	19	22	26	29	32	48	64
198	90	7	10	14	17	20	24	27	30	34	51	68
209	95	7	11	14	18	21	25	29	32	36	53	71
220	100	8	11	15	19	23	26	30	34	38	56	75
231	105	8	12	16	20	24	28	32	35	39	59	79
242	110	8	12	17	21	25	29	33	37	41	62	83
253	115	9	13	17	22	26	30	35	39	43	65	86

Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

Sepsis/Shock Management

Sepsis is a potentially life-threatening condition that occurs when the body's response to an infection damages its tissues. When the infection-fighting process turns on the body, they cause organs to function poorly and abnormally. Sepsis may progress to septic shock. This is a dramatic drop in blood pressure that can lead to severe organ problems and death. Early treatment with antibiotics and aggressive intravenous fluids improve chances for survival.

Indications of Sepsis

- Patient has a suspected infection
  - **AND AT LEAST 2 OF THE FOLLOWING:**
    - Altered mentation (Any GCS < 15, confused, agitated, etc.) Baseline changes.
    - Respiratory rate $\geq$ 20 breaths per minute.
    - Heart rate > 90 beats per minute.
    - Systolic Blood Pressure $\leq$ 100 mmHg.
    - Fever or hypothermia
    - $EtCO_2 \leq 25$ mmHg.

Initial Treatment

- Work to minimize scene time to LESS than 10 minutes.
- Notify the closest appropriate facility of **SEPSIS ALERT.**
- Establish a minimum of two (2) IV sites if possible.
- Consider early IO placement for volume resuscitation.
- Administer 500 ml bolus of Lactated Ringers or Normal Saline.
  - Consider repeating the bolus if patient improves clinically.
  - Monitor all vital signs and lung sounds after each bolus.
  - May give up to 1000 ml bolus.

Secondary Assessment and Treatment(s)

- Consider Vasopressors if unresponsive to IV fluid therapy.
  - ***Levophed*** – Start at 4 mcg/min, titrate to a systolic blood pressure of 100 mmHg. Maximum of 10 mcg/min.

OR

- ***Epinephrine*** **– 1:100,000** 0.5 – 2 ml every 3 -5 minutes
  - Mixing Instructions for Epinephrine 1:100,000
    - Mix 1 ml of 1:10,000 Epinephrine in 9 ml of Normal Saline.
    - Onset 1 minute
    - Duration 5 – 10 minutes
- Also consider ***ANCEF*** **– Contact Medical Control Prior to Administration.**
    - **2 gm** if > 176 pounds. Mix in 100 ml of Normal Saline and give over 10 minutes via 60 gtts.
    - **1 gm** if < 176 pounds. Mix in 100 ml of Normal Saline and give over 10 minutes via 60 gtts.

Hemorrhage Control

Apply manual direct pressure or pressure bandage to the wound

Hemorrhage Controlled ?

YES

NO

- Apply tourniquet for extremities. Use second tourniquet between initial tourniquet and wound if needed.
- Consider wound packing if appropriate.
- Consider JETT if available and appropriate.
- Consider use of ***Tranexamic Acid*** for non-compressible hemorrhage on age-appropriate patients (≥ 16 years old) with a Shock index ≥ 1.0.
- If hypoperfusion is present, consider aeromedical transport or rendezvous with whole blood provider.

- Monitor and re-assess treatments and bandaging frequently.
  - **DO NOT** remove blood-soaked bandages. Apply fresh bandage over old bandage.

Shock Index Formula

- **SI = Heart Rate(HR)/Systolic Blood Pressure(SBP)**
  - Normal Range – 0.5 – 0.7

TXA Mixing Instructions

- **2 gm** (2000 mg) mixed into 100 ml bag of Normal Saline and infused over 10 minutes via a 60 gtts drip set.

Document:

- Vital signs
- Mechanism of Injury
- Time of Injury
- Treatment(s)
- Communication with Medical Control

Bradycardia

This protocol applies to adult patients with a heart rate < 60 beats per minute. This protocol is not

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
    - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
    - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See* ***Shock Protocol***
    - Take measures to prevent hypothermia.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis – Treat if BGL is < 60 mg/dl.
- Initiate cardiac monitoring.
    - Record and evaluate 12-Lead ECG.
    - **DO NOT** delay treatments/therapy for acquisition of 12-Lead.
    - Treat dysrhythmias per the appropriate protocol.
- Consider $EtCO_2$ monitoring
- Physical Exam and OPQRST/SAMPLE history
- Initiate IV/IO of normal saline at a KVO rate.
    - 2nd IV access can be established if time permits.
- If patient is asymptomatic (i.e. no hypotension, no altered mental status, no difficulty in breathing, no signs of shock, no ischemic chest pain/discomfort, no acute heart failure)
    - Monitor patient closely.
- If signs of cardiopulmonary compromise:
    - If IV/IO access has been achieved, administer ***Atropine*** **– 0.5 – 1 mg**
        - May repeat every 3 – 5 minutes. **Max Dose – 3 mg.**
    - If IV/IO access is delayed or the bradycardia is a high degree block (2nd Degree Type II, or 3rd Degree) or atropine is ineffective, consider ***Transcutaneous Pacing.***
        - If time permits sedate patient prior to Transcutaneous Pacing
            - **DO NOT** delay pacing for administration of medications or IV/IO access.
            - Consider ***Midazolam*** **2 – 5 mg** IV/IO/IM/IN slowly.
            - After sedation, monitor respirations closely.
- May also consider:
        - ***Levophed IV Infusion*** **– 1 – 10 mcg/min.**
            - 1 mg in 250 ml of Normal Saline = 4 mcg/ml.
            - Monitor Patient for deterioration into cardiac arrest.
            - *See* ***Levophed Dosing Chart***
            - **OR**
- If no response, initiate *a* ***Dopamine*** **IV Infusion 2 – 20 mcg/kg/min.** *See Dopamine Infusion Chart.*
            - **OR**
- Initiate an ***Epinephrine*** **IV infusion – 2 – 10 mcg/min.**

STOP

Contact Medical Control For Additonal Orders

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Vital signs
- Communication with Medical Control

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	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

Tachycardia

This protocol applies to adult patients who present with a palpable pulse rate > 150 beats per minute.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See* ***Shock Protocol***
  - Take measures to prevent hypothermia.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis – Treat if BGL is < 60 mg/dl.
- Initiate cardiac monitoring.
  - Record and evaluate 12-Lead ECG.
  - **DO NOT** delay treatments/therapy for acquisition of 12-Lead.
  - Treat dysrhythmias per the appropriate protocol.
- Consider $EtCO_2$ monitoring
- Physical Exam and OPQRST/SAMPLE history
- Initiate IV/IO of normal saline at a KVO rate.
  - 2nd IV access can be established if time permits.
- Advanced airway/ventilatory management as needed.
- **Attempt to rule out sinus tachycardia as a potential cause of the symptoms. 220 minus the patient's age is the upper limit of sinus tachycardia.**
- If sinus tachycardia, search and treat causes. (i.e. hypovolemia, dehydration, trauma, etc.)

For Regular Narrow-Complex Tachycardia

- If no hypertension, acute altered mental status, signs of shock, ischemic chest pain/discomfort, and/or acute heart failure.
  - For narrow complex tachycardia, attempt Valsalva and/or other vagal maneuvers.
  - If regular, consider ***Adenosine*** IV/IO.
    - 1st Dose – **6mg** IV rapid push followed by 10ml saline flush.
    - 2nd Dose – **12 mg** IV rapid push followed by 10 ml saline flush. (If needed)

OR

- If signs of cardiopulmonary compromise, perform immediate ***Synchronized Cardioversion***.
  - If conscious, consider ***Midazolam*** **2 – 5 mg** IV/IO/IN or ***Valium*** **5 – 10 mg** IV/IO slowly prior to cardioversion.
    - Monitor respirations after sedation administration.

CONTINUED ON NEXT PAGE

Tachycardia (Continued)

For irregular narrow-complex tachycardia (Atrial Fibrillation/Atrial Flutter)

- If no hypotension, acute altered mental status, signs of shock, ischemic chest pain/discomfort, and/or acute heart failure:
  - Administer ***Amiodarone*** **150 mg** IV over 10 minutes.
    - Withhold with Wolff-Parkinson White (WPW)
      - Evidenced by short PR interval and Delta Wave
  - If unstable consider cardioversion.

For wide-complex tachycardia (QRS $\geq$ 0.12 seconds)

- If no hypotension, acute altered mental status, signs of shock, ischemic chest pain/discomfort, and/or acute heart failure:
  - If regular **and** monomorphic consider ***Adenosine*** IV/IO
    - 1st dose **6 mg**, administer rapidly followed by 10 ml flush.
    - 2nd dose **12 mg** administer rapidly followed by 10 ml flush. (if required)
  - If signs of cardiopulmonary compromise:
    - Perform immediate **Synchronized Cardioversion**
      - If conscious, consider administration of ***Midazolam*** **2 – 5 mg** IV/IO/IN slowly, or ***Valium*** **5 – 10 mg** IV/IO prior to cardioversion. Monitor Respirations.
- If, at any time, a cardiac rhythm other than a tachycardia is noted, treat based on the appropriate protocol.

- If tachycardia persists, despite synchronized cardioversion and/or adenosine administration:
  - If narrow-complex tachycardia, consider additional anti-dysrhythmics.
  - If wide-complex tachycardia, consider ***Amiodarone*** **150 mg** IV/IO over 10 minutes.
    - May repeat dose once.
    - Consider following **150 mg** ***Amiodarone*** bolus with maintenance infusion at **1 mg/min**.
- If polymorphic, wide-complex tachycardia (Torsade's), administer ***Magnesium Sulfate*** **1 – 2 grams** in 100 ml D_5W over 5 – 60 minutes IV. Titrate to control Torsade's.

Synchronized Cardioversion: Recommended Doses	
Narrow-Regular	**50 – 100 joules**
Narrow-Irregular	**120 – 200 joules biphasic, 200 joules monophasic**
Wide-Regular	**100 Joules**

Note: for wide-irregular rhythms utilize the defibrillation dose (not synchronized).

Document:

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

	Magnesium Drip	Amiodarone Drip
0.9% NaCl	100 ml	100 ml
Medication	2 gm Mag Sulfate	150 mg Amiodarone
Concentration	20 mg/ml	1.5 mg/ml

****Lidocaine*** may be utilized in replacement of ***Amiodarone***. Administer ***Lidocaine*** at 1 – 1.5 mg/kg. **Max Dose – 3 mg/kg.**

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Pulseless Arrest

Resuscitation

- Assess patient for respiratory and cardiac arrest.
- Initiate CPR and AED/Defibrillator using most current ***AHA Guidelines***.
- Provide high-quality compressions, minimizing interruptions.
  - Compressions should be at a rate of 100 – 120 per minute.
  - Apply Automated Compression Device ASAP.
- Ventilate with bag valve mask and 100% O_2, consider Oropharyngeal/Nasopharyngeal airway
- Consider advanced airway management.
  - Do not delay resuscitation for advanced airway placement.
  - If advanced airway is utilized, initiate $EtCO_2$ monitoring.

Shockable Rhythm?

YES

V-fib/Pulseless VT

- Defibrillate: 120 – 200 joules (biphasic) or 360 joules (monophasic)
- Resume CPR for 2 minutes
- If patient remains in V-fib/Pulseless VT, Deliver 2nd shock at equivalent or higher dose.
- Continue CPR
- Initiate IV/IO of normal saline.
- ***1:10,000*** **Epinephrine** **1 mg** IV/IO, repeat every 3 – 5 minutes
- Consider advanced airway and capnography
- Re-assess rhythm every 2 minutes, if rhythm is organized, check pulse.
- Defibrillate
- Resume CPR for 2 minutes.
- ***Amiodarone*** 1st Dose **300 mg** IV/IO; 2nd Dose **150 mg** IV/IO
- Continue CPR/treatment(s) as indicated.
- Consider and treat reversible causes.

NO

Asystole/PEA

- Initiate IV/IO of normal saline
- ***1:10,000*** **Epinephrine** **1 mg** IV/IO repeat every 3 – 5 minutes.
- Consider advanced airway and capnography.
- Re-assess rhythm every 2 minutes. If rhythm is organized, check pulse.
- Continue CPR/treatment as indicated.
- Consider and treat reversible causes.

Hypovolemia	Tension Pneumothorax
Hypoxia	Tamponade
Hydrogen Ion	Toxins
Hypoglycemia	Thrombosis
Hypokalemia	
Hyperkalemia	
Hypothermia	

Transport to the closest appropriate facility.

If cardiac rhythm change is noted, treat based on appropriate Protocol.

Lidocaine may be used in place of ***Amiodarone***. Administer ***Lidocaine*** at 1 – 1.5 mg/kg. **Max Dose 3 mg/kg**.

Document:

- Vital Signs
- Events preceding arrest
- Code Summary
- Treatment(s)
- Communication with Medical Control

Post Resuscitation

This protocol applies to adult patients with history of cardiac arrest and return of spontaneous circulation (ROSC).

Primary Survey

- Assess level of consciousness – AVPU.
- Ensure airway – ensure patency and positioning.
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain SpO_2 ≥ 94%.**
  - Assist with bag valve mask if respiratory effort is ineffective.
  - **DO NOT** hyperventilate.
- Assess circulation – control bleeding and manage shock appropriately. *See* ***Shock Protocol***
  - Check distal pulses, capillary refill time, skin color and temp.
  - Take measures to prevent hypothermia.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis. Treat if BGL is **< 60 mg/dl**.
- Cardiac monitor – record and evaluate 12 Lead ECG.
  - **DO NOT** delay treatments/therapy for 12-Lead acquisition.
- Consider $EtCO_2$ monitoring – Ideally > 20 mmHg.
- Physical Exam and OPQRST/SAMPLE history.
- Initiate IV/IO of normal saline
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See* ***Shock Protocol***.
  - Titrate fluid resuscitation to a systolic blood pressure of ≥ 90 mmHg.
  - Second IV line can be established, if time permits.
- If systolic blood pressure is < 90 mmHg, despite fluid administration, consider one of the following:
  - ***Epinephrine*** **2 – 10 mg/min.** IV Infusion

    OR

  - ***Levophed*** **1 – 10 mcg/min.** IV Infusion *See* ***Levophed Dosing Chart***.
  - ***Dopamine*** **2 – 20 mcg/kg/min** IV Infusion. *See* ***Dopamine Infusion Chart***.

- **The condition of post-resuscitation patients fluctuates rapidly and continuously. They require continuous close monitoring.**
- **Treat any presenting non-perfusing dysrhythmias in accordance with the appropriate protocol.**
- **Treat any perfusing dysrhythmias in accordance with the appropriate protocol.**

Document:

- Vital Signs
- Time of ROSC
- Onset/duration of event
- Treatment(s)
- Communication with Medical Control.

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Left Ventricular Assist Device (LVAD)

This protocol applies to adult patients who have a left ventricular assist device (LVAD) implanted. A ventricular assist device is a mechanical pump that is used to support heart function and blood flow in people who have weak hearts.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Assist with bag valve mask if respiratory effort is ineffective.
  - **DO NOT** hyperventilate.
- Assess circulation – control bleeding and manage shock appropriately. *See* ***Shock Protocol***
  - Take measures to prevent hypothermia.
- **In the majority of LVAD patients a pulse will not be palpable. This occurs because the LVAD unloads the left ventricle in a continuous fashion. Mental status and skin color are the best indicators of oxygenation and perfusion status.**

Secondary Assessment and Treatment

- **Locate the emergency contact sheet per the patient's hospital/physician. Call LVAD coordinator if device fails.**
- Listen over pump pocket – A hum should be detected if LVAD is running.
- Check for specific alarms
  - If alarms show red buttons – critical status
- Perform blood glucose analysis Treat if BGL is **< 60 mg/dl**.
- Cardiac monitor
  - No pads over the pump pocket.
  - **DO NOT** stop LVAD to assess.
- Physical Exam and OPQRST/SAMPLE history.
  - These patients, along with their family members, have been well trained in the care of themselves and their device. **LISTEN TO THEM !**
  - Assess for evidence of poor perfusion and/or acute congestive heart failure.
- Initial LVAD management
  - Check LVAD percutaneous lead connection(s)
  - Make sure the driveline and power sources (battery and/or AC power) are connected to the system controller.
  - Change battery if < 2 lights are showing. **Change batteries one at a time.**
  - Transport with 4 – 6 backup batteries and back up control unit.
- If in cardiac arrest, **NO CPR**.
- Initiate IV/IO of normal saline.
  - Drip rate is dependent upon perfusion status.
- If signs and symptoms of acute CHF, withhold fluid bolus. *See* ***CHF in the Respiratory Distress Protocol.***
- Transport to the appropriate facility.

Document:

- Vital signs
- Pertinent assessment findings
- Onset/duration of event
- Treatment(s)
- Communication with Medical Control.

Left Ventricular Assist Device (Continued)

Treating patients with Left Ventricular Assist Device (LVAD)

24/7 UR Medicine Advanced Heart Failure Team:
1-800-892-4964
585-273-3760

UR Medicine Transfer Center:
1-800-499-9298
585-275-4999

A Left Ventricular Assist Device (LVAD) is an internal heart pump used to treat heart failure.

The pump provides continuous blood flow from the left ventricle to the aorta. It requires an external controller and batteries.

When caring for a person with an LVAD, follow these steps:

- Listen for a humming sound where the heart is
- Find the percutaneous lead (the driveline exiting the abdomen and connected to the controller)
- Verify controller is connected to adequate power
- Identify and treat life-threatening events as you would with any other patient
- **Reversal of INR is reserved for life-threatening injuries and should be discussed with the attending cardiac surgeon**
- Cardioversion/defibrillation can be performed on an LVAD patient

TYPES OF CONTINUOUS FLOW LEFT VENTRICULAR ASSIST DEVICES

HeartMate II™
Axial Flow Pump

Speed: 8000-10000 rpms
Flow: 4 – 6 lpm
Power: 4 – 6 watts
Pulsatility Index: 4 – 7
MAP: 70 – 90 mm Hg
Warfarin/Aspirin
Pair of Batteries = 10 – 12 hours
Emergency Battery in Controller

Heartware HVAD™
Centrifugal Flow Pump

Speed: 2400-3200 rpms
Flow: 4 – 6 lpm
Power: 3 – 5 watts
Pulsatility Index: Not applicable
MAP: 70 – 90 mm Hg
Warfarin/Aspirin
Pair of Batteries = 8 – 12 hours
No Emergency Battery in Controller

HeartMate 3™
Centrifugal Flow Pump

Speed: 4800-6500 rpms
Flow: 4 – 6 lpm
Power: 3 – 5 watts
Pulsatility Index: 2 – 6
MAP: 70 – 90 mm Hg
Warfarin/Aspirin
Pair of Batteries = 15 hours
Emergency Battery in Controller

MEDICINE *of* THE HIGHEST ORDER

UR MEDICINE

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Left Ventricular Assist Device (Continued)

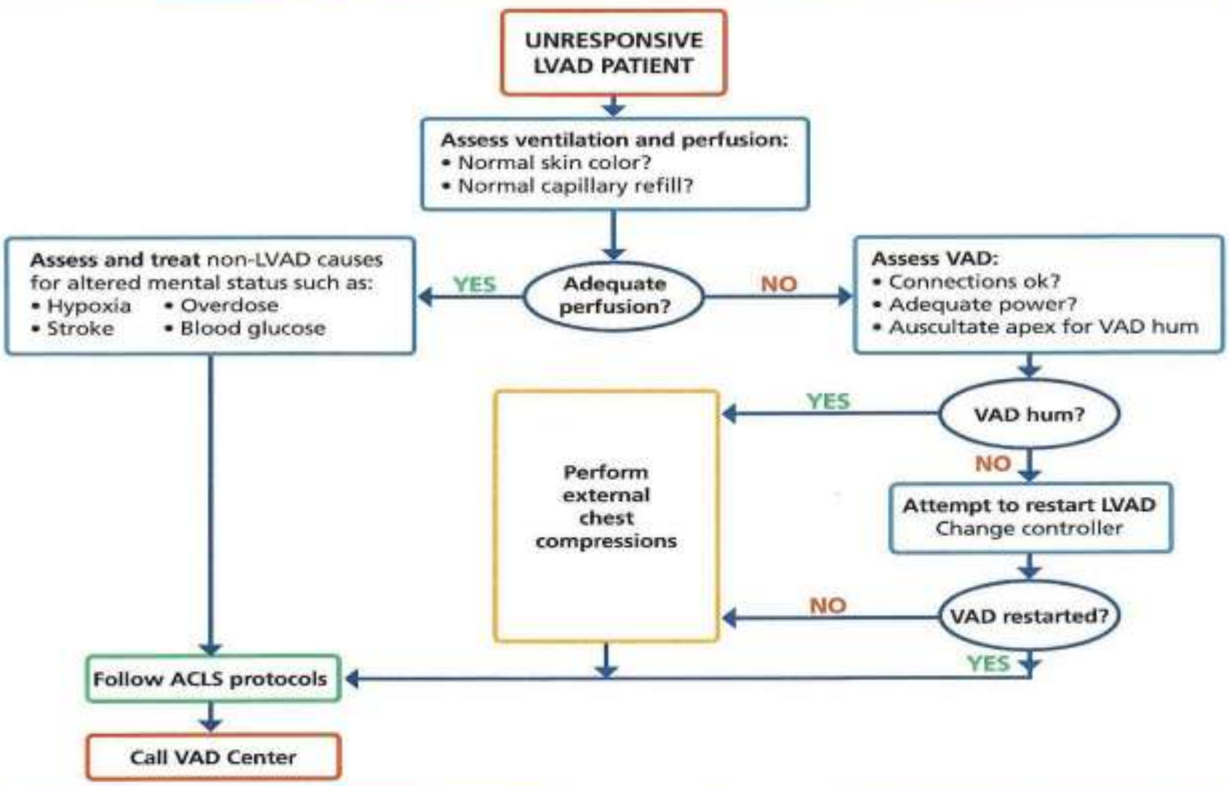
Understanding LVAD Vital Signs

- The patient may have a weak or absent radial pulse
- Assess adequate perfusion based on mentation, skin color and capillary refill
- Measure blood pressure by Doppler if the patient does not have a palpable radial pulse
  - Find arterial flow with Doppler
  - Increase cuff pressure until signal goes away
  - Decrease pressure until signal returns
  - Opening sound is considered the MAP
  - Goal MAP <90 mm Hg
- May utilize manual cuff and calculate a MAP if patient has a palpable radial pulse

Patient *without* LVAD:
MAP 84, pulse pressure **67**

Patient *with* LVAD:
MAP 83, pulse pressure **12**

ASSESSMENT OF THE UNRESPONSIVE LVAD PATIENT



LVAD Resource Materials for Community Providers

UR Medicine VAD Program educational resources can be found at www.vadresources.urmc.edu
For questions, please email our VAD Educator Lisa Fowler, lisa_chemotti@urmc.rochester.edu

v4/2019

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Abdominal Discomfort

This protocol applies to adult patients with pain/discomfort presenting in the abdomen or the flanks with no history or signs of trauma.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
    - Have suction equipment ready.
    - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
    - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See* ***Shock Protocol***
    - Take measures to prevent hypothermia.
- **Altered LOC and signs of poor perfusion could indicate shock and should be managed appropriately.**

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis. Treat if BGL is **< 60 mg/dl**.
- Cardiac monitor
    - Record and evaluate 12-Lead ECG.
- Physical exam and OPQRST/SAMPLE history.
    - History of blood in vomit or stool.
    - Prior abdominal surgery?
    - Last meal?
    - Assess for signs of dehydration and/or shock.
- Consider possible causes (GI, GU, cardiac, aneurysm, medications, toxic exposure, pregnancy, etc.)
- Save emesis or other drainage for signs of GI bleed etc.
- Keep the patient NPO
- Initiate IV of normal saline.
    - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See* ***Shock Protocol***.
    - Titrate to a systolic blood pressure of $\geq$ 90 mmHg.
- For pain management, *See* ***Pain Management Protocol***.
- If nausea or vomiting are present, *See* ***Nausea/Vomiting Protocol***.

Document:

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Agitated/Behavioral Patient

This protocol applied to adult patients who have become agitated or is having a behavioral emergency that has the potential to cause harm to the EMS crew or themselves. This protocol applies to patients 18-years-old or older.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq$ 94%.**
  - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See* ***Shock Protocol***
  - Take measures to prevent hypothermia.
- **Altered LOC and signs of poor perfusion could indicate shock and should be managed appropriately.**

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Initiate cardiac monitoring – treat dysrhythmias per appropriate protocol.
- Physical exam and OPQRST/SAMPLE history.
- Initiate IV of normal saline.
- Attempt to identify cause and treat in accordance with the appropriate protocol.
  - **Respiratory Distress (Hypoxia)**
  - **Trauma**
  - **Hypoglycemia**
  - **Drug Use**
- Request law enforcement for assistance as needed. If patient is handcuffed, law enforcement **MUST ACCOMPANY THE PATIENT TO THE HOSPITAL.**
- Combative patient:
  - Apply soft restraints on all extremities using a quick-release knot.
  - Law enforcement must be notified if patient has been restrained.
- Initiate transport as soon as possible.
- Frequently reevaluate ABCs and Mental Status during transport.
- **DO NOT** secure the patient to the stretcher in the prone position.
- If the patient continues to be a threat to their own safety or EMS crew safety:
  - Administer ***Valium*** ***5 – 10 mg*** IV/IO/IM.
    - **DO NOT** administer Valium Intra-nasal.

OR

  - Administer ***Versed*** **2 – 5 mg** slow IV/IM/IN. **Max Dose – 5 mg.**

CONTINUED ON NEXT PAGE

Agitated/Behavioral Patient (Continued)

STOP

Contact Medical Control for Administration of Ketamine

For severe cases of Excited Delirium/Psychosis

- Administer ***Ketamine*** **1 – 4 mg/kg**. IM or **0.2 – 1 mg/kg** IV. **Max Dose – 400 mg.**
    - Monitor respiratory drive, systolic blood pressure and O_2 saturation closely.
- **For any additional dosing or if patient becomes combative after initial treatment, contact medical control or additional orders.**
- **Use Law Enforcement to restrain the patient, if possible, for provider safety. Have all airway equipment ready.**

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Allergic Reaction/Anaphylaxis

This protocol applies to adult patients presenting with rash, hives, shortness of breath, or other signs and symptoms, up to and including shock, possibly due to an allergic reaction.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See* ***Shock Protocol***
  - Take measures to prevent hypothermia.
- **If respiratory compromise and/or signs of shock, treat immediately with epinephrine. All EMS provider levels are authorized to utilize epinephrine auto-injectors. AEMT, CT and Paramedic providers may give 1:1,000 Epinephrine SQ or IM.**

Secondary Assessment and Treatment

- **ISOLATE THE PATIENT FROM THE SOURCE OF THE ALLERGEN**.
- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis. Treat if BGL is **< 60 mg/dl**.
- Cardiac monitor – record and evaluate 12 Lead ECG.
  - **DO NOT** delay treatments/therapy for 12-Lead acquisition.
- Consider $EtCO_2$ monitoring – Ideally > 20 mmHg.
- Physical Exam and OPQRST/SAMPLE history.
- Initiate IV/IO of normal saline
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See* ***Shock Protocol***.
- If localized reaction (hives)
  - Administer ***Diphenhydramine*** **25 – 50 mg** IV slowly or deep IM.
  - Administer ***Methylprednisolone*** **125 mg** IVP.
- If respiratory distress, along with above treatments:
  - Administer ***1:1,000 Epinephrine*** **0.3 – 0.5 mg** SQ.
  - Nebulize ***DuoNeb*****.**
    - May repeat once 20 minutes post initial ***DuoNeb*****.**
- If anaphylactic shock:
  - Do not delay epinephrine administration attempting IV/IO access.
  - ***1:1,000 Epinephrine*** **0.3 – 0.5 mg** IM (preferred) or SQ.
- **All levels may repeat 1:1,000 *Epinephrine* IM/SQ/auto-injector (in accordance with their scope of practice) every 5 minutes as needed.**
- If no response within 10 minutes to the IM or SQ ***Epinephrine*** and fluid bolus, administer ***1:10,000 Epinephrine*** **0.1 – 0.2 mg** IV/IO.

Anaphylaxis – a systemic response to an antigen involving two (2) or more organ systems OR any involvement of the upper and/or lower respiratory systems OR any derangement of vital signs.

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

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Altered Level of Consciousness

This protocol applies to adult patients who are disoriented, weak, dizzy, confused, agitated, exhibit bizarre behavior, have had a syncopal episode, or are unconscious.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq$ 94%.**
  - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See* ***Shock Protocol***
  - Take measures to prevent hypothermia.
- **Altered LOC and signs of poor perfusion could indicate shock and should be managed appropriately.**

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis. Treat if BGL is **< 60 mg/dl**.
  - If patient can protect and maintain own airway:
    - Administer ***Oral Glucose*** **15 grams** PO.
  - If patient cannot protect their own airway:
    - administer ***$D_{50}W$*** **25 g** IV/IO OR ***$D_{10}W$*** **250 ml** IV/IO.
  - If IV/IO cannot be established:
    - Administer ***Glucagon*** **1 mg** IM/SQ/IN.
- Cardiac monitor – record and evaluate 12 Lead ECG.
  - **DO NOT** delay treatments/therapy for 12-Lead acquisition.
- Consider $EtCO_2$ monitoring.
- Physical Exam and OPQRST/SAMPLE history.
- Initiate IV/IO of normal saline.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See* ***Shock Protocol***.
- Consider possible causes: (AEIOUTIPS) alcohol, electrolytes, insulin, opiates, uremia trauma, infection, poison, psychogenic, seizure and/or shock.
- If the patient appears intoxicated, drug overdose, or toxic ingestion, *See* ***Toxic Ingestion Protocol***.
- If patient requires restraint, apply soft restraints for crew and patient's protection.
- If patient shows continued respiratory depression, consider administration of ***Naloxone*** **0.4 mg** IV/IO/IN slowly. May repeat x 2 (titrate to patient's respirations)
- If patient becomes agitated or violent, *See* ***Agitated/Behavioral Protocol***.

STOP

Contact Medical Control for additional orders

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Chest Pain/STEMI

This protocol applies to adult patients presenting with chest pain/discomfort suspected to be ischemic in nature. This may include classic presentations or anginal equivalents (i.e. epigastric pain, neck and/or jaw pain, and indigestion).

Contact Medical Control for all pediatric care under this protocol.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See* ***Shock Protocol***
  - Take measures to prevent hypothermia.

Secondary Assessment and Treatment

- Place patient in position of comfort and minimize patient exertion.
  - If patient is hypotensive, place in supine position and manage shock appropriately. *See* ***Shock Protocol***.
- Monitor vital signs and O_2 saturation
- Cardiac Monitor
  - Obtain and evaluate 12-Lead ECG as soon as possible.
  - Transmit ECG to appropriate facility.
  - Monitor continuously until patient is in care of ER/Cath lab staff.
  - Treat arrhythmias under the appropriate protocol.
- Perform blood glucose analysis. Treat if BGL is **< 60 mg/dl**.
- Initiate $EtCO_2$ monitoring.
- Physical exam and OPQRST/SAMPLE history.
- Consider possible causes: (acute myocardial infarction, angina, aneurysm, pulmonary embolus, gastro-intestinal issues, etc.)
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See* ***Shock Protocol***.
  - Establish 2nd IV only if time permits.
- If patient is not allergic, administer ***Aspirin*** **324 mg** orally. (Four 81 mg baby aspirin)
  - Instruct the patient to chew before swallowing
  - Administer regardless of patient has taken aspirin prior to EMS arrival.

CONTINUED ON NEXT PAGE

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- 12-Lead ECG
- Treatment(s)
- Communication with Medical Control.

Chest Pain/STEMI (Continued)

Secondary Assessment and Treatment (Continued)

- **If the patient presents with an inferior AMI (S-T elevation in Leads II, III, and aVF). Both nitroglycerin and opiate administration may cause fatal hypotension in these patients. Obtain a right-sided 12-Lead ECG.**
- Administer ***Nitroglycerin*** **0.4 mg** SL. May repeat every 5 minutes as long as SBP remains > 100 mmHg and chest pain continues. **Max Dose 1.2 mg**
  - If the systolic blood pressure falls below 100 mmHg in response to nitroglycerin therapy:
    - Position the patient flat.
    - **DO NOT** administer additional nitroglycerin.
    - Administer 250 ml fluid bolus IV, repeat up to **1000 ml.** (if no pulmonary edema), titrate to keep systolic blood pressure > 110 mmHg.
  - **DO NOT administer nitroglycerin to any patient who has taken erectile dysfunction medication in the last 24 hours.**
- A report should be given to the Cardiac Facility ASAP with any patient presenting with S-T elevation.
  - Administer ***Morphine Sulfate*** **2 mg** IV slow push in 5 – 10 minute intervals. **Max Dose 10 mg.**
    - Medical Control may authorize even when systolic blood pressure is < 110 mmHg in certain circumstances.
- If nausea develops:
  - ***Ondansetron*** **4 mg** IV/IO/IM/IN
    - Repeat every 15 minutes if no relief. **Max Dose 8 mg.**
- Fluid Resuscitation
  - Additional fluid boluses for inferior wall MI only (beyond original 1000 ml) in 250 ml bolus increments then reassess for pulmonary edema between each bolus.
- If patient presents with Cardiogenic Shock:
  - Ensure rate and rhythm have been treated accordingly.
  - If the systolic blood pressure is < 90 mmHg, consider a ***Dopamine*** ***Infusion.***
    - Dopamine Infusion 2-10 mcg/kg/min, titrating to effect. *See* ***Dopamine Infusion Chart.***
- If the chest pain is thought to be stimulant-induced (cocaine, amphetamine, ecstasy) and pulse rate is > 120 BPM consider:
  - ***Midazolam*** **2.5 mg** slow push IV/IO/IN.
    - May repeat 2.5 mg IV/IO/IN. **Max Dose 5 mg.**
    - Cool patient passively but **DO NOT** allow patient to shiver.
- **Early notification and rapid transport to a cardiac facility capable of reperfusion therapy is paramount.**

Childbirth-OB/GYN Emergencies

This protocol applies to adult women whose chief complaint is related to labor and/or impending delivery.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq$ 94%.**
  - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See* ***Shock Protocol***
  - Take measures to prevent hypothermia.
- **Altered LOC and signs of poor perfusion could indicate shock and should be managed appropriately.**
- **If patient is unstable, initial resuscitation/stabilization must precede any action specified in this protocol. Resuscitation of the mother is the key to survival of both the mother and fetus.**

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis. Treat if BGL is **< 60 mg/dl**.
- Consider cardiac monitoring.
- Obtain OPQRST/SAMPLE, last menstrual period, obstetric, and gynecological history.
- **Determine how many previous deliveries, due date, onset of contractions, if membranes have ruptured, if bleeding or vaginal discharge is present, if patient has the urge to push or move bowels, and if pregnancy is high-risk.**
- Time contractions – frequency and duration.
- Physical exam – assess for signs of shock.
- IV/IO access with normal saline **–** initiate 20 ml/kg bolus.
  - If fluid resuscitation is needed, 20 ml/kg bolus. *See* ***Shock Protocol.***
- Place patient in position of comfort, **EXCEPT** for 3rd trimester patient who should be transported on their left side.
- Keep the patient NPO.
- If bleeding is present, apply ABD pad to the external vaginal area. Consider possible causes:
  - Ruptured ectopic pregnancy.
  - Spontaneous abortion.
  - Placenta abruption.
  - Trauma.
  - Abnormal menstrual flow.
- Bring any products of conception to the hospital with the patient.
- If active labor, inspect the perineum for crowning.
  - If crowning is present, apply gentle pressure with your gloved hand to the infant's head and prepare for delivery.
  - If no crowning is present, monitor and reassess frequency and duration of contractions.
- **Priority symptoms: Crowing < 36 weeks gestation, prolapsed cord, abnormal presentation, severe vaginal bleeding, multiple gestation or seizure.**
- **If noted, expedite transport and notify Medical Control as early as possible.**

CONTINUED ON NEXT PAGE

Childbirth-OB/GYN Emergencies (Continued)

Secondary Assessment and Treatment (Continued)

Eclampsia

- ***Magnesium Sulfate* – 4 gm** mixed in 100 ml of Normal Saline, administered over 10 – 20 minutes.
  - Should be titrated until seizure activity stops.
  - Any seizure refractory to ***Magnesium Sulfate*** should be treated following the ***Seizure Protocol***. *See also **Toxemia Protocol**.*

Pre-Eclampsia and Post-Partum Hypertension

- Strongly consider alternative cause for hypertension prior to treatment.
  - **You must contact Medical Control for authorization prior to blood pressure treatment. BP must remain elevated for two consecutive sets of vital signs. Discuss target blood pressure with Medical Control Physician.**
  - **Systolic Blood Pressure $\geq$ 140/90** with any upper right quadrant/epigastric or abdominal pain, blurred vision, or severe headache, consider ***Labetalol***.
  - **Systolic Blood Pressure $\geq$ 160/110,** consider ***Labetalol***.

Post-Partum Hemorrhage

- Firmly massage the uterine fundus after the placenta has delivered.
- If perineal (external vaginal area) lacerations are present, apply direct pressure and wound care, but **DO NOT** place dressings inside the vagina.
- Consider ***Tranexamic Acid***.

Birthing Complications

- **Breach Birth**
  - If feet or buttocks presentation – **DO NOT** pull on the infant.
  - Support the infant's presenting part(s).
  - Place a gloved hand inside the vagina and form a "V" with the first two fingers, place over the infant's face to keep the vaginal wall away from the infant's face.
- **Prolapsed Cord**
  - Place the mother in a knee chest position to relieve pressure on the umbilical cord.
  - To further reduce pressure on the umbilical cord, place your gloved hand inside the vagina and push upward on the presenting part.
  - Cover the umbilical cord with moist sterile dressing and avoid manipulation of the cord.
- **Shoulder Dystocia**
  - Position the mother on her left side in a dorsal-knee-chest position to increase the diameter of the pelvis or position the mother with buttocks off the edge of the stretcher with thighs flexed upward as much as possible.
  - Apply firm open hand pressure above the symphysis pubis.
  - Attempt to guide the infant's head downward to allow the anterior shoulder to slip under the symphysis pubis.
  - Gently rotate the fetal shoulder girdle into the wider oblique pelvic diameter. The posterior shoulder usually delivers without resistance.
  - Complete the delivery as above.
  - If delivery does not occur, maintain airway patency as best as possible and immediately transport to an appropriate facility.

CONTINUED ON NEXT PAGE

Childbirth-OB/GYN Emergencies (Continued)

Delivery and Post Delivery Care of Mother and Infant

- Maintain gentle pressure on the infant's head and allow it to deliver in a controlled gradual manner. **Routine suctioning of the oropharynx and nasopharynx as soon as the head is delivered is NO LONGER RECOMMENDED.**
- Check around the infant's neck for the umbilical cord.
  - If the cord is looped around the infant's neck, use your finger to hook the cord and pull it over the infant's head.
  - If unable to free the cord, clamp the cord in two places and cut the cord between the clamps.
- Gently direct the infant's head and body downward to deliver the anterior shoulder and support the rest of the body as it delivers.
- Keep the infant at the level of the vagina and use a sterile gauze pad to wipe any secretions around the mouth and nose.
- Vigorously dry the infant and provide warmth (increasing ambient temperature, cover with blankets).
- If needed, stimulate breathing by flicking the soles of the infant's feet or rubbing the infant's back.
- Clamp the cord at 4 and 6 inches and cut the cord between the clamps.
- Wrap the infant in dry, clean towels or blankets
- Note time of delivery. Obtain an APGAR Score at 1 and 5 minutes after delivery. (See below)
  - Score $\leq$ 3 – Critical
  - Score $\geq$ 7 – Good
- If excessive secretions AND signs of compromise are present, clear airway with bulb syringe.
- Once the placenta delivers, place it in a clean container and transport it to the hospital with the mother and infant.
- After delivery, keep the mother warm and watch for signs of shock.
- If excessive blood loss (> 500 ml):
  - Apply ABD pad to the external vaginal area.
  - Consider an additional fluid bolus
  - Massage uterus to promote uterine contraction.
  - Consider allowing mother to breast feed infant.
- **Transport to a facility capable of treating and managing an obstetrical patient and infant.**
- **If the newborn fails to respond to initial stimulation and are in need of resuscitation efforts, initiate resuscitation and refer to the *Newborn Resuscitation Protocol*.**

SCORE	0 points	1 point	2 points
Appearance (Skin color)	Cyanotic / Pale all over	Peripheral cyanosis only	Pink
Pulse (Heart rate)	0	<100	100-140
Grimace (Reflex irritability)	No response to stimulation	Grimace or weak cry when stimulated	Cry when stimulated
Activity (Tone)	Floppy	Some flexion	Well flexed and resisting extension
Respiration	Apneic	Slow, irregular breathing	Strong cry

Document:

- Vital Signs
- Obstetric History
- Frequency/duration of contractions
- Treatment(s)
- Communication with Medical Control

CONTINUED ON NEXT PAGE

OB/GYN Pearls

- **Eclampsia may occur up to 2 months post-partum. Always consider in pregnant/recently pregnant seizing patients.**
- **Severe headache, vision changes, edema, RUQ pain may indicate pre-eclampsia.**
- **In the setting of pregnancy, hypertension is defined as a systolic blood pressure > 140 mmHg or a diastolic blood pressure > 90 mmHg, or a relative increase of 30 mmHg systolic, or a 20 mmHg increase of diastolic pressure above the patient's pre-pregnancy blood pressure.**
- **Ask the patient to quantify bleeding – number of pads used per hour.**
- **Any pregnant patient involved in a MVC should be seen immediately by a physician for evaluation and fetal monitoring in a TRAUMA CENTER !!!!!!!**
- **Magnesium Sulfate may cause hypotension and decreased respiratory drive.**
- **If > 20 weeks pregnant, consider left lateral position for transport.**
- **Due to the following physiological changes during pregnancy, it is often difficult to assess for shock:**
  - **Mother's heart rate increases:**
    - **By 3rd trimester, heart rate can be 15 – 20 beats per minute above normal.**
    - **Both systolic and diastolic pressures drop by 5 – 15 mmHg during 2nd trimester.**
  - **The mother's cardiac output and blood volume increase:**
    - **The pregnant patient may lose 30 – 35% of their blood volume before the signs and symptoms of shock become apparent.**
  - **Supine hypotension usually occurs in the 3rd trimester.**

Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

Cold Related Emergencies

This protocol applies to adult patients having a body temperature below 95° F (35° C) secondary to environmental exposure.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
    - Use least invasive means possible to secure airway.
        - Intubate only, if necessary, as gently as possible.
    - Have suction equipment ready.
    - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq$ 94%.**
    - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – check for pulse – if no pulse, begin CPR.
- **It may be necessary to assess pulse and respirations for up to 30 – 45 seconds to confirm arrest.**
- If no pulse, initiate CPR and AED/Defibrillator using most current ***AHA Guidelines***.
    - If severe hypothermia (< 86° F/30° C) is strongly suspected, limit defibrillation attempts to 1 and **withhold** medications.
    - If body temperature is > 86° F/30° C treat in accordance with the ***Pulseless Arrest Protocol.***
    - Resuscitation efforts should continue until core temperature approaches normal.
- If pulse is present, **DO NOT** initiate CPR, no matter how slow the rate.
    - Treat bradycardia only if the patient is hypotensive.
- Carefully move patient to a warm environment, remove all wet clothing, dry the patient and cover with blankets.
- Avoid any rough movement that may cause cardiac dysrhythmias. It may be beneficial to immobilize the patient on a backboard.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Initiate cardiac monitoring.
- Physical Exam and OPQRST/SAMPLE history.
- Initiate IV/IO of **warm** normal saline at KVO rate.
    - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See **Shock Protocol**.*
- Apply hot packs to groin, axilla, neck and chest.

CONTINUED ON NEXT PAGE

Cold Related Emergencies (Continued)

Secondary Assessment and Treatment (Continued)

- Protect injured, frostbitten areas, **DO NOT** rub or place on heated surface.
    - Remove clothing and jewelry from injured parts.
    - **DO NOT** attempt to thaw injured part with localized heat.
    - Severe frostbite injuries should be transported to a trauma center.
- Consider ***Morphine Sulfate*** for pain relief when the patent is conscious, alert, is not hypotensive and is complaining of severe pain. *See* ***Pain Management Protocol*****.**
    - Administer ***Morphine Sulfate*** **2 mg** IV slow push in 5 – 10 minute intervals. **Max Dose 10 mg.**

Document:

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control

Dystonic Reaction

This protocol applies to adult patients who are experiencing a reaction to Phenothiazine (Compazine or Phenergan) or Butyrophene (Haldol) compound. The reaction is usually non-life threatening and results in no long-term effects. Extrapyramidal symptoms are a possible side effect of antipsychotic medications.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See **Shock Protocol***
  - Take measures to prevent hypothermia.
- **If respiratory compromise and/or signs of shock, treat immediately with Allergic Reaction/Anaphylaxis or Shock Protocol.**

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Monitor capnography
- Perform blood glucose analysis. Treat if BGL is **< 60 mg/dl**.
- Initiate cardiac monitoring.
- Physical exam and OPQRST/SAMPLE history.
- **Frequently, no clear history or medication use is obtainable; therefore, this reaction should be treated based on its clinical presentation. Although, Diphenhydramine (Benadryl) is used to treat this reaction, the reaction is not allergenic in nature. The patient should be informed of this distinction. Dystonic reactions may require field treatment to alleviate patient discomfort.**
- Signs and symptoms may present as the following:
  - Oculogyric crisis – deviation of eyes in all directions
  - Buccolingual crisis – protruding or pulling sensation of the tongue.
  - Trismus – clenching jaws.
  - Forced jaw opening
  - Difficulty speaking.
  - Facial grimace.
  - Torticollis – usually associated with oculogyric and buccolingual crisis.
  - Opisthotonic crisis – condition characterized by rigidity and severe arching of the back with the head thrown backwards.
  - Tortipelvic crisis – Typically involves the hips, pelvis and abdominal wall muscles.
    - Causes difficulty with ambulation.

CONTINUED ON NEXT PAGE

Dystonic Reaction (Continued)

Secondary Assessment and Treatment (Continued)

- Mental status is unaffected; symptoms may cause the patient to panic or have a sense of impending doom.
- Vital signs are usually normal.
- Remaining physical exam findings are normal.
- Advanced airway/ventilatory management as needed.
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See **Shock Protocol**.*
- **Use caution if administering Diphenhydramine (Benadryl) to patients taking MAO inhibitors. MAO inhibitors prolong and intensify the anticholinergic (drying) effects of Diphenhydramine. Common MAO inhibitors include Nardial, Marplan, Eutonyl, and Parnate.**
- ***Diphenhydramine* *(Benadryl)* 50 mg** IV slow push (preferred) or deep IM.

Document:

- Vital signs
- OPQRST/SAMPLE
- Signs/Symptoms
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Heat Related Emergencies

This protocol applies to adult patients with fatigue or altered level of consciousness secondary to environmental heat exposure.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See* ***Shock Protocol***
  - Remove the patient from the environment.
- **Altered LOC and signs of poor perfusion could indicate shock and should be managed appropriately.**

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Cardiac monitor
  - Record and evaluate 12-Lead ECG.
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Physical exam and OPQRST/SAMPLE history.
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See* ***Shock Protocol.***
- If conscious and not vomiting or extremely nauseous provide oral fluids.
- If heat stroke is suspected, initiate active cooling with cold packs water and fan.
- **Signs and symptoms of heat stroke may include hot dry skin (25% of patients will still be moist), seizures, altered mental status, dilated pupils, rapid heart rate or arrythmia.**
- Prepare for seizure activity. *See* ***Seizure Protocol*** for management of seizures.

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Acute Hypertensive Crisis

This protocol applies to adult patients demonstrating an acute, potentially life-threatening elevation of blood pressure with evidence of end-organ perfusion damage. Defined as a Systolic blood pressure greater than 190 mmHg and a Diastolic greater than 120 mmHg.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq$ 94%.**
  - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See* ***Shock Protocol***.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Cardiac monitor
  - Record and evaluate 12-Lead ECG.
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Physical exam and OPQRST/SAMPLE history.
  - Assess for evidence of end-organ perfusion damage (i.e. stroke, acute coronary syndrome/congestive heart failure, renal failure).
- **Signs and symptoms may include elevated blood pressure, headache, dizziness, nausea/vomiting, blurred vision, dyspnea, pulmonary/peripheral edema, etc.**
- Consider possible causes: chest pain, CHF, overdose, increased intra-cranial pressure, tachycardia.
- Initiate IV/IO of normal saline at KVO rate.
- If suspected overdose of cocaine or amphetamine use, you may consider **ONE** of the following medications.
  - ***Midazolam*** **2.5 mg** IV/IO/IN. May repeat 2.5 mg IV/IO/IN once. **Max Dose 5 mg.**
  - ***Valium*** **2 10 mg** IV/IO
  - ***Labetalol*** **20 mg** IV over 20 minutes.
- Any neurological defect(s) noted ?
  - If no neurological defects are noted, **DO NOT** attempt to lower blood pressure without contacting Medical Control.
    - Orders may include:
      - ***Nitroglycerin*** **0.4 mg** sublingual.

STOP

Contact Medical Control for Addition Orders

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Nausea/Vomiting

This protocol applies to adult patients presenting with acute onset of nausea and/or vomiting.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq$ 94%.**
  - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See* ***Shock Protocol***.
- **Altered LOC and signs of poor perfusion could indicate shock.**

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Cardiac monitor
  - Record and evaluate 12-Lead ECG.
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Physical exam and OPQRST/SAMPLE history.
  - Assess for signs of dehydration.
- Consider possible causes (Gastrointestinal, GU, cardiac, meds/toxic ingestion, pregnancy, etc.)
- Save emesis for signs of gastrointestinal bleed, etc.
- Keep the patient NPO
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See* ***Shock Protocol.***
  - Titrate to a systolic blood pressure of $\geq$ 90 mmHg.
- Consider ***Ondansetron*** **4 mg** IV/IO/IM/IN – repeat in 15 minutes if no relief. **Max Dose 8 mg**

OR

- ~~***Phenergan***~~ **12.5 – 25 mg** diluted in 100 ml of normal saline over 10 minutes, OR 25 mg IM for patients > 15 years-old.
  - ***Phenergan* is NOT to be administered direct IV push.**

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Respiratory Distress

This protocol applies to patients presenting with difficulty breathing with no history or signs of trauma.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - If known COPD, start with 2 – 6 lpm via nasal cannula and increase as needed
  - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See* ***Shock Protocol***.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Monitor capnography
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Cardiac Monitor
  - Record and evaluate 12-Lead ECG
- Physical exam and OPQRST/SAMPLE history.
- Consider possible causes (anaphylaxis, pulmonary edema, COPD, asthma, TB, etc.)
- **If patient is coughing, apply a surgical mask to the patient (if tolerated) and providers should don a N-95 mask**

Pulmonary Edema/CHF

- ***NTG*** **0.4 mg** SL every 5 minutes with SBP > 100mmHg.
- ***Lasix*** **1 mg/kg** up to a max of 100 mg IV, slowly titrated to SBP > 100 mmHg
- ***Morphine*** **1-4 mg** IV slowly titrated to a SBP > 100 mmHg
- Apply ***CPAP*** ASAP -*See* ***CPAP Protocol.***

COPD

- Reassure/calm patient
- Allow patient to assume position of comfort
- Assist patient in taking their own bronchodilators
- IV normal saline at KVO
- Nebulize **3 ml** ***DuoNeb*** breathing treatment. (May repeat once after 20 mins.)
- ***Albuterol*** **2.5 mg** after 2nd DuoNeb treatment (if needed)
- ***Methylprednisolone*** 125 mg
- Apply ***CPAP – See CPAP Protocol***

Asthma

- Reassure/calm patient
- Allow patient to assume position of comfort
- Assist patient in taking their own bronchodilators
- Consider humidified O_2
- IV normal saline at KVO
- Nebulize **3 ml** ***DuoNeb*** breathing treatment. (May repeat once after 20 mins.)
- ***Albuterol*** **2.5 mg** after 2nd DuoNeb treatment (if needed)
- ***Methylprednisolone*** 125 mg

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Respiratory Distress (Continued)

If inadequate ventilatory effort is present, assist ventilations with bag valve mask and consider advanced airway placement.

Pulmonary Edema/CHF

- Nebulize **3 ml** ***DuoNeb*** breathing treatment. (May repeat once)

Administer DuoNeb breathing treatment as initial dose. May repeat DuoNeb breathing treatment once, 20 minutes after initial DuoNeb treatment.

COPD

- ***Brethine*** **0.25 mg** SQ

Administer DuoNeb breathing treatment as initial dose. May repeat DuoNeb breathing treatment once, 20 minutes after initial DuoNeb treatment.

Asthma

- If status asthmatics, ***1:1,000*** ***Epinephrine*** **0.3 – 0.5 mg** SQ
- ***Magnesium Sulfate*** **2 gm** in 100 ml of normal saline given over 10 minutes
- ***Brethine*** **0.25 mg** SQ

Administer DuoNeb breathing treatment as initial dose. May repeat DuoNeb breathing treatment once, 20 minutes after initial DuoNeb treatment.

STOP

Contact Medical Control for additional orders

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Magnesium Sulfate is to be given IV, 2 gm in 100 ml of normal saline over 10 minutes. Magnesium is contraindicated in 2nd and 3rd degree heart blocks and hypotension. _Magnesium Sulfate_ should not be given to renal failure or suspected renal failure patients.

CPAP

- **Patients must be alert and able to maintain their own airway,**
- **With CPAP, most patients will improve in 5 – 10 minutes. If no improvement is noted within this timeframe, consider ventilations with BVM and advanced airway intervention.**

Continuous Positive Airway Pressure (CPAP)

This protocol applies to adult patients who are in respiratory distress and require positive pressure to maintain their airway.

Indications for CPAP

- Consider CPAP for patients in either severe or moderate respiratory distress in the setting of the following conditions:
  - Pneumonia
  - Asthma/COPD
  - Pulmonary Edema
  - CHF
- Presence of **two or more** of the following:
  - Respiratory rate > 25 breaths per minute
  - Pulse oximetry < 90%
  - Retractions or accessory muscle use are present
  - Lung exam reveals wheezes, rales or diminished breath sounds
  - Respiratory distress from any etiology.
- **Altered LOC and signs of poor perfusion could indicate shock and should be managed appropriately.**

Contraindications for CPAP

- Unable to follow commands
- APNEA
- Vomit or GI bleed
- Major trauma
- Signs and symptoms of a pneumothorax
- Age < 12 years of age or mask doesn't fit
- Cardiac or respiratory arrest
- Agonal respirations
- Inability to maintain a patent airway
- Hypotensive (systolic BP < 90 mmHg)
- Major Trauma (especially with face, neck, chest and/or abdomen involvement)
- Nausea/Vomiting
- Inability to sit upright.

Preparation

- Prepare for application of O-Two CPAP device or AHP 300 Ventilator
- Set pressure to 5 cmH_2O CPAP
- Prepare the head strap
- Attach system to O_2 supply
- Ensure the availability of suction, airway and ventilation assist devices
- Prepare to slowly increase and titrate pressure in 2.5 cmH_2O increments as clinically indicated.

CONTINUED ON NEXT PAGE

Continuous Positive Airway Pressure (Continued)

Application of CPAP

- Apply the oxygen charged O-Two CPAP device or AHP 300 Ventilator system to the patient without securing the head strap.
- Coach the patient to:
    - Hold the mask firmly to their own face.
    - Inhale through the nose and exhale through the mouth.
    - Continue to breathe evenly against the increasing pressure.
- Troubleshoot for leaks and adjust the mask to maintain a complete seal while applying the head strap.
- Confirm the integrity of the circuit, mask seal, and oxygen delivery to the patient
- Immediately re-assess the patient:
    - Pulmonary assessment including lung sounds, respiratory rate, and work of breathing.
    - Cardiovascular assessment including blood pressure and heart rate.
- Consider the need to medicate for anxiety associated with CPAP.
- Slowly and sequentially increase CPAP in 2.5 cmH_2O increments, every 5 – 10 minutes as clinically indicated by the patient's response to therapy. **Max CPAP 15 cmH_2O.**
- **Immediately** re-assess the patient following each change in pressure.
    - Pulmonary assessment including lung sounds, respiratory rate and work of breathing.
    - Cardiovascular assessment including blood pressure and heart rate.
- Titrate CPAP in 2.5 cmH_2O increments to relief of dyspnea while continuously assessing patient's tolerance of CPAP. **Max CPAP 15 cmH_2O.**
- If patient experiences increased anxiety and/or inability to maintain mask seal, consider **ONE** of the following:
    - ***Midazolam*** **1 – 5 mg** IV/IO **Max Dose 5 mg**

      OR
    - ***Valium*** **2 – 10 mg** IV/IO **Max Dose 10 mg.**

Critical Success Targets for CPAP

- **Improvement in patient's perceived work of breathing**
- **Resolution of, or improvement in patient's dyspnea**
- **Improved signs of perfusion**
- **$SpO_2 \geq 94\%$**

STOP

Contact Medical Control for additional orders

O-Two CPAP Mask	
5 lpm	5cmH_2O
10 lpm	10cmH_2O
15 lpm	15cm H_2O

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- CPAP settings
- Treatment(s)
- Improvements
- Communication with Medical Control.

Adult Sickle Cell Crisis

This protocol applies to adult patients presenting with sickle cell crisis. The typical sickle cell EMS call in adults is severe pain in the abdomen and chest, jaundice, swelling of hands, feet and joints, and/or difficulty breathing with hypoxia. Many of these patients are dehydrated.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq$ 94%.**
  - If known COPD, start with 2 – 6 lpm via nasal cannula and increase as needed
  - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See **Shock Protocol***.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Cardiac Monitor
  - Record and evaluate 12-Lead ECG
- Physical exam and OPQRST/SAMPLE history.
  - Obtain description of seizure activity
    - Duration and severity
  - Note any history of illness or trauma.
- Advanced airway/ventilatory management as needed.
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See **Shock Protocol***.

Pain Management

- If pain scale is $\geq$ 6, consider one of the following:
  - ***Morphine*** **0.1 mg/kg** IV/IM/IO. **Maximum Dose – 10 mg.**

    OR
  - ***Fentanyl*** **1 – 2 mcg/kg IV/IO/IM/IN. Maximum Dose – 200 mcg.**
- If respirations become depressed:
  - Administer ***Naloxone*** **0.01 mg/kg** IV/IO/IN.
    - If no clinical improvement, **0.1 mg/kg. Maximum Dose – 2 mg.**
    - Titrate Naloxone administration to patient's respirations.
- If patient becomes nauseated or vomits, consider administering:
  - ***Ondansetron*** **(Zofran) 0.1 mg/kg** IV/IM/ODT. **Maximum Dose – 4 mg.**

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Seizure

This protocol applies to adult patients actively seizing or those that have a history of seizures prior to EMS arrival.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - If known COPD, start with 2 – 6 lpm via nasal cannula and increase as needed
  - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See* ***Shock Protocol***.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Cardiac Monitor
  - Record and evaluate 12-Lead ECG
- Physical exam and OPQRST/SAMPLE history.
  - Obtain description of seizure activity
    - Duration and severity
  - Note any history of illness or trauma.
- Advanced airway/ventilatory management as needed.
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See* ***Shock Protocol.***
- If the patient is actively seizing, consider **ONE** of the following:
  - ***Midazolam*** **2 – 5 mg** IV/IO/IM/IN. May repeat once at 2.5 mg if seizing continues. **Max Dose 5 mg.**

    OR
  - ***Diazepam*** **2 – 5 mg** IV/IO/IM/IN. May repeat once at 5 mg if seizing continues. **Max Dose 5 mg.**

STOP

Contact Medical Control for Additional Orders if Seizure Activity Continues

Document:

- Vital signs
- History
- Description of seizures
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

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Stroke/CVA

This protocol applies to adult patients presenting with full or one-sided body weakness, facial droop, difficulty speaking, and/or altered mental status, occurring separately or in conjunction with each other.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
    - Have suction equipment ready.
    - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
    - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See **Shock Protocol***.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation
    - **DO NOT** attempt to lower blood pressure.
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Cardiac Monitor
    - Record and evaluate 12-Lead ECG
- Physical exam and OPQRST/SAMPLE history.
    - If possible, determine the time the patient was last seen normal.
    - Obtain the name and contact information of the witness.
- **Remember F.A.S.T. Use the Cincinnati Stroke Scale**
    - **Face: Check for facial droop.**
    - **Arms: Assess for extremity weakness.**
    - **Speech: Assess for slurred speech.**
    - **Time: Note when patient was last seen normal.**
- Rule out stroke mimics such as hypoglycemia, seizures, and head injury.
    - Consult the appropriate guidelines for treatment options.
- Reassure and calm the patient.
- If no trauma, place the patient in a position of comfort or in a left lateral recumbent position.
- Protect paralyzed extremities
- Limit on-scene time to < 10 minutes.
- Initiate IV/IO of normal saline at KVO rate.
    - Avoid primary IV access in paralyzed extremities.
- **Rapid transport to the closest Stroke Center or Hospital capable of TPA administration.**

Document:

- Vital signs, GCS, BGL
- Last time seen normal
- OPQRST/SAMPLE
- Cincinnati Stoke Scale
- Treatment(s)
- Communication with Medical Control.

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Toxemia

This protocol applies to the adult obstetrical patient experiencing hypertension and/or eclampsia activity. (Seizures, swelling/edema, visual hallucination or coma)

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq$ 94%.**
  - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See **Shock Protocol***.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Cardiac Monitor
  - Record and evaluate 12-Lead ECG
- Physical exam and OPQRST/SAMPLE and obstetric history.
- **Signs and symptoms may include elevated blood pressure, severe headache, dizziness, nausea/vomiting, blurred vision, dyspnea, edema, etc.**
- Advanced airway/ventilatory management as required
- Position patient on left side.
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See **Shock Protocol**.*
- **Expedite transport** (as gently as possible, no lights or sirens

For Active Seizures

- ***Magnesium Sulfate*** **2 – 4 g** IV mixed in 100 ml of normal saline administered over 10-20 minutes.
  - Should be titrated to cessation of seizure activity.

  OR
- ***Midazolam*** **2.5 mg** IV/IO slowly.
  - Contact Medical Control to repeat 2.5 mg dosing)

For Systolic blood pressure > 160 mmHg on two or more consecutive readings:

- ***Magnesium Sulfate*** **4 g** IV mixed in 100 ml of normal saline administered over 10-20 minutes.

For Systolic blood pressure > 110 mmHg and < 160 mmHg not responding to *Magnesium Sulfate*:

- Contact Medical Control

- **Monitor patient for hypotension, respiratory depression, and heart block when administering Magnesium Sulfate and/or benzodiazepines.**

Document:

- Vital signs/history
- Description of seizures
- Cardiac rhythm
- Treatment(s)
- Communications with Medical Control.

Toxic Exposure

This protocol applies to adult patients with toxic exposure secondary to ingestion, inhalation, contact or intravenous administration of a potentially toxic substance.

Scene Safety and Initial Management

- **Prevent exposure of EMS personnel ! Assess and ensure scene security prior to proceeding with this guideline.**
- If toxic environment, have patient moved to safety by appropriately trained personnel using the proper level of personal protective equipment (PPE).
- If signs of hazardous materials incident, call for HazMat team, keep patient(s) isolated in contaminated zone until HazMat team arrives.
  - Coordinate efforts with HazMat personnel.
- Identify agent and mechanism/route of exposure (Inhaled, contact, etc.)
- Decontaminate as appropriate
  - EMS personnel must be wearing the appropriate level of PPE prior to helping with the decontamination process.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See **Shock Protocol***.
  - Take measures to prevent hypothermia, especially following decontamination.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- **Pulse oximetry may not be accurate for toxic inhalation victims.**
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Cardiac Monitor
  - Record and evaluate 12-Lead ECG
- Physical exam and OPQRST/SAMPLE and obstetric history.
  - Identify substance/toxin and amount of exposure
  - Determine mechanism, time, and duration of exposure.
  - If ingested, *See **Poisoning/Overdose Protocol**.*
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See **Shock Protocol**.*

CONTINUED ON NEXT PAGE

Toxic Exposure (Continued)

Secondary Assessment and Treatment (Continued)

- If known or suspected carbon monoxide poisoning:
  - Provide 100% high flow O_2.
  - Consult Medical Control for destination choice, including consideration of medical facilities equipped with hyperbaric capability.
- If organophosphate, carbamate, or nerve agent poisoning:
  - Administer ***Atropine*** **2 – 5 mg/kg** IV/IO/IM every 10-15 minutes, titrate to clinical symptoms (drying of secretions)
  - Contact **Georgia Poison Control** 1-800-222-1222 for consultation and/or **CHEMPACK** Deployment. *See* ***CHEMPACK*** in resources.
- If patient is asymptomatic, monitor for delayed effects.
- Frequently re-assess patient, mange any presenting respiratory distress, seizures, and/or dysrhythmias in accordance with the appropriate Protocol.

- **All suspected suicide attempts must be reported to law enforcement before leaving scene.**

- **EMS personnel may contact Poison Control directly. EMS personnel are directed to follow the advice offered by Poison Control Center as if it came directly from Medical Control. Georgia Poison Control: 1-800-222-1222.**

Document:

- Vital signs
- Agent/mechanism of exposure
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control/Poison Control

Toxic Ingestion

This protocol applies to adult patients with an acute overdose or toxic ingestion.

Primary Survey

- Assess level of consciousness – AVPU
- Ensure airway – ensure patency and positioning
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- Assess breathing – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Assist with bag valve mask if respiratory effort is ineffective.
- Assess circulation – control bleeding and manage shock appropriately. *See **Shock Protocol***.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Cardiac Monitor
  - Record and evaluate 12-Lead ECG
- Physical exam and OPQRST/SAMPLE history.
  - Identify substance/toxin and amount of exposure.
- Keep the patient NPO.
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See **Shock Protocol**.*

If calcium-channel blocker or beta blocker overdose:

- ***Glucagon*** **1 mg** IM/SQ/IN

If tricyclic antidepressant overdose with wide complex tachycardia:

- ***Sodium Bicarbonate*** **1 mEq/kg** IV/IO slowly.

If narcotic overdose:

- ***Naloxone*** **0.4 mg** IV/IO/IM/IN may repeat x 2 (titrated to patient's respirations)

If stimulant/hallucinogen overdose (cocaine, amphetamine, ecstasy, etc.):

- ***Midazolam*** **2.5 mg** IV/IO/IM/IN
  - Contact Medical Control to repeat 2.5 mg dose.
  - Cool patient passively but **DO NOT** allow the patient to shiver.

All suspected suicide attempts must be reported to law enforcement before leaving scene.

EMS personnel may contact Poison Control directly. EMS personnel are directed to follow the advice offered by Poison Control Center as if it came directly from Medical Control. Georgia Poison Control: 1-800-222-1222.

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

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Multiple System Trauma

This protocol applies to adult patients presenting with injury to more than one body system.

Primary Survey

- **Massive Hemorrhage**
  - Treat all **life-threatening** hemorrhage. *See* ***Hemorrhage Control Protocol.***
    - Tourniquets for massive extremity hemorrhage
    - Wound packing or junctional tourniquet for junctional hemorrhage
- **Airway** – ensure patency and positioning
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- **Respiratory/breathing** – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Manage any injuries impairing ventilation immediately.
    - Chest wounds: Open/Sucking, Impaled objects, bruising
    - Signs of Pneumothorax
    - Flail Segments
  - Assist with bag valve mask if respiratory effort is ineffective.
- **Circulation** – Assess radial pulses and heart rate.
  - Consider Pelvic Binder if appropriate.
- **Head/Hypothermia**
  - Initial GCS/AVPU
  - Pupils
  - Prevent hypothermia

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Cardiac Monitor
  - Record and evaluate 12-Lead ECG
- Physical exam and OPQRST/SAMPLE history.
  - Expose and rapidly assess the head, chest abdomen, pelvis and extremities for injury.
    - Evaluate the patient's posterior when possible.
- Perform spinal motion restriction, rigid C-collar and secure to a long spine board if indicated.
- Initiate patient transport to a trauma center as soon as possible.
- Advanced airway/ventilatory support as needed. *See* ***Sedation Assisted Intubation Protocol.***
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See* ***Shock Protocol.***
- Re-evaluate ABCs and perform detailed/focused assessment of the head, neck, chest, abdomen, pelvis, and all extremities.
  - Repeat neurological exam.
- Continue all resuscitation efforts and evaluations enroute.
- Manage any presenting respiratory distress, seizures, and/or dysrhythmias in accordance with the appropriate guideline.

Document:

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control

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Head and Spine Injury

This protocol applies to adult patients presenting with injuries to the head and/or spine.

Primary Survey

- **Massive Hemorrhage**
  - Treat all **life-threatening** hemorrhage. *See* ***Hemorrhage Control Protocol.***
    - Tourniquets for massive extremity hemorrhage.
    - Wound packing or junctional tourniquet for junctional hemorrhage.
- **Airway** – ensure patency and positioning.
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- **Respiratory/breathing** – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Manage any injuries impairing ventilation immediately.
    - Chest wounds: Open/Sucking, Impaled objects, bruising.
    - Signs of Pneumothorax.
    - Flail Segments.
  - Assist with bag valve mask if respiratory effort is ineffective.
- **Circulation** – Assess radial pulses and heart rate.
  - Consider Pelvic Binder if appropriate.
- **Head/Hypothermia**
  - Initial GCS/AVPU.
  - Pupils.
  - Prevent hypothermia.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation.
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Cardiac Monitor
  - Record and evaluate 12-Lead ECG.
- Physical exam and OPQRST/SAMPLE history.
  - Expose and rapidly assess the head, chest abdomen, pelvis and extremities for injury.
    - Evaluate the patient's posterior when possible.
- Perform spinal motion restriction, rigid C-collar and secure to a long spine board if indicated.
- Initiate patient transport to a trauma center as soon as possible.
- Advanced airway/ventilatory support as needed. *See* ***Sedation Assisted Intubation Protocol.***
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See* ***Shock Protocol.***
  - If suspected Traumatic Brain Injury, titrate fluids to maintain a systolic blood pressure **≥ 110 mmHg.**

A single incident of hypotension in an adult with a traumatic brain injury may increase the mortality rate by 150%.

- Perform a detailed physical assessment of the patient.
  - Re-evaluate ABCs.
  - Repeat neurological examination.

CONTINUED ON NEXT PAGE

Head and Spine Injury (Continued)

Secondary Assessment and Treatment (Continued)

- **Frequently reassess for clinical signs of cerebral herniation:**
  - **One or both pupils dilated and/or unreactive to light.**
  - **Abnormal posturing (Decorticate or Decerebrate).**
  - **No motor response.**
  - **Decrease in GCS > 2 points when their GCS was initially < 9.**
  - **Monitor for Cushing's Triad (Severe hypertension, bradycardia, irregular respirations).**
- Hyperventilation therapy titrated to clinical effect may be necessary for brief periods in cases of cerebral herniation or acute neurological deterioration.
  - Hyperventilation is administered as:
    - 20 breaths per minute in an adult.
    - Maintain an $EtCO_2$ of 30-35 mmHg.
- Manage any presenting seizures in accordance with the ***Seizure Protocol.***
- If patient presents with bradycardia secondary to increased ICP or Neurogenic Shock, consult with Medical Control regarding management.

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Eye Trauma

This protocol applies to adult patients with blunt or penetrating trauma to the eye(s) or who have chemical substances in the eye(s).

Primary Survey

- **Massive Hemorrhage**
  - Treat all **life-threatening** hemorrhage. *See* ***Hemorrhage Control Protocol.***
    - Tourniquets for massive extremity hemorrhage.
    - Wound packing or junctional tourniquet for junctional hemorrhage.
- **Airway** – ensure patency and positioning.
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- **Respiratory/breathing** – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Manage any injuries impairing ventilation immediately.
    - Chest wounds: Open/Sucking, Impaled objects, bruising.
    - Signs of Pneumothorax.
    - Flail Segments.
  - Assist with bag valve mask if respiratory effort is ineffective.
- **Circulation** – Assess radial pulses and heart rate.
  - Consider Pelvic Binder if appropriate.
- **Head/Hypothermia**
  - Initial GCS/AVPU.
  - Pupils.
  - Prevent hypothermia.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Cardiac Monitor
  - Record and evaluate 12-Lead ECG.
- Physical exam and OPQRST/SAMPLE history.
  - Expose and rapidly assess the head, chest abdomen, pelvis and extremities for injury.
    - Evaluate the patient's posterior when possible.
- Perform spinal motion restriction, rigid C-collar and secure to a long spine board if indicated.
- Initiate patient transport to a trauma center as soon as possible.
- Advanced airway/ventilatory support as needed. *See* ***Sedation Assisted Intubation Protocol.***
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See* ***Shock Protocol.***
- **Assess vision, if possible, with the injured eye.**
  - **Can the patient count the number of fingers you hold up, if not, can the patient perceive light.**
  - **Never apply pressure to the eyeball.**

CONTINUED ON NEXT PAGE

Eye Trauma (Continued)

Secondary Assessment and Treatment (Continued)

- **If the eye has been avulsed or if the globe has been ruptured,**
    - Carefully cover the injured eye to protect it.
    - Prevent conjugated eye movements – also cover the uninjured eye.
    - **DO NOT** apply any pressure.
    - **DO NOT** apply any absorbent dressing.
- **If a foreign body is embedded in the eye,**
    - **DO NOT** attempt to remove the object.
    - Attempt to stabilize the object.
    - Carefully cover both eyes.
- **If the eyes are injured by chemical exposure, pepper spray or mace,**
    - Responders should protect themselves with appropriate personal protective equipment.
    - Remove the victim from the source of exposure.
    - Remove contaminated clothing and seal in plastic bags.
    - Irrigate eyes with copious amounts of sterile water or normal saline.
- Transport the patient with head elevated about 30 degrees, and **BOTH** eyes closed or loosely patched (Unless irrigating).
- Treat pain as needed. *See* ***Pain Management Protocol.***
    - **DO NOT** treat eye injury pain with NSAIDs (**Toradol**).

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Chest Trauma

This protocol applies to adult patients presenting with chest trauma.

Primary Survey

- **Massive Hemorrhage**
  - Treat all **life-threatening** hemorrhage. *See* ***Hemorrhage Control Protocol.***
    - Tourniquets for massive extremity hemorrhage.
    - Wound packing or junctional tourniquet for junctional hemorrhage.
- **Airway** – ensure patency and positioning.
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- **Respiratory/breathing** – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Manage any injuries impairing ventilation immediately.
    - Chest wounds: Open/Sucking, Impaled objects, bruising.
      - Seal with a gloved hand.
      - Apply an occlusive dressing.
      - Monitor for signs of tension pneumothorax.
    - Signs of Pneumothorax.
      - Perform needle decompression on affected side.
    - Flail Segments.
      - Stabilize with bulky dressing, gentle pressure. **DO NOT** impair ventilation.
  - Assist with bag valve mask if respiratory effort is ineffective.
- **Circulation** – Assess radial pulses and heart rate.
  - Consider Pelvic Binder if appropriate.
- **Head/Hypothermia**
  - Initial GCS/AVPU.
  - Pupils.
  - Prevent hypothermia.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Cardiac Monitor
  - Record and evaluate 12-Lead ECG.
- Physical exam and OPQRST/SAMPLE history.
  - Expose and rapidly assess the head, chest abdomen, pelvis and extremities for injury.
    - Evaluate the patient's posterior when possible.
- Perform spinal motion restriction, rigid C-collar and secure to a long spine board if indicated.
- Initiate patient transport to a trauma center as soon as possible.
- Advanced airway/ventilatory support as needed. *See* ***Sedation Assisted Intubation Protocol.***
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See* ***Shock Protocol.***

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Chest Trauma (Continued)

Secondary Assessment and Treatment (Continued)

- Reevaluate ABCs and perform detailed/focused assessment.
  - Frequently reevaluate patient's respiratory and perfusion status.
  - Auscultate breath sounds.
  - Apply capnography as needed.
    - Maintain $EtCO_2$ of 35-45 mmHg.
- Continue resuscitation and evaluation enroute.

IMPORTANT – NEEDLE CHEST DECOMPRESSION

Indications

- Peri-arrest, PEA, shock with hypotension and **at least one of the following:**
  - Neck Vein distention
  - Tracheal deviation away from the injured side.
  - Increased resistance when ventilating.
  - Hyper-expanded chest with little movement on respiration.

- **If cardiac arrest is present, and ACLS has been initiated, DO NOT terminate efforts unless needle chest decompression is performed.**

- **Needle chest decompression should never be utilized based solely on the presence of poor or absent breath sounds on one side of the chest. The procedure has complications and should not be used lightly. However, when used appropriately, it can be life-saving.**

- **Overly aggressive positive pressure ventilation may cause a pneumothorax or exacerbate and existing pneumothorax.**

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Abdominal and Pelvic Trauma

This protocol applies to adult patients presenting with injury to the abdomen and/or pelvis.

Primary Survey

- **Massive Hemorrhage**
  - Treat all **life-threatening** hemorrhage. *See* ***Hemorrhage Control Protocol.***
    - Tourniquets for massive extremity hemorrhage.
    - Wound packing or junctional tourniquet for junctional hemorrhage.
- **Airway** – ensure patency and positioning.
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- **Respiratory/breathing** – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq$ 94%.**
  - Manage any injuries impairing ventilation immediately.
    - Chest wounds: Open/Sucking, Impaled objects, bruising.
    - Signs of Pneumothorax.
    - Flail Segments.
  - Assist with bag valve mask if respiratory effort is ineffective.
- **Circulation** – Assess radial pulses and heart rate.
  - Consider Pelvic Binder if appropriate.
- **Head/Hypothermia**
  - Initial GCS/AVPU.
  - Pupils.
  - Prevent hypothermia.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Cardiac Monitor
  - Record and evaluate 12-Lead ECG.
- Physical exam and OPQRST/SAMPLE history.
  - Expose and rapidly assess the head, chest abdomen, pelvis and extremities for injury.
    - Evaluate the patient's posterior when possible.
    - Note any abdominal rigidity, distention, tenderness, etc.
    - Note any pelvic instability.
- Perform spinal motion restriction, rigid C-collar and secure to a long spine board if indicated.
- Initiate patient transport to a trauma center as soon as possible.
- Advanced airway/ventilatory support as needed. *See* ***Sedation Assisted Intubation Protocol.***
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See* ***Shock Protocol.***

CONTINUED ON NEXT PAGE

Abdominal and Pelvic Trauma (Continued)

Secondary Assessment and Treatment (Continued)

- **For evisceration:**
  - **DO NOT** attempt to replace protruding organs.
  - Apply a moistened sterile dressing directly to the site.
  - Cover the moistened sterile dressing with an occlusive dressing.
  - Place the patient on their back, with legs flexed at the knees to reduce pain.
- **For Impaled objects:**
  - Carefully cut away any clothing that is around the object.
  - Manually stabilize the object avoid applying any pressure to the object.
  - Use bulky dressings and cravats to stabilize the object.
  - Minimize patient movement.
- **If impaled object has been removed prior to EMS arrival, make every attempt to transport the object with the patient.**
- Continue resuscitation and evaluation enroute.

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Extremity Trauma

This protocol applies to adult patients presenting with extremity trauma.

Primary Survey

- **Massive Hemorrhage**
  - Treat all **life-threatening** hemorrhage. *See* ***Hemorrhage Control Protocol.***
    - Tourniquets for massive extremity hemorrhage.
    - Wound packing or junctional tourniquet for junctional hemorrhage.
- **Airway** – ensure patency and positioning.
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- **Respiratory/breathing** – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Manage any injuries impairing ventilation immediately.
    - Chest wounds: Open/Sucking, Impaled objects, bruising.
    - Signs of Pneumothorax.
    - Flail Segments.
  - Assist with bag valve mask if respiratory effort is ineffective.
- **Circulation** – Assess radial pulses and heart rate.
  - Consider Pelvic Binder if appropriate.
- **Head/Hypothermia**
  - Initial GCS/AVPU.
  - Pupils.
  - Prevent hypothermia.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation
- Perform blood glucose analysis – Treat if BGL is **< 60 mg/dl**.
- Cardiac Monitor
  - Record and evaluate 12-Lead ECG.
- Physical exam and OPQRST/SAMPLE history.
  - Expose and rapidly assess the head, chest abdomen, pelvis and extremities for injury.
    - Evaluate the patient's posterior when possible.
- Perform spinal motion restriction, rigid C-collar and secure to a long spine board if indicated.
- Initiate patient transport to a trauma center as soon as possible.
- Advanced airway/ventilatory support as needed. *See* ***Sedation Assisted Intubation Protocol.***
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See* ***Shock Protocol.***
- **For fractures or dislocations:**
  - Assess distal pulse motor and sensation before and after splinting and throughout transport.
  - If open fracture, control bleeding and cover with dry sterile dressing.
  - If the extremity is severely angulated **AND** pulses are absent, apply gentle traction in an attempt to restore distal pulses.
    - If pulses are present or if resistance is encountered, splint the extremity in the angulated position.
  - Apply appropriate splinting device(s).
  - To reduce swelling, elevate extremity and apply cold pack(s).

CONTINUED ON NEXT PAGE

Extremity Trauma (Continued)

Secondary Assessment and Treatment (Continued)

- **For amputation:**
  - If located, initiate care for both the patient as well as the amputated part.
    - Remove gross contaminants by rinsing with saline.
    - Wrap in saline moistened gauze and place in a plastic bag or container.
    - Seal the bag or container tightly and place in solution of ice water, if available.
    - Transport the amputated part to the hospital regardless of condition.
    - If the amputated part cannot be located immediately, transport the patient and other field providers search for and transport the part as soon as possible.
- Continue resuscitation and evaluation enroute.
- For pain control, *See* ***Pain Management Protocol.***

Document:

- Vital signs
- OPQRST/SAMPLE
- Vital signs
- Treatment(s)
- Communication with Medical Control.

Trauma Arrest

This protocol applies to adult trauma patients with absent vital signs.

Patients with injuries incompatible with life are covered under the *Withholding or Termination of Resuscitation* Protocol.

Primary Survey

- In the event of multiple patients, ensure you have the appropriate resources and have triaged all other patients before committing to the treatment of a traumatic arrest patient.
- Assess for signs of life.
- Control life-threatening hemorrhage.
  - Tourniquets for massive extremity injuries.
- Begin high quality CPR and restrict interruptions of compressions as much as possible. Apply automated compression device if available.
- Ensure airway/ventilatory support.
  - A blind insertion airway device (BIAD) or supra-glottic airway (SGA) may be inserted early, otherwise ventilate with a bag valve mask and 100% O_2.
- **DO NOT** attempt insertion of an endotracheal tube for the first 5 minutes of the resuscitation attempt, except in the presence of stridor.
- Ventilate with 100% O_2 only until the chest rises. Ventilate at a rate of 6-8 breaths per minute. **DO NOT over ventilate**.
- **Airway management, bleeding control and rapid transport are the most important interventions for victims of traumatic arrest.**
- **Minimize scene time to 10 minutes or less, barring extrication time, and perform only critical interventions before transport.**

Secondary Assessment and Treatment

- Physical exam and OPQRST/SAMPLE history.
  - Expose and rapidly assess the head, chest abdomen, pelvis and extremities for injury.
    - Evaluate the patient's posterior when possible.
- Perform spinal motion restriction, rigid C-collar and secure to a long spine board if indicated.
- Initiate patient transport as soon as possible.
- Continue protocol(s) enroute.
- Initiate cardiac monitoring.
  - Manage dysrhythmias per appropriate protocol.
- Advanced airway/ventilatory support as needed.
  - Initiate $EtCO_2$ monitoring.
    - Maintain $EtCO_2$ of **30-35 mmHg**.
- Continue with chest compressions until return of adequate pulses.
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See **Shock Protocol**.*
  - If palpable pulse is restored, titrate the IV infusion to a systolic of blood pressure of 80-90 mmHg.
- If mechanism of injury, symptoms and physical exam suggest a tension pneumothorax, consider needle decompression to the affected side(s).

CONTINUED ON NEXT PAGE

Trauma Arrest (Continued)

Secondary Assessment and Treatment (Continued)

- Transport patient to the closest appropriate facility.
- Contact Medical Control with patient status and treatment(s) as soon as possible.
- If the patient remains in persistent Asystole, contact Medical Control for ***Termination of Resuscitation*** protocol.

IMPORTANT – NEEDLE CHEST DECOMPRESSION

Indications

- Peri-arrest, PEA, shock with hypotension and **at least one of the following:**
    - Neck Vein distention
    - Tracheal deviation away from the injured side.
    - Increased resistance when ventilating.
    - Hyper-expanded chest with little movement on respiration.

If cardiac arrest is present, and ACLS has been initiated, DO NOT terminate efforts unless needle chest decompression is performed.

Needle chest decompression should never be utilized based solely on the presence of poor or absent breath sounds on one side of the chest. The procedure has complications and should not be used lightly. However, when used appropriately, it can be life-saving.

Overly aggressive positive pressure ventilation may cause a pneumothorax or exacerbate and existing pneumothorax.

Document:

- Vital signs
- Mechanism of Injury
- ROSC/vital signs changes if relevant
- Cardiac rhythm(s)
- Treatment(s)
- Advanced airway procedures with $EtCO_2$.
- Medications administered
- Available history
- Communication with Medical Control.

Burns

This protocol applies to adult patients who have sustained thermal, chemical or electrical burns and/or have sustained inhalation injuries.

- **Hypotension is not normally seen with prehospital burn patients. Hypotension suggests other trauma. Refer to the trauma protocol(s) as needed.**
- Ensure scene safety
- Remove patient from the burning process, only if properly trained and equipped to do so.

Primary Survey

- **Massive Hemorrhage**
  - Treat all **life-threatening** hemorrhage. *See* ***Hemorrhage Control Protocol.***
    - Tourniquets for massive extremity hemorrhage.
    - Wound packing or junctional tourniquet for junctional hemorrhage.
- **Airway** – ensure patency and positioning.
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- **Respiratory/breathing** – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Manage any injuries impairing ventilation immediately.
    - Look closely for evidence of inhalation injury (hoarseness, stridor, sooty sputum, facial burns, or singed nasal or facial hair).
    - Airway burns may require aggressive airway management.
    - Consider false pulse oximetry readings due to possible carbon monoxide exposure.
  - Assist with bag valve mask if respiratory effort is ineffective.
- **Circulation** – Assess radial pulses and heart rate.
  - Consider Pelvic Binder if appropriate.
- **Head/Hypothermia**
  - Initial GCS/AVPU.
  - Pupils.
  - Prevent hypothermia.

Secondary Assessment and Treatment

- Remove and secure any jewelry, belts, shoes, etc. from burned areas.
  - Remove any burned or singed clothing not stuck to skin.
- Initial care for burn wounds:
  - **Chemical Injury** – brush off chemical, flush with water to remove any residual chemical.
    - Ensure chemical **IS NOT** water reactive before flushing with water.
  - **Electrical Injury** – treat dysrhythmias per appropriate cardiac dysrhythmia protocol.
    - Electrical Injuries are very difficult to estimate TBSA.
  - **Thermal Injury** – dry sterile dressings and/or burn sheets.
- Begin transport as soon as possible. Consider aeromedical transport.
  - If no other trauma mechanism, transport to a burn center.
  - If trauma mechanism exists, consider transport to a trauma center.
  - Transport patients with an unmanageable airway, uncontrolled hemorrhage, and/or hemodynamic instability to the closest hospital emergency department.

CONTINUED ON NEXT PAGE

Burns (Continued)

Secondary Assessment and Treatment (Continued)

- Advanced airway/ventilatory support as needed.
  - Initiate $EtCO_2$ monitoring.
  - SpO_2 readings may not be accurate due to carbon monoxide exposure.
- Initiate cardiac monitoring.
  - Manage dysrhythmias per appropriate protocol.
- Assess for the following;
  - Possible carbon monoxide poisoning. Signs and symptoms may include any of the following:
    - Headache.
    - Nausea and/or vomiting.
    - Confusion or disorientation.
    - Blurred Vision.
    - Loss of Consciousness.
    - Dizziness.
    - Weakness and fatigue.
    - Shortness of breath.
    - Chest pain.
  - Heat inhalation injury. Signs and symptoms may include any of the following:
    - Hoarseness or loss of voice.
    - Burns to the lips, mouth or tongue.
    - Swelling or redness inside the oropharynx.
    - Persistent coughing or wheezing.
    - Stridor.
    - Singed facial hair, nasal hair.
    - Dark gray or black sputum.
    - Difficulty swallowing/pain.
    - Shortness of breath or tachypnea.
  - Total Body Surface Area Estimation.
    - Rule of 9's, or Palmar Method
      - Estimate total body surface area burned.
        - TBSA calculations - only use Partial Thickness burns (2nd degree) and higher (3rd and 4th degree) for fluid resuscitation calculations.
    - Depth of burns
      - Superficial – 1st Degree
      - Partial Thickness – 2nd Degree
      - Full Thickness – 3rd Degree
      - Fourth Degree – extends beyond skin and into muscle and/or bone.
- Initiate IV/IO of normal saline at KVO rate.
  - ***Lactated Ringers*** is preferred for burn resuscitation.
  - Use the Parkland Formula, Consensus Formula, or USAISR Rule of Ten for calculating flow rates. (See formulas on next page)
  - **DO NOT** delay transport to establish IV access.
- For pain management, *See **Pain Management Protocol.***

Document:

- Vital signs
- Burn type, location, size and depth.
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Burns (Continued)

Parkland Formula

4 ml/kg x weight(Kg) x % TBSA Burned

2

This will give you the total milliliters of crystalloid needed to be administered in the first 8 hours.

Consensus Formula

2 – 4 ml/kg x weight (Kg) x % TBSA Burned

2

This will give you the total milliliters of crystalloid needed to be administered in the first 8 hours.

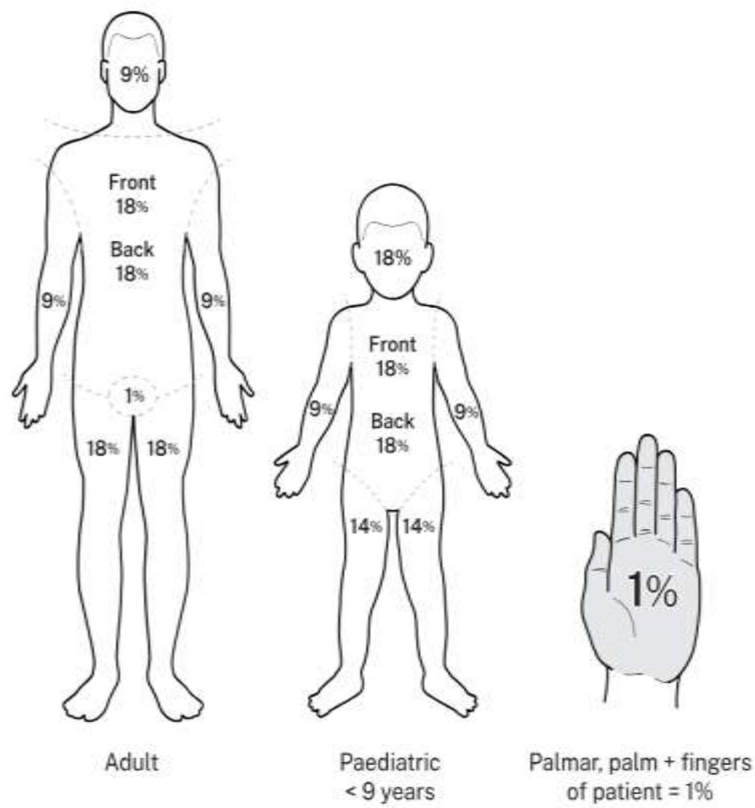
USAISR Rule of Ten

- **TBSA calculated and rounded to the nearest 10. (i.e. 37% TBSA would be rounded to 40.)**
- **That rounded percentage is multiplied by 10.**
- **This quick calculation will give you the number of milliliters per hour of crystalloid to be administered.**
- **This rule only applies to patients weighing 88-154 lbs. (40-70 kg). For any patient exceeding this weight range, for each 10kg in body weight over 70 kg, an additional 100 ml/hr. is given.**

Palmar Method

The patient's palm + fingers = 1%

TBSA Estimation Methods



- **DO NOT exceed 2 liters of crystalloid IV solution unless authorized by Medical Control.**
- **Contact Medical Control for fluid orders in patients with congestive heart failure or cardiac disease.**

Snakebite

Special Note: Safety of rescue personnel is top priority ! Ensure scene safety and determine location of snake. DO NOT transport snake. (A picture will suffice.) DEAD SNAKES ARE STILL DANGEROUS.

Primary Survey

- **Massive Hemorrhage**
  - Treat all **life-threatening** hemorrhage. *See **Hemorrhage Control Protocol.***
- **Airway** – ensure patency and positioning.
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- **Respiratory/breathing** – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Manage any injuries impairing ventilation immediately.
  - Assist with bag valve mask if respiratory effort is ineffective.
- **Circulation** – Assess radial pulses and heart rate.
  - Consider Pelvic Binder if appropriate.
- **Head/Hypothermia**
  - Initial GCS/AVPU.
  - Pupils.
  - Prevent hypothermia.

Secondary Assessment and Treatment

- Place patient in position of comfort and minimize patient exertion.
  - If patient is hypotensive, place in supine position and manage shock appropriately. *See **Shock Protocol**.*
- Monitor vital signs and O_2 saturation
- Cardiac Monitor
  - Obtain and evaluate 12-Lead ECG as soon as possible.
  - Treat arrhythmias under the appropriate protocol.
- Perform blood glucose analysis. Treat if BGL is **< 60 mg/dl**.
- Initiate $EtCO_2$ monitoring.
- Physical exam and OPQRST/SAMPLE history.
  - Assess for swelling, skin color changes, and/or shock.
  - Mark on skin the leading edge of swelling and erythema and record time.
  - Repeat if leading edge is progressing.
  - If able, safely determine type, size and length of snake.
- Advanced airway/ventilatory support as needed.
  - Initiate $EtCO_2$ monitoring.
  - Maintain $EtCO_2$ of **30-35 mmHg**
- **DO NOT** place bitten extremity in an elevated or lowered position.
- Clean wound, apply light dressing, unless wound is bleeding profusely.
  - **No ice, No constricting bands, No cutting of bite.**
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See **Shock Protocol**.*
- For pain management, *See **Pain Management Protocol.***

Document:

- Vital Signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Submersion Event

Applies to any adult patient that has been submerged under water for any period of time.

Special Note: Safety of rescue personnel is top priority! Enter water only if trained to do so, only if equipped and as a last resort.

Primary Survey

- **Massive Hemorrhage**
  - Treat all **life-threatening** hemorrhage. *See* ***Hemorrhage Control Protocol.***
- **Airway** – ensure patency and positioning.
  - Have suction equipment ready.
  - Consider spinal motion restriction if indicated.
- **Respiratory/breathing** – Give supplemental O_2 if signs of compromise. **Maintain $SpO_2 \geq 94\%$.**
  - Manage any injuries impairing ventilation immediately.
  - Assist with bag valve mask if respiratory effort is ineffective.
- **Circulation** – Assess radial pulses and heart rate.
  - Consider Pelvic Binder if appropriate.
- **Head/Hypothermia**
  - Initial GCS/AVPU.
  - Pupils.
  - Prevent hypothermia.

Secondary Assessment and Treatment

- Monitor vital signs and O_2 saturation
- Cardiac Monitor
  - Obtain and evaluate 12-Lead ECG as soon as possible.
  - Treat arrhythmias under the appropriate protocol.
- Perform blood glucose analysis. Treat if BGL is **< 60 mg/dl**.
- Initiate $EtCO_2$ monitoring.
- Physical exam and OPQRST/SAMPLE history.
- Advanced airway/ventilatory support as needed.
  - Initiate $EtCO_2$ monitoring.
  - Maintain $EtCO_2$ of **30-35 mmHg**
- Initiate IV/IO of normal saline at KVO rate.
  - If fluid resuscitation is needed, 20 ml/kg fluid bolus. *See* ***Shock Protocol***.
- If patient is hypothermic, *See* ***Cold Related Emergencies Protocol*****.**

All submersion victims should be transported, even if they appear uninjured or appear to have recovered.

Document:

- Vital signs
- OPQRST/SAMPLE
- Cardiac rhythm
- Treatment(s)
- Communication with Medical Control.

Medications

EMS Drug Formulary Table of Contents

Medications

EMS Drug Formulary (List)

Medications listed here and described in this section include all medications referenced in these clinical protocols. Each local EMS medical director had the responsibility of approving a formulary consistent with the State scope of practice and appropriate for local EMS resources and needs.

Please note that the current scope of practice prohibits use of paralytics to initiate advanced airway management/endotracheal intubation.

Acetaminophen Pediatric Liquid (Tylenol®)

Indication:

- Used for fevers >100.2 to prevent increase of fever and to lower body temperature. Can be used post-febrile seizure as long as patient is responsive. Pain relief of minor pediatric injuries.

Pediatric dose range:

- 0-10 yrs.: 15mg/kg PO if > 4 hours since last antipyretic.

Time to onset:

- PO: 20 minutes

Contraindications:

- Known hypersensitivity (rare)

What side effects/potential complications are expected?

- Abdominal Pain
- Nausea / Vomiting

How is it given?

- Use a disposable syringe to inject liquid medication into cheek area of mouth. (No sharp attached). Then dispose of syringe.

Are there any special instructions/considerations?

- Use with caution by mouth if noted altered mental status or lethargy.

Adenosine (Adenocard®)

Indications:

- PSVT and undifferentiated regular wide-complex tachycardia.

Adult Dose Range:

- 6 mg; if not effective within 1 – 2 minutes, 12 mg may be given; may repeat 12 mg bolus if needed.

Pediatric Dose Range:

- **(Given only after orders from Medical Control)** 0.1 mg/kg; if not effective, administer 0.2 mg/kg. Maximum initial dose: 6 mg/ Maximum additional single dose: 12 mg.

Time to Onset

- Rapid

Contraindications:

- 2^{nd} or 3^{rd} degree AV block, or sick sinus syndrome, or any bradycardic rhythm (except in patients with a pacemaker).
- Known hypersensitivity.

How is it given?

- Rapid IV push over 1 – 2 seconds via peripheral IV line with at least 20 ml Normal Saline flush.

What should be monitored?

- ECG
- Heart rate
- Blood Pressure

Major drug interactions:

- Theophylline and caffeine (may require increased dose of adenosine).
- Dipyridamole (may require reduced dose of adenosine)
- Carbamazepine (may increase heart block)

What side effects/potential complications are expected?

- Facial flushing.
- Palpitations.
- Chest pain.
- Hypotension.
- Headache.
- Shortness of breath/dyspnea.
- Sweating.

Are there any special instructions/considerations?

- Use large proximal vein.
- Follow medication immediately with syringe flush of normal saline (not just open line; not fast enough).
- Use two syringe technique if possible.
- Consider other arrythmias such as Atrial Flutter, Atrial Fibrillation, or Ventricular Tachycardia before administration.

Albuterol (Proventil®, ProAir®, Xopenex®)

Indications:

- Bronchodilator in reversible airway obstruction due to reactive airway disease, asthma, COPD, CHF, anaphylaxis or other respiratory conditions causing bronchospasm.

Adult Dose Range:

- 2.5 – 5 mg as needed. Pre-hospital personnel may assist patient with self-administration of their metered dose inhaler.

Pediatric Dose Range:

- If < 15 kg – 2.5 mg; If > 15 kg up to 5 mg.

Time to onset:

- 5 – 15 minutes (if inhaled)

Contraindications:

- Hypersensitivity to albuterol.
- Adult heart rate above 180 beats per minute without contacting Medical Control.
- Pediatric heart rate above 220 beats per minute without contacting Medical Control.

How is it given?

- Via nebulization.

What should be monitored?

- Heart rate.
- CNS stimulation.
- Respiratory status.

Major drug interactions:

- Beta blockers (decreased effect)
- MAO inhibitors and TCA's (may increase cardiovascular effects)
- Other sympathomimetic aerosol bronchodilators or epinephrine should not be used concomitantly with Albuterol, including over-the-counter aerosols.

What side effects/potential complications are expected?

- Tachycardia, palpitations, pounding heartbeat.
- GI upset, nausea.
- CNS stimulation.
- Tremors.

Are there any special instructions/considerations?

- Patient may need assistance and coaching with treatment(s).

Amiodarone (Cordarone®)

Indications:

- Recurring ventricular fibrillation, pulseless ventricular tachycardia, unstable ventricular tachycardia.

Adult Dose Range:

- V-Fib/Pulseless VT: 300 mg rapid IV/IO bolus, repeat dose of 150 mg can be given in 3 – 5 minutes.
- V-Tach with Pulse: 150 mg slow IV/IO over 10 minutes.
- Atrial Fibrillation: 150 mg slow IV/IO over 10 minutes.

Pediatric Dose Range:

- V-Fib or VT: 5 mg/kg rapid IV/IO bolus can repeat up to two times.
- Perfusing tachycardia: Loading dose of 5 mg/kg IV/IO over 20 – 60 minutes.

Time to onset:

- Immediate.

Contraindications:

- Hypersensitivity.
- Severe sinus node dysfunction.
- 2nd and 3rd degree AV block.
- Cardiogenic shock.
- Relative Asthma – Contact Medical Control.
- Sinus bradycardia, except if pacemaker is placed.
- Pregnancy.

How is it given?

- IV/IO

What should be monitored?

- ECG.
- Heart rate.

Major drug interactions:

- Beta blockers.
- Calcium channel blockers.
- Digoxin.

What side effects/potential complications are expected?

- Hypotension.
- CNS effects.
- Myocardial depression.
- Nausea/vomiting.
- Arrythmias.
- Flushing.
- Visual disturbances.

Are there any special instructions/considerations?

- Do not give rapid IV/IO push to a patient with a pulse.

Ancef

Indication:
- Severe Wounds (deep, crushed, or exposed tendon; open fracture; heavy contamination) Also consider for severe Sepsis as per *Sepsis Protocol*.

Adult dose range:
- 2 gm > 176 pounds in 100 ml NS over 10 minutes.
- 1 gm < 176 pounds over 10 minutes.

Reconstitute:
- **USE STERILE WATER ONLY**
- Draw up 2.5 ml of Sterile Water, inject into Ancef vial, shake until mixed well.
- Draw into a 3cc syringe, inject into a 100 ml bag of NS with a **60** gtts and let run wide open.

Time to onset:
- IV Immediate

Contraindications:
- Allergic
- Cephalosporin Allergies

How is it given?
- IV Drip

What should be monitored?
- Vital Signs
- Cardiac monitoring

Major drug interactions:
- Blood thinners

What side effects/potential complications are expected?
- Swelling
- Redness
- Pain
- Nausea / Vomiting
- Diarrhea
- Headache

Are there any special instructions/considerations?
- Caution in Kidney Disease / Liver Failure / Stomach Disease
- Contact Medical Control prior to treatment of Sepsis.

Aspirin

Indications:

- Onset of chest pain suggestive of myocardial infarction.

Adult Dose Range:

- 160 – 324 mg chewable tablets.

Pediatric Dose Range:

- Not recommended.

Time to Onset:

- 15 – 30 minutes.

Contraindications:

- Hypersensitivity to aspirin.
- Stomach ulcers.
- Gastrointestinal bleeding.

How is it given?

- Orally.

What should be monitored?

- Heart rate.
- Respiratory rate.

Major drug interactions:

- Blood thinners.

What side effects/potential complications are expected?

- Gastrointestinal upset, nausea.
- Vomiting.
- Wheezing.

Are there any special instructions/considerations?

- Do not give large amounts of water to drink, as vomiting may occur.
- Relatively contraindicated in persons with asthma.

Atropine

Indications:
- Bradycardia, per cardiac arrythmia protocol.
- Symptomatic organophosphate exposure: Nerve gas (terrorism), or pesticides (industrial/farming)

Adult Dose Range:
- Bradycardia: 0.5 mg IV/IO bolus every 3 – 5 minutes as needed. Not to exceed total dose of 3 mg.
- Organophosphate Poisoning/Nerve Agents: 1 – 2 mg IV/IO bolus every 5 minutes, until bronchial secretions and bradycardia are controlled.

Pediatric Dose Range:
- Bradycardia: 0.02 mg/kg bolus IV/IO. Minimum Dose is 0.1 mg. Maximum Single Dose is 0.5 mg. Maximum Total Dose is 1 mg. For ET Dosing – 0.04 – 0.06 mg/kg followed by 5 ml flush and 5 ventilations.
- Organophosphate Poisoning/Nerve Agents: 0.02 mg/kg IV/IO bolus every 5 minutes until bronchial secretions and bradycardia are controlled.

Time to Onset:
- Immediate.

Contraindications:
- Absence of bradycardia.
- Absence of signs of organophosphate poisoning/nerve agent exposure.

How is it given?
- IV/IO: administered undiluted by IV bolus.
- IM: Only if IV is not established.
- ET Dose: 2 – 2.5 normal dosing followed by flush and ventilations.
- Auto-injector: e.g. MARK I kits.

What should be monitored?
- Airway secretions.
- Heart rate.
- Mental status.

Major drug interactions:
- Phenothiazines (Promethazine, prochlorperazine)
- Antihistamines.

What side effects/potential complications are expected?
- Dry, hot skin and mouth.
- Tachycardia.
- Urinary retention.
- Decreased gastrointestinal motility.

Are there any special instructions/considerations?
- For large-scale exposures to organophosphates/nerve agents, access CHEMPACK caches. *See **CHEMPACK FACT SHEET*** in the Resources section of these protocols or call **Georgia Poison Control Center** directly at **1-800-222-1222**.
- ET dosing should only be performed if IV/IO attempts have been made and are unsuccessful.

Brethine

Indications:

- Asthma.
- COPD exacerbation.
- CHF/Pulmonary edema.

Adult Dose Range:

- 0.25 mg SQ.
- May repeat once in 15 minutes if no relief. Maximum Dose – 0.5 mg over 4 hours.

Pediatric Dose Range:

- > 12 years old: 0.25 mg SQ. May repeat once after 15 minutes if no relief.
- < 12 years old: Contact Medical Control.

Time to Onset:

- 6 – 15 minutes.

Contraindications:

- Patients under 12 years of age
- Hypersensitivity to Brethine.

How is it given?

- SQ.

What should be monitored?

- Vital signs.
- Cardiac rhythm.

Major drug interactions:

- Additive effect with other adrenergic drugs.
- MAOI – Tranylcypromine risk of severe hypertension.
- Beta blockers – reducing bronchodilation and increasing cardiovascular risk.
- Tricyclic Antidepressants – may increase sympathetic effects and risk of arrythmias.

What side effects/potential complications are expected?

- Tachycardia.
- Palpitations.
- Nervousness.
- Tremors.
- Dizziness.
- Nausea/Vomiting.

Are there any special instructions/considerations?

- Exercise caution when administering to patients with hypertension, cardiac history, and/or diabetes.

Calcium Chloride 10%

Indications:

- Hypocalcemia.
- Magnesium intoxication due to overdose of magnesium sulfate.
- Cardiac resuscitation, particularly after open heart surgery, parenteral calcium can be used when epinephrine has failed to improve weak or ineffective myocardial contractions.
- Hyperkalemia, calcium chloride injection may aid in antagonizing cardiac toxicity as measure by electrocardiographic (ECG) changes, provided the patient is not receiving digitalis therapy.

Adult Dose Range:

- Cardiac Resuscitation: 500 mg – 1 gram (5 – 10 ml) IV.
- Magnesium intoxication: 500 mg (5 ml).

Pediatric Dose Range:

- 5 – 7 mg/kg of a 10% solution.

Time to Onset:

- Immediate.

Contraindications:

- Cardiac resuscitation in the presence of ventricular fibrillation.
- Digitalized patients.
- Hypercalcemia and hypercalciuria (e.g. hyperparathyroidism, vitamin D overdose, decalcifying tumors such as plasmacytoma, bone metastases).
- Severe renal disease.

How is it given?

- IV use only.
- **DO NOT** inject IM or SQ.

What should be monitored?

- Cardiac rhythm changes.
- Heart rate.

What side effects/potential complications are expected?

- Rapid injection of calcium chloride may cause vasodilation, decreased blood pressure, bradycardia, cardiac arrythmias, syncope, and cardiac arrest. Parenteral calcium may cause flushing, nausea, vomiting, drowsiness, sweating, and hypotension. Vasomotor collapse may ensure if IV injection is too rapid. Injections of calcium chloride are accompanied by peripheral vasodilation as well as local burning sensation.

Are there any special instructions/considerations?

- Pregnancy: Animal reproduction studies have not been conducted with Calcium Chloride. It also is not known if Calcium Chloride can cause fetal harm when administered to a pregnant woman or can affect reproduction capacity. Calcium Chloride should be given to a pregnant female only if it is clearly needed.
- Calcium Chloride injection is irritating to veins and must not be injected into tissues due to severe necrosis and sloughing that could occur. Great care should be taken to avoid extravasations or accidental injection into perivascular tissues.

Dextrose in Water

10%, 25%, 50%

Indications:

- Hypoglycemia.
- Hyperkalemia.

Adult Dose Range:

- D_{50} W: 25 – 50 g IV as needed.

Pediatric Dose Range:

- D_{10} W: < 6 months – 0.5 g/kg (5 ml/kg)
- D_{25} W: 6 months – 2 years – 0.5 g/kg (2ml/kg)
- D_{50} W: > 2 years – 0.5 g/kg (1ml/kg)

Time to Onset:

- Immediate.

Contraindications:

- None in the emergency setting for hypoglycemic patients.

How is it given?

- Slow IV bolus.
- For pediatric use:
  - Dilute dose with Normal Saline in a 1:1 ratio to create 25% (D_{25} W)
  - Dilute dose with Normal Saline in a 2:1 ratio to create 10% (D_{10} W)

D_{50} to D_{10}	D_{50} to D_{25}
Expel 40 ml of 50% solution and draw up 40 ml of Normal Saline. = 10% solution	Expel 25 ml of 50% solution and draw up 25 ml of Normal Saline. = 25% solution

What should be monitored?

- Blood glucose.
- Level of consciousness.
- IV site.

Major drug interactions:

- No major drug interactions.

What side effects/potential complications are expected?

- Hyperglycemia.
- Vein irritation.
- Cerebral edema in stroke patients.

Are there any special instructions/considerations?

- For concentrations above 25%, administer by patent peripheral vein or IO route.
- May cause tissue necrosis.
- May precipitate neurologic symptoms in Thiamine deficient patients. Consider administration of 100 mg of Thiamine IV in malnourished patients.
- Use caution in the setting of the following:
  - Acute stroke.
  - Diabetic coma and hyperglycemia.
  - Delirium tremens in dehydrated patients.

Diazepam (Valium®)

Indications:

- Seizure activity
- Excited delirium
- Acute agitation

Adult Dose Range:

- 2.5 – 5 mg slow IV/IO/IM up to a maximum dose of 10 mg.
- Contact Medical Control for further dosing.

Pediatric Dose Range:

- 0.1 – 0.3 mg/kg slow IV/IO to a maximum dose of 5mg for pediatric patients < 5 years old. For pediatric patients > 5 years old a maximum of 10 mg can be administered.
- Contact Medical Control or advisement on use.

Time to Onset:

- 1 – 5 minutes.

Contraindications:

- Hypersensitivity to diazepam.
- Pre-existing CNS depression.
- Respiratory depression.

How is it given?

- Slow IV push. **DO NOT** exceed 1 – 2 mg/minute in children, 5 mg/minute in adults.

What should be monitored?

- Airway.
- Level of consciousness.
- Heart rate.
- Respiratory rate.
- Blood pressure.

Major drug interactions:

- Any opiates.
- Any medication for mood disorder.
- Seizure medications.
- Antihistamines.

What side effects/potential complications are expected?

- Decreased level of consciousness.
- Inability to maintain airway.
- Respiratory depression/apnea.
- Hypotension.

Are there any special instructions/considerations?

- Contact Medical Control for patients with any of the following:
  - Neurologic
  - Geriatric
  - Pregnancy or lactating patient.
- Respiratory depression last longer than seizure activity. Be prepared to support respirations.

Diphenhydramine (Benadryl®)

Indications:
- Allergic reaction, anaphylaxis, dystonic reactions.

Adult Dose Range:
- 25 – 50 mg/dose. PO/IM/IV/IO. Maximum dose of 50 mg.

Pediatric Dose Range:
- 1 mg/kg. Maximum dose of 50 mg.

Time to Onset:
- 15 – 30 minutes.

Contraindications:
- Hypersensitivity to diphenhydramine (Benadryl®).
- Newborns or infants.
- Nursing mothers.

How is it given?
- PO/IM/IV/IO. Can be given undiluted at a rate of 25 mg/minute.

What should be monitored?
- Pulse oximetry.
- Blood pressure.
- Improvement of symptom(s) being treated.
- Sedation level.

Major drug interactions:
- Additive CNS depression with alcohol, sedatives, narcotics.

What side effects/potential complications are expected?
- Sedation, dizziness, lightheadedness, altered mental status.
- Hypotension, tachycardia.
- Blurred vision.

Are there any special instructions/considerations?
- For respiratory depression and signs of shock secondary to anaphylaxis, epinephrine should be administered early and often.

Dopamine (Inotropin®)

Indications:

- Cardiogenic shock, hemodynamically significant hypotension not resulting from hypovolemia. Symptomatic bradycardia.

Adult Dose Range:

- 2 – 20 mcg/kg/min titrated to desired response.
    - 2 – 4 mcg/kg/min: Renal dose.
    - 5 – 10 mcg/kg/min: Inotropic dose.
    - 10 – 20 mcg/kg/min: Pressor dose.

Pediatric Dose Range:

- 2 – 20 mcg/kg/min, titrated to desired response.

Time to Onset:

- Less than 5 minutes.

Contraindications:

- Hypersensitivity to sulfites.
- Hypovolemic shock.
- Tachyarrhythmias.
- Ventricular fibrillation.

How is it given?

- Administered as a continuous infusion, titrating to effect. Gradually increasing dosage until optimum response occurs.
- Direct intravenous push is **NOT** recommended.
- See infusion charts on the next page.

What should be monitored?

- Blood pressure.
- Heart rate.
- Peripheral pulses.
- IV site.

Major drug interactions:

- Deactivated by sodium bicarbonate.
- Hypotension and/or bradycardia occurs with phenytoin.
- Reduced side effects with Beta-adrenergic blockers.
- Potentiated effects with MAO inhibitors.

What side effects/potential complications are expected?

- Hypertension.
- Tachycardia, palpitations, arrythmias.

Are there any special instructions/considerations?

- Tissue necrosis is associated with extravasation.
- Medical Control should be consulted before initiating Dopamine on any patient taking MAO inhibitors.

Dopamine (Inotropin®) (Continued)

DOPAMINE DRIP RATES				
Dopamine concentration = 1600 mcg/ml solution = 400 mg in 250 ml D5W or NS				
Drops per minute based on microdrip tubing (60 gtt/ml)				
Patient Weight (kg)	**5 mcg / kg / min**	**10 mcg / kg / min**	**15 mcg / kg / min**	**20 mcg / kg / min**
40	8	15	23	30
45	8	17	25	34
50	9	19	28	38
55	10	21	31	41
60	11	23	34	45
65	12	24	37	49
70	13	26	39	53
75	14	28	42	56
80	15	30	45	60
85	16	32	48	64
90	17	34	51	68
95	18	36	53	71
100	19	38	56	75
105	20	39	59	79
110	21	41	62	83

Pediatric Dopamine Infusion Chart

Dopamine Infusion: Standard 1600 mcg/ml Concentration						
Broselow Color	**Weight (kg)**	**Milliliters/hr. or drops per minute with micro drip tubing (60gtt/ml)**				
		5mcg/kg/hr.	7.5mcg/kg/hr.	10mcg/kg/hr.	15mcg/kg/hr.	20mcg/kg/hr.
Gray	3	0.6*	0.8*	1.1	1.7	2.3
Gray	4	0.8*	1.1	1.5	2.3	3.0
Gray	5	0.9*	1.4	1.9	2.8	3.8
Pink	6-7	1.2	1.9	2.4	3.7	4.9
Red	8-9	1.6	2.4	3.2	4.8	6.4
Purple	10-11	2	3	3.9	5.9	7.9
Yellow	12-14	2.4	3.7	4.9	7.3	9.8
White	15-18	3.1	4.7	6.2	9.3	12.4
Blue	19-23	3.9	5.9	7.9	11.8	15.8
Orange	24-29	5	7.5	9.9	14.9	19.9
Green	30-36	6.2	9.3	12.4	18.6	24.8
***For rates < 1ml/hr., consider using 800mcg/ml concentration**						

Dopamine Infusion: 800 mcg/ml Concentration						
Broselow Color	**Weight (kg)**	**Milliliters/hr. or drops per minute with micro drip tubing (60gtt/min)**				
		5mcg/kg/hr	7.5mcg/kg/hr.	10mcg/kg/hr.	15mcg/kg/hr.	20mcg/kg/hr.
Gray	3	1.1	1.7	2.3	3.4	4.5
Gray	4	1.5	2.3	3.0	4.5	6
Gray	5	1.9	2.8	3.8	5.6	7.5

DuoNeb

Indications:

- Asthma, COPD, Allergic Reaction, Bronchospasm.

Adult Dose Range:

- 3ml via nebulizer. Maximum 2 treatments.

Pediatric Dose Range:

- < 13 years of age – 1.5 – 3 ml nebulizer treatment every 20 minutes. Maximum 2 treatments.
- > 13 years of age – 3 ml nebulizer treatment every 20 minutes. Maximum 2 treatments.

Time to Onset:

- 5 – 15 minutes.

Contraindications:

- Hypersensitive to albuterol/ipratropium

How is it given?

- Nebulized.

What should be monitored?

- Vital signs.
- Cardiac rhythm/rate.

Major drug interactions:

- Diuretics.
- Beta receptor.
- Beta adrenergic agents.

What side effects/potential complications are expected?

- Chills.
- Headache.
- Sore Throat.
- Cough.
- Runny nose.
- Tachycardia.
- Hypertension.

Are there any special instructions/considerations?

- Use caution in patients with cardiac arrhythmias, hypertension, seizures, and/or diabetes.

Epinephrine (Adrenalin®)

Indications:

- Anaphylaxis, cardiac arrest, croup, severe bronchospasm, symptomatic bradycardia.

See note under *Special Instructions/Considerations* on the following page in reference to different concentrations of epinephrine.

Adult Dose Range:

- Cardiac Arrest: 1 mg (1:10,000) every 3 – 5 minutes IV/IO.
- Symptomatic Bradycardia not relieved by atropine or transcutaneous pacing: 1 mg (1:1,000) in 250 ml or normal saline or D5W administered at 2 – 10 mcg/min.

Epinephrine Infusion: 4 mcg/ml Concentration									
Mix 1mg of Epinephrine 1:1,000 in 250 ml of NS or D5W (4mcg/ml Concentration)									
mcg/min	2	3	4	5	6	7	8	9	10
micro drops/min	30	45	60	75	90	105	120	135	150

- Bronchospasm/Anaphylaxis:
  - 0.3 – 0.5 (0.3 - 0.5 ml 1:1,000) SQ or IM using auto-injector.
  - 0.3 – 0.5 (3 – 5 ml 1:10,000) IV if severe or no response to SQ/IM injection.
  - Repeat every 5 – 10 minutes as needed for respiratory and hemodynamic support.

Pediatric Dose Range:

- Cardiac Arrest: 0.01 mg/kg (1:10,000) every 3 – 5 minutes IV/IO. Endotracheal tube dose 0.1 mg/kg followed by a 5 ml flush and 5 ventilations.
- Symptomatic Bradycardia: 0.01 mg/kg (1:10,000) every 3 – 5 minutes IV/IO.
- Bronchospasm/Anaphylaxis: 0.01 mg/kg (1:1,000) SQ to a maximum dose of 0.3 mg/dose.
  - > 30kg: 0.3 mg IM using an auto-injector.
  - 10 – 30 kg: 0.15 mg IM using a junior auto-injector.
- Croup:
  - < 15 kg: 2.5 ml (1:1,000) in 3 ml of normal saline nebulized.
  - > 15 kg: 5 ml (1:1,000) nebulized.

Time to Onset:

- < 1 minute IV. 3 – 10 minutes IM/SQ.

Duration of effect: 3 – 5 minutes IV.

Contraindications:

- None in cardiac arrest.
- Hypersensitivity to epinephrine.
- Hypertension and/or tachycardia.

How is it given?

- IV/IO/IM/SQAuto-injector.

CONTINUED ON NEXT PAGE

Epinephrine (Adrenalin®) (Continued)

What should be monitored?

- Blood pressure.
- Heart rate.
- Pulmonary function.

Major Drug interactions:

- Deactivated by sodium bicarbonate.
- Reduced effects with Beta-adrenergic blocker.
- Potentiated effects with MAO inhibitor.
- Increased arrythmias with sympathomimetics (i.e. caffeine, cocaine) and phosphodiesterase inhibitors (i.e. Viagra®, Cialis®, Levitra®)

What side effects/potential complications are expected?

- Tachycardia, palpitations, angina.
- Flushing, hypertension.

Are there any special instructions/considerations?

- **DO NOT** confuse concentration strengths of Epinephrine 1:1,000 and 1:10,000.
- Epinephrine 1:1,000 is **NOT** for IV/IO use.
- **DO NOT** mix with sodium bicarbonate.
- For anaphylaxis, give epinephrine early and often.
- ET dosing should not be administered until attempts have been made at IV/IO insertions without success.

Etomidate (Amidate®)

Indications:

- Etomidate is indicated by intravenous injection for the induction of general anesthesia. When considering use of Etomidate, the usefulness of its hemodynamic properties should be weighed against the high frequency of transient skeletal muscle movements. It is utilized in EMS as an agent to decrease the pain and/or stress associated with procedures such as endotracheal intubation.
    - *See **Etomidate Dosing Chart***

Adult Dose Range:

- 0.3 mg/kg IVP. Maximum dose 40 mg over 30 – 60 seconds.

Pediatric Dose Range:

- 0.3 mg/kg IVP. Maximum dose 40 mg over 30 – 60 seconds.
- Not to be administered to the pediatric patient under 10 years of age.

Time to Onset:

- Within 60 seconds.

Duration:

- 3 – 5 minutes.

Contraindications:

- Hypersensitivity to Etomidate.
- Use caution in elderly patients.

How is it given?

- IVP over 30 – 60 seconds.

What should be monitored?

- Cardiac monitoring including $EtCO_2$.
- Airway.
- Vital signs.

Major drug interactions:

- None in the EMS setting.

What side effects/potential complications are expected?

- Apnea.
- Bradycardia.
- Decreased blood pressure.
- Arrythmias.
- Nausea/Vomiting.

Are there any special instructions/considerations?

- There is inadequate data to make dosage recommendations for induction of anesthesia in patients below the age of 10 years of age. Therefore, such use is not recommended.
- Geriatric patients may require reduced doses of Etomidate.

Fentanyl (Sublimaze®)

Indications:

- Moderate to severe pain, Sedation Assisted Intubation.

Adult Dose Range:

- 25 – 200 mcg.

Pediatric Dose Range:

- 2 – 12 years: 1 mcg/kg. ***Consult Medical Control for advisement on use.***

Time to Onset:

- Rapid.

Contraindications:

- Severe hemorrhage.
- Signs of shock.
- Respiratory depression.

How is it given?

- Slow IV/IO/IM.

What should be monitored?

- Level of consciousness.
- Airway.
- Respirations
- Monitor for dysrhythmias (bradycardia).

Major drug interactions:

- Other CNS depressants may potentiate the effect of Fentanyl (i.e.narcotics, barbiturates, tranquillizers).
- MAO Inhibitor use within the previous 14 days. ***Contact Medical Control for advisement.***

What side effects/potential complications are expected?

- Inability to maintain airway.
- Decreased level of consciousness.
- Respiratory depression.
- Bradycardia.

Are there any special instructions/considerations?

- Use caution in patients with liver and kidney dysfunction.
- Narcotic antagonist such as Naloxone should be readily available to manage apnea.
- Reduced dosages may be necessary for high-risk or geriatric patients.

Furosemide (Lasix®)

Indications:

- CHF, Pulmonary Edema.

Adult Dose Range:

- Dose equal to twice the patient's normal daily dose. Maximum dose of 80 mg.
- If daily dose is unknown or if the patient is not currently prescribed Lasix, administer 40 mg.

Pediatric Dose Range:

- 1 – 2 mg/kg. Not to exceed 6 mg/kg. ***Contact Medical Control for advisement.***

Time to Onset:

- 5 – 10 minutes IV.

Contraindications:

- Pregnancy.
- Dehydration.
- Hypovolemic.
- Hypokalemia.

How is it given?

- IV/IO

What should be monitored?

- Respiratory status.
- Vital signs.
- Monitor for dysrhythmias.

Major drug interactions:

- Not applicable.

What side effects/potential complications are expected?

- Dehydration.
- Dysrhythmias.
- Nausea/Vomiting.

Are there any special instructions/considerations?

- Withhold if patient's systolic blood pressure is less than 90 mmHg.

Glucagon

Indications:

- Hypoglycemia, Beta blocker overdose.

Adult Dose Range:

- 1 mg.

Pediatric Dose Range:

- 0.1 mg/kg.

Time to Onset:

- 5 – 20 minutes.

Contraindications:

- Known hypersensitivity to Glucagon.

How is it given?

- IM/SQ/IV

What should be monitored?

- Level of consciousness.
- Blood pressure.
- Pulse rate.
- Blood glucose.

Major drug interactions:

- No acute interactions in the emergency setting.

What side effects/potential complications are expected?

- Hypotension.
- Dizziness.
- Headache.
- Nausea/Vomiting.

Are there any special instructions/considerations?

- Glucagon will work only if there are sufficient glucose stores in the liver and may be ineffective for poorly malnourished patients.
- Effects are slower than Dextrose IV administration and should only be considered if an IV line cannot be established.
- Positive inotropic effects may be seen with administration.
- Glucagon for parenteral administration is derived from pork or beef pancreas.

Ketalar (Ketamine®)

Indications:
- Sedation Assisted Intubation; this would be a beneficial addition and/or substitute for Etomidate in our Sedation Assisted Intubation protocol when the patient is hypotensive or if other parameters are present indicating its use. Ketamine is also more preferrable as a maintenance sedative than the additional doses of Etomidate.
- Sedation for the Excited Delirium Patient: Ketamine has a rapid onset and a safer hemodynamic/respiratory profile than use of benzodiazepines.
- Analgesia for adult and pediatric patients.

Adult Dose Range:
- For Sedation Assisted Intubation Induction: 1 – 2 mg/kg. Administered over at least 1 minute.
- For Sedation Maintenance: 0.5 – 1 mg/kg. Administered over at least 1 minute.
- For Analgesic Effect: 0.1 – 0.2 mg/kg. Administered over at least 1 minute.
    - May repeat every 15 minutes x 2.
- For Agitated or Psychotic Patients: 1 – 4 mg/kg. Administered IM.

Pediatric Dose Range:
- For Sedation Assisted Intubation Induction: 0.5 – 2 mg/kg. Administered over at least 1 minute.
- For Sedation Maintenance: 0.5 – 1 mg/kg. Administered over at least 1 minute.
- For Analgesic Effect: 0.1 – 0.2 mg/kg. Administered over at lease 1 minute.

Time to Onset:
- Rapid.

Contraindications:
- Hypertension.
- Patients who are < 3 months old.
- Patients with open eye injuries. (Ketamine increase intraocular pressure)
- Alcohol intoxication.
- Benzodiazepine intoxication. (Can increase respiratory depression)
- Patients with cardiac disease. (Increases cardiac demand by releasing epinephrine and norepinephrine)

How is it given?
- IV/IO/IM

What should be monitored?
- Vital signs.
- Respirations.
- Cardiac rhythm.

Major drug interactions:
- No major drug interactions in EMS environment.

What side effects/potential complications are expected?
- Tachycardia.
- Visual Hallucinations.
- Tonic/Clonic movements.

Are there any special instructions/considerations?
- **DO NOT** give 100 mg/ml preparation undiluted.
- Slow IVP administration is necessary to reduce the pressor effects of Ketamine.

Ketorolac (Toradol®)

Indications:

- Moderate to severe pain relief. (NSAID)

Adult Dose Range:

- < 65 years old: 15 mg IV/IM.
- > 65 years old: 15 mg IV/IM.
- < 50 kg (110 lbs.) or renal impairment: 15 mg IV/IM.

Pediatric Dose Range:

- 0.5 mg/kg: Maximum dose is 15 mg.

Time to Onset:

- 10 – 15 minutes.

Contraindications:

- Severe kidney disease.
- Bleeding or blood clotting disorder.
- Possible closed head injury or brain bleed.
- History of stomach ulcers, or gastrointestinal bleeding.

How is it given?

- IV/IM. Given over 15 seconds.

What should be monitored?

- General patient assessment.
- Signs of bleeding.

Major drug interactions:

- Pentoxifylline (Trental)
- Probenecid (Benemid)
- Aspirin or other NSAIDs

What side effects/potential complications are expected?

- Stomach pain.
- Bruising at injection site.
- Bleeding.
- Nausea.

Are there any special instructions/considerations?

- Administration at a rapid rate may precipitate the onset of adverse reactions, particularly nausea and vomiting.

Labetalol

Indications:

- Hypertension when a brain infarct is not occurring.

Adult Dose Range:

- 20 mg slow IV over 2 minutes.

Pediatric Dose Range:

- Not recommended.

Time to Onset:

- 5 minutes.

Contraindications:

- Active Bronchial Asthma.
- Cardiac failure.
- Cardiogenic shock.
- Severe bradycardia.
- Hypersensitivity to Labetalol.

How is it given?

- Slow IVP. Given over 2 minutes.

What should be monitored?

- Vital signs.
- Cardiac rhythm.

Major drug interactions:

- Lasix.
- Norvasc.
- Xanax.
- Zoloft.

What side effects/potential complications are expected?

- Hypotension.
- Sweating.
- Flushing.
- Dizziness.
- Nausea/Vomiting.
- Diarrhea.
- Abdominal pain.
- Headache.

Are there any special instructions/considerations?

- Caution must be taken when reducing severely elevated blood pressure.
- Use caution in patients with liver dysfunction.

Lactated Ringers

Indications:

- Primary fluid replenishment as a source of water and electrolytes. Treatment of metabolic alkalosis in the presence of fluid loss, extracellular fluid replacement, and hypernatremia.

Adult Dose Range:

- 20ml/kg for bolus/25 ml/hr. for KVO.

Pediatric Dose Range:

- 20 ml/kg for bolus/25 ml/hr. for KVO.

Time to Onset:

- Rapid.

Contraindications:

- Fluid overload.
- Hypersensitivity to Lactated Ringers.
- **DO NOT use with blood products.**
- **NOT compatible with most medications.**

How is it given?

- IV/IO

What should be monitored?

- Blood pressure.
- Heart rate.
- Lung sounds.

Major drug interactions:

- Blood products.

What side effects/potential complications are expected?

- Fluid overload.

Are there any special instructions/considerations?

- Use with caution in patients with renal impairment, hepatic impairment, hyperkalemia, and hyperglycemia.

Levophed® (Norepinephrine)

Therapeutic Effects:

- Levophed functions as a peripheral vasoconstrictor (alpha-adrenergic action), as an inotropic stimulator of the cardiac muscle and vasodilator of the coronary arteries (beta-adrenergic action).

Indications:

- For blood pressure control in certain acute hypotensive states (e.g. pheochromocytomectomy, sympathectomy, poliomyelitis, spinal anesthesia, myocardial infarction, septicemia, blood transfusion, and drug reactions). As an adjunct in the treatment of cardiac arrest and profound hypotension.

Contraindications:

- Levophed should not be given to patients who are hypotensive from blood volume deficits except as an emergency measure to maintain coronary and cerebral artery perfusion until blood volume replacement therapy can be completed. If Levophed is continuously administered to maintain blood pressure in the absence of blood volume replacement, the following may occur: severe peripheral and visceral vasoconstriction, decreased renal perfusion and urine output poor systemic blood flow despite "normal" blood pressure, tissue hypoxia, and lactate acidosis.

Adverse reactions:

- Cardiac arrythmias may result from the use of Levophed in patients with profound hypoxia or hypercarbia. Should be used with extreme caution in patients receiving monoamine oxidase inhibitors (MAOI) or antidepressants of the triptyline or imipramine types, because severe, prolonged hypertension may result. Levophed Bitartrate injection contains sodium metabisulfite, a sulfite that may cause allergic-type reactions, including anaphylactic symptoms and life-threatening or less severe asthmatic episodes in certain susceptible people.

Dosage and Administration:

- **Dosage should be very carefully monitored. Mix 4 mg in 250 ml of Normal Saline. Your concentration is now 16 mcg/ml. Start at the lowest dosage possible of 1 mcg/min and slowly titrate in 1 mcg/min increments over 5 – 10 minutes per increase. Typical range for the drug is 1 – 10 mcg/min.** *See **Levophed Dosing Chart** for drip rates.*

Other information:

- Because of the potency of Levophed and because of varying response to pressor substances, the possibility always exist that dangerously high blood pressure may be produced with overdoses of this pressor agent. It is desirable, therefore, to record the blood pressure every 2 minutes from the time administration is started until the desired blood pressure is obtained, then every 5 minutes if the administration is to be continued. Headache may be the first symptom of hypertension due to overdosage. The rate of flow must be watched constantly, and the patient should never be left unattended. Infusions of Levophed should be given into a large vein, particularly the antecubital vein because when administered into this vein, the risk of necrosis of the overlying skin from prolonged vasoconstriction is apparently very slight. The infusion site should be checked frequently for free flow. Care should be taken to avoid extravasation of Levophed into the tissues.

How is it given?

- IV/IO

What should be monitored?

- Level of consciousness.
- Blood pressure.
- Pulse.
- Cardiac rhythm.

Lidocaine (Xylocaine®)

Indications:
- Used as an anti-arrhythmic for ventricular tachycardia, ventricular fibrillation and malignant PVCs.

Adult Dose Range:
- 1 – 1.5 mg/kg for the initial dose. 0.5 – 0.75 mg/kg for subsequent doses. Maximum dose is 3 mg/kg.
- For pulseless rhythms, repeat dosing every 3 – 5 minutes.
- For rhythms with a pulse, dosing may be repeated every 5 – 10 minutes.
- If conversion occurs, begin Lidocaine infusion at 2 – 4 mg/min.
- If premixed bag is not available, place 1 gram of Lidocaine in a 250 ml bag of D_5W or Normal Saline (or 2 grams in a 500 ml bag of D_5W or Normal Saline) for a concentration of 4mg/ml.

Lidocaine Infusion: 4mg/ml concentration			
mg/min	2	3	4
micro drops/min	30	45	60

Pediatric Dose Range:
- 1 mg/kg IV/IO. Infusions at 20 – 50 mcg/kg/min.

Time to Onset:
- 1 – 3 minutes.

Contraindications:
- Second-degree Mobitz type II and third-degree AV blocks.
- Not to be given in bradycardic rhythms as first line treatment.

How is it given?
- IV/IO

What should be monitored?
- Level of consciousness.
- Blood pressure.
- Pulse.
- Cardiac rhythm.

Major drug interactions:
- Use with caution when administering concomitantly with any of the following:
  - Procainamide.
  - Phenytoin.
  - Quinidine.
  - Beta-blockers.

What side effects/potential complications are expected?
- Altered level of consciousness.
- Seizures.
- Hypotension.
- Bradycardia/heart blocks.
- Nausea/Vomiting.

Are there any special instructions/considerations?
- Dosage of Lidocaine should be reduced by 50% in patients over the age of 70 years old, patients with liver disease, patients with heart failure.
- Lidocaine 2% can be used to reduce discomfort of IO insertion on conscious patients. Dosage is given through IO line after insertion. Adult dose for this procedure is 20 – 40 mg. Pediatrics – 0.5 mg/kg with a maximum dose of 40 mg.

Lorazepam (Ativan®)

Indications:

- Seizure activity, excited delirium, acute agitation.

Adult Dose Range:

- Sedation – 1 – 4 mg. Maximum Dose 4 mg.
- Anxiety/Agitation – 0.04 mg/kg. Maximum Dose 2 mg.

Pediatric Dose Range:

- 0.05 mg/kg. Maximum Dose 2 mg. (0.02 – 0.1 mg/kg range)

Time to Onset:

- 1 – 5 minutes, IV.

Contraindications:

- Hypersensitivity to Lorazepam.
- Pre-existing central nervous system depressi
- Respiratory depression.

How is it given?

- IV/IM

What should be monitored?

- Airway.
- Level of consciousness.
- Heart rate.
- Respiratory rate.
- Blood pressure.

Major drug interactic

- Any other
- Seizure
- Antih
- Ar

What ... ons are expected?

- ... ousness.
- ... apnea.
- H

Are there an ... structions/considerations?

- Contact ... cal Control for patients with any of the following:
    - Neurologic disorders.
    - Geriatric.
    - Pregnancy or lactating patients.

REMOVED

Magnesium Sulfate

Indications:

- Torsades de pointes.
- Treatment of cardiac arrythmias caused by hypomagnesemia.
- Seizure activity associated with toxemia/eclampsia of pregnancy.

Adult Dose Range:

- 1 – 2 grams IV push for abnormal ventricular rhythms.
- 2 – 4 grams IV for seizure activity related to toxemia – if no IV is available, give 2 grams IM.

Pediatric Dose Range:

- 20 – 50 mg/kg. Maximum single Dose – 2 grams.

Time to Onset:

- Immediate when given IV.

Contraindications:

- Heart block.

How is it given?

- IV/IM

What should be monitored?

- Blood pressure.
- Respiratory and central nervous system depression during rapid IV administration.
- Magnesium levels.
- Monitor for arrythmias.

Major drug interactions:

- No major drug interactions.

What side effects/potential complications are expected?

- Central nervous system depression.
- Respiratory depression.
- Complete heart block.

Are there any special instructions/considerations?

- Hypotension and asystole may occur with rapid administration.

Methylprednisolone (Solu-Medrol®)

Indications:

- Anti-inflammatory medication.
- Asthma.
- Exacerbation of COPD.
- Anaphylaxis.

Adult Dose Range:

- 125 mg.

Pediatric Dose Range:

- 1.0 – 2.0 mg/kg. Maximum Dose – 125 mg.

Time to Onset:

- Rapid.

Contraindications:

- Known hypersensitivity to Methylprednisolone.

How is it given?

- IV/IM.

What should be monitored?

- Blood pressure.
- Blood glucose.
- Electrolytes.

Major drug interactions:

- Decreased by Phenytoin. Phenobarbital, and Rifampin (Anti-tuberculosis).

What side effects/potential complications are expected?

- Fluid retention.
- Congestive heart failure.
- Hypertension.
- Vertigo.
- Headache.
- Projectile vomiting with rapid IV push.
- Hiccups.

Are there any special instructions/considerations?

- Dosing should be based on the lesser of ideal body weight or actual body weight.
- Long-term use may cause gastrointestinal bleeding, prolonged wound healing.
- Watch for possible problems if patient is on home therapies. Consider lower dosing (80 mg).
- Recent administration of live vaccines may cause reduced effects.

Midazolam (Versed®)

Indications:
- Seizure activity.
- Sedation Assisted Intubation (SAI)
- Excited delirium.
- Acute Agitation.

Adult Dose Range:
- 1 – 5 mg. Titrate to effect and respiratory effort.
    - Contact Medical Control for repeat dosing.

Pediatric Dose Range:
- 0.05 – 0.1 mg/kg. Maximum Dose – 5 mg.
    - Contact Medical Control for advisement on use.
    - ***See Versed Administration Chart***.

Time to Onset:
- 1 – 3 minutes IV.

Contraindications:
- Hypersensitivity to Midazolam
- Pre-existing central nervous system depression.
- Respiratory depression.

How is it given?
- IV/IO/IM/IN.

What should be monitored?
- Airway.
- Level of consciousness.
- Heart rate.
- Respiratory rate.
- Blood pressure.

Major drug interactions:
- Any opiates.
- Seizure medications.
- Antihistamines.
- Any medication for mood disorder.

What side effects/potential complications are expected?
- Inability to maintain airway.
- Decreased level of consciousness.
- Respiratory depression/apnea.
- Hypotension.

Are there any special instructions/considerations?
- Lower dosing is recommended when administered for sedation prior to cardioversion or transcutaneous pacing. 1 – 2.5 mg titrated to desired effect.
- Contact Medical Control for patients with any of the following:
    - Neurologic disorders.
    - Geriatric.
    - Pregnancy or lactating patients.

Morphine Sulfate

Indications:

- Moderate to severe pain control.

Adult Dose Range:

- 2 mg increments. Titrate to pain relief. Maximum Dose – 10 mg.

Pediatric Dose Range:

- 0.1 mg/kg. Titrate to pain relief. Maximum Dose – 10 mg.
  - **Not recommended in pediatric patients under the age of 2 years.**

Time to Onset:

- 1 – 3 minutes.

Contraindications:

- Head Injury.
- Volume depletion.
- Respiratory depression.
- Hypotension.
- Caution in patient with acute inferior MI.

How is it given?

- IV/IO/IM.

What should be monitored?

- Level of consciousness.
- Respiratory rate.
- Blood pressure.

Major drug interactions:

- Use caution with other vasodilators or central nervous system depressants.

What side effects/potential complications are expected?

- Altered level of consciousness.
- Inability of patient to maintain airway.
- Respiratory depression.
- Hypotension.
- Nausea/vomiting.

Are there any special instructions/considerations?

- Have airway management equipment and Naloxone ready and available for respiratory depression.

Naloxone (Narcan®)

Indications:

- Antidote for narcotic agonist.

Adult Dose Range:

- 0.4 – 2 mg. May repeat at 2 – 3 minute intervals.

Pediatric Dose Range:

- Initial dose of 0.01 mg/kg IV/IO.
    - If no clinical improvement is noted, administer 0.1 mg/kg IV/IO. Maximum dose – 2 mg.

Time to Onset:

- 2 minutes.

Contraindications:

- Hypersensitivity to Naloxone.
- Caution in patient known to be narcotic dependent.

How is it given?

- IV/IO/IM

What should be monitored?

- Blood pressure.
- Heart rate.
- Respiratory rate.

Major drug interactions:

- Decreased effect of narcotic analgesia.
- May precipitate acute narcotic withdrawal in patient who is narcotic dependent.

What side effects/potential complications are expected?

- Hypertension.
- Hypotension.
- Tachycardia.
- Ventricular arrythmias.
- Cardiac Arrest.
- Nausea/vomiting.
- Dyspnea.
- Pulmonary edema.
- Sneezing.
- Diaphoresis.

Are there any special instructions/considerations?

- Naloxone's effectiveness is due to narcotic reversal, not to an effect on opiate receptors. Adverse events occur secondary to reversal (withdrawal) of narcotic analgesia and sedation, which can cause severe reactions.

Nitroglycerin (Nitroquick®,Nitrostat®)

Indications:
- Angina pectoris.
- Pulmonary/systemic hypertension.

Adult Dose Range:
- Sublingual: 0.4 mg tablet.
  - May repeat once every 5 minutes to a maximum of 3 doses.

Pediatric Dose Range:
- Contraindicated in pediatric patients.

Time to Onset:
- 1 – 3 minutes sublingual.

Contraindications:
- Withhold from any patient taking erectile dysfunction medications within the last 72 hours.
  - Consult with Medical Control.
- Use caution and contact Medical Control for patient with ECG signs of acute inferior myocardial infarction or right ventricular myocardial infarction.
- Hypersensitivity to nitroglycerin.
- Increased intracranial pressure.
- Systolic blood pressure less than 110 mmHg.

How is it given?
- Sublingual.

What should be monitored?
- Level of consciousness.
- Blood pressure.
- Heart rate.

Major drug interactions:
- Alcohol
- Beta blockers
- Calcium channel blockers
- Sildenafil and other erectile dysfunction medications my increase vasodilatory effects and result in severe irreversible hypotension.

What side effects/potential complications are expected?
- Headache.
- Dizziness.
- Hypotension/orthostasis.
- Postural syncope.
- Tachycardia.

Are there any special instructions/considerations?
- Do not chew or swallow sublingual dosage forms.
- Keep patient supine when possible and monitor blood pressure frequently.
- Use with caution in the following:
  - Hypovolemia.
  - Hypotension.
  - Right ventricular infarctions.

Normal Saline

Indications:

- Parenteral fluid replenishment as a source of water and electrolytes. Treatment of metabolic alkalosis int the presence of fluid loss, extracellular fluid replacement, hypernatremia, increased ICP and mild sodium depletion. Dilution/mixing agent for other medications. As a priming solution for hemodialysis procedures. External flushing agent for eyes, nasal passages, wounds etc.

Adult Dose Range:

- 20 ml/kg for bolus/25 ml/hr. for KVO rate.

Pediatric Dose Range:

- 20 ml/kg for bolus/25 ml/hr. for KVO rate.

Time to Onset:

- Rapid

Contraindications:

- Fluid overload.
- Hypersensitivity to Normal Saline.
- Severe renal impairment.
- CHF.
- Severe edema.

How is it given?

- IV/IO

What should be monitored?

- Blood pressure.
- Heart Rate.
- Lung Sounds.

Major drug interactions:

- Corticosteroids may cause sodium retention.

What side effects/potential complications are expected?

- Rapid volume expansion.
- Electrolyte imbalances.
- Hypervolemia.
- Hypernatremia.
- Central nervous system manifestations to include:
  - Seizures.
  - Coma.
  - Cerebral edema and death.

Are there any special instructions/considerations?

- Use with caution in patients with CHF.

Ondansetron (Zofran®)

Indications:

- Anti-emetic for vomiting or severe nausea.

Adult Dose Range:

- 4 mg IV/IM. Slow push over 2 – 3 minutes undiluted. Maximum Dose – 8mg.
  - Contact Medical Control for dosing greater than 4 mg.

Pediatric Dose Range:

- > 15 years old – 0.15 mg/kg.
- < 15 years old – Contact Medical Control for orders.

Time to Onset:

- 3 – 5 minutes.

Contraindications:

- Hypersensitivity to Ondansetron.
- Hypotension.

How is it given?

- Slow IV/IO/IM/PO/ODT

What should be monitored?

- Blood pressure.
- Heart rate.
- Sedation.
- ECG (Prolonged QT)

Major drug interactions:

- Phenytoin (Dilantin®)
- Phenobarbital (Luminal®)
- Carbamazepine (Carbatrol®)
- Rifampin (Rifadin®, Rimactane®, Rifater®)
- Apomorphine (Apokyn®, Uprima®, Spontane®)

What side effects/potential complications are expected?

- Blurring of vision.
- Dizziness.
- Headache.
- Constipation.
- Chest pain.
- Hypotension.

Are there any special instructions/considerations?

- As with other anti-emetics, routine prophylaxis is not recommended for patients in whom there is little expectation of nausea and/or vomiting.
- Hepatic impairment. Maximum Dose – 8 mg IV.

Oral Glucose (Glutose®, Insta-Glucose®)

Indications:

- Conscious patient with suspected hypoglycemia.

Adult Dose Range:

- 15 grams PO.

Pediatric Dose Range:

- 7.5 grams PO.

Time to Onset:

- 5 – 10 minutes.

Contraindications:

- Decreased level of consciousness.
- Inability to swallow.
- Nausea/vomiting.

How is it given?

- PO.

What should be monitored?

- Level of consciousness.
- Blood glucose level.

Major drug interactions:

- None in the emergency setting.

What side effects/potential complications are expected?

- Nausea/vomiting.
- Improvement in blood sugar levels.

Are there any special instructions/considerations?

- Must be swallowed.
- Check glucose readings before and at least 10 minutes after administration.
- With altered level of consciousness, start an IV/IO and administer dextrose solution.
  - In the event an IV cannot be established, administer glucagon IM/IN.

Oxygen

Indications:

- Hypoxia.
- Carbon Monoxide toxicity.

Adult Dose Range:

- 24 – 100 percent FiO_2 as required.

Pediatric Dose Range:

- 24 – 100 percent FiO_2 as required.

Time to Onset:

- Rapid.

Contraindications:

- None in the emergency setting.

How is it given?

- Inhalation.
- Positive pressure assist.

What should be monitored?

- Level of consciousness.
- Pulse oximetry.

Major drug interactions:

- None in the emergency setting.

What side effects/potential complications are expected?

- Drying of mucus membranes without humidification.
- Improvement of a hypoxic event as indicated by patient presentations, pulse rate, and SpO_2 readings.

Are there any special instructions/considerations?

- In most situations, oxygen is administered to maintain an SpO_2 reading of $\geq$ 94%.
- Pulse rates are good indicators of oxygen administration's effectiveness.
    - Bradycardia, especially in the pediatric patient, indicates severe hypoxic conditions.
- Closely monitor COPD patients treated with oxygen. These patients may rapidly become sedated from loss of hypoxic drive.
- Humidify whenever possible when providing high flow volumes.
- Cold oxygen may worsen asthma or create hypothermic conditions in some patients.

Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

Phenergan

Indications:

- Prevention or control of nausea/vomiting.
- Treatment for motion sickness/vertigo.
- Alternative for patients with hypersensitivity to Benadryl®.

Adult Dose Range:

- 12.5 – 25 mg diluted in 100 ml of normal saline over 5 – 10 minutes.
- 25 mg deep IM for patients > 15 years old.

Pediatric Dose Range:

- 0.5 mg/kg
  - Contraindicated for children < 2 years of age.

Time to Onset:

- 3 – 5 minutes.

Contraindications:

- Do not use in comatose patients or in the r… central nervous system depressant (e.g. Alcohol, and/or drugs)…
- Do not use in children < 2 years of ag…
- Do not use in patients with a know… iazine class of drugs.

How is it given?

- IV/IM

What should be monitored?

- Vital signs
- Respiratory statu…
- Signs of extrap…
- Swelling or …

Major drug inte…

- Patie…

What sid… …ations are expected?

- …
- …l.
- …

Are ther… …cial instructions/considerations?

- Drug …eractions, including an increased incidence of extrapyramidal effects, have been reported when some MAO inhibitors and phenothiazines are used concomitantly. The treatment of choice for resulting hypotension is administration of IV fluids, accompanied by repositioning if indicated. In the event that vasopressors are considered for the management of severe hypotension which does not respond to IV fluids and repositioning, the administration of norepinephrine or phenylephrine should be considered. **EPINEPHRINE SHOULD NOT BE USED,** since its use in patients with partial adrenergic blockade may further lower the blood pressure. Extrapyramidal reactions may be treated with anticholinergic antiparkinsonian agents, diphenhydramine or barbiturates.

Sodium Bicarbonate

Indications:
- Metabolic Acidosis.
- Hyperkalemia.
- Tricyclic antidepressant overdose with wide QRS complex.

Adult Dose Range:
- 1 mEq/kg (8.4%) when appropriate.
    - May repeat with 0.5 mEq/kg.

Pediatric Dose Range:
- Age < 2 years: 1 mEq/kg (4.2%)
    - May repeat with 0.5 mEq/kg in 10 minutes x 1 or as indicated by patient's acid-base status.
- Age > 2 years: 1 mEq/kg (8.4%)
    - May repeat with 0.5 mEq/kg in 10 minutes x 1 or as indicated by patient's acid-base status.

Time to Onset:
- Rapid.

Contraindications:
- Alkalosis.
- Hypocalcemia/hypernatremia.
- Inadequate ventilation during cardiopulmonary resuscitation.

How is it given?
- IV/IO

What should be monitored?
- Vein patency.
- Blood pH.
- PO_2.
- PCO_2.
- Cardiac arrhythmias.

Major drug interactions:
- Inhibits
    - Tetracycline.
    - Chlorpropamide.
    - Lithium carbonate.
    - Methotrexate.
    - Salicylates.
- Potentiates
    - Anorexiants.
    - Sympathomimetics.
    - Quinidine.

What side effects/potential complications are expected?
- Alkalosis.
- Hypernatremia.
- Hypokalemia.
- Local site irritation.

Are there any special instructions/considerations?
- Extravasation causes tissue necrosis.
- Patients should be adequately ventilated before administration during cardiac arrest.

Tranexamic Acid (TXA)

Indications:

- Severe, non-compressible and/or uncontrollable hemorrhage.
  - < 1 hour from onset of injury to administration of medication.
  - Signs and symptoms of shock.
    - Mean arterial pressure < 60 mmHg.
    - Shock Index > 1.0.
    - Age $\geq$ 16 years of age.

Adult Dose Range (> 16 years of age):

- 2 grams IV/IO
  - 2 grams mixed in 100 ml bag of normal saline infused over 10 minutes.

Pediatric Dose Range (< 16 years of age):

- Contact Medical Control for Advisement on use.

Time to Onset:

- 5 – 15 minutes after IV administration.
  - Peak effect – 4 minutes, all routes.
  - Duration – 1 to 2 hours.

Contraindications:

- Isolated traumatic brain injury without external uncontrollable hemorrhage and a GCS $\leq$ 8.
- > 1 hour from injury.
- Hypersensitivity to Tranexamic Acid.
- History of venous or arterial thromboembolism (i.e. DVT, PE, etc.).
- Renal Failure.

How is it given?

- IV infusion via 100 ml normal saline infused over 5 – 10 minutes.

What should be monitored?

- Vital signs

What side effects/potential complications are expected?

- Seizure.
- Diarrhea.
- Color vision change/vision loss.
- Renal impairment.
- Myalgia.
- Headache.
- Abdominal pain.

Are there any special instructions/considerations?

- Rapid infusion of tranexamic acid will cause profound hypotension.
- May rarely contribute to coagulopathies.
- The risk of tranexamic acid administration is likely to outweigh any benefit, if administered more than 3 hours after injury, after full activation of endogenous fibrinolysis begins.

RESOURCES

Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

Resources Table of Contents

Content

Abbreviations and Definitions

AC power	*Alternating Current Power*
ACS	*Acute Coronary Syndrome*
AED	*Automated External Defibrillator*
AEIOUTIPS	*Alcohol, Electrolytes, Insulin, Opiates, Uremia, Trauma, Infection, Poison, Psychogenic, Seizure, Shock*
AHA	*American Heart Association*
AIDS	*Acquired Immune Deficiency Syndrome*
ALS	*Advanced Life Support*
ALTE	*Apparent Life Threatening Event*
AMI	*Acute Myocardial Infarction*
AMS	*Altered Mental Status*
APGAR	*Appearance, Pulse Rate, Grimace, Activity, Respiration*
ASA	*Aspirin*
AVPU	*Response level: Alert, responsive to verbal stimuli, responsive to painful stimuli only, unresponsive*
BGA	*Blood Glucose Analysis*
BIAD	*Blind Insertion Airway Device*
BLS	*Basic Life Support*
BP	*Blood Pressure*
BSA	*Body Surface Isolation*
BSA	*Body Surface Area*
BVM	*Bag Valve Mask*
°C	*Degrees Celsius*
Cardiac monitoring	*Using electrodes to identify rhythm with continuous readout*
CDC	*Centers for Disease Control*
CHF	*Congestive Heart Failure*
cm	*Centimeters*
CO2	*Carbon Dioxide*
COPD	*Chronic Obstructive Pulmonary Disease*
CPAP	*Continuous Positive Airway Pressure*
CPR	*Cardiopulmonary Resuscitation*
CRT	*Capillary Refill Time*
D5W	*5% Dextrose in water*
D10W	*10% Dextrose in water*
D25W	*25% Dextrose in water*
D50W	*50% Dextrose in water*
DIB	*Difficulty in Breathing*
dL	*Deciliter*
DNR	*Do Not Resuscitate*
DPH	*Department of Public Health*
ECG	*Electrocardiogram*
ET	*Endotracheal Intubation*

Abbreviations and Definitions (Continued)

ETA……*Estimated Time of Arrival*
ETCO2……*End-Tidal CO2*
°F……*Degrees Fahrenheit*
FAST……*Stroke Assessment; Facial droop, Arm drift, Speech, Time*
FBAO……*Foreign Body Airway Obstruction*
g……*Gram*
GCS……*Glascow Coma Scale*
GI……*Gastrointestinal*
gtt/min……*drops per minute (with* ***micro drip*** *tubing, equivalent to milliliters per hour)*
gtt……*Drop*
GU……*Gastrourinary*
HEPA……*High Efficiency Particulate Air (HEPA mask)*
HR……*Heart Rate*
ICP……*Intracranial Pressure*
IM……*Intramuscular*
IN……*Intranasal*
IV……*Intravenous*
J……*Joules*
KG……*Kilogram*
KVO……*Keep vein open*
L……*Liter*
LMP……*Last Menstrual Period*
LOC ……*Level of Consciousness*
LPM……*Liter Per Minute*
LSB……*Long Spine Board*
LVAD……*Left Ventricular Assess Device*
mcg……*Microgram*
MDI……*Metered Dose Inhaler*
mEq ……*Milliequivalent*
mg……*Milligram*
ml……*Milliliter*
mmHg……*Millimeters of Mercury*
MVC……*Motor Vehicle Collision*
NPA……*Nasopharyngeal Airway*
NPO……*Nothing by mouth*
NS……*Normal saline*
NV……*Nausea and Vomiting*
O2……*Oxygen*
ODT……*Oral Dissolving Tablet*
OCGA……*Official Code of Georgia*

Abbreviations and Definitions (Continued)

OPA…… *Oropharyngeal Airway*
OPQRST…… *History of present illness; Onset, Provocation, Quality, Region and Radiation, Severity, Time*
Pallor…… *Pale skin*
PE…… *Pulmonary Embolism*
PEA…… *Pulseless Electrical Activity*
PO…… *Per Oral; by mouth*
PPE…… *Personal Protective Equipment*
PPMHx…… *Past Pertinent Medical History*
PPV …… *Positive pressure ventilation* PR
Interval…… *The period from the beginning of the P wave to the beginning of the QRS complex*
q…… *Each, every*
QRS…… *Ventricular depolarization complex*
ROSC…… *Return of Spontaneous Circulation*
SAI…… *Sedation Assisted Intubation*
SAMPLE…… Symptoms, Allergies, Medications, Past medical history, Last oral intake, Events
SBP…… *Systolic Blood Pressure*
SGA…… *Supraglottic Airway*
SL…… *Sublingual; under tongue*
SMR…… *Spinal Motion Restriction*
SpO2…… Oxygen Saturation (Ideally greater than or equal to 94%)
SQ…… *Subcutaneous (beneath skin)*
SVT…… *Supraventricular Tachycardia*
TB…… *Tuberculosis*
TBI…… *Traumatic Brain Injury*
TBSA…… *Total Body Surface Area*
TCP…… *Transcutaneous Pacing*
Tourniquet…… *Device used to control venous/arterial bleeding to and extremity*
URI…… *Upper Respiratory Infection*
USAISR…… *United States Army Institute of Surgical Research*
VF…… *Ventricular Fibrillation*
VT…… *Ventricular Tachycardia*

Important Numbers

State and National Resources
Adult Protective Services ……………………………………………………………..1-888-774-0152
CHEMPACK Request…………………………...………………………………………1-800-222-1222
CHEMTREC…...……………………………………………………………………..1-800-424-9300
Georgia Child Protective Services.…………………………….……1-404-657-3400 or 1-855-422-4453
Georgia Crisis and Access Line (Mental Health)…………………………………………1-800-715-4225
Georgia Critical Incident Stress Foundation Crisis Hotline……………………………...1-404-419-6506
Georgia Division of Aging Services……………………………………………………..1-866-552-4464
Georgia Office of Emergency Medical Services…………………………...…………..1-404-679-0547
Georgia Poison Control..1-800-222-1222
Mental Health Hotline…………………………………………………………………..1-800-715-4225
National Domestic Violence Hotline……………………………………………………1-800-799-7233
Trauma Communications Center...….....1-866-556-3314 or FREE mobile to mobile 1-404-229-6405

Office of Emergency Medical Services and Trauma
Georgia Office of EMS……………………………………...……………...…………..1-404-679-0547
Northwest Georgia Region I EMS………………………………………………………1-706-295-6176
North Georgia Region II EMS……………………………………………………………1-770-535-5743
Metro Atlanta Region III EMS…………………………...………………………………1-404-248-8995
West Georgia Region IV EMS…………………………………...……………………1-706-845-4035
Central Georgia Region V EMS…………………………………………………………1-478-993-4990
East Central Georgia Region VI EMS…………………………………………………..1-706-667-4336
West Central Georgia Region VII EMS…………………...……………………………1-706-321-6150
Southwest Georgia Region VIII EMS…………………...………………………………1-404-989-5173
Southeast Georgia Region IX EMS…………………..…………………………………1-912-262-3035
Northeast Georgia Region X EMS…………………...…………………………………1-706-583-2862

Georgia Air Ambulance Providers
Air Evac Lifeteam………………………………………………………….……………1-800-247-3822
Carrolton, Cordele, Dublin, Jesup, Lagrange, Statesboro, Vidalia, Waycross
Air Methods Georgia…………………………………………………………………..… 1-888-763-1010
Augusta, Carrolton, Conyers, Griffin, Gainesville, Jasper, Kennesaw, Newnan, (Air Life Georgia); Springfield and Vidalia (Life Star)
Children's Healthcare (Children's 1)………………………..……………………..……1-404-785-6540
Atlanta
Med Trans (Life Force)……………………………………………1-800-523-6723 or 1-423-778-5433
Blue Ridge, Calhoun

Transcutaneous Pacing

Indications

- Monitored heart rate less than 60 per minute with signs and symptoms of inadequate cerebral or cardiac perfusion such as:
  - Chest pain
  - Hypotension
  - Pulmonary edema
  - AMS, disorientation, confusion, etc.
  - Ventricular ectopy.

Procedure

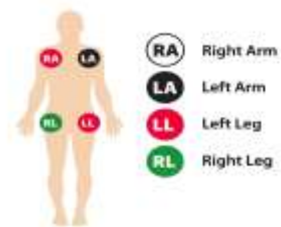
- Attach standard four-lead monitor.
- Apply defibrillation/pacing pads:
  - **Anterior / Posterior**: Anterior electrode on left precordium below the left nipple. Avoid placing on the nipple. Posterior electrode below left scapula, lateral to spine at heart level.
  - **Anterior / Lateral**: Lateral (apex) electrode lateral to left nipple with the center of the electrode on the midaxillary line. Anterior electrode below the right clavicle lateral to sternum.
- Press pacer button and observe for sensor markers on each QRS complex.
- Press rate or slowly rotate selector knob and adjust rate to 80 BPM for an adult and 100 BPM for pediatric.
- Press current or slowly rotate selector knob until capture is obtained.
- Slowly increase output until capture of electrical rhythm on the monitor.
- If unable to capture while at maximum current output, stop pacing immediately.
- If capture observed on monitor, check for corresponding pulses and assess vital signs, skin color and capillary refill for improved perfusion.
- Consider the use of sedation or analgesia if patient is uncomfortable.
- Document the dysrhythmia and the response to external pacing with EKG strips on the patient care report.

Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

12 Lead ECG

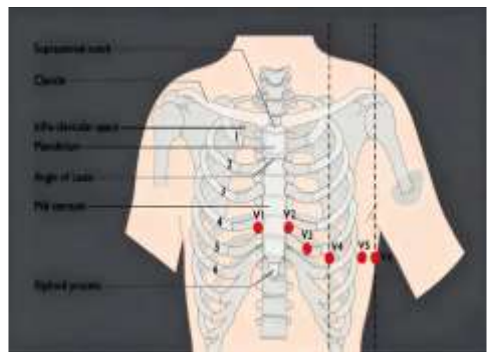
Limb Lead Placement

- LA – Left Arm, anywhere below the shoulder.
- LL – Left Leg, anywhere below the torso.
- RA – Right Arm, anywhere below the shoulder.
- RL – Right Leg, anywhere below the torso.



Precordial Lead Placement

- V1 – Right side of sternum, 4th intercostal space.
- V2 – Left side of sternum, 4th intercostal space.
- V3 – Directly between V2 and V4.
- V4 – 5th intercostal space, left mid clavicular line.
- V5 – 5th intercostal space, anterior axillary line, directly between V4 and V6.
- V6 – 5th intercostal space, left mid axillary line.



Continuous Leads

- II, III, aVF – Inferior leads. Reciprocal changes in I, aVL, V2 – V4.
- V1,V2 – Septal Leads. Reciprocal changes in Leads II, III, aVF.
- V3,V4 – Anterior Leads. Reciprocal changes in Leads II, III, aVF.
- V5,V6, I, aVL – Lateral Leads. Reciprocal changes in aVR.

I Lateral left ventricle	**AVR** Reciprocal Changes Only	**V1** Septal	**V4** Anterior
II Inferior portion of the left ventricle	**aVL** Lateral left ventricle	**V2** Antero-Septal	**V5** Lateral left ventricle
III Inferior portion of the left ventricle	**aVF** Inferior portion of the left ventricle	**V3** Antero-Septal	**V6** Lateral left ventricle

Medication Administration

Assisting Patient with taking OTC and/or own prescription medications

- Georgia Scope of Practice allows all levels of EMT to assist patients in taking their own prescribed medications and/or over-the-counter medications. Medications must be approved by your local Medical Director.

Auto-Injector Drug Delivery

- All levels of EMS licensure are allowed to administer ***Epinephrine*** parentally to patients experiencing anaphylaxis. EMT, and EMT-I may administer by auto-injector only.
- All levels of licensure are allowed to administer unit dose commercially pre-filled containers or auto-injectors for the administration of life-saving medications intended for self, peer, or patient rescue in **hazardous materials situations.**

Nasal Drug Delivery

- Medications administered via the IN route require a higher concentration of drug in a smaller volume of fluid than typically used in the IV route. Administer **NO MORE THAN 1 ml** of volume per nostril.
- Do not administer medications via the IN route if the patient's nose is bleeding or if nasal congestion or nasal discharge is present. Nasal administration does not always work for every patient and is less likely to be effective if the patient has been abusing vasoconstrictors, such as cocaine.
- Medications commonly delivered via the IN route are:
  - ***Benadryl***
  - ***Epinephrine 1:1,000***
  - ***Fentanyl***
  - ***Glucagon***
  - ***Narcan***
  - ***Versed***
  - ***Zofran***

Nebulized Drug Delivery

- All levels of EMS licensure are allowed to deliver inhaled medications through a nebulizer or through use of metered-dose-inhaler to patients with difficulty breathing.
- Treatment should continue until medication in reservoir is depleted.
- Nebulized medication may be used with CPAP.
- Patient monitoring should include pulse, respiratory rate, and breath sounds.

Intra-osseous Drug Delivery

- IO placement or removal is approved for EMT-I, AEMT, CT, and Paramedic. This includes placement in both adults and children.
- Local medical directors should specify approved anatomic site of insertion and indications for insertion (e.g. if peripheral IV access should be attempted first, how many peripheral IV attempts)
- All fluids and medications that can be administered via IV may be given IO, unless specified by local medical director.
  - If the patient experiences pain during infusion, inject ***Lidocaine*** into marrow cavity.
  - Adult: 2 – 5 ml (20 – 40 mg) 1% or 2% ***Lidocaine***. **(Paramedic only)**
  - Pediatric: 0.5 mg/kg 1% or 2% ***Lidocaine*****. (Paramedic only)**
- Contraindications include:
  - Placement in or distal to a fractured bone.
  - Placement through an area with infection or a burn.

Pre-existing in-dwelling catheters or other implanted ports

- Scope of Practice allows CT and Paramedics to access these ports if approved by local medical director.
- Implanted access ports and devices in children should not be accessed without contacting Medical Control.

AEMT Scope

- Medications allowed for AEMT, in addition to those allowed for EMT-I are:
  - ***Glucagon*** for hypoglycemia, via IM, SQ, IV, IO, or IN.
  - ***Nitroglycerin*** SL for chest pain of suspected cardiac origin.
  - ***Naloxone*** to a patient with suspected narcotic overdose, via IM, SQ, IV, IO, or IN.
  - Administration of nitrous oxide (50/50 mix) for pain relief.
  - ***Epinephrine*** for anaphylaxis, prepared by AEMT, via IM or SQ.

Spinal Motion Restriction

Neuro Exam: Any focal deficit? — YES → Apply Cervical Immobilization Device / NO ↓

Significant mechanism of injury? High energy events such as ejection, high falls and abrupt deceleration crashes — YES → / NO ↓

Alertness: Alteration in mental status? — YES → / NO ↓

Intoxication: Any evidence? Includes Drugs and/or Alcohol. — YES → / NO ↓

Distracting Injury: Any painful injury that might distract the patient from the pain of a C-spine injury. — YES → / NO ↓

Spinal Exam: Point tenderness over the spinous process(es) or pain to ROM. — YES → / NO ↓

Spinal Motion Restriction Not Required → **Exit to appropriate protocol(s).**

Apply Cervical Immobilization Device

Restrict Spinal Movement

Follow Spinal Precautions Procedure

AMBULATORY

Bring stretcher to patient, assist patient onto stretcher with minimal spinal movement, and then secure patient to stretcher.

NON-AMBULATORY

Use Long Spine Board (or any of the multiple equivalent devices) to TRANSFER patient to stretcher with minimal spinal movement, remove the device, then secure to stretcher.

May use multiple providers to transfer patient to stretcher using in-line spinal techniques such as log roll/straddle slide to maintain spinal precautions without a device, then secure to stretcher

- **Recommended Exam: Mental Status, Skin, Neck, Heart, Lungs, Abdomen, Back, Extremities, Neuro.**
- **Patients meeting all the above criteria do not require spinal motion restriction. However, patients who fail one or more criteria above, require spinal motion restriction, but does NOT require the use of long spine board for immobilization.**
- **Long Spine Boards are NOT considered standard of care in most cases of potential spinal injury. Spinal motion restriction with cervical collar and securing patient to cot, while padding all void areas is appropriate.**
- **True spinal immobilization is not possible. Spine protection and spinal motion restriction do not equal long spine board.**
- **Spinal motion restriction is always utilized in at-risk patients. These include cervical collar, securing to stretcher, minimizing movement/transfers and maintenance of in-line spine stabilization during any necessary movement/transfers. This includes the elderly or others with body or spine habitus preventing them from laying flat.**
- **Consider spinal motion restriction in patients with arthritis, cancer, dialysis, underlying spine or bone disease.**
- **Range of motion (ROM) is tested by touching chin to chest (look down), extending neck (look up) and turning head from side to side (shoulder to shoulder) without posterior cervical mid-line pain. ROM should NOT be assessed if patient has mid-line spinal tenderness. Patient's ROM should NOT be assisted.**
- **Immobilization on a long spine board is not necessary when:**
  - **Penetrating trauma to the head, neck, or torso with no signs/symptoms of spinal injury.**
- **Concerning mechanisms that may result in spinal column injury:**
  - **Fall from feet and/or stairs or steps.**
  - **MVC high speed, rollover, and/or ejection.**
  - **Motorcycle, bicycle, other mobile device, or pedestrian-vehicle crash.**
  - **Diving or axial load to spine.**
  - **Electric shock.**

Standard Precautions

Indications

- Standard precautions are guidelines for reducing the risk of transmission of blood-borne and other pathogens that apply to all patients receiving care regardless of their diagnosis or presumed infection status.

Type and Use of Personal Protective Equipment

- **Gloves** – for any patient contact, and when cleaning/disinfecting contaminated equipment. Puncture resistant gloves will be worn in situations where sharp or rough edges are likely to be encountered, i.e. auto extrication.
- **Face mask and eye protection** – facial protection will be used in any situation where splash contact with the face is possible. This protection may be afforded by using both a face mask and eye protection, or by using a full-face shield. When treating a patient with suspected or known airborne transmissible disease, particulate facemasks should be used. For respiratory illness (TB, SARS) it is beneficial to mask the patient.
- **Coverall/fluid resistant gowns** – designed to protect clothing from splashes, gowns may interfere with or present a hazard to the member in some circumstances. The decision to use gowns to protect clothing will be left to the member. Structural firefighting gear also protects clothing from splashes and is preferrable in fire, rescue or vehicle extrication activities.
- **Shoe/Head Coverings** – fluid barrier protection will be used if suspected contamination is possible.

General Precautions Against Disease

- If it is wet, it is infectious – Use gloves.
- If it could splash onto your face – Use eye shields and mask or full-face shield.
- If it is airborne – Mask yourself and the patient.
- If it can splash on your clothes – Use gown or structural firefighting gear.
- If it could splash on your head or feet – Use appropriate barrier protection.

Post Exposure Management

- **Provide first aid**
  - Secure area to prevent further contamination. (Stop bleeding with direct pressure)
  - Remove contaminated clothing and flush.
  - Wash the contaminated area well with soap and water or waterless hand cleanser.
  - If in the eyes, nose, or mouth are involved, flush them well with large amounts of water.
- **Notification and relief of duty** – the worker's supervisor should be immediately notified if a worker experiences an exposure involving potentially infectious source material. The supervisor should determine if the worker needs to be relieved of duty.
- **Report the exposure** – the worker or immediate supervisor should promptly complete an exposure report, appropriate for your agency, and submit the report to the designated Infection Control Officer.
- **Seek medical attention, counseling, consent and testing per protocol.**

APGAR Scoring

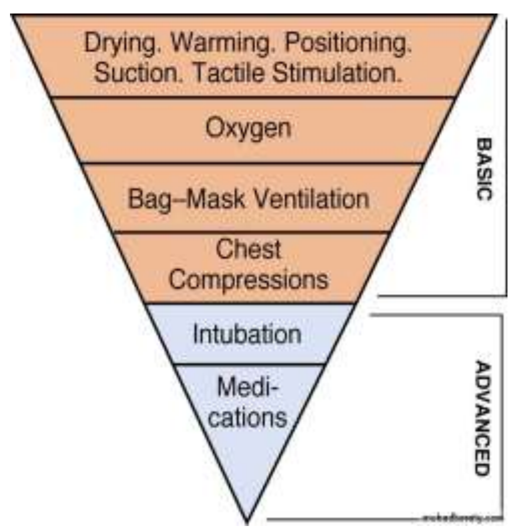
The APGAR score was devised in 1952 by the eponymous Dr. Virginia Apgar as a simple and repeatable method to quickly and summarily assess the heath of newborn children immediately after birth. The APGAR score is determined by evaluating the newborn baby on five simple criteria on a scale from zero to two, then summing up the five values obtained. The resulting APGAR score ranges from zero to ten. The APGAR score should be calculated at the 1 and 5 minutes after delivery.

A score of $\leq$ 3is considered critical. A score of $\geq$ 7 is good to excellent.

SCORE	0 points	1 point	2 points
Appearance (Skin color)	Cyanotic / Pale all over	Peripheral cyanosis only	Pink
Pulse (Heart rate)	0	<100	100-140
Grimace (Reflex irritability)	No response to stimulation	Grimace or weak cry when stimulated	Cry when stimulated
Activity (Tone)	Floppy	Some flexion	Well flexed and resisting extension
Respiration	Apneic	Slow, irregular breathing	Strong cry

Inverted Pyramid of Neonatal Resuscitation

The Neonatal Resuscitation Pyramid is a stepwise approach for treatment of a newborn. The American Academy of Pediatrics and the American Heart Association have developed standards and guidelines. The care that may be required is depicted as an inverted pyramid. The interventions, which are most commonly required, are at the top of this pyramid. As you progress down the pyramid, the interventions indicated become less commonly required. Most neonates transition to post-natal life without difficulty. About 10% of infants require some assistance to begin breathing at birth. Less than 1% require extensive resuscitative measures.



Burn Fluid Resuscitation

Fluid Resuscitation Formulas:

- **Parkland Formula**
  - $\frac{\text{Patient weight in kg x TBSA x 4ml}}{2}$
  - This formula gives you the amount of fluid to be administered in the first 8 hours post burn insult.
    - The initial portion of this formula (Patient weight in kg x TBSA x 4ml) gives you the total amount of fluid to be administered in the first 24 hours post burn insult.
    - When this number is divided by 2, this amount is the total milliliters to be administered in the first 8 hours.
    - Once you have arrived at that number, divide this total by 8 to determine ml/hr. flow rate.
- **Consensus Formula**
  - $\frac{\text{Patient weight in kg x TBSA x 3ml}}{2}$
  - This formula gives you the amount of fluid to be administered in the first 8 hours post burn insult.
    - The initial portion of this formula (Patient weight in kg x TBSA x 3ml) gives you the total amount of fluid to be administered in the first 24 hours post burn insult.
    - When this number is divided by 2, this amount is the total milliliters to be administered in the first 8 hours.
    - Once you have arrived at that number, divide this total by 8 to determine ml/hr. flow rate.
- **Brooks Formula**
  - $\frac{\text{Patient weight in kg x TBSA x 2ml}}{2}$
  - This formula gives you the amount of fluid to be administered in the first 8 hours post burn insult.
    - The initial portion of this formula (Patient weight in kg x TBSA x 2ml) gives you the total amount of fluid to be administered in the first 24 hours post burn insult.
    - When this number is divided by 2, this amount is the total milliliters to be administered in the first 8 hours.
    - Once you have arrived at that number, divide this total by 8 to determine ml/hr. flow rate.
- **USAISR Rule of Ten**
  - To simplify the process of calculating fluid requirements for burn patients in the prehospital setting researchers developed the Rule of 10.
    - The calculated TBSA is rounded to the nearest 10 (i.e. the TBSA is calculated at 37%, this is then rounded to 40).
    - This new rounded calculation is then multiplied by 10 to get the number of milliliters per hour of crystalloid (in this example 40 x 10 = 400 ml/hr).
    - This formula is used for adults weighing between 88 – 154 lbs. (40 – 70 kg). If the patient exceeds this weight range, for each 10 kg in body weight over 70 kg, an additional 100 ml/hr. is given.
- **Do Not withhold fluids while calculating fluid resuscitation formula. Initiate LR @ 250 ml/hr. then adjust accordingly after calculation is completed.**
- **The fluid of choice for burn fluid resuscitation is *Lactated Ringers*.**
- **Contact Medical Control for fluid orders in patients with CHF or cardiac disease.**
- **In addition to burn fluid resuscitation, in pediatric burns, you should consider also administering D_5W IV maintenance, using the 4-2-1 rule.** ***See Pediatric Burn Protocol.***

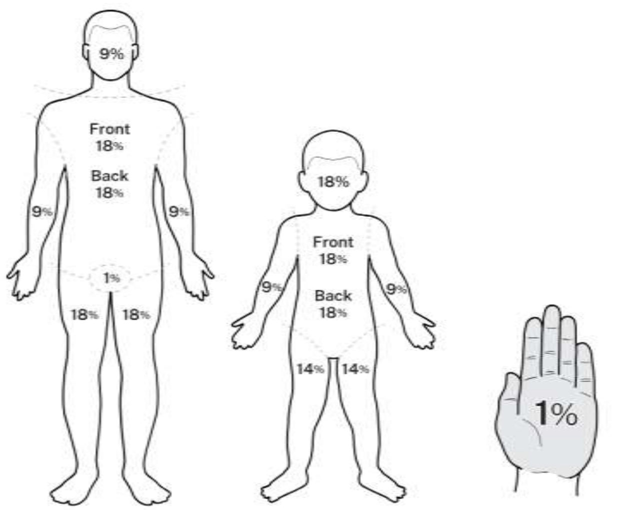
Burn Estimation

Total Body Surface Area (TBSA) is an assessment measure of burns of the skin. In adults the "Rule of Nines" is used to determine the total percentage of area burned for each major section of the body. In some cases, the burns may cover more than one body part or may not fully cover a part. In these cases, burns are measured using the casualty's palm and fingers as a reference for 1% of the body.

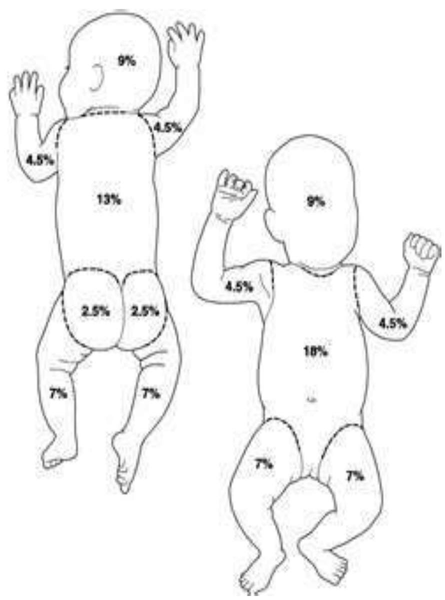
Burn Depth

- *Superficial* – 1st Degree – red and painful. Does not form blisters
- *Partial Thickness* – 2nd Degree – painful with blisters.
- *Full Thickness* – 3rd Degree – thick, dry, white, leathery **–** Painless in 3rd Degree areas but surrounding areas can still be painful and blistered
- *Subdermal Burns* – Burns below the dermis, these burns go into fat, muscle, bone and organs.

NOTE: When estimating total body surface area, only 2nd, 3rd, and 4th degree burns are included.



Adult — Paediatric < 9 years — Palmar, palm + fingers of patient = 1%



Adult	
Anatomic structure	Surface area
Anterior head	4.5%
Posterior head	4.5%
Anterior torso	18%
Posterior torso	18%
Anterior leg, each	9%
Posterior leg, each	9%
Anterior arm, each	4.5%
Posterior arm, each	4.5%
Genitalia/perineum	1%

Child	
Anatomic structure	Surface area
Anterior head	9%
Posterior head	9%
Anterior torso	18%
Posterior torso	18%
Anterior leg, each	7%
Posterior leg, each	7%
Anterior arm, each	4.5%
Posterior arm, each	4.5%
Genitalia/perineum	1%

Canadian C-Spine Rule

1. **Any one High-Risk factor which mandates immobilization**
   - Age 65 years
   - Dangerous mechanism*
   - Numbness or tingling in extremities

YES

2. **Any one Low-Risk factor which allows Safe Assessment of Range of Motion.**
   - Simple rear end MVC**
   - Ambulatory at any time at scene.
   - No neck pain at scene when asked. (answer “yes” if no pain.)
   - No pain during mid-line C-spine palpation. (answer “yes” if no pain.)

YES

3. **Patient voluntarily able to actively rotate neck 45° left and right when requested, regardless of pain.**
   - Patient voluntarily able to actively rotate neck 45° to right and left.
     - If unable, immobilize.

ABLE

NO C-Spine Immobilization

C-Spine Immobilization

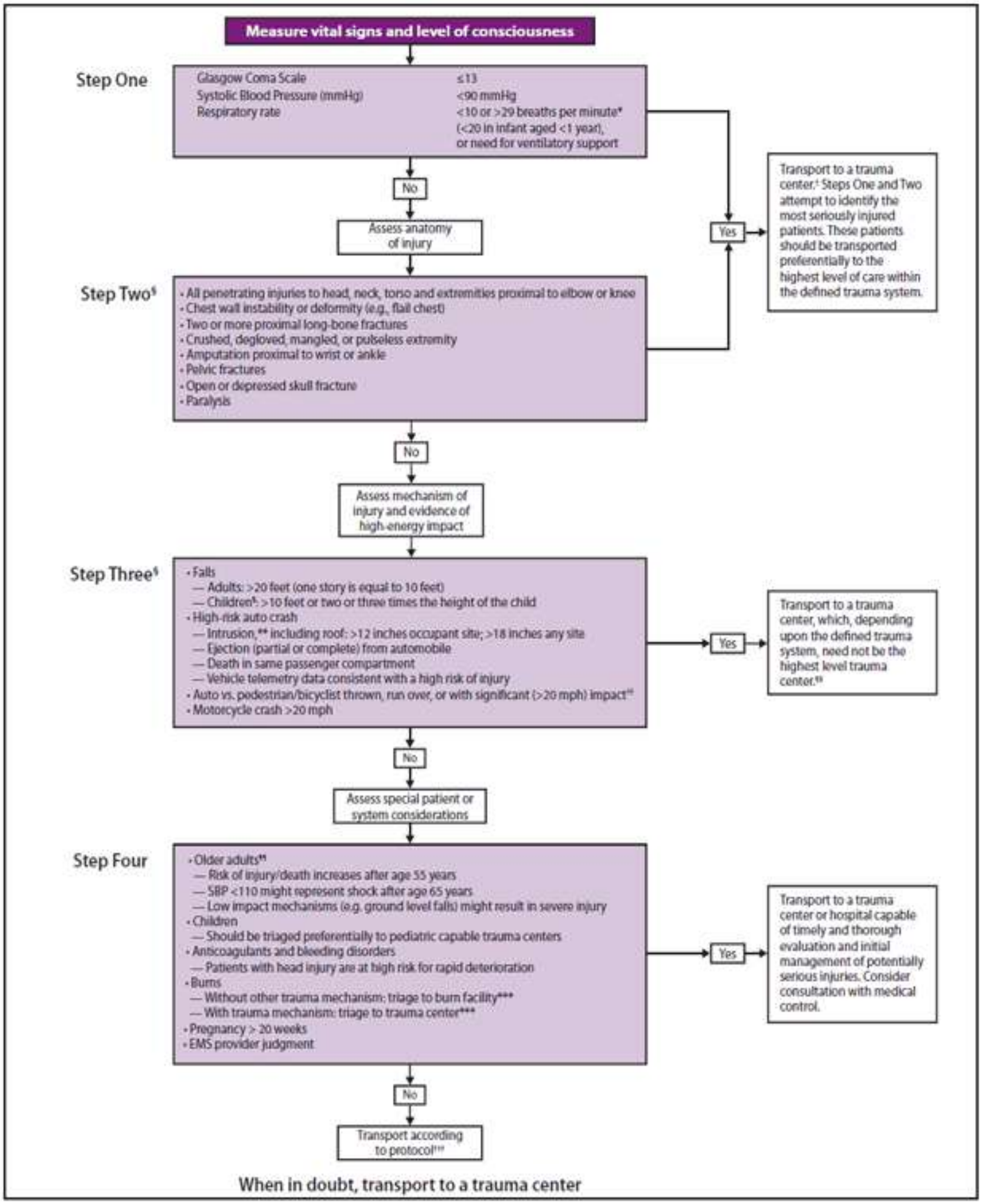
***Dangerous Mechanism**

- Fall from elevation > 3 feet or 5 stairs.
- Axial load to head, (e.g. diving)
- MVC high speed, rollover, ejection.
- Motorized recreational vehicle, (e.g. ATV)
- Bicycle collision with object, (e.g. post, car)

****Simple rear-end MVC excludes:**

- Pushed into oncoming traffic.
- Hit by bus/large truck.
- Rollover.
- Hit by high-speed vehicle.

Field Triage of Injured Patients



CHEMPACK Fact Sheet

What is CHEMPACK?

- The CHEMPACK is a cache of antidotes for nerve agent or organophosphate exposure/toxicity.

What is CISM?

- The Centers for Disease Control and Prevention's (CDC) CHEMPACK Program is a nationwide initiative for the placement of CHEMPACK caches in local communities.
- The CHEMPACK program fills a void in emergency preparedness by placing timely, critical, and life-saving antidotes in communities where they will be readily available to EMS, Hospitals, Fire, Law Enforcement and other CHEMPACK response partners.

Georgia's CHEMPACK Program

- Georgia's CHEMPACK program consist of more than 40 CHEMPACK sites pre-positioned throughout the state.
- CHEMPACK assets can be requested and received quickly to support emergency response.
- Training and additional support materials are available for EMS, Hospitals, Fire, Law Enforcement, and other CHEMPACK response system partners.

Georgia's Poison Center

- The center provides clinical guidance for potential nerve agent or organophosphate exposure/toxicity.
- The center also receives all request for CHEMPACK assets and coordinates delivery.

CHEMPACK Container

- CHEMPACK requests are scalable and will be configured based upon need.
- The CHEMPACK is for use by first responders and hospital staff.

CHEMPACK Container Formulary Components

- Mark I Nerve Agent Antidote Kit
- Diazepam Auto-Injectors
- Atropine, Pralidoxime, Diazepam multi-dose vials
- Atropine Pediatric Doses

CHEMPACK REQUESTS?
CALL
1-800-222-1222

Georgia Critical Incident Stress Foundation

Crisis Hotline: (404) 419-6506

What is a Critical Incident?

- A critical incident or traumatic event is defined as an event so stressful that it overwhelms the existing coping skills of the individual or group. After being exposed to the critical incident, an individual may experience a range of reactions, which are manifested physically, cognitively, behaviorally, and/or emotionally, and may interfere with one's ability to function at work and at home. The stress reaction may include, but is not limited to fatigue, muscle tremors, rapid heart rate, confusion, poor attention, poor problem solving, nightmares, anxiety, grief, fear, depression, inappropriate emotional responses, withdrawal, changes in activity, etc. A critical incident often leads to increased absenteeism and poor work performance.
- **Critical Incidents:**
    - Suicide
    - Violent Crimes
    - Homicide
    - Death or violence to a child
    - Traffic Accidents.
    - Mass Casualty Incidents.
    - Unexpected Death.
    - School related crisis.
    - Robbery.
    - Life-threatening injury.
    - Natural disasters.
    - Workplace violence.

What is the Critical Incident Stress Program?

- Critical Incident Stress Management or CISM is a comprehensive, multi-component crisis intervention approach. CISM is considered to be comprehensive because it consists of multiple crisis intervention components, which functionally span the entire temporal spectrum of a crisis. CISM interventions range from the pre-crisis phase through the acute crisis phase, and into the post-crisis phase. CISM consist of interventions which may be applied to individuals, small functional groups, large groups, families, organizations and communities.
- The seven core components include:
    1. Pre-crisis preparation.
    2. Disaster or large-scale incident.
    3. Defusing.
    4. Critical Incident Stress Debriefing (CISD).
    5. One-on-one crisis intervention/counseling.
    6. Family crisis intervention and organizational consultation.
    7. Follow-up and referral mechanisms for assessment and treatment.

Georgia Critical Incident Stress Foundation

- GCISF is dedicated to the prevention and mitigation of disabling stress through the provision of education, training, crisis response support services, and coordination for all at risk populations. They provide and promote consistent crisis response training relevant to the needs of Emergency Services, Law Enforcement, Critical Healthcare professionals, School systems, Mental Health professionals, and lay persons who are referred to as peer support providers. GCISF offers consultation and direction in the establishment of comprehensive crisis response programs to organizations and communities throughout the State of Georgia. Local GCISF networked CISM teams actively pursue the concept that no one should ever face the harmful effects of critical incident stress without appropriately trained, well qualified assistance. GCISF maintains a 24/7/365 hotline for the coordination of crisis response activities.

Emergency Medical Treatment and Active Labor Act (EMTALA)

EMTALA is a U.S. Act of Congress passed in 1986 as part of the Consolidated Omnibus Budget Reconciliation ACT (COBRA). It requires hospitals and ambulance services to provide care to anyone needing emergency healthcare treatment regardless of citizenship, legal status, or ability to pay. There are no reimbursement provisions. As a result of the Act, patients needing emergency treatment can be discharged only under their own informed consent or when their condition requires transfer to a hospital that is better equipped to administer the treatment they need.

EMTALA was passed to combat the practice of "Patient dumping", i.e. refusal to treat people because of their inability to pay or having insufficient insurance, transferring, or discharging emergency patients on the basis of highly anticipated diagnosis and treatment cost. The law applies when an individual with medical emergency "comes to the emergency department" regardless of whether the condition is visible to others or is simply stated by the patient with no external evidence.

- Unstable patients may occasionally be transferred if essential services are not available at the sending hospital – "a higher level of care transfer".
- Patients being transferred from one acute care facility to another **MUST** have been accepted by the receiving facility. Ambulance crews should ensure that the appropriate arrangements have been made prior to loading the patient.
- The ambulance crew should ensure that **ALL** transfer paperwork accompanies the patient.
- If paramedics suspect that a transfer has the risk of being possible EMTALA violation, they should attempt to tactfully discuss the matter with the hospital personnel. The On Duty Field Supervisor or Operations Manager should be a resource if the crew is in doubt about the appropriateness of the transfer.

Glascow Coma Score

The Glascow Coma Score (GCS) is the most widely used scoring system used in quantifying the level of consciousness following a traumatic brain injury. It is used primarily because it is simple, has a relatively high degree of interobserver reliability and because it correlates well with outcome following severe brain injury.

To obtain a GCS, first determines the best eye-opening response, the best verbal response and the best motor response using the chart below. The GCS represents the sum of the numeric scores of each of the categories.

Total GCS = E+V+M

Glascow Coma Score – Adults and Children		
Eye Opening	**Verbal Response**	**Motor Response**
4 – Spontaneous	5 – Oriented and converses	6 – Obeys verbal commands
3 – To Verbal Commands	4 – Disoriented and converses	5 – Localizes Pain
2 – To Pain	3 – Inappropriate Words	4 – Withdraws from Pain
1 – No Response	2 – Incomprehensible Sounds	3 – Flexion
	1 – No Verbal Response	2 – Extention
		1 – No Motor Response
Modified Glascow Coma Score – Infants		
Eye Opening	**Verbal Response**	**Motor Response**
4 – Spontaneous	5 – Coos, babbles	6 – Spontaneous
3 – To Verbal Commands	4 – Irritable cries	5 – Localizes Pain
2 – To Pain	3 – Cries to Pain	4 – Withdraws from Pain
1 – No Response	2 – Moans, grunts	3 – Flexion
	1 – No Response	2 – Extension
		1 – No Response

Values:

- A score between 13 and 15 may indicate a mild head injury.
- A score between 9 and 12 may indicate a moderate head injury.
- A score of 8 or less may indicate a severe head injury.

HIPPA Fact Sheet: Emergency Services

Public Health Activities Protected by HIPPA

The comments to the preamble of the Privacy Rule explicitly protect state public health laws by making it clear that “nothing in this [Rule] shall be construed to invalidate or limit the authority, power, or procedures established under any law providing for the reporting of disease or injury, child abuse, birth or death, public health surveillance, or public health investigation or intervention.

HIPPA Does Not Preempt State Public Health Laws

The Privacy Rule specifically states that it does not preempt contrary state public health laws, including state procedures established under such laws that provide for the reporting of disease or injury, child abuse, birth or death, or for the conduct of public health surveillance, investigation, or intervention. ***[45 CFR 160.203 (a)(1)(iv)&(c)]***

Public Health Authorities Defined

Public health authorities include state public health agencies (e.g. state public health departments, division, districts, or regions), local public health agencies, and anyone performing public health functions under a grant of authority from a public health agency. ***[45 CFR 164.501]***

Disclosures Required by Law

The Privacy Rule permits covered entities to disclose protected health information, without authorization, to public health authorities who are authorized by law to receive such reports for the purpose of preventing or controlling disease, injury or disability and for conducting public health surveillance, investigations or interventions. This includes federal, tribal, local or state laws (or state procedures established under such law) that provide for receiving reporting of disease, injury, or conducting public health surveillance, investigation or intervention. ***[45 CFR 164.512 (a)&(b)]***

Public Health Authorities are not Business Associates of Covered Entities

Public health authorities receiving information from covered entities as required or authorized by law ***[See 45 CFR 164.512 (a)&(b)]*** are not business associates of the covered entities and therefore are not required to enter into business associate agreements. ***[CDC MMWR, Vol. 52, Pg 8 (May 2003)]***

Minimum Necessary Rule

Generally, a covered entity must make reasonable efforts to limit protected health information to the minimum necessary to accomplish the intended purpose of the use, disclosure, or request. ***[45 CFR 164.502 (b).*** However, covered entities are not required to make a minimum necessary determination for public health disclosures that are required by law. ***[45 CFR 164.502 (b)].*** For disclosures to a public health authority, covered entities may reasonably rely on a requested disclosure, as the minimum necessary if the public health authority represents that the information requested is the minimum necessary for the stated purpose. ***[45 CFR 164.514 (d)(3)(iii)]***

Accounting for Public Health Disclosures

The Privacy Rule provides for a simplified means of accounting because the vast amount of data exchanged between covered entities and public health authorities is made through ongoing regular reporting. For example, ambulance service providers are required by law regularly submit copies of pre-hospital care reports to regional offices that are part of the state public health authority. (continued)

CONTINUED ON NEXT PAGE

HIPPA Fact Sheet: Emergency Medical Services (Continued)

Accounting for Public Health Disclosures (Continued)

In such cases, the covered entity need only identify the recipient of such repetitive disclosures (regional public health authority), the purpose of the disclosure (required for injury control and prevention), and describe the protected health information routinely disclosed. The date of each disclosure need not be tracked. Rather, the accounting may include the date of the first and last such disclosure during the accounting period (June 1, 2003 – July 1, 2003), and a description of the frequency or periodicity (monthly) of such disclosures. Therefore, the covered entity would not need to annotate each patient's medical record whenever a routine public health disclose was made. ***[CDCMMWR, Vol 52, Pg 9 (May 2003)]***

Relevant State Laws

O.C.G.A. § 31-11-5; Rules and Regulations for Ambulance Services

O.C.G.A. § 31-12-6; Records of Ambulance Services

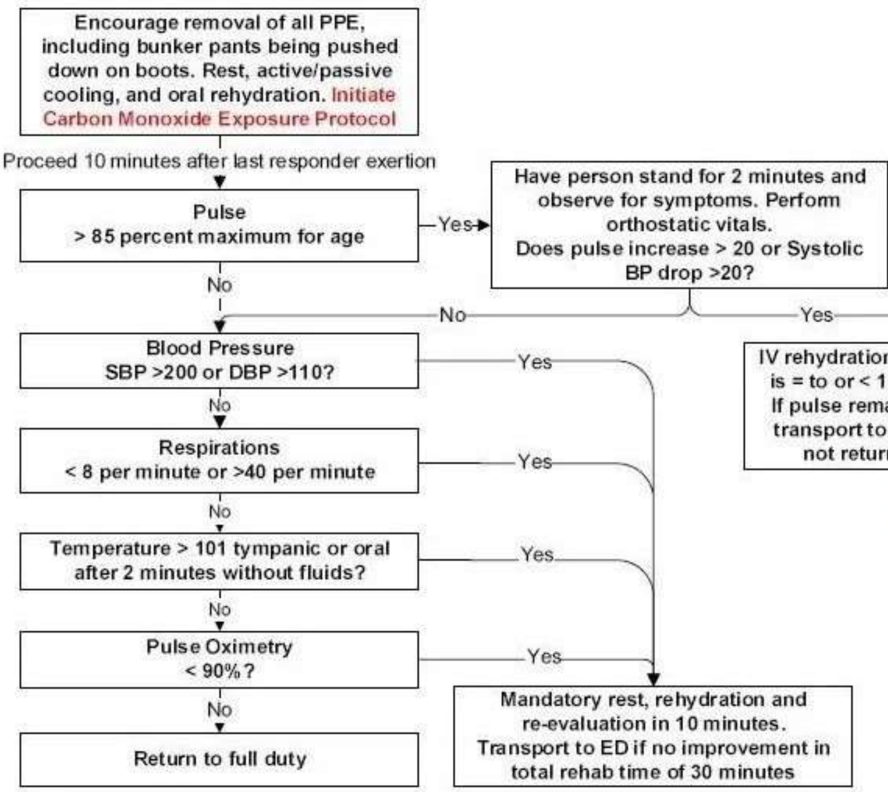
DHR Rules and Regulations Chapter 290-5-30; Emergency Medical Services

Sources

U.S. Department of Health and Human Services
Office of Civil Rights
HIPPA Privacy – Disclosure for Public Health Activities (Revised April 3, 2003)
Summary of the HIPPA Privacy Rule (May 2003)
http://www.hhs.gov/ocr/hippa/privacy.html

U.S. Department of Health and Human Services
Centers for Disease Control and Prevention
Morbidity and Mortality Weekly Report
Vol. 52 Supplement (May 2003)
http://www.cdc.gov/mmwr/preview/ind2003.su.html

Firefighter Rehabilitation Algorithm



NFPA Age-Predicted 85% Maximum Heart Rate	
Age	85 Percent
20 – 25	170
25 – 30	165
30 – 35	160
35 – 40	155
40 – 45	152
45 – 50	148
50 – 55	140
55 – 60	136
60 – 65	132

Pearls:

- This protocol may be applied to adult patients on fire scenes and any gathering approved by Medical Director
- Exam: Mental Status, Skin, Neck, Heart, Lungs, Abdomen, Back, Extremities, Neuro
- When used for firefighter rehab, firefighters should report to rehab for evaluation after 45 minutes (2 thirty minute or 1 sixty minute cylinder), or earlier if the firefighter or incident commander desires.
- Any person complaining of shortness of breath, confusion, combativeness or headache will be treated with highflow oxygen and be transported to the hospital.
- Automatic Transport Criteria:
  - Chest Pain
  - Shortness of breath unresolved by 10 minutes on high flow oxygen.
  - Heart rhythm other than NSR or sinus tach
  - A syncopal episode, disorientation, or confusion
  - Vital signs that have not returned to normal after 30 minutes of rest
  - Inability to hold fluids down or an episode of vomiting
  - Request for transport for any reason
- Cooling Techniques
  - Expose by full gear removal
  - Immerse forearms in water in rehab chairs when available (most effective technique)
  - Do not apply wet towels to neck or head of firefighters- risk of steam burns is increased
  - Cooling fans, ambient evaporative cooling
- Medical Control contact is not required to discontinue IV therapy to adult Emergency Services personnel treated under this protocol. For other approved gatherings, contact requirements are determined at approval.

Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

Firefighter Rehabilitation Roster

Fire Site:	**Name & Unit**						
Date:							
Evaluation Time:							
Initial Evaluation	**Pulse Rate**						
	O$_2$ Sat						
	CO Level						
	Injuries?	**Y N**	**Y N**	**Y N**	**Y N**	**Y N**	**Y N**
	Ill?	**Y N**	**Y N**	**Y N**	**Y N**	**Y N**	**Y N**
	Other						

Deny Return to Duty

- Vomiting, diarrhea, heat exhaustion last 72 hrs.
- Large open skin wounds or rash.
- Insulin-using diabetic has not eaten in past 4 hrs.
- Wheezing or Congested Lungs.
- Pulse over 120 bpm or irregular.
- CO level over 5 – 8%

All firefighters hydrated 8 – 16 oz Water or Electrolyte Solution

Evaluation Time:							
2nd Evaluation	**Pulse Rate**						
	O$_2$ Sat						
	CO Level						
	Injuries?	**Y N**	**Y N**	**Y N**	**Y N**	**Y N**	**Y N**
	Ill?	**Y N**	**Y N**	**Y N**	**Y N**	**Y N**	**Y N**

All firefighters hydrated 8 – 16 oz Water or Electrolyte Solution

Evaluation Time:							
3rd Evaluation	**Pulse Rate**						
	O$_2$ Sat						
	CO Level						
	Injuries?	**Y N**	**Y N**	**Y N**	**Y N**	**Y N**	**Y N**
	Ill?	**Y N**	**Y N**	**Y N**	**Y N**	**Y N**	**Y N**

To Hospital

- Symptoms of Heat Stroke
- Shortness of Breath
- Abnormal Lung Sounds
- Altered Mental Status
- Irregular Pulse
- Persistent Pulse
- Significant Injury
- Chest Pain or Severe Headache

Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
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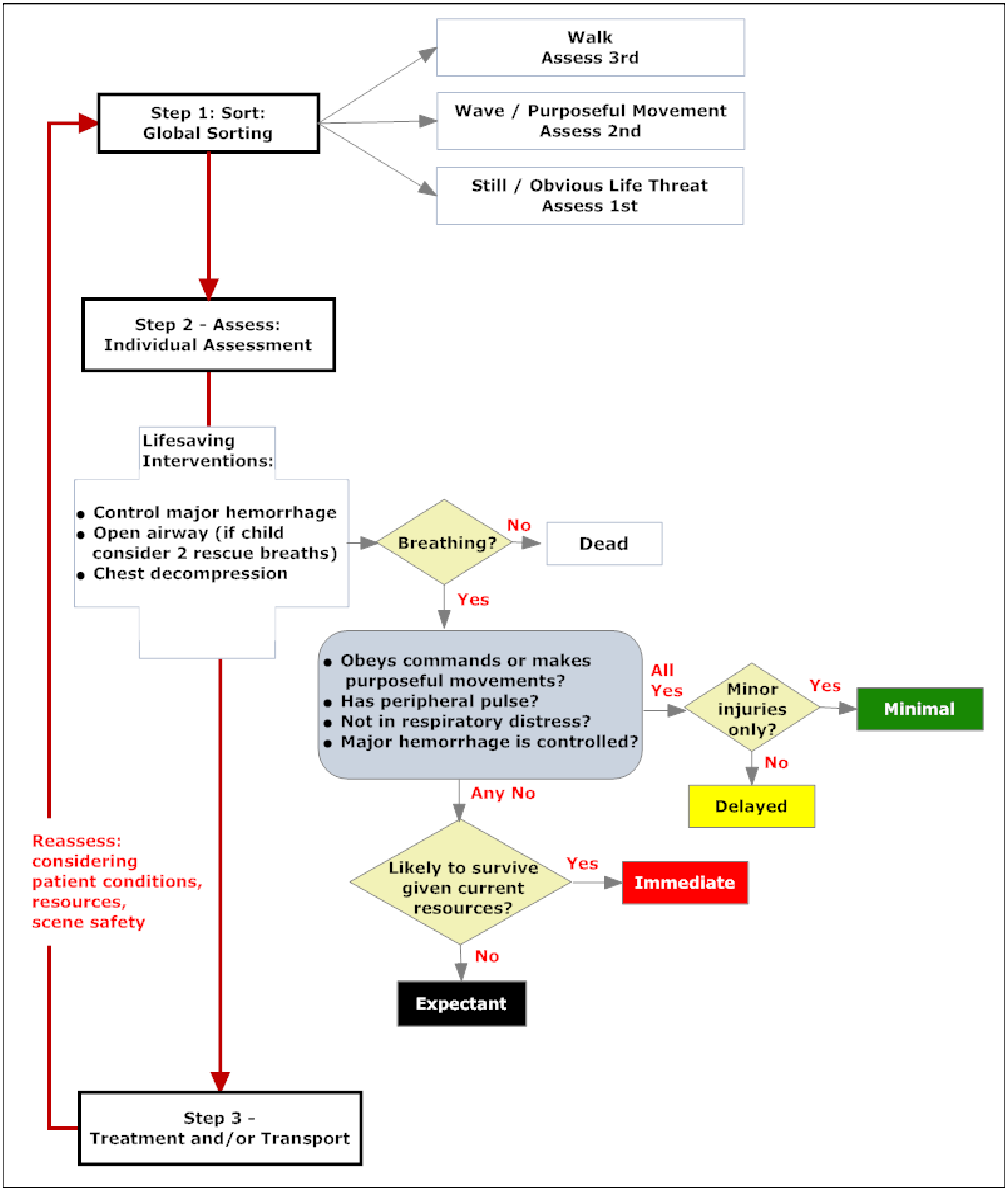
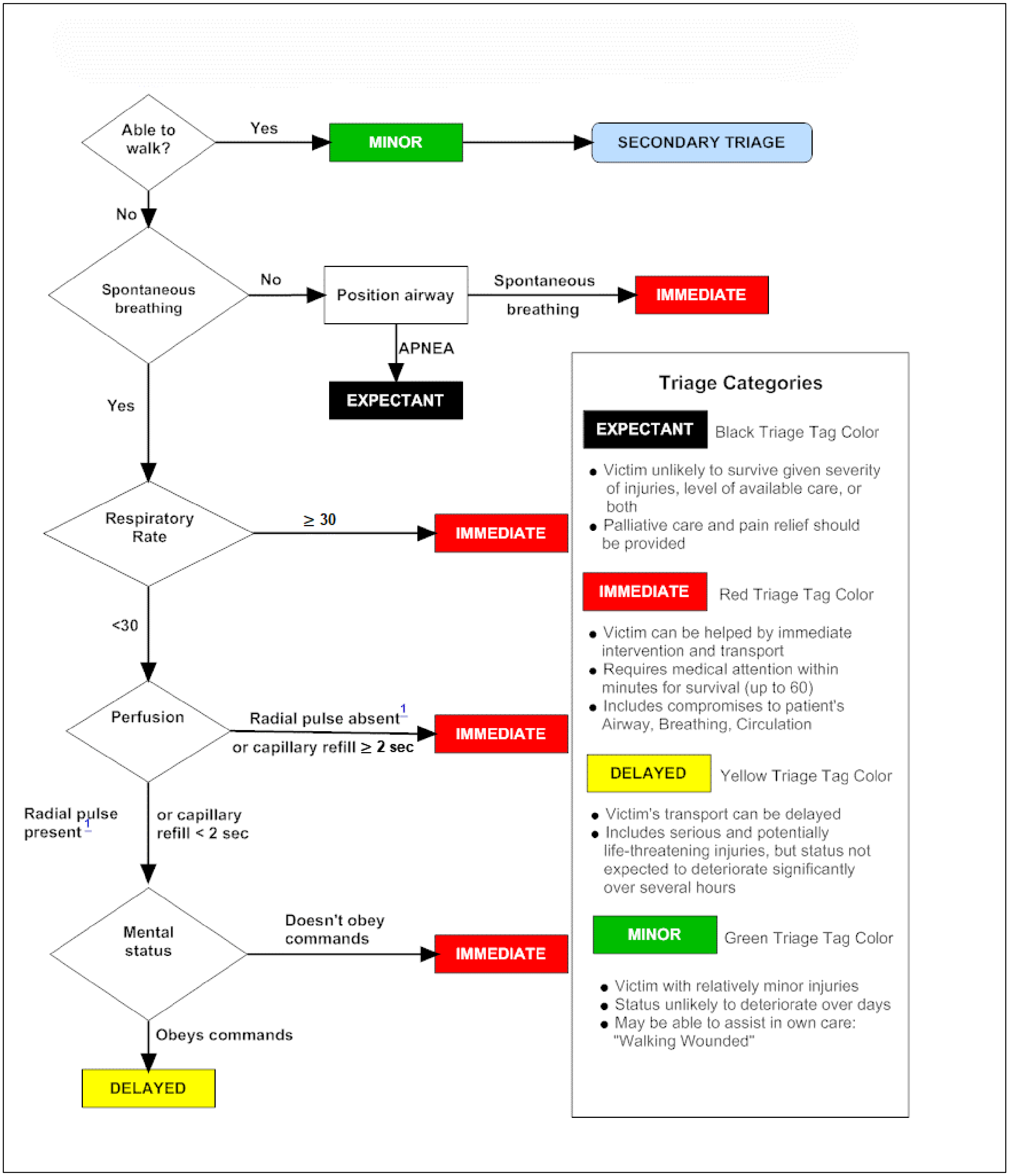
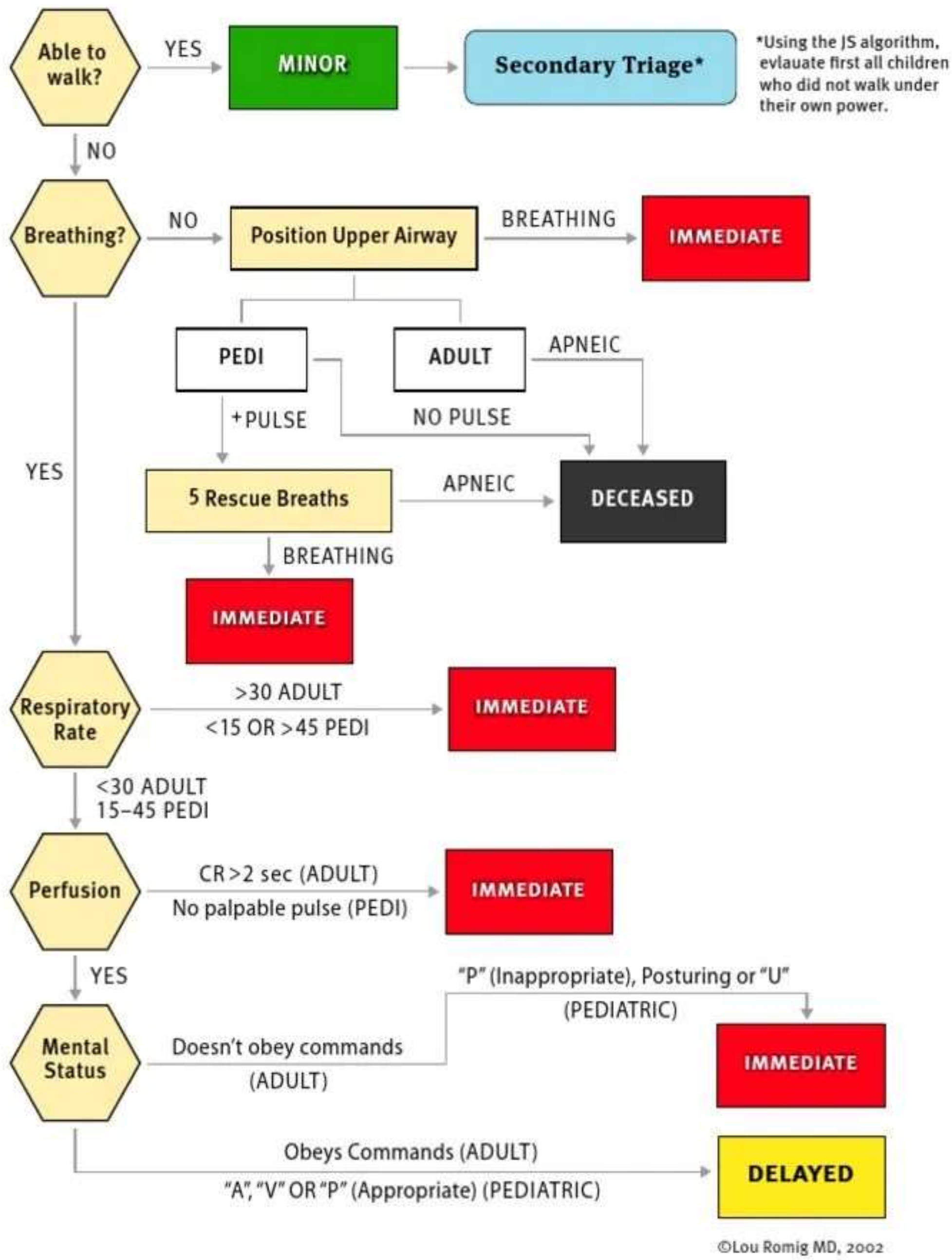
SALT MCI Triage Algorithm



START MCI Triage Algorithm



Combined START/JumpSTART Triage Algorithm

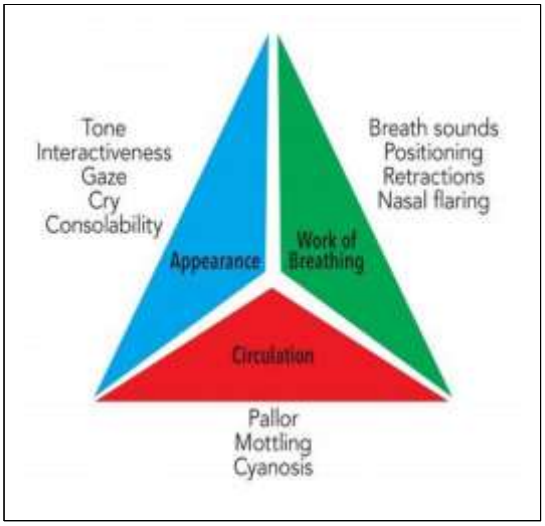


Pediatric Assessment Triangle (PAT)

The Pediatric Assessment Triangle (PAT) is a tool for medical professionals to rapidly assess a pediatric patient on sight and obtain a “first impression” of the child’s condition. Using the PAT, an EMS provider can determine if a child is in immediate need of rapid transport or emergency treatment before a full assessment.

Below are the parameters to be assessed using the PAT.

Appearance	
Characteristics:	**Features to Look For:**
Tone	Good muscle tone OR limp, listless, flaccid.
Interactivity	Alert, will reach for toy, light OR is uninterested in playing or interacting.
Consolability	Can be consoled OR crying or agitation is unrelieved.
Look/Gaze	Fixes on face, object OR glassy eyed stare.
Speech/Cry	Cry strong and spontaneous OR weak or high-pitched. IS Speech age appropriate OR confused, garbled?



Breathing	
Characteristics:	**Features to Look For:**
Abnormal Airway Sounds	Snoring, muffled or hoarse speech, stridor, grunting, wheezing.
Abnormal Positioning	Sniffing position, tripoding, refusing to lie down.
Retractions	Supraclavicular, intercostal, and/or sternal retractions of chest wall, head bobbing in infants.
Flaring	Flaring of the nares on inspiration.

Circulation/Skin Color	
Characteristics:	**Features to Look For:**
Pallor	White or pale skin or mucous membranes
Mottling	Patchy/lacey skin discoloration due to vasoconstriction or vasodilation
Cyanosis	Bluish discoloration of skin/mucous membranes
Flaring	Flaring of the nares on inspiration.

Normal Pediatric Vital Signs by Age and Weight

Age	Weight in kg.	Pulse	Respirations	Systolic BP	Diastolic BP
Premature	1	145	< 60	42 +/- 10	21 +/- 8
Premature	1 – 2	135	< 60	50 +/- 10	28 +/- 8
Newborn	2 – 3	125	< 60	60 +/- 10	37 +/- 8
1 Month	4	120	24 – 35	80 +/- 16	46 +/- 16
6 Months	7	120	24 – 35	89 +/- 29	60 +/- 10
1 Year	10	120	20 – 30	96 +/- 30	66 +/- 25
2-3 Years	12 – 14	115	20 – 30	99 +/- 25	64 +/- 25
4-5 Years	16 – 18	100	20 – 30	99 +/- 20	65 +/- 20
6-9 Years	20 – 26	100	12 – 25	100 +/- 20	65 +/- 15
10-12 Years	32 – 42	75	12 – 25	112 +/- 20	68 +/- 15
Over 14 Years	> 50	70	12 – 18	120 +/- 20	75 +/- 15

Abnormal Vital Signs by Age

Age	Pulse	Respirations	Systolic BP	Temperature
0 Days - < 1 month	< 80 > 205	< 30 > 60	< 60	< 36 > 38
≥ 1 month - 3 months	< 80 > 205	< 30 > 60	< 70	< 36 > 38
≥ 3 months - < 1 year	< 75 > 190	< 30 > 60	< 70	< 36 > 38.5
≥ 1 year - < 2 years	< 75 > 190	< 24 > 40	< 70 + (age x 2)	< 36 > 38.5
≥ 2 years - < 4 years	< 60 > 140	< 24 > 40	< 70 + (age x 2)	< 36 > 38.5
≥ 4 years - < 6 years	< 60 > 140	< 22 > 34	< 70 + (age x 2)	< 36 > 38.5
≥ 6 years - < 10 years	< 60 > 140	< 18 > 30	< 70 + (age x 2)	< 36 > 38.5
≥ 10 years - < 13 years	< 60 > 100	< 18 > 30	< 90	< 36 > 38.5
≥ 13 years - < 18 years	< 60 > 100	< 12 > 18	< 90	< 36 > 38.5

GEORGIA DEPARTMENT OF PUBLIC HEALTH

GEORGIA OFFICE OF EMS AND TRAUMA

SCOPE OF PRACTICE FOR EMS PERSONNEL

LAST UPDATE: 01/23/2026

EFFECTIVE DATE: 01/23/2026

Table of Contents

Introduction

Broadly defined, a Scope of Practice outlines the parameters of duties or services that an individual with specific credentials is authorized to perform. This description, as defined by the Department of Public Health, specifies what a Licensee is legally permitted or prohibited from doing based on their level of licensure. It serves as a legal framework that distinguishes between licensed health care personnel and the general public, as well as among various licensed health care professionals. The Scope of Practice for EMS Personnel, for instance, does not include certain skills like chest tube insertion or surgical procedures, as these are reserved for individuals with higher levels of healthcare licensure.

As noted in the 2019 National EMS Scope of Practice Model (available at www.ems.gov), an individual may only perform a skill or role for which that person is:

- **EDUCATED** (has been trained to perform the skill or role), AND
- **CERTIFIED** (has demonstrated competence in the skill or role), AND
- **LICENSED** (has legal authority issued by the State to perform the skill or role), AND
- **CREDENTIALED** (has been authorized by the medical director to perform the skill or role).

In Georgia, EMS personnel are authorized to perform skills and administer medications within the scope of practice for their licensure level following standing, verbal, or written orders. This authorization is only valid when the provider:

1. has received proper training and certification for those skills and medications; AND
2. has been credentialed for those skills and medications by their local EMS Agency Medical Director; AND
3. is working as a member of a licensed EMS agency.

Alternatively, a licensed EMS provider may perform skills and administer medications within their scope of practice in settings such as hospitals, emergency departments, medical facilities, or physician offices under the direct supervision of a licensed physician. Direct supervision requires the supervising physician to be present in the same physical location and be readily available to assist and guide the EMS provider throughout the patient encounter.

When EMS personnel are authorized to perform skills or administer medications as outlined in this document, the decision to execute a skill or administer medications for a specific patient in a given situation should be based on the patient's clinical presentation, the standing, verbal, or written orders of a duly licensed physician, and current evidence-based practice. The ability to perform a skill or administer a medication does not automatically imply that a provider should do so.

Statutory Authority Regarding Scope of Practice

The following Georgia statutes contain the enabling legislation for the Scope of Practice for EMS Personnel.

- O.C.G.A. § 31-11-53. Services which may be rendered by certified emergency medical technicians and trainees
- O.C.G.A. § 31-11-54. Services which may be rendered by paramedics and paramedic trainees
- O.C.G.A. § 31-11-55. Services which may be rendered by certified cardiac technicians and trainees

Delegation of Skills

Medics are authorized to delegate only the skills they can perform themselves. Delegation to other licensees is permissible only if those skills fall within the respective Scope of Practice of the delegate. For instance, a Paramedic can delegate the administration of amiodarone (an antiarrhythmic) to a Cardiac Technician, subject to approval by the local Medical Director. However, a Paramedic is not allowed to delegate the administration of amiodarone to an EMT-R, EMT, EMT-I, or AEMT.

Scope of Practice Approval

This document is hereby approved as of the date listed below.

Approval:		1/23/2026
	Michael B. Johnson, Director, Office of EMS and Trauma	Date

Approval:	P McDougal, MD	1/23/2026
	Patrick McDougal, Medical Director, Office of EMS and Trauma	Date

Scope of Practice Guide for EMT-R, EMT, EMT-I, AEMT, CT, PMDC, CCP

Key to Provider Levels		
EMT-R	R	Emergency Medical Technician - Responder
EMT	E	Emergency Medical Technician
EMT-I	I	Emergency Medical Technician - Intermediate
AEMT	A	Advanced Emergency Medical Technician
CT	C	Cardiac Technician
PMDC	P	Paramedic
CCP	CC	Critical Care Paramedic

NOTE: If a provider code (the single letter code from the table above) is listed for a particular skill, then that level of EMS provider is permitted to perform that skill. If a skill does not have the letter code shown for a provider level, then personnel licensed at that level are NOT permitted to perform that skill. EMS providers performing skills outside their scope of practice may be subject to disciplinary action under DPH Rules and Regulations 511-9-2. Interpretive guidelines serve to clarify or modify the skill listed. If an asterisk (*) appears with the letter code for a specific provider level, then the interpretive guidelines may modify the skill for that provider level.

If the letter code includes a "GA" below it, this means that individual skill for the respective license level is not included in the National EMS Scope of Practice Model but is included in the Georgia Scope of Practice for EMS Personnel. For a licensee to perform these skills, the EMS Agency and the Medical Director must ensure that the skill has been appropriately taught and verified for each individual EMS Agency Medic licensed at the respective level.

Airway and Breathing Skills	Levels							Interpretive Guidelines
1. Supplemental oxygen therapy								
a. Oxygen delivery devices	R*	E	I	A	C	P	CC	*EMT-Rs are limited to the use of nasal cannulas and non-rebreather masks. This does **not** include High Flow Nasal Cannulas (HFNC) which is a system that provides oxygen at a flow rate that exceeds 15 LPM and requires a specially designed nasal canula.
b. Humidified oxygen		E	I	A	C	P	CC	
c. High flow oxygen via nasal cannula						P*	CC	*Paramedics can only perform this skill with PLS approval.
2. Basic airway management								
a. Manual maneuvers to open and control the airway	R	E	I	A	C	P	CC	
b. Manual maneuvers to remove obstructions from the airway	R	E	I	A	C	P	CC	
c. Insertion of airway adjuncts intended to go into the oropharynx	R	E	I	A	C	P	CC	
d. Insertion of airway adjuncts intended to go into the nasopharynx		E	I	A	C	P	CC	
3. Ventilation management								
a. Manual ventilation by mouth	R	E	I	A	C	P	CC	
b. Bag-valve mask	R	E	I	A	C	P	CC	
c. Manually triggered ventilators		E	I	A	C	P	CC	
d. Automatic transport ventilators capable of rate and tidal volume adjustments only		E*	I*	A*	C	P	CC	*EMTs, EMT-Is and AEMTs are limited to the initiation of automatic transport ventilators during resuscitative efforts only.
e. Chronic-use home ventilators		E	I	A	C	P	CC	
f. Advanced transport ventilators						P*	CC	*Paramedics can only perform this skill with PLS approval.
4. Suctioning								
a. Upper airway suctioning	R	E	I	A	C	P	CC	
b. Tracheobronchial suctioning				A*	C	P	CC	*AEMTs are limited to tracheobronchial suctioning of patients with pre-established airways.

SCOPE OF PRACTICE FOR EMS PERSONNEL
GEORGIA OFFICE OF EMS AND TRAUMA
POLICY: SOP-2024 (effective 1/23/2026)

Airway and Breathing Skills	Levels							Interpretive Guidelines
5. Advanced airway management								
a. CPAP administration and management		E	I	A	C	P	CC	
b. BiPAP administration and management			I	A	C	P	CC	
c. Supraglottic airway device/BIAD (Blind Insertion Airway Device) insertion/removal			I*	A*	C	P	CC	**EMT-Is and AEMTs are limited to the insertion of devices not intended to be placed into trachea.*
d. Endotracheal intubation					C	P	CC	
e. Airway obstruction removal by direct laryngoscopy					C	P	CC	
f. Percutaneous cricothyrotomy						P*	CC	**This would include devices that puncture the skin and/or cricothyroid membrane. Paramedics are not permitted to make a surgical incision of the cricothyroid membrane. Paramedics may perform skin incisions with a surgical blade for percutaneous cricothyrotomy.*
g. Gastric decompression						P	CC	
h. Pleural decompression via needle thoracostomy						P	CC	
i. Chest tube monitoring and management						P	CC	
j. Rapid sequence intubation							CC	
k. Surgical cricothyrotomy							CC	

Assessment Skills	Levels							Interpretive Guidelines
1. Basic assessment skills								
a. Perform simple patient assessments	R	E	I	A	C	P	CC	*Includes determination of chief complaint, mechanism of injury/nature of illness, associated signs/symptoms, assessment of pain, and performing a rapid full body scan.*

Assessment Skills	Levels							Interpretive Guidelines
1. Basic assessment skills								
b. Perform comprehensive patient assessments		E	I	A	C	P	CC	*Includes investigation of chief complaint, past medical history, pertinent negatives, and more detailed assessments of major body systems and anatomical regions.*
c. Manual blood pressure	R	E	I	A	C	P	CC	
2. Advanced assessment skills/Monitoring devices								
a. Non-invasive (automated) blood pressure measurement	R* GA	E	I	A	C	P	CC	**EMT-Rs may only use an automated blood pressure device under the direction of an EMT or higher licensed Medic, RN, PA, or MD/DO who is present with the EMT-R and the patient.*
b. Pulse oximetry measurement	R* GA	E	I	A	C	P	CC	**EMT-Rs may only use a pulse oximetry device in preparation for the arrival of the responding EMS vehicle (with an EMT or higher licensed Medic), or under the direction of an EMT or higher licensed Medic, RN, PA, or MD/DO who is present with the EMT-R and the patient.*
c. Pulse oximetry interpretation		E	I	A	C	P	CC	
d. CO-oximetry measurement		E	I	A	C	P	CC	*Georgia specific skill.*
e. CO-oximetry interpretation		E	I	A	C	P	CC	*Georgia specific skill.*
f. Blood glucose measurement	R* GA	E	I	A	C	P	CC	**EMT-Rs may only obtain a capillary blood glucose level in preparation for the arrival of the responding EMS vehicle (with an EMT or higher licensed Medic), or under the direction of an EMT or higher licensed Medic, RN, PA, or MD/DO who is present with the EMT-R and the patient.*
g. Blood glucose interpretation		E	I	A	C	P	CC	

Assessment Skills	Levels							Interpretive Guidelines
2. Advanced assessment skills/Monitoring devices								
h. End-tidal CO2 monitoring and interpretation of waveform capnography				A	C	P	CC	
i. Blood chemistry analysis						P	CC	
j. Telemetric monitoring devices and transmission of clinical data, including video data		E	I	A	C	P	CC	
k. Vascular Doppler monitoring and interpretation			I	A	C	P	CC	*EMS personnel may only utilize Vascular Dopplers to detect peripheral pulses and blood flow. It is not permitted to assess fetal heart tones.*
l. Point of Care Ultrasound (POCUS)						P*	CC	**Paramedics may only use POCUS to establish peripheral IV access in an extremity.*
3. Specimen Collection								
a. Perform specimen collection for infectious diseases		E*	I	A	C	P	CC	**EMTs are not permitted to perform venipuncture for specimen collection.*

Pharmacological Interventions/Skills	Levels							Interpretive Guidelines
1. Fundamental pharmacological skills								
a. Use of unit dose commercial pre-filled containers or auto-injectors for the administration of life saving medications for chemical/hazardous material	R	E	I	A	C	P	CC	
b. Assist patients in taking their own prescribed medications as approved by the local EMS Medical		E	I	A	C	P	CC	*Georgia specific skill.*
c. Administration of over-the-counter medications with appropriate medical direction		E	I	A	C	P	CC	*Includes oral glucose for hypoglycemia, aspirin for chest pain of suspected ischemic origin, and analgesics for pain or fever.*
2. Advanced pharmacological skills: Venipuncture/Vascular access								
a. Obtaining peripheral venous blood specimens			I	A	C	P	CC	*This is either through direct venipuncture or through an existing IV catheter.*

	Pharmacological Interventions/Skills	Levels							Interpretive Guidelines
2. Advanced pharmacological skills: Venipuncture/Vascular access									
b.	Transport of a patient with a pre-established peripheral INT/saline lock		E*	I	A	C	P	CC	*EMTs are not permitted to access the INT/saline lock, nor are they permitted to remove it.
c.	Peripheral IV insertion and maintenance; includes removal as needed			I	A	C	P	CC	Peripheral lines include external jugular veins but does not include placement of umbilical catheters.
d.	Intraosseous device insertion; includes removal as needed			I	A	C	P	CC	
e.	Access indwelling catheters and implanted central IV ports for fluid and medication administration					C*	P*	CC*	*CTs, Paramedics and CCP are NOT permitted to place or remove central venous catheters.
f.	Central line monitoring					C	P	CC	
g.	Accessing central lines							CC	
h.	Obtaining arterial blood gasses							CC	
i.	Prehospital lab analysis							CC	
j.	Arterial line monitoring							CC	
k.	Umbilical vein catheterization							CC	
3. Advanced pharmacological skills: Medication/Fluid administration									
a.	Administration of crystalloid IV solutions			I*	A*	C	P	CC	*EMT-Is and AEMTs are limited to the initiation of crystalloid solutions that do not have added pharmacological agents.
b.	Maintenance of non-medicated IV fluids			I	A	C	P	CC	
c.	Maintenance of medicated IV fluids				A*	C*	P	CC	*AEMTs are permitted to administer TXA and Cyanokits via an infusion. *CTs are authorized to maintain only the following: antiarrhythmics, vagolytic agents, chronotropic agents, alkalizing agents, analgesic agents and vasopressor agents.
d.	Administration of hypertonic dextrose solutions for hypoglycemia			I	A	C	P	CC	

Pharmacological Interventions/Skills	Levels							Interpretive Guidelines
3. Advanced pharmacological skills: Medication/Fluid administration								
e. Administration of glucagon for hypoglycemia		E*	I*	A	C	P	CC	*EMT and EMT-I may only administer Glucagon via a commercially available auto-injector.
f. Administration of SL nitroglycerin to a patient experiencing chest pain of a suspected ischemic origin		E*	I*	A	C	P	CC	*EMTs and EMT-Is may only administer SL nitroglycerin using the patient's own prescribed medication.
g. Administration of epinephrine for cardiac arrest				A	C	P	CC	
h. Administration of epinephrine via auto-injector or Intranasal Atomizer for anaphylaxis	R GA	E	I	A	C	P	CC	*EMT-Rs and EMTs may administer epinephrine from a vial/syringe from a commercially or pharmacy pre-assembled and pre-measured kit, via the IM or IN route.
i. Parenteral administration of epinephrine for anaphylaxis		E*	I*	A*	C	P	CC	*EMTs and EMT-Is may administer epinephrine from a vial/syringe from a commercially or pharmacy pre-assembled and pre-measured kit, via the IM route. *AEMTs may prepare and administer epinephrine via the IM route only.
j. Administration of inhaled (nebulized) beta agonist/ bronchodilator and anticholinergic agents for dyspnea and wheezing		E*	I*	A	C	P	CC	*EMTs and EMT-Is may only administer pre-measured unit doses of nebulized medications.
k. Administration of a narcotic antagonist to a patient with a suspected narcotic overdose	R*	E*	I*	A	C	P	CC	*EMT-Rs, EMTs and EMT-Is may only administer narcotic antagonists via the intranasal route or via an auto-injector.
l. Adminstration of non-narcotic/non-controlled analgesics				A*	C	P	CC	*AEMTS may only administer non-narcotic/non-controlled analgesics
m. Adminstration of antiemetics				A*	C	P	CC	*AEMTS may only administer Zofran via the parenteral or ODT route
n. Administration of nitrous oxide (50% mixture) for pain relief				A	C	P	CC	Nitrous oxide is required to be patient self-administered.

Pharmacological Interventions/Skills	Levels							Interpretive Guidelines
3. Advanced pharmacological skills: Medication/Fluid administration								
o. Vaccine administration		E* GA	I*	A*	C*	P	CC	*EMTs, EMT-Is, AEMTs, and CTs may only administer vaccinations during public health emergency (as defined in O.C.G.A. § 31-12-1.1) and then only after approved training. *EMTs and EMT-Is may only administer vaccines that have been prepared by a higher-level provider. See SB46/O.C.G.A. § 31-11-53 (a)(3) & O.C.G.A. § 31-11-55 (a)(2)(E).
p. Administration of other physician-approved medications					C*	P*	CC	*CTs are authorized to give only the following: antiarrhythmics, vagolytic agents, chronotropic agents, alkalizing agents, analgesic agents and vasopressor agents. (See O.C.G.A. § 31-11-55). *In addition to the medications with respective interpretive guidelines in previous sections, Paramedics are authorized to give any additional medication via approved enteral or parenteral routes.
q. Paralytic administration						P*	CC	***Administration of paralytics by Paramedics for DAI/RSI is NOT permitted*** unless an agency has obtained ***written*** approval from the Office of EMS and Trauma. *Paramedics are only authorized to use non-depolarizing paralytics to maintain the paralysis of already intubated patients during interfacility transports. CC are permised to give paralytics for RSI.

Pharmacological Interventions/Skills	Levels							Interpretive Guidelines
3. Advanced pharmacological skills: Medication/Fluid administration								
r. Sedative/Hypnotic agents						P	CC	*Administration of sedative/hypnotic agents for the purpose of intubating a patient is generally not recommended and should be utilized only by EMS systems that, in the judgment of the local EMS Medical Director(s), have a specific need for the procedure and possess adequate resources to develop and maintain a prehospital drug-assisted intubation (DAI) protocol. EMS providers performing DAI should possess training, knowledge, and experience in the techniques and in the use of pharmacologic agents used to perform DAI. (adapted from the NAEMSP position statement on DAI).*
s. Blood or blood products						P*	CC	**Paramedics may maintain a blood/blood product infusion started at the sending facility.* **Paramedics can initiate initial or an additional unit of blood or blood products with PLS approval*
t. Cyanokit administration				A	C	P	CC	
u. Administration of TXA				A	C	P	CC	*Georgia specific skill.*

Pharmacological Interventions/Skills	Levels							Interpretive Guidelines
3. Advanced pharmacological skills: Medication/Fluid administration								
v. Administration of intramuscular hydrocortisone sodium succinate						P	CC	*In addition to Paramedics being permitted to administer hydrocortisone sodium succinate based on physician orders in the pre-hospital setting, per O.C.G.A. § 31-11-55.2, Paramedics are authorized to administer intramuscular hydrocortisone sodium succinate to a patient who:* *1. Has congenital adrenal hyperplasia; or any adrenal insufficiency;* *2. Is believed to be in adrenal crisis; and* *3. Has on his or her person or in his or her belongings hydrocortisone sodium succinate in packaging that clearly states the appropriate dosage and has an unbroken seal.* *Within a reasonable period of time, all Paramedics who administer hydrocortisone sodium succinate pursuant to O.C.G.A. § 31-11-55.2 shall make available a printed or electronically stored report to the licensed ambulance service which transports the patient.*
4. Medication administration: Approved routes of administration								
a. Aerosolized/nebulized		E	I	A	C	P	CC	
b. Endotracheal tube					C	P	CC	
c. Inhaled		E	I	A	C	P	CC	
d. Intradermal						P	CC	
e. Intramuscular				A	C	P	CC	
f. Intramuscular auto-injector	R	E	I	A	C	P	CC	
g. Intranasal				A	C	P	CC	
h. Intranasal unit-dosed,	R	E	I	A	C	P	CC	
i. Intraosseous			I	A	C	P	CC	
j. Intravenous			I	A	C	P	CC	
k. Mucosal/Sublingual		E	I	A	C	P	CC	*Includes buccal.*
l. Nasogastric						P	CC	
m. Oral		E	I	A	C	P	CC	
n. Rectal						P	CC	

SCOPE OF PRACTICE FOR EMS PERSONNEL
GEORGIA OFFICE OF EMS AND TRAUMA
POLICY: SOP-2024 (effective 1/23/2026)

Pharmacological Interventions/Skills	Levels							Interpretive Guidelines
4. Medication administration: Approved routes of administration								
o. Subcutaneous				A	C	P	CC	
p. Topical						P	CC	
q. Transdermal						P	CC	
r. Ocular						P*	CC	*Paramedics are only permitted to perform irrigation.
s. Intrathecal								Not permitted at any provider level.
t. Vaginal								Not permitted at any provider level.
u. Otic								Not permitted at any provider level.
v. Intraurethral								Not permitted at any provider level.
w. Penile Injection								Not permitted at any provider level.
x. Other routes not listed above								Not permitted at any provider level.

Cardiac/Medical Skills	Levels							Interpretive Guidelines
1. Fundamental cardiac skills								
a. Manual external CPR	R	E	I	A	C	P	CC	
b. Use of an automated or semi-automated external defibrillator	R	E	I	A	C	P	CC	
2. Advanced cardiac skills								
a. Use of mechanical CPR assist		E	I	A	C	P	CC	
b. ECG acquisition and transmission		E*	I*	A*	C	P	CC	*EMTs, EMT-Is, and AEMTs may only obtain and transmit a 12-lead ECG.
c. ECG monitoring and interpretation					C	P	CC	
d. Manual cardiac defibrillation					C	P	CC	
e. Emergency cardioversion; includes vagal maneuvers					C	P	CC	
f. Transcutaneous cardiac pacing					C	P	CC	
g. Transvenous cardiac pacing						P*	CC	*Paramedics can only perform this skill with PLS approval.
h. Percutaneous ventricular assist device monitoring						P*	CC	*Paramedics can only perform this skill with PLS approval.
i. Intra-aortic balloon pump maintenance and monitoring						P*	CC	*Paramedics can only perform this skill with PLS approval.
j. Extracorporeal membrane oxygenation (ECMO) monitoring						P*	CC	*Paramedics can only perform this skill with PLS approval.
3. Emergency childbirth management								
a. Assist in the normal (uncomplicated) delivery of a newborn	R	E	I	A	C	P	CC	

GEORGIA DEPARTMENT OF PUBLIC HEALTH

SCOPE OF PRACTICE FOR EMS PERSONNEL
GEORGIA OFFICE OF EMS AND TRAUMA
POLICY: SOP-2024 (effective 1/23/2026)

Cardiac/Medical Skills		Levels							Interpretive Guidelines
3. Emergency childbirth management									
b.	Assist in the complicated delivery of a newborn		E	I	A	C	P	CC	*This includes external fundal massage for post-partum bleeding but does not include internal fundal massage.*
c.	Fetal monitoring							CC	
4. Behavioral emergency skills									
a.	Manual and mechanical patient restraints for behavioral		E	I	A	C	P	CC	
b.	Chemical restraints of combative patients						P	CC	

Trauma Care Skills		Levels							Interpretive Guidelines
1. Managing injuries									
a.	Manual cervical stabilization and cervical collar use	R	E	I	A	C	P	CC	
b.	Manual stabilization of orthopedic trauma	R	E	I	A	C	P	CC	
c.	Assist with the application of Spinal Motion Restriction (SMR) for supine or seated patients	R* GA	E	I	A	C	P	CC	**EMT-Rs may only assist an EMT or higher licensed Medic with the application of SMR upon the direction of the EMT or higher licensed Medic present on the scene.*
d.	Application of Spinal Motion Restriction (SMR) for supine or seated patients		E	I	A	C	P	CC	
e.	Extremity splinting	R	E	I	A	C	P	CC	*Does not include traction splints.*
f.	Traction splints		E	I	A	C	P	CC	
2. Managing other trauma injuries									
a.	Progressive bleeding control	R	E	I	A	C	P	CC	
b.	Fundamental eye irrigation	R	E	I	A	C	P	CC	
c.	Complex eye irrigation						P	CC	*Hands-free irrigation using a sterile eye irrigation device.*
d.	Management of soft tissue injuries	R	E	I	A	C	P	CC	*This does not include suturing.*
3. Movement/Extrication of patients									
a.	Emergency moves for endangered patients	R	E	I	A	C	P	CC	
b.	Rapid extrication of patients		E	I	A	C	P	CC	

Crew Member Response Role	Levels							Interpretive Guidelines
1. Patient caregiver								
a. Serve as the **primary** patient caregiver during transport	R*	E	I	A	C	P	CC	**EMT-Rs licensed as a RN, NP, PA, or Physician may function as a primary patient caregiver if approved by agency protocol.*
b. Serve as a **secondary** patient caregiver during transport	R*	E	I	A	C	P	CC	**EMT-Rs may only be present in the patient compartment during transport on ground ambulances when a Medic with an EMT license or higher or an EMT-R licensed as an RN, NP, PA, or Physician is serving as the primary patient caregiver in the patient compartment during the patient transport.*
2. Disposition determination								
a. Determine the patient disposition, including patient refusals		E	I	A	C	P	CC	*Determination should be based on assessment findings, patient's wishes, and Medical Director protocols.*

EMT-R	R	EMT	E	EMT-I	I	AEMT	A	CT	C	PMDC	P	CCP	CC

Changelog

Date of Change	Summary of Change
4/23/2020	Extended the mandatory date of compliance with the Post-Licensure Skills for Paramedics to January 1, 2021. Also separated the Post-Licensure Skills for Paramedics from the main Scope of Practice.
4/23/2020	Added "Perform specimen collection for infectious diseases." to Assessment Skills. This was added to help the COVID-19 specimen collection and testing efforts in Georgia.
10/27/2021	Added EMT-R level and updated document to include medication administration routes and crew member response roles. Added clarification to vaccine administration (only during public health emergencies, and not mass vaccination clinics). Added reference to O.C.G.A. § 31-11-55.2. Administration of hydrocortisone sodium succinate intramuscular by emergency medical services personnel; training; reporting; immunity.
12/7/2022	Added "Initiated of the administration of blood/blood products in the pre-hospital environment." Changed EMS agency PLS approval to a 2-year period.
1/1/2024	Added ETC02 monitoring and interpretation to the AEMT level
1/1/2024	Added Vascular Monitoring and Interpretation (Doppler)
1/1/2024	Added Glucagon via Auto-Injector to EMT and EMT-I level
1/1/2024	Added POCUS for peripheral (extremity) IV insertion
1/1/2024	Separated CPAP/BiPAP and added CPAP to EMT
1/1/2024	Added Cyanokit to AEMT
1/1/2024	Added non-narcotic/non-controlled analgesics to AEMT
1/1/2024	Added antiemetics (Zofran) via parenteral and OTD route to AEMT
6/3/2025	Added CCP level and updated document to include advanced ventilators, rapid sequence intubation, surgical cricothyrotomy, accessing central lines, obtaining arterial blood gasses, prehospital lab analysis, arterial line monitoring, ventricular assist device monitoring, intra-aortic balloon pump maintenance and monitoring, extracorporeal membrane oxygenation (ECMO) monitoring, fetal monitoring, blood product administration, point of care ultrasound (POCUS), and umbilical vein catheterization

6/3/2025	Updated preamble to clarify where a medic can work under the direction of a physician
6/3/2025	Added administration of TXA to AEMT and administration of epinephrine in cardiac arrest to AEMT
1/23/2026	Added that EMT-R and EMT can administer Epi for Anaphylaxis via IM or IN route
1/23/2026	Amended Maintenance of Medicated IV Fluids to add *AEMTs are permitted to administer TXA and Cyanokits via an infusion.

Search and Rescue Activation

Georgia Search and Rescue

Georgia Search and Rescue (GSAR) Teams provide the knowledge and resources to safely and effectively implement technical rescue services. GSAR is on of the few state-initiated heavy search and rescue programs in the nation and receives funding through the U.S. Department of Homeland Security Office of Domestic Preparedness. The state's eight teams, made up of local firefighters, law enforcement and other emergency management personnel can respond to incidents anywhere in Georgia within two hours. Each GSAR team member undergoes approximately 400 hours of search and rescue training to receive certification.

Search and Rescue Activation

To request a search and rescue resource, the on-scene commander should contact their local 911 and have the county EMA Director contact GEMA to request the resource.

Endotracheal Tube Insertion/Sedation Assisted Intubation

General Guidelines

1. For the spontaneous breathing patient requiring intubation:
   a. ***Nasotracheal Intubation*** may be attempted.
2. For the patient with no gag reflex and no spontaneous breathing or in SAI:
   a. ***Orotracheal Intubation:***
      i. Via direct manual laryngoscopy.
      ii. Via video laryngoscopy.
      iii. King Airway.

Sedation Assisted Intubation – INITIAL INDUCTION

Conscious and/or spontaneously breathing patients, who require immediate airway control to prevent severe deterioration (especially airway burns), may require the paramedic to sedate and use an advanced airway to secure and maintain the airway. This protocol should be employed with extreme caution, and a minimum of 2 paramedics is preferred, if available and (1) being CCEMTP certified. The most experienced provider should perform the insertion of the advanced airway. At no time should the patient's SpO_2 drop below 92% during the procedure. **If unable to ventilate (and the patient would be unlikely to survive after performing this procedure), perform needle cricothyrotomy.**

Pre-medicate the patient to assist with blunting the effects of increased ICP

****Draw ALL medications prior to intubation attempt****

1. ***Fentanyl***: 2 – 3 mcg/kg PRN **Maximum Dose – 250 mcg**
   a. Onset: 1 – 3 minutes.
   b. Duration: 15 – 30 minutes.
   c. Used for blunting of circulatory responses to intubation or suspected/known increased intracranial pressure. Use with caution in Multisystem trauma patients.

Sedation – Initial

1. ***Etomidate:*** 0.3 mg/kg IV **Maximum Dose – 40 mg**
   a. Onset: 15 – 45 seconds.
   b. Duration: 3 – 12 minutes.

AND/OR

2. ***Ketamine:*** 2 mg/kg IV **Maximum Dose – 500 mg Adult**
   a. Onset: < 30 seconds.
   b. Duration: 5 – 15 minutes.

Consider benzodiazepine – Third tier agent for initial sedation: (PRN).

1. ***Versed:*** 0.05 mg/kg slow IV – if patient is not hypotensive.

Consider Push Dose *Epinephrine* in hypotensive or presumed peri-arrest.

1. ***Epinephrine 1:100,000:*** 0.5 – 2 ml every 3 – 5 minutes.
   Take a 10 ml flush, expel 1 ml of the saline flush. Draw up 1 ml from a pre-filled ***Epinephrine 1:10,000*** into 9ml of normal saline. This gives a concentration of 1:100,000.
   a. Onset: 1 minute.
   b. Duration: 5 – 10 minutes.

CONTINUED ON NEXT PAGE

Endotracheal Tube Insertion/Sedation Assisted Intubation (Continued)

INSERT AND SECURE ET TUBE AND CONFIRM WITH AT LEAST 4 METHODS, ESPECIALLY $EtCO_2$.

CONTINUED SEDATION OF THE INTUBATED PATIENT

Once the patient is successfully intubated, placement is confirmed and the patient requires continued sedation, the following protocol will assist the treating paramedic(s) with direction. **Strict patient monitoring must be continuous.**

1. *Fentanyl*: **2 – 3 mcg/kg** IV every 10 – 15 minutes as needed.

   AND/OR

2. *Ketamine*: **1 mg/kg** slow IV over 1 minute. **Maximum Dose – 250 mg**
   Consider Benzodiazepine – third tier agent for continued sedation as needed.
3. *Midazolam*: **0.05 mg/kg** every 5 – 10 minutes as needed. (Monitor BP closely.)

****CONTACT MEDICAL CONTROL****

Sedation Assisted Intubation Best Practices

In response to frequent request to EMSMDAC and the State Office of EMS and Trauma for clarification regarding the use of paralytics and/or induction agents in the field to facilitate endotracheal intubation, a subcommittee of EMSMDAC has developed a best-practices guideline. This guideline was unanimously accepted by EMSMDAC on October 23, 2012.

The guideline document follows. Please direct comments or questions to Dr. Robert Cox, EMSMDAC Chair (rjcox@aol.com or Dr. Jill Mabley, deputy Medical Director, Office of EMS and trauma (jamabley@dhr.state.ga.us.)

Georgia EMS Medical Directors Advisory Council

Sedation Assisted Intubation Best Practices Guidelines

October 2012

Introduction

Airway management by prehospital providers is currently an area of active research and many important questions remain unanswered. This statement is particularly true regarding the safety and efficacy of paralytics and sedatives during endotracheal intubation (ETI) by paramedics. Because of the concerns of increased morbidity and mortality demonstrated in several studies, **rapid sequence intubation (RSI) by paramedics remains prohibited in Georgia,** however, sedation assisted intubation (SAI) is an allowable procedure. Although research evaluating SAI is not as robust as RSI, it has not been proven to be superior to ETI without pharmacological assistance. In light of these findings, the Georgia EMS Medical Directors Advisory Council (EMSMDAC) recommends developing and following a rigorous training and quality assurance system if the service and local Medical Director choose to perform SAI in the field. To assist in this process, EMSMDAC has developed some best practices recommendations which are detailed below.

Best Practices for SAI:

1. SAI should only be attempted if other less invasive airway management procedures have failed.
2. The procedure should be limited to those paramedics that the local medical director has personally evaluated and documented as capable of performing SAI. Credentialling of paramedics for SAI should be done on an annual basis.
3. SAI medications should be issued only to those paramedics credentialed by the local Medical Director to perform SAI.
4. Paramedics must have a mandatory minimum number of successful intubations per year to continue to perform SAI, as determined by the local Medical Director.
5. Continuous pulse oximetry throughout the event and waveform capnography post intubation must be used. Paramedics should fully understand the association between hypoxia and hyper/hypocapnia and poor outcomes, especially in patients with traumatic brain injury. Diligent efforts should be taken to prevent and manage hypoxia and hyper/hypocapnia.
6. If SAI is attempted, a back-up airway device (e.g. a super-glottic device) must be immediately available.
7. The local Medical Director should frequently review the quality assurance data to identify and address problems that could affect the health and safety of patients undergoing this procedure.
8. The Medical Director should culture a collaborative relationship between him/herself and the facilities that are receiving their patients so that outcome data can be utilized to improve and/or modify SAI practices. Such outcome measures should include data such as malposition airways, evidence of aspiration, hospital acquired pneumonia, prolonged mechanical ventilation and death.

CONTINUED ON NEXT PAGE

Sedation Assisted Intubation Best Practices (Continued)

Additionally, we recommend the collection of these specific data for each event of SAI:

1. Indication(s) for intubation.
2. Reason for use of sedative/induction agent.
3. Outcome of attempt. (i.e. successful, number of attempts, complications, etc.)
4. Pre and post vital signs including SpO_2.
5. Highest and lowest SpO_2 during the event.
6. Continuous waveform capnography.
7. Any complications surrounding the procedure.
8. Documented confirmation of endotracheal tube placement by receiving physician.

Finally, for more in depth and specific guidance in the development and maintenance of an SAI program, we recommend the National Association of EMS Physicians (NAEMSP) position paper on Drug Assisted Intubations in the Prehospital Setting:

http://www.naemsp.org/Documents/Drug%20Assisted%20Intubation%20New.pdf

Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

BE FAST Stroke Screening Tool

Blood samples for performing blood glucose analysis should be obtained through a finger stick (heel stick for infants). Venous blood samples may produce artificially high glucose values and should be avoided.

IV access is preferred 18g or 20g with AC placement.

Start Here ⬇ Is the patient having a stroke?	Check if Yes	
Balance • Perform bilateral index finger-to-nose test and bilateral heel-to-shin test • Does the patient have sudden loss of balance or coordination, trouble walking or dizziness?		**B**
Eyes • Assess 4 quadrants of visual field by having patient locate your index finger. • Does the patient have trouble seeing out of one or both eyes or sudden double vision?		**E**
Face • Ask the patient to smile or show their teeth. • Does the patient's face look uneven, have sudden drooping or numbness on one side?		**F**
Arms • Ask the patient to raise and extend both arms with their palms up. • Does one arm drift downward? • Does the patient have sudden numbness or weakness of the arm on one side of the body?		**A**
Speech • Ask the patient to say, "*You can't teach an old dog new tricks".* • Does the patient have slurred speech, trouble speaking, understanding or seem confused ?		**S**
Time • What time did the symptoms start? ______:_______ • What time was the patient last known well (last appear normal) ______:_______		**T**
Glucose level = ______________		

FAST-ED Stroke Severity Tool

Ask if the patient is on any anticoagulant medications such as:

- Coumadin/Warfarin
- Pradaxa/Dabigatran
- Eliquis/Apixaban
- Xarelto/Rivaroxaban
- Savaysa/Edoxaban
- Heparin/Enoxaparin

Time anticoagulant last taken:

- Any other anticoagulants ? (Please list):

F	**Facial Palsy** (Ask the patient to show their teeth or smile). • Both sides of the face move equally or not at all. • One side of the face droops or is clearly asymmetric.	**Score** 0 1
A	**Arm Weakness** (with eyes closed, ask the patient to hold arms out with their palms up and hold them there for 10 seconds). • Both arms remain up for > 10 seconds or slowly move down equally. • Patient can raise arms but one arm drifts down in < 10 seconds. • One or both arms fall rapidly, can be lifted, or no movement occurs at all.	**Score** 0 1 2
S	**Speech Changes** *Expressive Aphasia* – ask the patient to name 3 common items. • Names 2 to 3 items correctly. • Names 0 to 1 item correctly. *Receptive Aphasia* – ask the patient to perform a simple command. (Example: “show me two fingers”) • Normal – patient can follow the simple command. • Unable to follow simple command.	**Score** 0 1 0 1
T	**Time** • What time did the symptom start ? • What time was the patient’s last known well (Last appear normal).	____:____ ____:____
E	**Eye Deviation** • No deviation; eyes move equally to both sides. • Patient has clear difficulty when looking to one side (left or right). • Eyes are deviated to one side and do not move to the other side.	**Score** 0 1 2
D	**Denial/Neglect** Denial - show the patient their affected arm and ask “Do you feel weakness in this arm?” • Patient **recognizes** the weakness in their weak arm. • Patient **does NOT recognize** the weakness in their weak arm. Neglect – show the patient their affected arm and ask “Who’s arm is this?” • Patient **recognizes** their weak arm. • Patient **does NOT recognize** their weak arm.	**Score** 0 1 0 1

CINCINNATI PREHOSPITAL STROKE SCALE

UNIFORM DOCUMENT FOR GEORGIA EMS PROVIDERS

DATE:		TIME OF ASSESSMENT:	
ASSESSED BY:		SERVICE:	

NAME OF PATIENT:	

FACIAL DROOP	
	NORMAL: BOTH SIDES OF FACE MOVE EQUALLY
	ABNORMAL: ONE SIDE OF FACE DOES NOT MOVE AT ALL

ARM DRIFT	
	NORMAL: BOTH ARMS MOVE EQUALLY OR NOT AT ALL
	ABNORMAL: ONE ARM DRIFTS COMPARED TO THE OTHER

SPEECH: HAVE THE PATIENT STATE THE FOLLOWING SENTENCE YOU CAN'T TEACH AN OLD DOG NEW TRICKS.	
	NORMAL: PATIENT USES CORRECT WORD WITH NO SLURRING
	ABNORMAL: SLURRED OR INAPPROPRIATE WORDS OR MUTE

Form Developed 11/02/2011

American Heart Association. Mission: Lifeline® Stroke

RACE
RAPID ARTERIAL OCCLUSION EVALUATION SCALE

A STROKE ASSESSMENT TOOL FOR EMS

EMS RACE Stroke Scale - Rapid Arterial oCclusion Evaluation Scale, used to predict large cerebral arterial occlusions.*

ITEM	INSTRUCTION		RACE Score
FACIAL PALSY	Ask the patient to show their teeth	**ABSENT** (symmetrical movement) **MILD** (slightly asymmetrical) **MODERATE TO SEVERE** (completely asymmetrical)	0 1 2
ARM MOTOR FUNCTION	Extending the arm of the patient 90 degrees (if sitting) of 45 degrees (if supine)	**NORMAL TO MILD** (limb upheld more than 10 seconds) **MODERATE** (limb upheld less than 10 seconds) **SEVERE** (patient unable to raise arm against gravity)	0 1 2
LEG MOTOR FUNCTION	Extending the leg of the patient 30 degrees (if supine)	**NORMAL TO MILD** (limb upheld more than 5 seconds) **MODERATE** (limb upheld less than 5 seconds) **SEVERE** (patient unable to raise leg against gravity)	0 1 2
HEAD AND GAZE DEVIATION	Observe eyes and cephalic deviation to one side	**ABSENT** (eye movements to both sides were possible and no cephalic deviation was observed) **PRESENT** (eyes and cephalic deviation to one side was observed)	0 1
APHASIA If right hemiparesis	Ask the patient two verbal orders: - *"close your eyes"* - *"make a fist"*	**NORMAL** (performs both tasks correctly) **MODERATE** (performs one task correctly) **SEVERE** (performs neither task)	0 1 2
AGNOSIA If left hemiparesis	Asking: - *"Who's arm is this?" while showing him/her the paretic arm (asomatognosia)* - *"Can you move your arm?" (anosognosia)*	**NORMAL** (no asomatognosia nor anosognosia) **MODERATE** (asomatognosia or anosognosia) **SEVERE** (both asomatognosia and anosognosia)	0 1 2
		RACE SCALE TOTAL: Any score above a "0" is a "Stroke Alert"	

*Chart adapted from Perez de la Ossa N, Carrera D, Gorchs M, et al. Design and validation of a prehospital stroke scale to predict large arterial occlusion: the rapid arterial occlusion evaluation scale. Stroke; a journal of cerebral circulation. Jan 2014;45(1):87-91.

ADDITIONAL INFORMATION

PRE-HOSPITAL MANAGEMENT OF AN ACUTE STROKE[1]

- Assess the airway, breathing and circulatory status
- Check blood glucose
- Obtain full set of vital signs
- Review patient medications
- Perform 12 lead ECG
- Establish IV access

ACUTE ISCHEMIC STROKE - IV t-PA CONTRAINDICATIONS[2]

- Active internal bleeding
- Recent intracranial or intraspinal surgery or serious head trauma
- Intracranial conditions that may increase the risk of bleeding
- Bleeding diathesis
- Current severe uncontrolled hypertension
- Current intracranial hemorrhage
- Subarachnoid hemorrhage

PATIENT REPORT TO ED - KEY ITEMS[3]

- Patients' age, sex, weight
- Mechanism of injury or medical problem
- Chief complaint with brief history of present illness
- Vital signs
- Level of consciousness
- General appearance, distress, cardiac rhythm
- Interventions by EMS (IV, medication administration)
- ETA (the more critical the patient, the earlier you need to notify the receiving facility)

** MECHANICAL THROMBECTOMY IS AN OPTION FOR ALL STROKE PATIENTS UNTIL PROVEN OTHERWISE**

Always follow your state, local or EMS agency/medical directors' protocols.

[1] Maggiore, W.A. (2012). 'Time is Brain' in Prehospital Stroke Treatment. Journal of Emergency Medical Services, 1-9.
[2] Genentech USA, Inc. Highlights of prescribing information, Activase (alteplase for injection, for intravenous use), http://www.gene.com/download/pdf/activase_prescribing.pdf. Accessed on February 15, 2018.
[3] Campbell S, Robinson MR. Paramedic Lab Manual. Upper Saddle River, N.J.: Pearson Prentice Hall; 2005.

LEARN MORE

To learn more and obtain free EMS resources from the **American Heart Association** - visit

www.heart.org/nebraskamissionlifeline

Table 1 Vision, aphasia, neglect emergent large vessel occlusion screening tool

Stroke VAN

How weak is the patient? Raise both arms	☐ Mild (minor drift) ☐ Moderate (severe drift - touches or nearly touches ground) ☐ Severe (flaccid or no antigravity) ☐ Patient shows no weakness. Patient is VAN negative

(exceptions are confused or comatose patients with dizziness, focal findings, or no reason for their altered mental status then basilar artery thrombus must be considered; CTA is warranted)

Visual disturbance	☐ Field cut (which side) (4 quadrants) ☐ Double vision (ask patient to look to right then left; evaluate for uneven eyes) ☐ Blind new onset ☐ None
Aphasia	☐ Expressive (inability to speak or paraphasic errors); do not count slurring of words (repeat and name 2 objects) ☐ Receptive (not understanding or following commands) (close eyes, make fist) ☐ Mixed ☐ None
Neglect	☐ Forced gaze or inability to track to one side ☐ Unable to feel both sides at the same time, or unable to identify own arm ☐ Ignoring one side ☐ None

Patient must have weakness plus one or all of the V, A, or N to be VAN positive. VAN positive patients had 100% sensitivity, 90% specificity, positive predictive value 74%, and negative predictive value 100% for detecting large vessel occlusion.
CTA, CT angiography; VAN, vision, aphasia, and neglect.

Source: Teleb MS, Ver Hage A, Carter J, et al J NeuroIntervent Surg Published Online First: doi:10.1136/ neurintsurg-2015-012131

Stroke Thrombolytic Checklist

This checklist is intended as a tool for the pre-hospital identification of patients who may benefit from the administration of thrombolytics for acute stroke.

Date:_______________ **Time:**___________ **Unit:** _____________ **PSS:**____________________

Patient Name: _______________________ **Age:** ______________ **Est. Wt:** _____________**lbs/kg**

Time last seen at baseline: ______________
Time of Symptom Onset: ________________
Onset Witnessed or reported by: _____________________

Symptoms (Circle abnormal findings)

A single finding = POSSIBLE STROKE

FACIAL DROOP	**R**	**L**	
ARM DRIFT	**R**	**L**	
SPEECH	**SLURRED**	**WRONG WORDS**	**MUTE**

Possible contraindications: (Check all that apply)

Current use of anticoagulants (e.g. Warfarin, Eliquis, etc.)?	Yes	No	?
Has blood pressure consistently over 180/110 mmHg?	Yes	No	?
Witnessed seizure at symptom onset?	Yes	No	?
History of intracranial hemorrhage?	Yes	No	?
History of GI or GU bleeding, ulcers, varices?	Yes	No	?
Is within 3 months of prior stroke?	Yes	No	?
Is within 3 months of serious head trauma?	Yes	No	?
Is within 21 days of acute myocardial infarction?	Yes	No	?
Is within 21 days of lumbar puncture?	Yes	No	?
Is within 14 days of major surgery or serious trauma?	Yes	No	?
Is pregnant?	Yes	No	?
Abnormal blood glucose level (< 50 or > 400): FSBS (if done) ___________	Yes	No	?

Have you identified any contraindications to thrombolytic therapy?(circle one) **YES** **NO**

Receiving Site/Physician: ________________________________ **Time**: ________________

EMT#: ______________________ **Signature**: ___________________________________

Specialty Care Centers

The Specialty Care Centers listed below are within transporting range for Hart Co. EMS if the need arises.

BURN CENTERS:

JMS Burn Center at Doctors Hospital (Augusta, GA)	706-651-6080
Grady Memorial (Atlanta, GA)	404-616-6178

Cardiac Care:

Georgia:

Level I

Northeast Georgia Medical Center - Gainesville Door Code: 1234	770-219-7911

Level II

Northeast Georgia Medical Center – Braselton	770-848-8000

South Carolina:

ANMED – Anderson SC	864-512-1984
Prisma Health – Greenville Memorial	864-455-7725

Comprehensive Stroke Centers:

Georgia:

Emory University – Atlanta	404-712-2000
Grady Memorial Hospital – Atlanta	404-616-1000
Northeast Georgia Medical Center – Gainesville Door Code: 1234	770-219-7911

South Carolina:

Prisma Health Upstate – Greenville	864-455-7725

Specialty Care Centers (Continued)

Primary Stroke Center

Georgia:

Piedmont Athens Regional – Athens	706-475-7000
St. Mary's Hospital – Athens	706-389-3000

South Carolina:

ANMED – Anderson SC	864-512-1323

Remote Stroke Centers

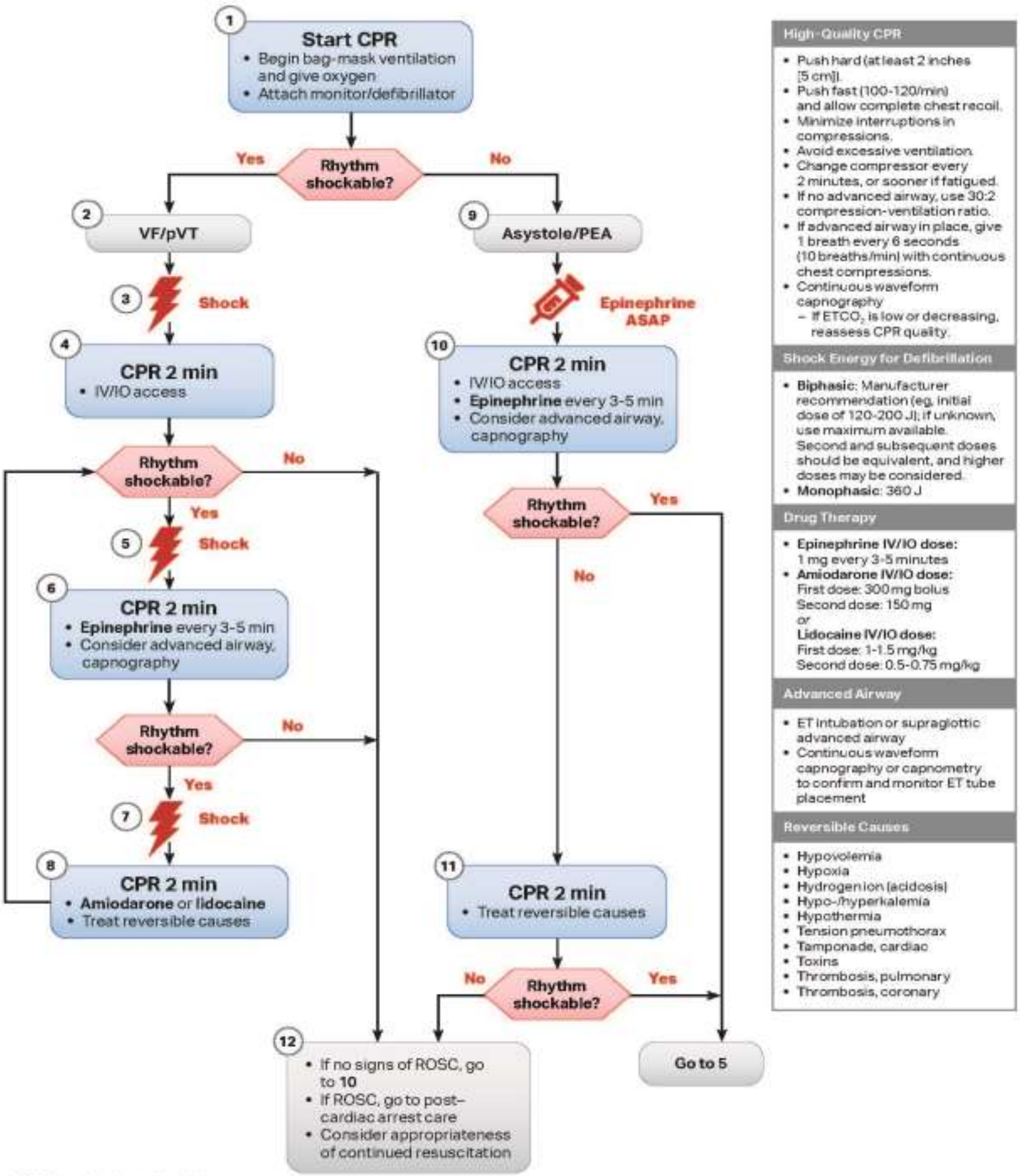
Georgia:

Northeast Georgia Medical Center _ Barrow	770-867-3400
Northeast Georgia Medical Center – Braselton	770-848-8000
St. Mary's Sacred Heart – Lavonia	706-356-7800

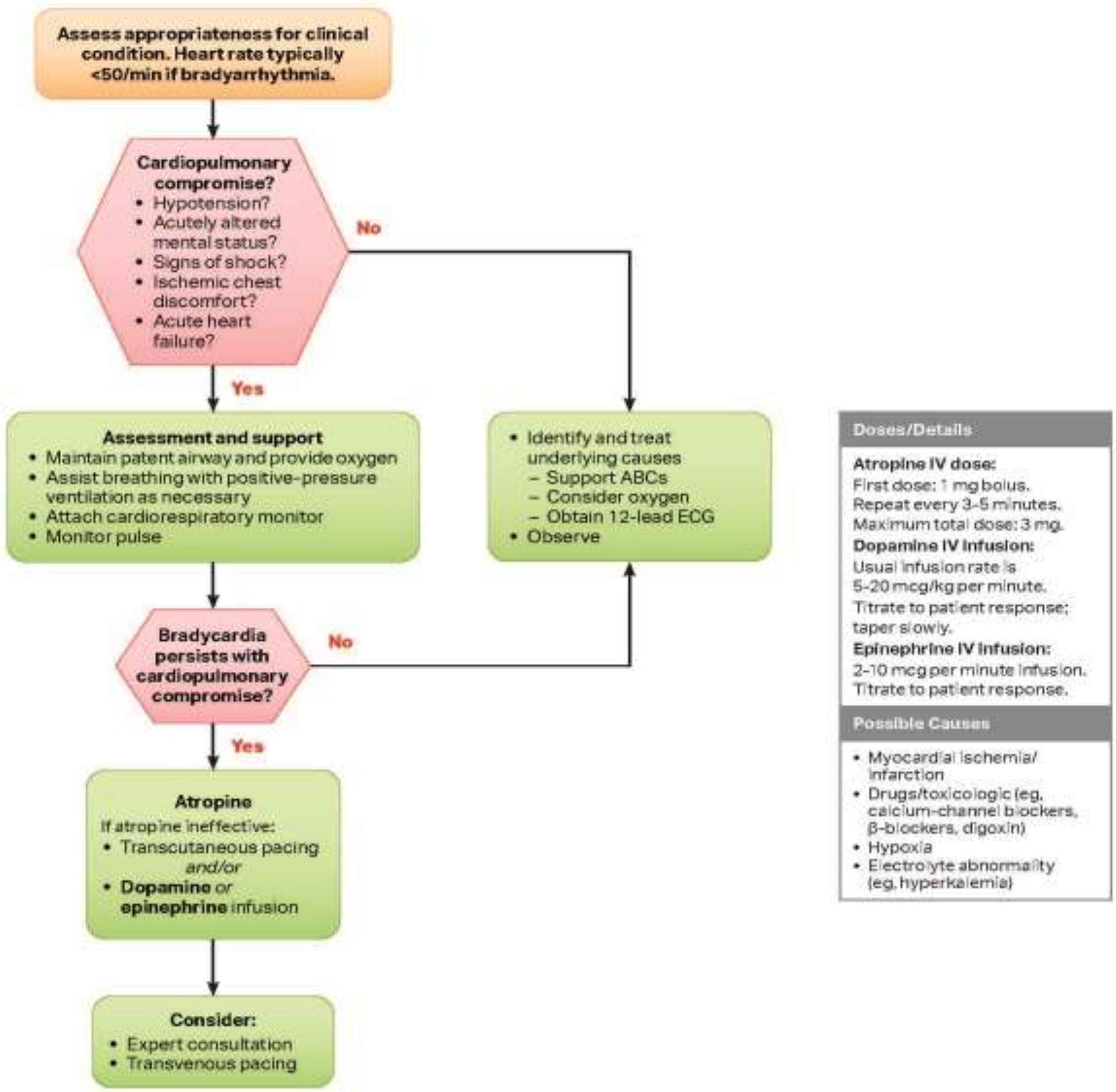
South Carolina:

St. Francis Hospital Eastside – (Greenville)

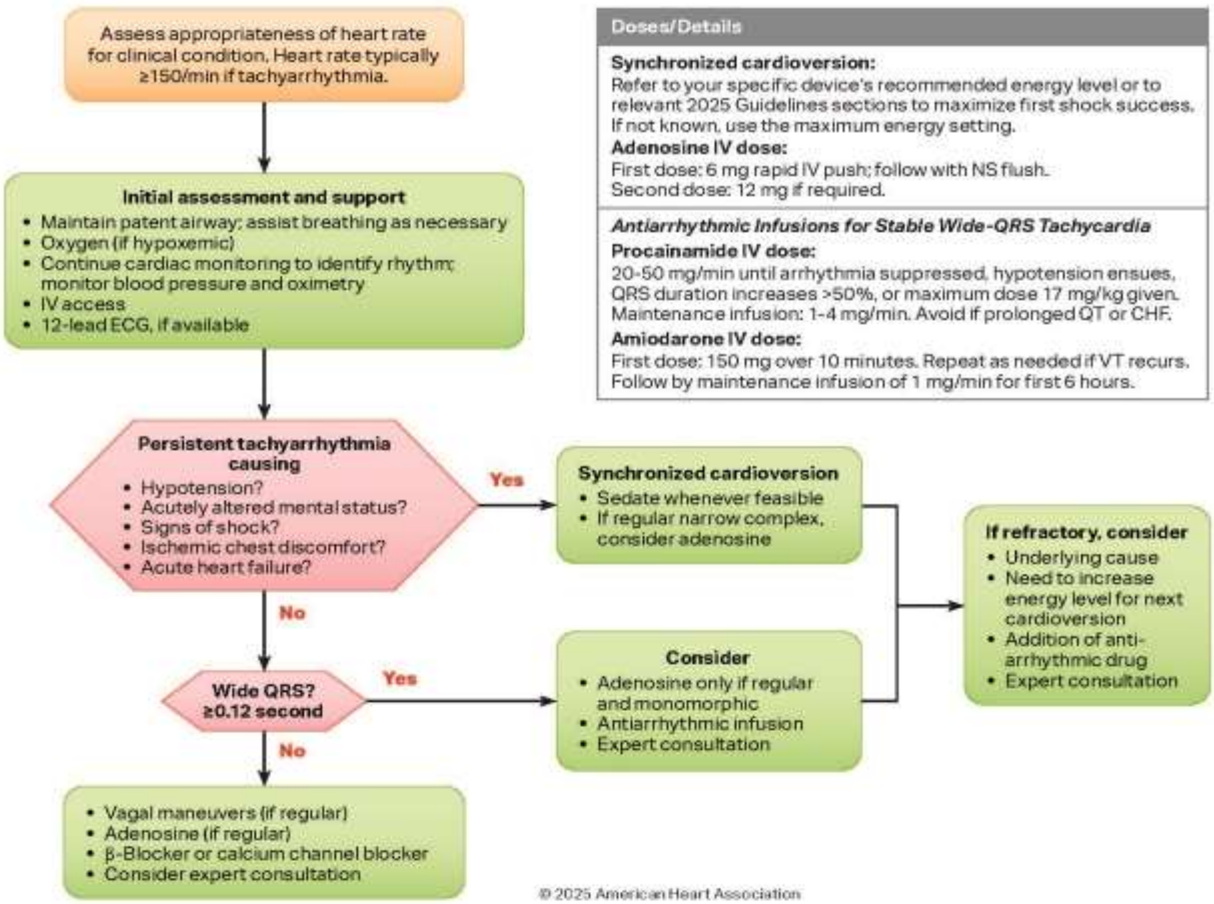
Adult Cardiac Arrest Algorithm (VF/pVT/Asystole/PEA)



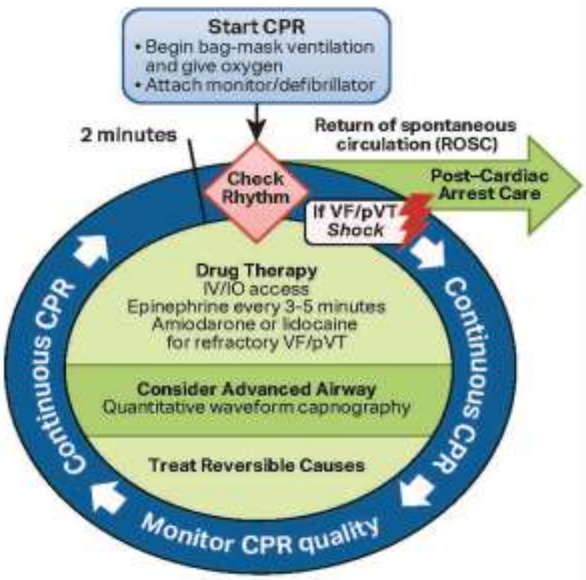
Adult Bradycardia With a Pulse Algorithm



Adult Tachyarrhythmia With a Pulse Algorithm



Adult Cardiac Arrest Circular Algorithm



High-Quality CPR

- Push hard (at least 2 inches [5 cm]).
- Push fast (100-120/min) and allow complete chest recoil.
- Minimize interruptions in compressions.
- Avoid excessive ventilation.
- Change compressor every 2 minutes, or sooner if fatigued.
- If no advanced airway, 30:2 compression-ventilation ratio.
- If advanced airway in place, give 1 breath every 6 seconds (10 breaths/min) with continuous chest compressions.
- Continuous waveform capnography
  – If $ETCO_2$ is low or decreasing, reassess CPR quality.

Shock Energy for Defibrillation

- **Biphasic:** Manufacturer recommendation (eg, initial dose of 120-200 J); if unknown, use maximum available. Second and subsequent doses should be equivalent, and higher doses may be considered.
- **Monophasic:** 360 J

Drug Therapy

- **Epinephrine IV/IO dose:** 1 mg every 3-5 minutes
- **Amiodarone IV/IO dose:** First dose: 300 mg bolus. Second dose: 150 mg.

  or
- **Lidocaine IV/IO dose:** First dose: 1-1.5 mg/kg. Second dose: 0.5-0.75 mg/kg.

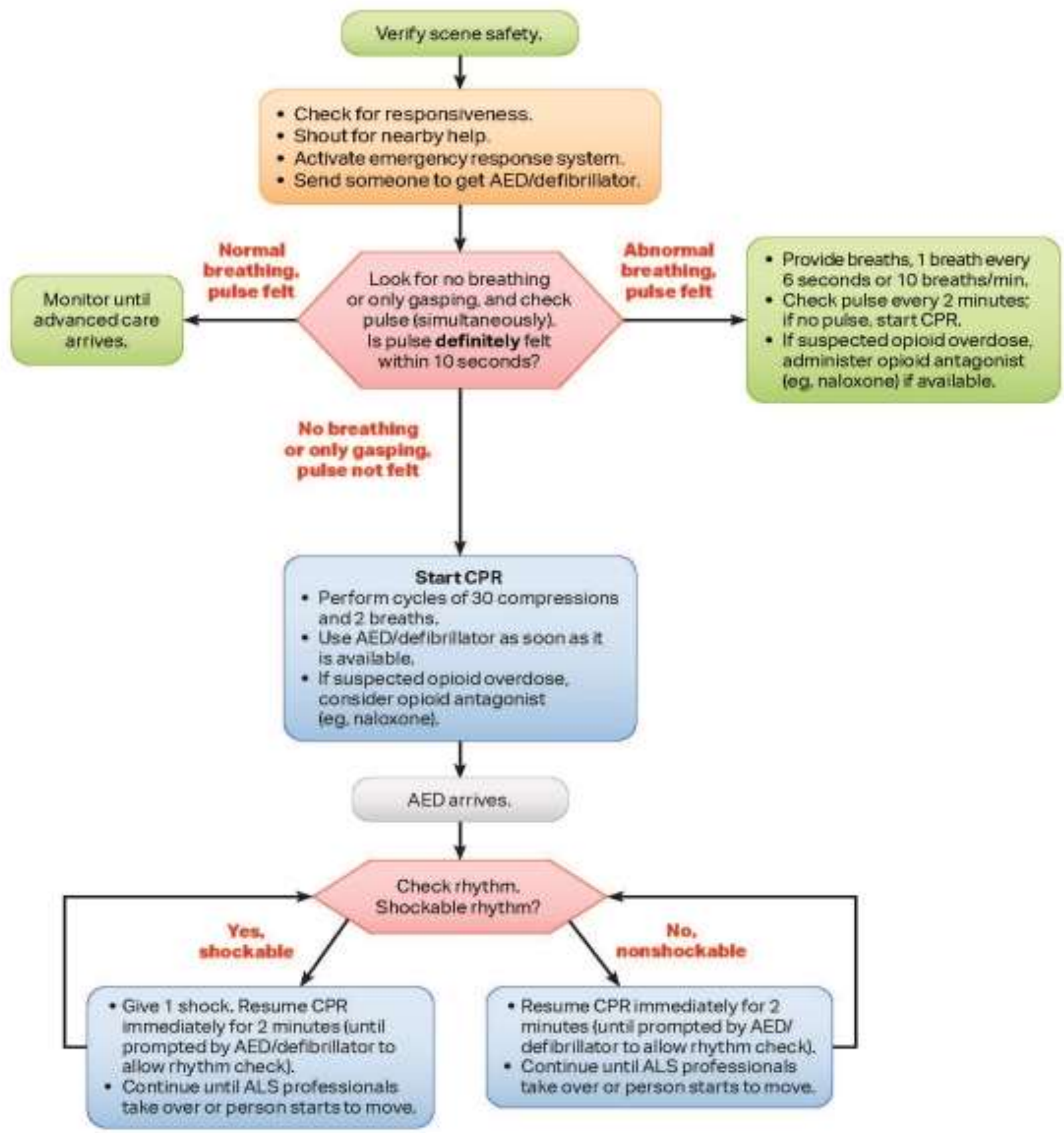
Advanced Airway

- ET intubation or supraglottic advanced airway
- Continuous waveform capnography or capnometry to confirm and monitor ET tube placement
- Once advanced airway in place, give 1 breath every 6 seconds (10 breaths/min) with continuous chest compressions

Reversible Causes

- **H**ypovolemia
- **H**ypoxia
- **H**ydrogen ion (acidosis)
- **H**ypo-/hyperkalemia
- **H**ypothermia
- **T**ension pneumothorax
- **T**amponade, cardiac
- **T**oxins
- **T**hrombosis, pulmonary
- **T**hrombosis, coronary

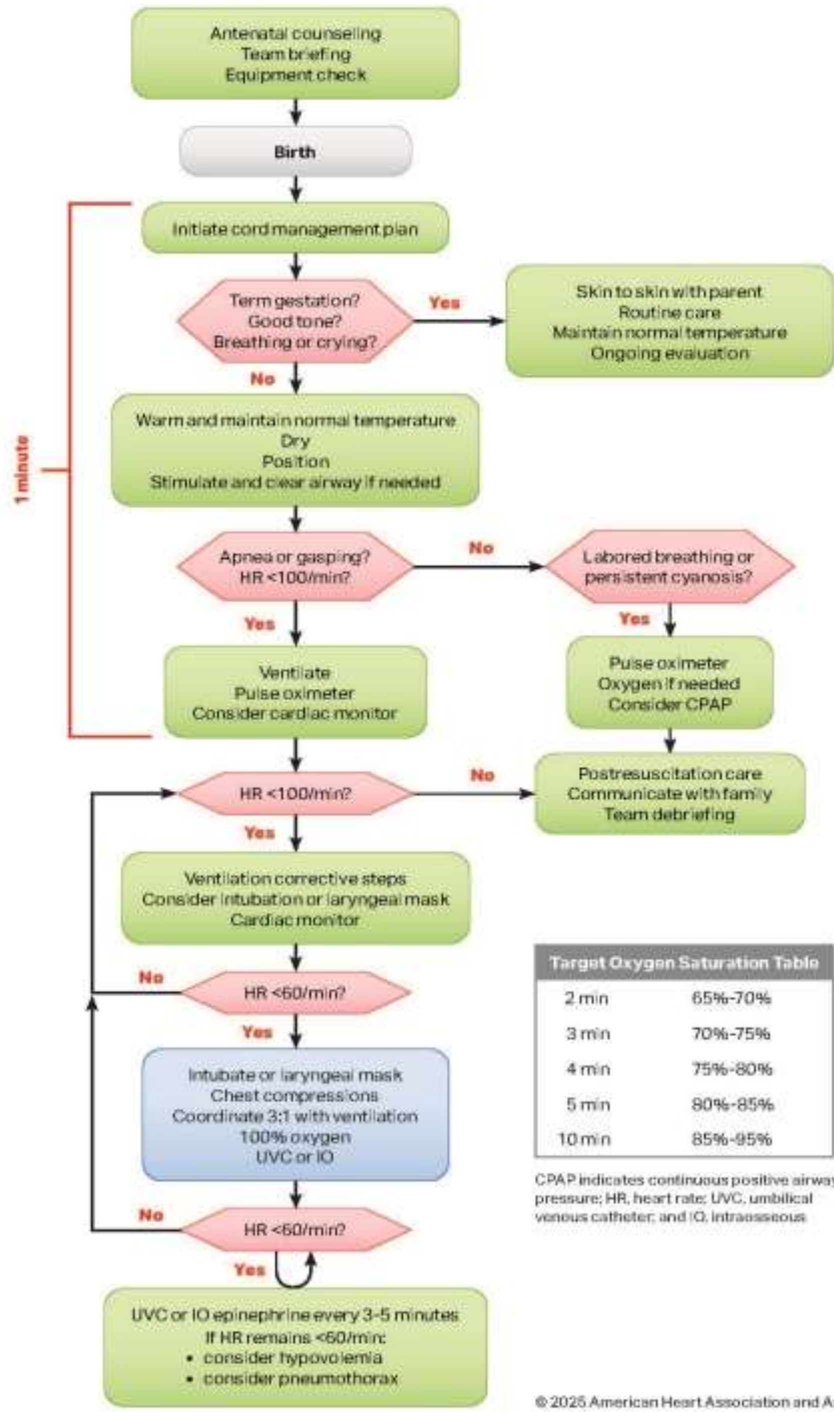
Adult* Basic Life Support Algorithm for Health Care Professionals



*If signs of puberty, treat as adult.

Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

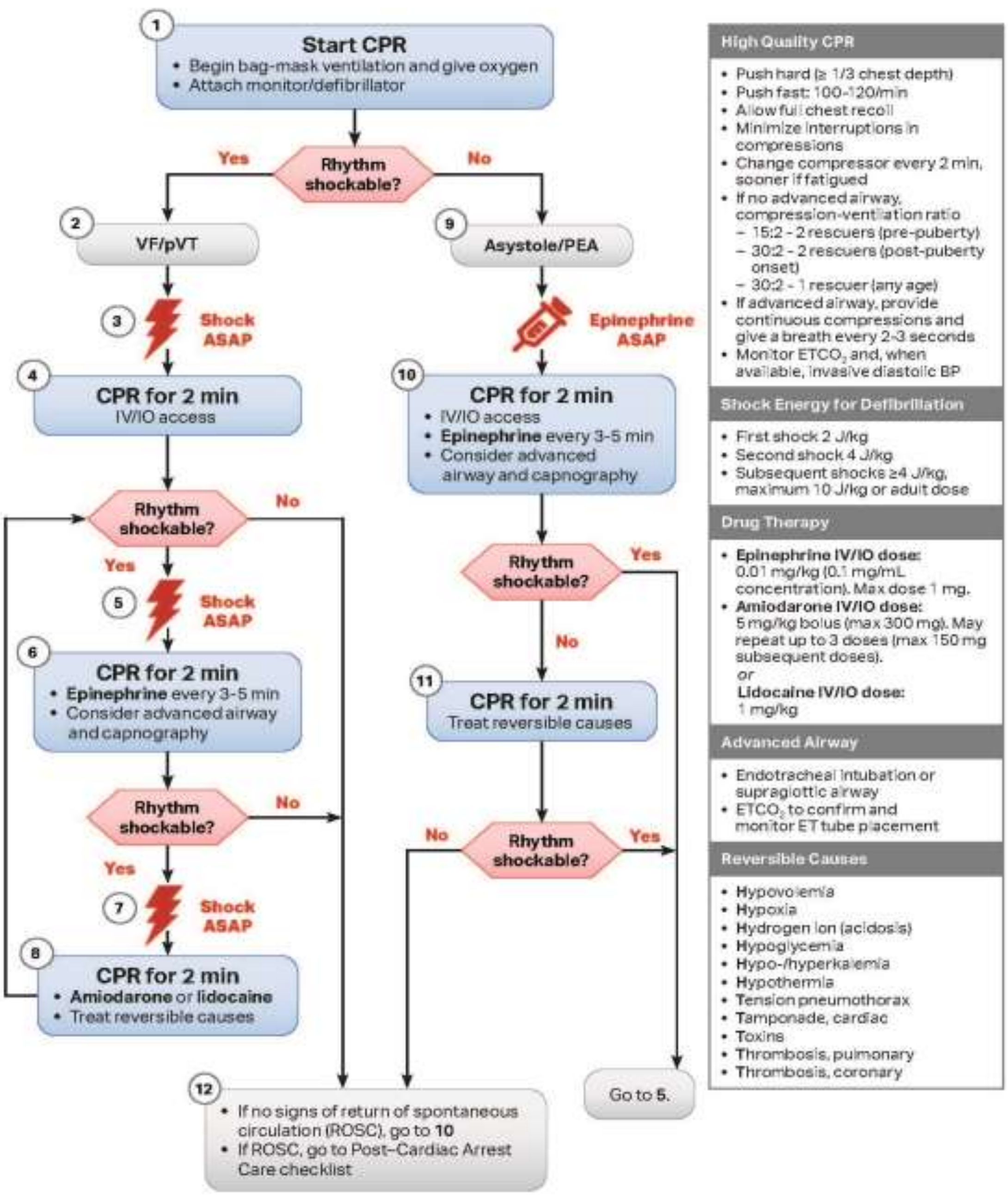
Neonatal Resuscitation Algorithm



Target Oxygen Saturation Table	
2 min	65%-70%
3 min	70%-75%
4 min	75%-80%
5 min	80%-85%
10 min	85%-95%

CPAP indicates continuous positive airway pressure; HR, heart rate; UVC, umbilical venous catheter; and IO, intraosseous.

Pediatric Cardiac Arrest Algorithm



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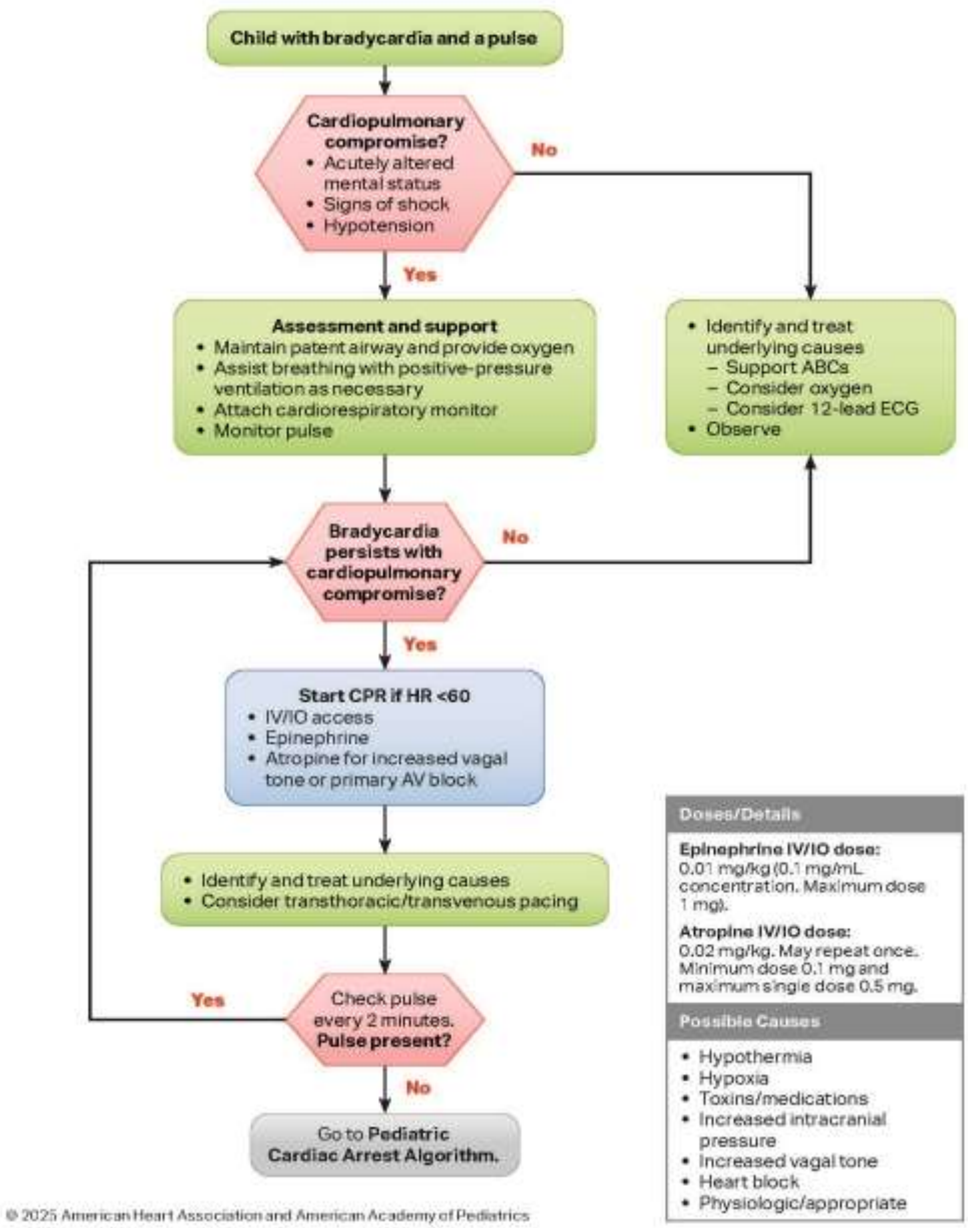
	Effective Date:	6/8/2026

Pediatric Bradycardia With a Pulse Algorithm



Hart County Emergency Medical Service		
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	EMS Director:	Mike Adams
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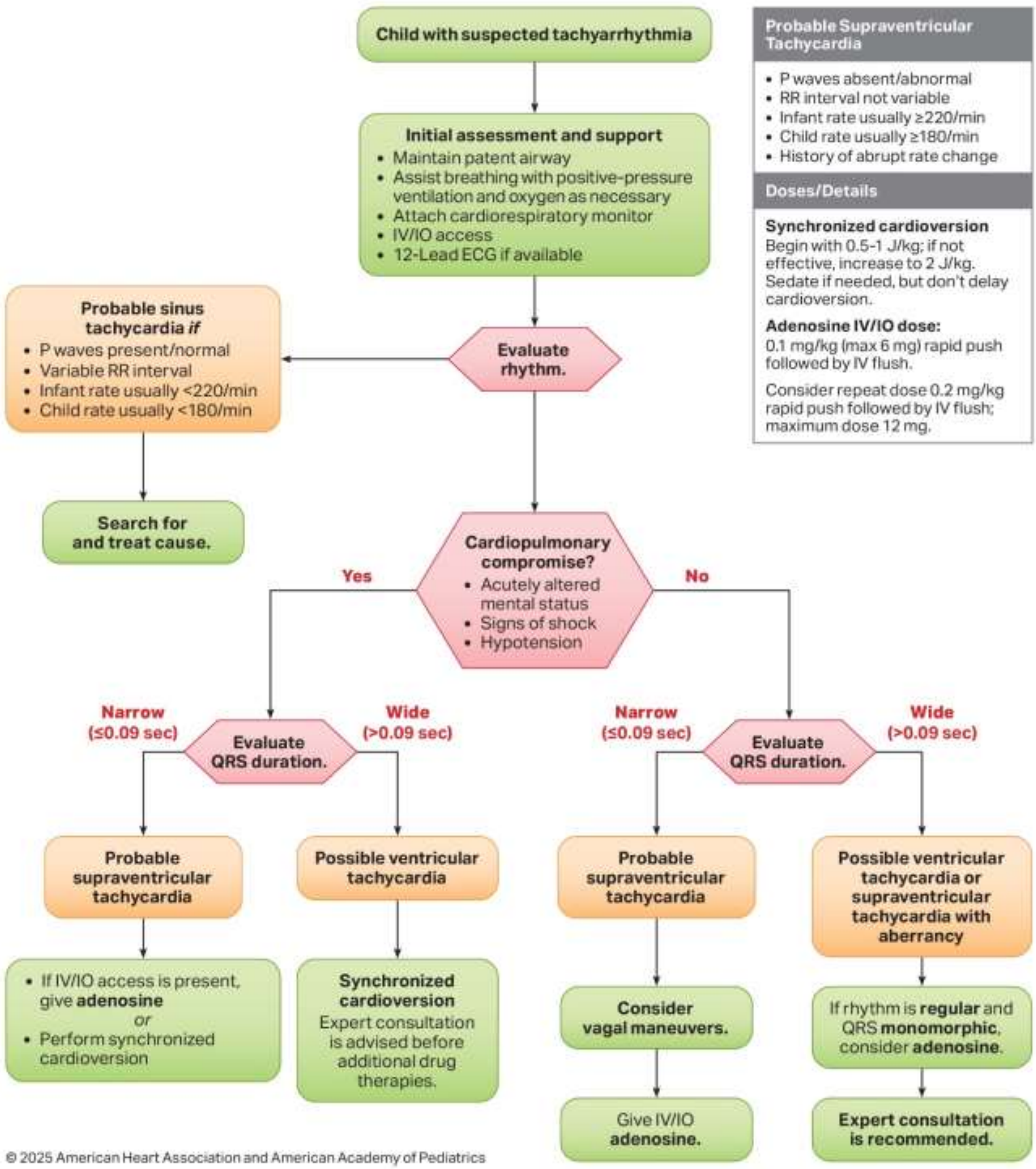
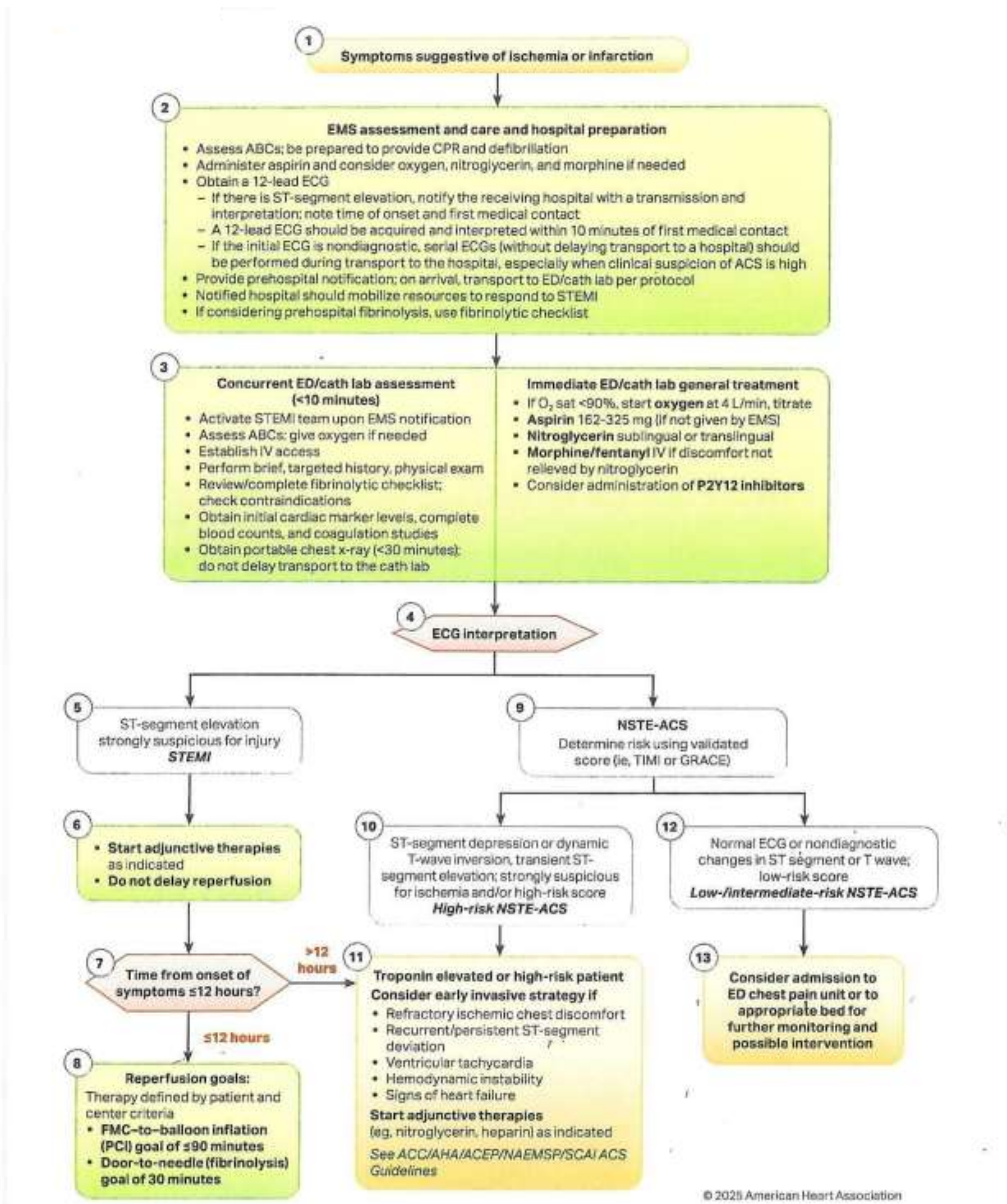
Pediatric Tachyarrhythmia With a Pulse Algorithm

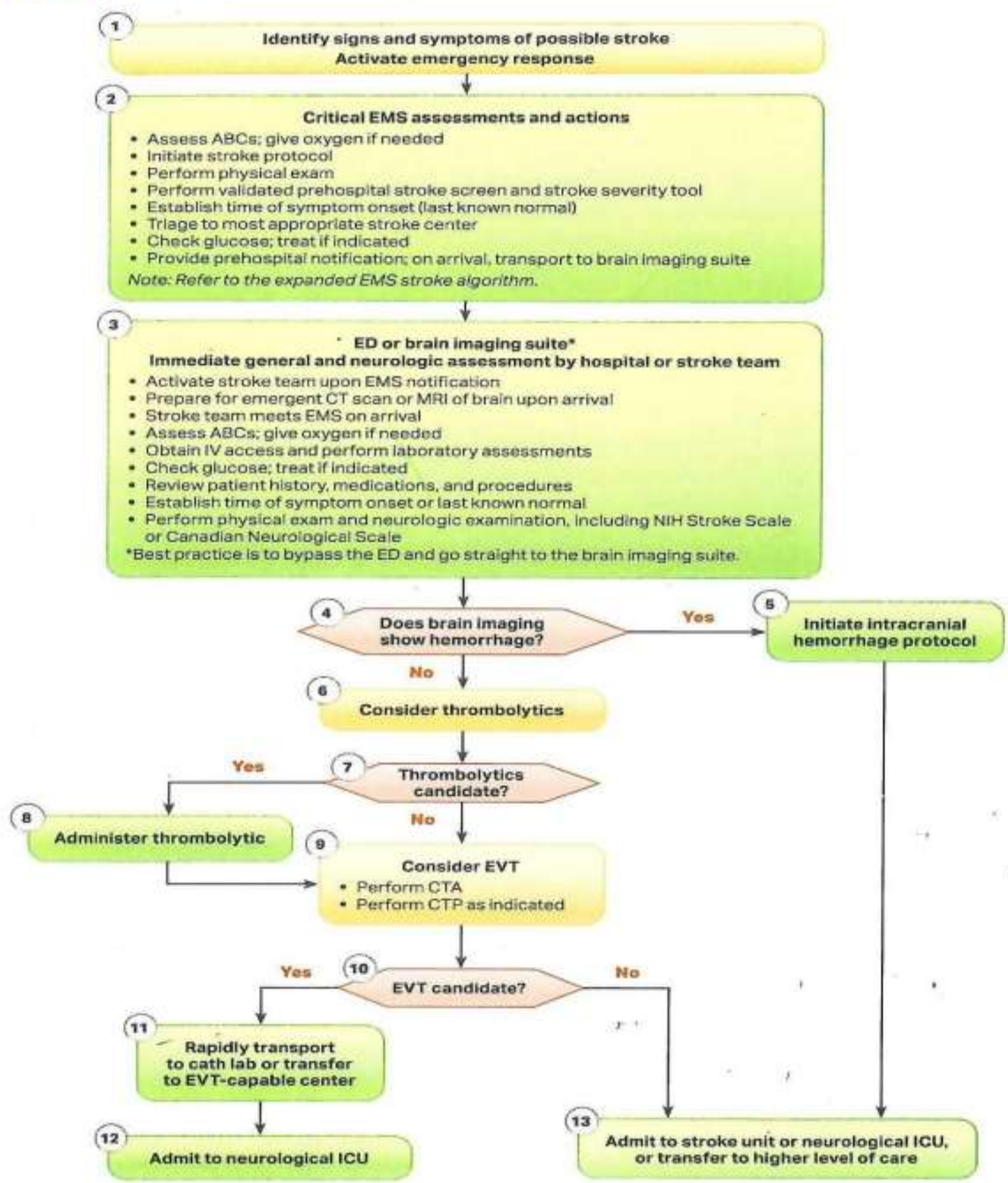


Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

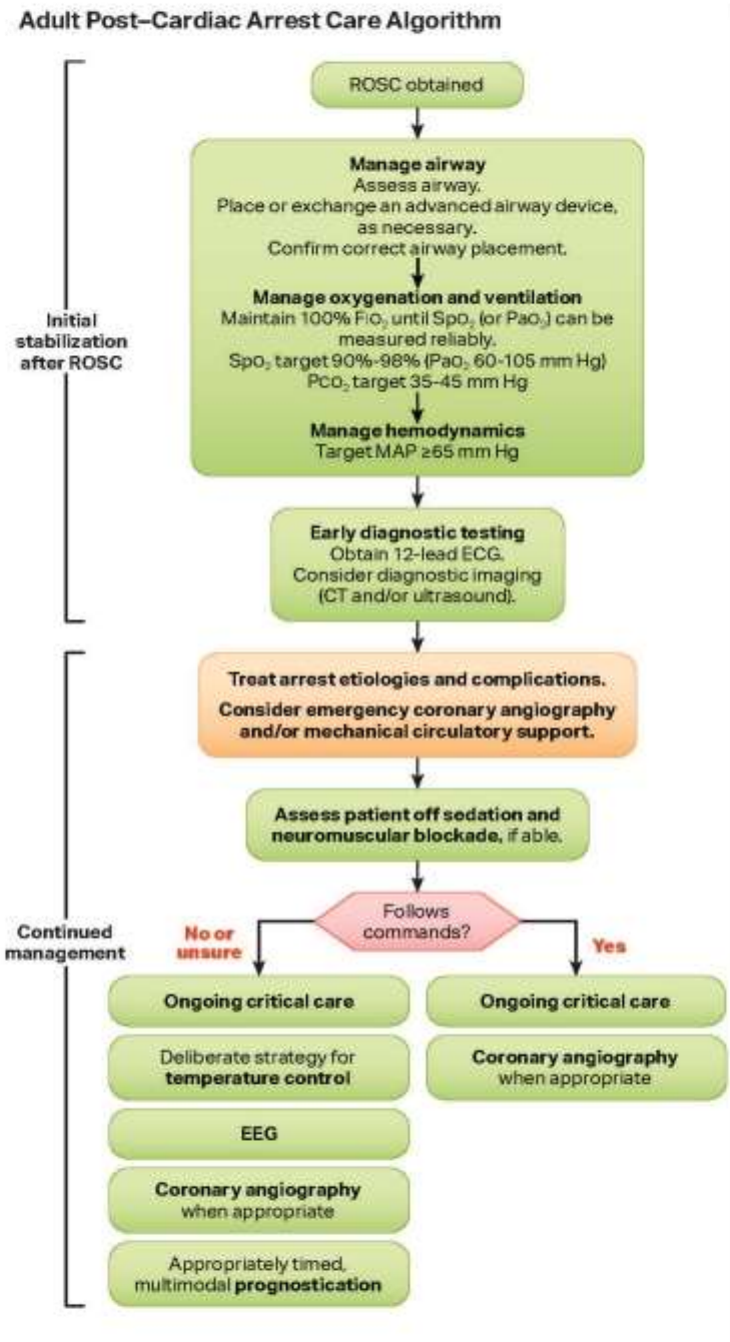
Acute Coronary Syndrome Algorithm



Adult Suspected Stroke Algorithm



Abbreviations: CTA, computed tomography angiography; CTP, computed tomography perfusion; NIH, National Institutes of Health.



Initial Stabilization After ROSC

Resuscitation is ongoing during the post-ROSC phase, and many of these activities can occur concurrently.

Manage airway: Assess and consider placement or exchange of an advanced airway device (usually endotracheal tube or supraglottic device). Confirm correct placement of an advanced airway. This generally includes the use of waveform capnography or capnometry.

Manage oxygenation and ventilation: Titrate FIO_2 for SpO_2 90%-98% (or PaO_2 60-105 mm Hg). Adjust minute ventilation to target PCO_2 35-45 mm Hg in the absence of severe acidemia.

Manage hemodynamics: Initiate or adjust vasopressors and/or fluid resuscitation as necessary for goal MAP ≥65 mm Hg.

Early diagnostic testing: Obtain 12-lead ECG to assess for ischemia or arrhythmia. Consider CT head, chest, abdomen, and/or pelvis to determine cause of arrest or assess for injuries sustained during resuscitation. Point-of-care ultrasound or echocardiography may be reasonable to identify clinically significant diagnoses requiring intervention.

Continued Management

Treat arrest etiologies and complications.

Consider emergency cardiac intervention:

- Persistent ST-segment elevation present
- Cardiogenic shock
- Recurrent or refractory ventricular arrhythmias
- Severe myocardial ischemia

Temperature control: If patient is not following commands off sedation and neuromuscular blockade or is unable to assess, initiate a deliberate strategy of temperature control with goal 32 °C-37.5 °C as soon as possible.

Evaluate for seizure: Evaluate for clinical seizure and obtain EEG to evaluate for seizure in patients not following commands.

Prognostication: Multimodal approach with delayed impressions (≥72 hours from ROSC or achieving normothermia).

Ongoing critical care includes the following:

- Target PaO_2 60-105 mm Hg, PCO_2 35-45 mm Hg (unless severe acidemia); avoid hypoglycemia (glucose <70 mg/dL) and hyperglycemia (glucose >180 mg/dL); target MAP ≥65 mm Hg.
- Consider antibiotics.

EMS Interfacility Ground Transport Protocol for Patients during/after IV Thrombolytic Administration for Acute Ischemic Stroke

- Strict NPO
- Obtain and record vital signs a minimum of every 15 minutes
- Obtain and record neuro checks per the Cincinnati Prehospital Stroke Scale a minimum of every 15 mins
- BP management per Medication Guide below
- HOB flat unless patient at risk for aspiration, airway obstruction, or is unable to maintain oxygenation

Acute Stroke Management

- Maintain BP<180/105
- If BP>180/105, follow BP protocol below
- If SBP<140 or DBP<80 and patient on antihypertensive drip, titrate down and/or DC
- For patient's receiving Alteplase:
    - Total Alteplase infusion time should be 60 minutes
    - Once Alteplase infusion completes, hang NS at existing rate with existing tubing to infuse remaining Alteplase

*****No other medications through Thrombolytic infusion line**

*****If SBP precipitously drops below 140, contact receiving facility for guidance.**

*****STOP Thrombolytic if the patient develops the following symptoms: worsening LOC, hemorrhage, severe headache, acute hypertension, nausea and vomiting, difficulty breathing or angioedema.**

Guide for controlling BP in patients during/after IV Thrombolytic administration for Acute Ischemic Stroke

If BP>180/105 and HR>60, give labetalol 10 mg IV x1 over 2 min; If no response after 10 minutes, may repeat x1.

OR

Initiate Nicardipine drip at 2.5 mg/hr and titrate by 2.5 mg/hr every 15 minutes up to a maximum of 15 mg/hr. Consider reduction to 3 mg/hour after response is achieved. Monitor and titrate to lowest dose necessary to maintain BP within parameters. May also consider Nicardipine if BP not responsive to labetalol.

POTENTIAL COMPLICATIONS

SYMPTOM	TREATMENT
Hypotension (SBP<90)	• HOB flat • D/C any antihypertensive drips • Administer 500cc NS fluid bolus • If major bleeding suspected, STOP Thrombolytic
Hypertension (BP>180/105)	• Per medication guide above
Neurologic Deterioration	• Assess circulation, airway, breathing (CAB) • Obtain full set of vitals and neurological check • Check glucose and treat if <60
Difficulty Breathing or Angioedema	• STOP Thrombolytic if infusing • Treat according to allergic reaction protocol** **Withhold Epi unless angioedema is causing impending airway compromise
Nausea and Vomiting	• Treat according to protocol
Bleeding	• Apply direct pressure • Treat according to protocol • If major bleeding suspected, STOP Thrombolytic

CONTACT SENDING OR RECEIVING FACILITY FOR QUESTIONS

We protect lives.

Levophed Dosing Chart

4 mg in 250 ml of Normal Saline
Your concentration is now 16 mcg/ml.

mcg/min	micro drops/min.
1 mcg/min	4
2 mcg/min	8
3 mcg/min	11
4 mcg/min	15
5 mcg/min	18
6 mcg/min	22
7 mcg/min	26
8 mcg/min	30
9 mcg/min	34
10 mcg/min	38
11 mcg/min	42
12 mcg/min	45

Alternative Mixing Chart

Mix 4mg in 1000ml of Normal Saline OR 1 mg in 250ml of Normal Saline
Your concentration is now 4 mcg/ml

mcg/min.	micro drops/min.
1 mcg/min	15
2 mcg/min	30
3 mcg/min	45
4 mcg/min	60
5 mcg/min	75
6 mcg/min	90
7 mcg/min	105
8 mcg/min	120
9 mcg/min	135
10 mcg/min	150
11 mcg/min	165
12 mcg/min	180

Etomidate Dosing Chart (0.3 mg/kg)

This reference applies to patients needing sedation prior to and post intubation.

- **Adult Initial Dose:** 0.3 mg/kg.
  - **Adult Maintenance Dose:** 0.1 mg/kg.
- **Pediatric Initial Dose:** 0.3 mg/kg. (Not to be administered to children under 10 years of age.)
  - **Pediatric Maintenance Dose:** 0.1 mg/kg.

Maximum Dose: 40 mg

LBS	kg	0.1 mg/kg	0.3 mg/kg
10	4.5	0.5	1.4
15	6.8	0.7	2.0
20	9.1	0.9	2.7
25	11.4	1.1	3.4
30	13.6	1.4	4.1
35	15.9	1.6	4.8
40	18.2	1.8	5.5
45	20.5	2.0	6.1
50	22.7	2.3	6.8
55	25.0	2.5	7.5
60	27.3	2.7	8.2
65	29.5	3.0	8.9
70	31.8	3.2	9.5
75	34.1	3.4	10.2
80	36.4	3.6	10.9
85	38.6	3.9	11.6
90	40.9	4.1	12.3
95	43.2	4.3	13.0
100	45.5	4.5	13.6
105	47.7	4.8	14.3
110	50	5	15
115	52.3	5.2	15.7
120	54.5	5.5	16.4
125	56.8	5.7	17.0
130	59.1	5.9	17.7
135	61.4	6.1	18.4
140	63.6	6.4	19.1
145	65.9	6.6	19.8
150	68.2	6.8	20.5

LBS	kg	0.1 mg/kg	0.3 mg/kg
155	70.5	7.0	21.1
160	72.7	7.3	21.8
165	75.0	7.5	22.5
170	77.3	7.7	23.2
175	79.5	8.0	23.9
180	81.8	8.2	24.5
185	84.1	8.4	25.2
190	86.4	8.6	25.9
195	88.6	8.9	26.6
200	90.9	9.1	27.3
205	93.2	9.3	28.0
210	95.5	9.5	28.6
215	97.7	9.8	29.3
220	100	10	30
225	102.3	10.2	30.7
230	104.5	10.5	31.4
235	106.8	10.7	32.0
240	109.1	10.9	32.7
245	111.4	11.1	33.4
250	113.6	11.4	34.1
255	115.9	11.6	34.8
260	118.2	11.8	35.5
265	120.5	12.0	36.1
270	122.7	12.3	36.8
275	125.0	12.5	37.5
280	127.3	12.7	38.2
285	129.5	13.0	38.9
290	131.8	13.2	39.5
295	134.1	13.4	40.2

Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

Fentanyl Dosing Chart for Airway Management

- **Adult Initial Dose:** 1 – 3 mcg/kg.
  - **Adult Maintenance Dose:** 0.5 – 1 mcg/kg. **Maximum Dose: 250 mcg.**
- **Pediatric Initial Dose:** 1 – 2 mcg/kg.
  - **Pediatric Maintenance Dose:** 0.5 – 1 mcg/kg. **Maximum Dose: 100 mcg.**

LBS	kg	0.5 mcg/kg	1 mcg/kg	2 mcg/kg	3 mcg/kg
10	4.5	2.3	4.5	9.1	13.6
15	6.8	3.4	6.8	13.6	20.5
20	9.1	4.5	9.1	18.2	27.3
25	11.4	5.7	11.4	22.7	34.1
30	13.6	6.8	13.6	27.3	40.9
35	15.9	8.0	15.9	31.8	47.7
40	18.2	9.1	18.2	36.4	54.5
45	20.5	10.2	20.5	40.9	61.4
50	22.7	11.4	22.7	45.5	68.2
55	25.0	12.5	25.0	50.0	75.0
60	27.3	13.6	27.3	54.5	81.8
65	29.5	14.8	29.5	29.1	88.6
70	31.8	15.9	31.8	63.6	95.5
75	34.1	17.0	34.1	68.2	102.3
80	36.4	18.2	36.4	72.7	109.1
85	38.6	19.3	38.6	77.3	115.9
90	40.9	20.5	40.9	81.8	122.7
95	43.2	21.6	43.2	86.4	129.5
100	45.5	22.7	45.5	90.9	136.4
105	47.7	23.9	47.7	95.4	143.1
110	50	25.0	50.0	100.0	150.0
115	52.3	26.1	52.3	104.5	156.8
120	54.5	27.3	54.5	109.1	163.6
125	56.8	28.4	56.8	113.6	170.5
130	59.1	29.5	59.1	118.2	177.3
135	61.4	30.7	61.4	122.7	184.1
140	63.6	31.8	63.6	127.3	190.9
145	65.9	33.0	65.9	131.8	197.7
150	62.8	34.1	68.2	136.4	204.5

LBS	kg	0.5 mcg/kg	1 mcg/kg	2 mcg/kg	3 mcg/kg
155	70.5	35.2	70.5	140.9	211.4
160	72.7	36.4	72.7	145.5	219.2
165	75.0	37.5	75.0	150.0	225.0
170	77.3	38.6	77.3	154.5	231.8
175	79.5	39.8	79.5	159.1	238.6
180	81.8	40.9	81.8	163.6	245.5
185	84.1	42.0	84.1	168.2	250.0
190	86.4	43.2	86.4	172.7	250.0
195	88.6	44.3	88.6	177.3	250.0
200	90.9	45.5	90.9	181.8	250.0
205	93.2	46.6	93.2	186.4	250.0
210	95.5	47.7	95.5	190.9	250.0
215	97.7	48.9	97.7	195.5	250
220	100.0	50.0	100.0	200.0	250.0
225	102.3	51.1	102.3	204.5	250.0
230	104.5	52.3	104.5	209.1	250.0
235	106.8	53.4	106.8	213.6	250.0
240	109.1	54.5	109.1	218.2	250.0
245	111.4	55.7	111.4	222.7	250.0
250	113.6	56.8	113.6	227.3	250.0
255	115.9	58.0	115.9	231.8	250.0
260	118.2	59.1	118.2	236.4	250.0
265	120.5	60.2	120.5	240.9	250.0
270	122.7	61.4	122.7	245.5	250.0
275	125.0	62.5	125.0	250.0	250.0
280	127.3	63.6	127.3	250.0	250.0
285	129.5	64.8	129.5	250.0	250.0
290	131.8	65.9	131.8	250.0	250.0
295	134.1	67.1	134.1	250.0	250.0

Ketamine Dosing Chart for Airway Management

- **Adult Initial Dose:** 1 – 2 mg/kg. **Maximum Dose: 500 mg.**
  - **Adult Maintenance Dose:** 0.5 – 1 mg/kg. **Maximum Dose: 250 mg.**
- **Pediatric Initial Dose:** 0.5 – 2 mg/kg. **Maximum Dose: 100 mg.**
  - **Pediatric Maintenance Dose:** 0.5 – 1 mg/kg. **Maximum Dose: 50 mg.**

LBS	kg	0.5 mg/kg	1 mg/kg	2 mg/kg
10	4.5	2.3	4.5	9.1
15	6.8	3.4	6.8	13.6
20	9.1	4.5	9.1	18.2
25	11.4	5.7	11.4	22.7
30	13.6	6.8	13.6	27.3
35	15.9	8.0	15.9	31.8
40	18.2	9.1	18.2	36.4
45	20.5	10.2	20.5	40.9
50	22.7	11.4	22.7	45.5
55	25.0	12.5	25.0	50.0
60	27.3	13.6	27.3	54.5
65	29.5	14.8	29.5	29.1
70	31.8	15.9	31.8	63.6
75	34.1	17.0	34.1	68.2
80	36.4	18.2	36.4	72.7
85	38.6	19.3	38.6	77.3
90	40.9	20.5	40.9	81.8
95	43.2	21.6	43.2	86.4
100	45.5	22.7	45.5	90.9
105	47.7	23.9	47.7	95.4
110	50	25.0	50.0	100.0
115	52.3	26.1	52.3	104.5
120	54.5	27.3	54.5	109.1
125	56.8	28.4	56.8	113.6
130	59.1	29.5	59.1	118.2
135	61.4	30.7	61.4	122.7
140	63.6	31.8	63.6	127.3
145	65.9	33.0	65.9	131.8
150	62.8	34.1	68.2	136.4

LBS	kg	0.5 mg/kg	1 mg/kg	2 mg/kg
155	70.5	35.2	70.5	140.9
160	72.7	36.4	72.7	145.5
165	75.0	37.5	75.0	150.0
170	77.3	38.6	77.3	154.5
175	79.5	39.8	79.5	159.1
180	81.8	40.9	81.8	163.6
185	84.1	42.0	84.1	168.2
190	86.4	43.2	86.4	172.7
195	88.6	44.3	88.6	177.3
200	90.9	45.5	90.9	181.8
205	93.2	46.6	93.2	186.4
210	95.5	47.7	95.5	190.9
215	97.7	48.9	97.7	195.5
220	100.0	50.0	100.0	200.0
225	102.3	51.1	102.3	204.5
230	104.5	52.3	104.5	209.1
235	106.8	53.4	106.8	213.6
240	109.1	54.5	109.1	218.2
245	111.4	55.7	111.4	222.7
250	113.6	56.8	113.6	227.3
255	115.9	58.0	115.9	231.8
260	118.2	59.1	118.2	236.4
265	120.5	60.2	120.5	240.9
270	122.7	61.4	122.7	245.5
275	125.0	62.5	125.0	250.0
280	127.3	63.6	127.3	254.5
285	129.5	64.8	129.5	259.0
290	131.8	65.9	131.8	263.6
295	134.1	67.1	134.1	268.2

Hart County Emergency Medical Service	EMS Medical Director:	Morgan Wood, MD
	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

Versed Dosing Chart for Airway Management

- **Adult Initial Dose:** 0.025 – 0.05 mg/kg (PRN if others unsuccessful) **Maximum Dose: 5 mg.**
  - **Adult Maintenance Dose:** 0.025 – 0.05 mg/kg.
- **Pediatric Initial Dose:** 0.025 – 0.05 mg/kg (PRN if others unsuccessful) **Maximum Dose: 5 mg.**
  - **Pediatric Maintenance Dose:** 0.025 – 0.05 mg/kg.

LBS	kg	0.025 mg/kg	0.05 mg/kg
10	4.5	0.1	0.2
15	6.8	0.2	0.3
20	9.1	0.2	0.5
25	11.4	0.3	0.6
30	13.6	0.3	0.7
35	15.9	0.4	0.8
40	18.2	0.5	0.9
45	20.5	0.5	1.0
50	22.7	0.6	1.1
55	25.0	0.6	1.3
60	27.3	0.7	1.4
65	29.5	0.7	1.5
70	31.8	0.8	1.6
75	34.1	0.9	1.7
80	36.4	0.9	1.8
85	38.6	1.0	1.9
90	40.9	1.0	2.0
95	43.2	1.1	2.2
100	45.5	1.1	2.3
105	47.7	1.2	2.4
110	50	1.3	2.5
115	52.3	1.3	2.6
120	54.5	1.4	2.7
125	56.8	1.4	2.8
130	59.1	1.5	3.0
135	61.4	1.5	3.1
140	63.6	1.6	3.2
145	65.9	1.6	3.3
150	62.8	1.7	3.4

LBS	kg	0.025 mg/kg	0.05 mg/kg
155	70.5	1.8	3.5
160	72.7	1.8	3.6
165	75.0	1.9	3.8
170	77.3	1.9	3.9
175	79.5	2.0	4.0
180	81.8	2.0	4.1
185	84.1	2.1	4.2
190	86.4	2.2	4.3
195	88.6	2.2	4.4
200	90.9	2.3	4.5
205	93.2	2.3	4.7
210	95.5	2.4	4.8
215	97.7	2.4	4.9
220	100.0	2.5	5.0
225	102.3	2.6	5.0
230	104.5	2.6	5.0
235	106.8	2.7	5.0
240	109.1	2.7	5.0
245	111.4	2.8	5.0
250	113.6	2.8	5.0
255	115.9	2.9	5.0
260	118.2	3.0	5.0
265	120.5	3.0	5.0
270	122.7	3.1	5.0
275	125.0	3.1	5.0
280	127.3	3.2	5.0
285	129.5	3.2	5.0
290	131.8	3.3	5.0
295	134.1	3.4	5.0

Pediatric Versed Dosing Chart

Commonly used medication dosages for the pediatric patient.

For Versed 1 mg/ml only.

0.05 – 0.1 mg/kg or 0.2 mg/kg IM or IN to a maximum of 5 mg.

0.05 mg/kg (IV)			0.1 mg/kg (IV)			0.2 mg/kg (IM/IN)		
Weight (kg)	**Dose (mg)**	**Volume (ml)**	**Weight (kg)**	**Dose (mg)**	**Volume (ml)**	**Weight (kg)**	**Dose (mg)**	**Volume (ml)**
2	0.1	0.1	**2**	0.2	0.2	**2**	0.4	0.4
4	0.2	0.2	**4**	0.4	0.4	**4**	0.8	0.8
6	0.3	0.3	**6**	0.6	0.6	**6**	1.2	1.2
8	0.4	0.4	**8**	0.8	0.8	**8**	1.6	1.6
10	0.5	0.5	**10**	1.0	1.0	**10**	2.0	2.0
12	0.6	0.6	**12**	1.2	1.2	**12**	2.4	2.4
14	0.7	0.7	**14**	1.4	1.4	**14**	2.8	2.8
16	0.8	0.8	**16**	1.6	1.6	**16**	3.2	3.2
18	0.9	0.9	**18**	1.8	1.8	**18**	3.6	3.6
20	1.0	1.0	**20**	2.0	2.0	**20**	4.0	4.0
22	1.1	1.1	**22**	2.2	2.2	**22**	4.4	4.4
24	1.2	1.2	**24**	2.4	2.4	**24**	4.8	4.8
26	1.3	1.3	**26**	2.6	2.6	**26**		
28	1.4	1.4	**28**	2.8	2.8	**28**		
30	1.5	1.5	**30**	3.0	3.0	**30**		
32	1.6	1.6	**32**	3.2	3.2	**32**		
34	1.7	1.7	**34**	3.4	3.4	**34**		
36	1.8	1.8	**36**	3.6	3.6	**36**		
38	1.9	1.9	**38**	3.8	3.8	**38**		
40	2.0	2.0	**40**	4.0	4.0	**40**		
42	2.1	2.1	**42**	4.2	4.2	**42**		
44	2.2	2.2	**44**	4.4	4.4	**44**		
46	2.3	2.3	**46**	4.6	4.6	**46**		
48	2.4	2.4	**48**	4.8	4.8	**48**		
50	2.5	2.5	**50**	5.0	5.0	**50**		

For IV/IN route:

If patient weight is 21 kg or greater, then 5 mg (5cc) is the maximum dose.

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	EMS Director:	Mike Adams
	Effective Date:	6/8/2026

Ativan Dosing Chart

- **Adult Initial Dose:** 0.1 mg/kg **Maximum Dose: 4 mg.**
- **Pediatric Initial Dose:** 0.05 – 0.1 mg/kg **Maximum Dose: 4 mg.**

LBS	kg	0.04 mg/kg	0.05 mg/kg	0.1 mg/
10	4.5	0.2	0.2	[illegible]
15	6.8	0.3	0.3	[illegible]
20	9.1	0.4	0.5	[illegible]
25	11.4	0.5	[illegible]	[illegible]
30	13.6	0.5	[illegible]	[illegible]
35	15.9	0.6	[illegible]	[illegible]
40	18.2	0.7	[illegible]	[illegible]
45	20.5	[illegible]	[illegible]	[illegible]
50	22.7	[illegible]	[illegible]	[illegible]
55	25.	[illegible]	[illegible]	[illegible]
60	[illegible]	[illegible]	[illegible]	2.7
65	[illegible]	[illegible]	[illegible]	3.0
[illegible]	[illegible]	[illegible]	[illegible]	3.2
[illegible]	[illegible]	[illegible]	[illegible]	3.4
[illegible]	[illegible]	[illegible]	1.8	3.6
[illegible]	[illegible]	[illegible]	1.9	3.9
[illegible]	[illegible]	[illegible]	2.0	4.0
[illegible]	[illegible]	[illegible]	2.2	4.0
[illegible]	[illegible]	1.8	2.3	4.0
[illegible]	[illegible]	1.9	2.4	4.0
[illegible]	[illegible]	2.0	2.5	4.0
[illegible]	52.3	2.1	2.6	4.0
[illegible]	54.5	2.2	2.7	4.0
125	56.8	2.3	2.8	4.0
130	59.1	2.4	3.0	4.0
135	61.4	2.5	3.1	4.0
140	63.6	2.5	3.2	4.0
145	65.9	2.6	3.3	4.0
150	62.8	2.7	3.4	4.0

REMOVED

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Mixing Medication Drips

A quick reference for commonly used medication drips.

Amiodarone:

- Rapid infusion: 150 mg in 100 ml of normal saline. Concentration is now 1.5 mg/ml. Drip at 100 drops per minute using a 10 gtts/ml IV administration set. This will administer the standard dose of 150 mg over 10 minutes.

Epinephrine:

- During CPR to maintain 1 mg every 3-5 minutes, mix 12 mg of 1: 1,000 in 250 ml of normal saline or D_5W. Drip at 72 gtts/min or 6 gtts every 5 seconds. This drip will run for 36 minutes.
- For vasopressor usage, mix 1 mg in 250 ml of normal saline or D_5W. Concentration is now 4 mcg/ml. Administer at 2 – 10 mcg/min.

Magnesium Sulfate:

- For use in cardiac arrest, mix 1 – 2 grams into 100 ml of normal saline and run wide open using a 10 gtts/ml IV set.
- For eclampsia, mix 2 – 4 grams in 250 ml of normal saline and run over 5 minutes with a 10 gtts/ml IV set. (100 gtts/min.
- For use in patients with perfusing rhythms, mix 1 gram in 250 ml of normal saline. Concentration Is 4mg/ml. Using a 10 gtts.ml IV set, administer 20 mg/min and monitor closely for respiratory depression, hypotension and other changes.

Sodium Bicarbonate:

- 50 mEq 50 ml pre-filled syringe in 100 ml of normal saline. Draw out 50 ml of normal saline and add the 50 mEq/50ml of Sodium Bicarbonate to the 50 ml remaining in the 100 ml bag of normal saline. Concentration is now 50 mEq in 100ml.

Making D-25: (Age 1 – 12 years of age) and **D-10** (Less than 1 year old)

D_{50} to D_{10}	D_{50} to D_{25}
Expel 40 ml of 50% solution and draw up 40 ml of Normal Saline. = 10% solution	Expel 25 ml of 50% solution and draw up 25 ml of Normal Saline. = 25% solution

Pediatric Medication Dosages

Commonly used medication dosages for the pediatric patient.

Medication	Dosage
Acetaminophen	15 mg/kg PO.
Adenosine	0.1 mg/kg 1st dose up to 6 mg. 0.2 mg/kg 2nd dose up to 12 mg.
Albuterol	2.5 mg unit dose.
Amiodarone	5 mg/kg mix with 100 ml if perfusing.
Atropine	0.02 mg/kg – 0.2 ml/kg.
Atrovent	0.25 – 0.5 mg unit dose.
Benadryl	2 – 5 mg/kg up to 25 – 50 mg Max
Calcium Chloride	20 mg/kg – 0.2 ml/kg.
50% Dextrose	0.5 g/kg up to 12.5 g. Consider dilution.
Dopamine	2 – 20 mcg/kg/min – 20 mg + 100 ml = 200 mcg/ml.
Epinephrine (Cardiac Arrest)	0.01 mg/kg – 0.1 ml/kg (1:10,000).
Epinephrine (Allergy or Asthma)	0.15 – 3.0 mg SQ – 0.15 – 3 ml (1:1,000).
Epinephrine Infusion	Mix 1 mg in 250 ml = 4mcg/ml. 1 – 10 mcg/kg/min.
Etomidate	0.3 mg/kg up to 20 mg.
Fentanyl	1 – 2 mcg/kg up to 50 mcg Max.
Glucagon	0.03 mg/kg up to 1 mg Max.
Lasix	1 mg/kg up to 40 mg Max.
Lidocaine	1 mg/kg – 0.05 ml/kg.
Lidocaine Infusion	Mix 100 mg in 100 ml. 20 – 50 mcg/kg/min.
Magnesium Sulfate	Mix in 100 ml. 25 -50 mg/kg.
Motrin	10 mg/kg PO
Phenergan	0.5 mg/kg up to 12.5 mg Max.
Sodium Bicarbonate	1 mEq/kg – 1 ml/kg.
Solu-Medrol	2 mg/kg up to 125 mg Max.
Toradol	0.5 mg/kg up to 15 mg Max.
Valium	0.2 mg/kg up to 5 mg Max.
Versed	0.1 mg/kg up to 2.5 mg Max.
Zofran	0.1 mg/kg.
Defibrillation	2 J/kg and repeats @ 4 J/kg.
Cardioversion	0.5 J/kg – 1 J/kg and repeats @ 2 J/kg.

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King LT Airway

King LT Size	Patient Height	Cuff Volume	Tip Color
3	4 – 5 feet	50 ml	Yellow
4	5 – 6 feet	70 ml	Red
5	Greater than 6 feet	80 ml	Purple

King LT Airway

When endotracheal intubation is unsuccessful after 2 attempts on patients over 4 feet tall and in respiratory or cardiac arrest. It is not necessary to attempt intubation if a difficult airway is anticipated or visualized. The King airway may be used as the first line airway in these cases. **Contraindications to King Airway use are as follows: < 4 feet in height, intact gag reflex, caustic substance ingestion, known esophageal disease and laryngectomy with stoma.**

King LT Airway Insertion:

1. Choose the correct size King Airway based on the patient's height.
2. Test the cuff inflation system by injecting the maximum recommended volume of air into the cuffs.
3. Remove all air from the cuffs prior to insertion.
4. Apply a water-based lubricant to the beveled distal tip and posterior aspect of the tube taking care to avoid the introduction of lubricant in or near the ventilator openings.
5. Pre-oxygenate the patient.
6. Position the head: The ideal position for the insertion of the King LT Airway is the sniffing position. The King LT can also be used with the head in a neutral position.
7. Hold the King LT at the connector end with the dominant hand. With the non-dominate hand hold the mouth open and apply a chin lift unless contraindicated due to suspected spinal injury.
8. With the King LT airway rotated laterally 45 – 90 degrees such that the blue orientation line is touching the corner of the mouth, introduce the tip of the tube into the mouth and advance behind the base of the tongue. Never force the tube into position.
9. As the tube tip passes under the tongue, rotate the tube back to midline. Blue orientation line faces the chin when in the proper position.
10. Without exerting excessive force, advance the King LT until the base of the connector aligns with the teeth or gums.
11. Fully inflate the cuffs using the maximum volume of the syringe included in the EMS kit. (see chart)
12. Attach the bag valve mask device to the 15 mm connector of the King LT and gently start bagging the patient to assess ventilation, simultaneously withdraw the airway until ventilation is easy and free flowing (large tidal volume with minimal airway pressure).
13. Note the depth markings to give an approximate distance in centimeters to the vocal cords.
14. Confirm proper position by auscultation, chest movement and verification of CO_2 using waveform capnography.
15. Readjust cuff inflation to just seal the airway.
16. Secure the King LT to the patient using an accepted method and bite block/oral airway. Use care not to place tape over the proximal opening of the gastric access device.

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EZ IO Drill

General Guidelines

Indications for use: For intraosseous access anytime in which vascular access is difficult to obtain in emergent, urgent or medically necessary cases. Two sites have been approved for adult and pediatric patients: Proximal humerus and proximal tibia.

Contraindications for use: Fracture in the target bone, previous significant orthopedic procedure at the site, prosthetic limb or joint, IO in the past 48 hours of the target bone, infection at the area of insertion, excessive tissue (severe obesity) and/or inability to clearly identify anatomical landmarks.

Intraosseous Access

1. Prepare all equipment and clean the insertion site.
2. Attach selected needle to the drill and remove the safety cap from the needle.
3. Remove the trigger guard.
4. Push the needle through the skin until the needle touches bone.
    a. At least one black line must be visible outside the skin.
5. Apply gentle to moderate pressure and squeeze the trigger.
6. Advance the needle and release the trigger.
    a. Pediatrics: release trigger when sudden “give” or “pop” is felt, indicating entry into the medullary space.
    b. Adult: Advance needle set approximately 1 – 2 cm after entry into medullary space. In proximal humerus for most adults Needle Set should be advanced 2 cm or until hub is flush against the skin.
7. Stabilize the Needle Set Hub, disconnect the driver and remove the stylet.
8. Attach a primed EZ-Connect® extension set to the hub, firmly secure by twisting clockwise.
9. Stabilize the site.
10. For patients responsive to pain, consider Lidocaine up to 40 mg in adults and 20 mg in pediatrics (0.5mg/kg).
11. Flush the EZ IO with normal saline (0.9 % Sodium Chloride) 5 – 10 ml for adults, 2- 5 ml for pediatrics)
    a. Prior to flush, aspirate slightly for visual confirmation of bone marrow.
    b. Failure to appropriately flush the EZ-IO catheter may result in limited or no flow. Repeat flush as needed.
12. Confirm catheter placement with the following recommendations.
    a. Stability of the catheter in the bone.
    b. Ability to aspirate after the flush.
    c. Adequate flow rate.
13. Document the date and time of insertion and apply EZ-IO® wristband. (if present)
    a. Caution: Monitor insertion site frequently for extravasation.
    b. Caution: Do not leave the catheter inserted for longer than 24 hours.

Proximal Tibia

Proximal Humerus

Advanced Transport Ventilators (AHP-300)

General Guidelines:

The mechanical ventilator shall be used on all intubated patients between the heights of 4 feet and 7 feet unless there is a direct physician's order not to use the ventilator, or other life-saving treatment priorities exist. In those rare cases, every effort should be made to place the patient on the ventilator as soon as the immediate life-threats are addressed. This is an ALS post-licensure skill that is to be performed only by a paramedic, and then only once the paramedic has been ***trained*** on those skills, ***certified*** as competent, ***credentialed*** to perform those skills by the agency's Medical Director, and only while operating and understanding, verbal or written orders from the agency's Medical Director, transferring physician or Medical Control Physician.

This skill has been approved by Hart County EMS Medical Director, Dr. Morgan Wood, for Paramedics ONLY.

Indications for Use:

1. To be used on patients between the heights of 4 feet and 7 feet tall.
   a. Experiencing impending/actual respiratory failure.
   b. Inadequate respiratory drive.
   c. Apnea.
   d. Poor gas exchange.
   e. Decreased respiratory drive.
2. To provide emergency respiratory support by means of a face mask or tube inserted into the patient's airway.
3. Can be used for prehospital use or interfacility transports.
4. If sedation is required, *See* ***Sedation Assisted Intubation Protocol.***

Contraindications for Use:

1. Less than 4 feet tall or over 7 feet tall.
2. Inability to maintain adequate oxygen saturation or $EtCO_2$ readings.
3. Paramedics not approved by Hart County Emergency Medical Service to utilize the AHP-300 ventilator.
4. Suspected Tension Pneumothorax.
5. Unresponsive or Altered Mental Status for CPAP or BiPAP.

Non-intra-facility transport/Scene Calls

- If the patient has just been intubated or on a ventilator previously, then set the ventilator as follows:
  - SIMV mode.
  - Tidal Volume: 6 – 8 ml/kg of patient's ideal body weight. *See **PBW chart**.*
  - FiO_2 of 100%, titrate to keep the SpO_2 between 94% and 99%.
  - PEEP – 3 cmH_2O. May need to increase to maintain SpO_2.

CONTINUED ON NEXT PAGE

Advanced Transport Ventilators (Continued)

Ventilator Setup/Ventilator Strategies

- After the ventilator is attached to the patient, the following devices will be attached to the patient to monitor their clinical status:
  - **Apply and monitor waveform capnography.**
    - Target $EtCO_2$ is 35 – 45 mmHg, unless otherwise established by transferring facility.
      - If $EtCO_2$ is < 30 mmHg:
        - Decrease respiratory rate.
        - Decrease tidal volume.
        - Decrease pressure control.
      - If $EtCO_2$ is > 45 mmHg:
        - Increase respiratory rate.
        - Increase tidal volume.
        - Increase pressure control.
  - **Apply and monitor SpO_2.**
    - Target SpO_2 is $\geq$ 94% unless otherwise established by transferring facility.
      - $SpO_2 \leq$ 93%:
        - increase PEEP.
        - Increase FiO_2.
      - SpO_2 100%:
        - Titrate FiO_2 down for target.
  - **Apply and monitor ECG.**
  - **Monitor and record vital signs in 5-minute intervals.**
    - Hypotension is an indication of increased intrathoracic pressure.
      - Discontinue/Decrease PEEP (if applied)
      - Consider reducing respiratory rate.
      - Consider reducing inspiratory time.
      - Consider initiating IV fluid therapy.
  - **Monitor chest rise, capillary refill time and skin color.**
  - **Monitor Advanced Transport Ventilator feedback continually.**
    - Peek Inspiratory Pressure (PIP)
      - Less than 25 cmH_2O:
        - Check tidal volume.
        - Check patient connections.
        - Check for airway obstructions.
      - Higher than 30 cm H_2O:
        - Check for displacement of ET tube.
        - Check for obstructions of ET tube (secretions, mucous plugs).
        - Check for Pneumothorax.
        - Check for equipment failure.
      - Delivered Tidal Volume – should closely match the tidal volume (Vt). For discrepancy > 50 ml of set tidal volume:
        - Rule out leak:
          - Ensure the pilot bulb is inflated.
          - Evaluate for fluids in the airway with bubbling following respirations. (Indicates leak)
        - Suspected Dead Loss:
          - Increase tidal volume setting by difference between current tidal volume setting and the delivered tidal volume feedback.
  - **Be prepared to suction secretions.**

Advanced Transport Ventilators (Continued)

Interfacility Transport

- Determine and record ventilator settings which have been established by the transferring facility to include:
  - Ventilation Mode (Volume AC, Volume SIMV, Pressure AC or Pressure SIMV).
  - Breaths per Minute (BPM).
  - Tidal Volume (V_t).
  - Inspiratory Time (T_i).
  - Inspiration to Expiration Ratio (I:E ratio).
  - Fraction of Inspired Oxygen (FiO_2).
  - Peak Inspiratory Pressure (PIP).
  - Positive End Expiratory Pressure (PEEP).
- Determine the transferring facility's target SpO_2 and $EtCO_2$.
- Adjust the ventilator settings to reflect the settings established by the transferring facility.
- Transfer the patient from the facility ventilator to the AHP-300 Transport Ventilator.
- Monitor the patient for no less than one minute after transferring the patient from the facility ventilator to the transport ventilator to ensure stabilization prior to transporting the patient.
- Allow respiratory therapist to make changes to the AHP-300 Transport Ventilator, if needed.
- Paramedic may need to adjust during transport to maintain target SpO_2 and $EtCO_2$. Additional sedation may be needed during transport. *See **Sedation Assisted Intubation Protocol** for sedation guidance.*
- Document the initial ventilator settings in the PCR and Ventilator Worksheet as well as any changes made.
- The settings may be adjusted at the determination of the clinician to maintain the comfort of the patient with a SpO_2 consistent with the baseline of the patient and an $EtCO_2$ between 35 – 45 mmHg. Ventilator settings should not be adjusted any more than once every 5 minutes unless a life-threatening situation occurs.
- If the patient is not sedated by a continuous IV drip, then sedation and analgesia shall be maintained by following the ***Sedation Assisted Intubation Protocol*.**

Advanced Transport Ventilators: Special Considerations

Special Considerations:

- **COPD**
  - COPD causes high lung compliance (difficult expiration). The COPD approach focuses on providing longer expiratory times and reduced breaths per minute. Increased CO_2 and hypotension are indications of possible breath stacking in COPD patients. Ensure that the flow is returning to zero on the feedback panel. If breath stacking is suspected, reduce the breaths per minute and reduce the inspiratory time to facilitate a longer expiratory phase. Consider temporary disconnection of the ventilator to allow for the patient to exhale.
- **ARDS**
  - ARDS causes low lung compliance (difficult inspiration). The ARDS approach focuses on lower tidal volume, higher PEEP, and increased respiratory rate.
- **CHF**
  - CHF is the inability of the heart to pump blood effectively. This will allow blood and fluid to back up in the lungs. The CHF approach focuses on an increase in PEEP in order to increase intrathoracic pressure, reducing venous return. An increase in PEEP also helps shift lung fluid from the alveoli to the interstitial space. Include any changes in settings in the patient care report.
- **Sepsis**
  - The sepsis patient may have low venous return. Increasing intrathoracic pressure will decrease an already low venous return, thereby lowering an already low blood pressure. Always have venous access for intubated sepsis patients. Be prepared for hypotension.
- **Metabolic Acidosis**
  - Metabolic acidosis patients typically need significant excess minute ventilation to assist in clearing excess acid via the respiratory system. The target $EtCO_2$ is often lower than for patients without metabolic acidosis. Be sure to determine the target $EtCO_2$ range of the transferring facility. DO NOT attempt to raise the $EtCO_2$ above the target range.

Documentation

- In addition to documentation of vital signs and all other assessments and interventions, also document the following on all ventilated patients.
  - Physician interventions and diagnosis
  - Reason for Transfer
- Transferring facility ventilator order:
  - Mode of ventilation
  - Breaths per minute (Rate)
  - Peak end expiratory pressure (PEEP)
  - Positive Inspiratory Pressure (PIP)
  - Tidal Volume (V_t)
  - Inspiratory Time (T_i)
  - Capnography
  - Patient's response
  - Any Complications

REMEMBER:

You are responsible for the breathing of the patient. The BVM should be utilized while troubleshooting the ventilator. DO NOT be so consumed with adjusting the vent settings that the patient does not receive adequate ventilation.

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Predicted Body Weight Chart: Male/Female

FEMALE

Predicted Body Weight and Tidal Volume (Female)

Height	PDW	4 ml	5 ml	6 ml	7 ml	8 ml
4’0”	*17.9*	72	90	107	125	143
4’1”	*20.2*	81	101	121	141	162
4’2”	*22.5*	90	113	135	158	180
4’3”	*24.8*	99	124	149	174	198
4’4”	*27.1*	108	136	163	190	217
4’5”	*29.4*	118	147	176	206	235
4’6”	*31.7*	127	159	190	222	254
4’7”	*34*	136	170	204	238	272
4’8”	*36.3*	145	182	218	254	290
4’9”	*38.6*	154	193	232	270	309
4’10”	*40.9*	164	205	245	286	327
4’11”	*43.2*	173	216	259	302	346
5’0”	*45.5*	182	228	273	319	364
5’1”	*47.8*	191	239	287	335	382
5’2”	*50.1*	200	251	301	351	401
5’3”	*52.4*	210	262	314	367	419
5’4”	*54.7*	219	274	328	383	438
5’5”	*57*	228	285	342	399	456
5’6”	*59.3*	237	297	356	415	474
5’7”	*61.6*	246	308	370	431	493
5’8”	*63.9*	256	320	383	447	511
5’9”	*66.2*	265	331	397	463	530
5’10”	*68.5*	274	343	411	480	548
5’11”	*70.8*	283	354	425	496	566
6’0”	*73.1*	292	366	439	512	585
6’1”	*75.4*	302	377	452	528	603
6’2”	*77.7*	311	389	466	544	622
6’3”	*80*	320	400	480	560	640
6’4”	*82.3*	329	412	494	576	658
6’5”	*84.6*	228	423	508	592	677
6’6”	*86.9*	348	435	521	608	695
6’7”	*89.2*	357	446	535	624	714
6’8”	*91.5*	366	458	549	641	732
6’9”	*93.8*	375	469	563	657	750
6’10”	*96.1*	384	481	577	673	769
6’11”	*98.4*	394	492	590	689	787
7’0”	*100.7*	403	504	604	705	806

MALE

Predicted Body Weight and Tidal Volume (Male)

Height	PBW	4 ml	5 ml	6 ml	7 ml	8 ml
4’0”	*22.4*	90	112	134	157	179
4’1”	*24.7*	99	124	148	173	198
4’2”	*27*	108	135	162	189	216
4’3”	*29.3*	117	147	176	205	234
4’4”	*31.6*	126	158	190	221	253
4’5”	*33.9*	136	170	203	237	271
4’6”	*36.2*	145	181	217	253	290
4’7”	*38.5*	154	193	231	270	308
4’8”	*40.8*	163	204	245	286	326
4’9”	*43.1*	172	216	259	302	345
4’10”	*45.4*	182	227	272	318	363
4’11”	*47.7*	191	239	286	334	382
5’0”	*50*	200	250	300	350	400
5’1”	*52.3*	209	262	314	366	418
5’2”	*54.6*	218	273	328	382	437
5’3”	*56.6*	228	285	341	398	455
5’4”	*59.2*	237	296	355	414	474
5’5”	*61.5*	246	308	369	431	492
5’6”	*63.8*	255	319	383	447	510
5’7”	*66.1*	264	331	397	463	529
5’8”	*68.4*	274	342	410	479	547
5’9”	*70.7*	283	354	424	495	566
5’10”	*73*	292	365	438	511	584
5’11”	*75.3*	301	377	452	527	602
6’0”	*77.6*	310	388	466	543	621
6’1”	*79.9*	320	400	479	559	639
6’2”	*82.2*	329	411	493	575	658
6’3”	*84.5*	338	423	507	592	676
6’4”	*86.8*	347	434	521	608	694
6’5”	*89.1*	356	446	535	624	713
6’6”	*91.4*	366	457	548	640	731
6’7”	*93.7*	375	469	562	656	750
6’8”	*96*	384	480	576	672	768
6’9”	*98.3*	393	492	590	688	786
6’10”	*100.6*	402	503	604	704	805
6’11”	*102.9*	412	515	617	720	823
7’0”	*105.2*	421	526	631	736	842

Prehospital Trauma Fluid Resuscitation

The prehospital phase of care is the time of injury where resuscitation practices have been shown to be associated with differential outcomes. Over usage of crystalloid solution during the resuscitation phase has been well documented with negative impacts such as decreased coagulability, increase in ARDS and Acute Kidney Injury (AKI). Current research is showing that decrease in access attempts, decrease in overall crystalloid solution administration and allowing permissive hypotension to occur has positive impacts.

Hemodynamically Stable Patients (Present radial pulses and appropriate mentation).

1. IV access attempts should be made during transport if at all. **DO NOT delay transport for procedures.**
2. If IV procedure is successful, keep IV rate at KVO and do not administer crystalloid or blood products.

Hemodynamically Unstable Patients (Weak or absent radial pulses, altered mental status, hypotensive, or concern for hemorrhage).

1. Distance to appropriate trauma center should be factored into transport decision.
    a. Recommended if greater than 30 minutes or risk for decompensation of the airway, notify aeromedical for rapid transport to trauma center.
    b. Early notification via direct report while on scene or allowing your dispatch center to notify the receiving facility to allow prompt availability and early mobilization of the Trauma Center Resources.
2. Blood products via ground or air transport should be the standard fluid resuscitation product if possible. (Refer to Region 2 Prehospital Blood Product Administration Guideline)
3. IV/IO attempts should be made during transport, waiting for aeromedical transport or in conjunction with the field extrication process. **The goal is not to increase scene time for procedures.**
4. If blood products are available, infuse to a systolic blood pressure of 100 mmHg.
5. If there is any concern for traumatic brain injury, infuse blood products or 250 ml crystalloid boluses to a target systolic blood pressure greater than 100 mmHg.
6. If crystalloid is the only solution available and there is no concern for traumatic brain injury:
    a. Short prehospital time (< 20 minutes) infuse targeting systolic blood pressure of 80 mmHg utilizing 250 ml boluses.
    b. Prolonged prehospital time (> 20 minutes) infuse targeting systolic blood pressure of 100 mmHg utilizing 250 ml boluses.

I-View Video Laryngoscopes

Description:

The I View video laryngoscope is a single use, fully disposable, device designed for video laryngoscopy, which is a technique used to intubate patients. It features a MacIntosh blade, making it suitable for both video laryngoscopy and direct laryngoscopy.

Indications:

The I-View is to be used as a secondary tool to facilitate endotracheal intubation when a difficult airway is encountered.

- Requirement for positive pressure ventilation.
- To maintain a patent airway, such as in patients with trauma to the face or neck.
- Airway protection from secretions, blood or gastric content.
- Maintenance of adequate oxygenation and/or ventilation.

Contraindications:

All contraindications are the same as direct laryngoscopy. In the controlled or elective setting the following contraindications should be considered:

- Anticipated difficult intubation due to multiple anatomic features or past history.
- Severe limitation of mouth opening.
- Unstable cervical spine or unstable atlanto-occipital joint (e.g., severe rheumatoid arthritis, trauma, Down syndrome).
- Major trauma, abnormal anatomy, or tumor of the mandible, maxilla, larynx, neck, mediastinum, or trachea.
- History of extensive radiation therapy of the airway with a fixed larynx.
- History of tracheotomy or tracheal stenosis.
- Active airway bleeding.

Instructions for use:

1. Turn the device on using the on/off switch located on the back side of the screen.
2. Insert the tip gently down on the tongue until you visualize and identify the epiglottis.
3. Move the blade into the vallecula.
4. Using minimal force, indirectly lift the epiglottis to expose the glottis. This can be achieved under direct or indirect visualization.
5. Pass an appropriately sized endotracheal tube through the vocal cords to the required depth.
6. Gently withdraw the video laryngoscope from the patient's oropharynx.
7. Confirm ET tube placement.
8. Turn off the video laryngoscope.